```
# ==== run.py ====
# -*- coding: utf-8 -*-
"""系统启动入口。
运行：
    python run.py                启动并自动打开浏览器
    python run.py --no-browser   只启动服务，不打开浏览器
    python run.py --dev          开发模式（开启代码热重载与调试器）
    python run.py --port 5001    指定端口
默认以 127.0.0.1:5000 启动，就绪后自动打开浏览器。
"""
import os
import sys
import threading
import time
import webbrowser
if hasattr(sys.stdout, "reconfigure"):
    sys.stdout.reconfigure(encoding="utf-8")   # 控制台中文正常显示
from app import create_app
from app.models import db
PORT = 5000
for i, a in enumerate(sys.argv):
    if a == "--port" and i + 1 < len(sys.argv):
        PORT = int(sys.argv[i + 1])
URL = "http://127.0.0.1:%d" % PORT
NO_BROWSER = "--no-browser" in sys.argv
DEV = "--dev" in sys.argv
app = create_app()
def open_browser_later(delay=1.2):
    """等服务工作起来再打开浏览器，避免打开时页面还没就绪。
    放在后台线程里，不用 sleep 阻塞服务启动；线程设为守护线程，
    用户 Ctrl+C 停止服务时不会残留。
    """
    def run():
        time.sleep(delay)
        try:
            webbrowser.open(URL)
        except Exception:
            pass            # 打开失败不影响服务本身
    t = threading.Thread(target=run, daemon=True)
    t.start()
if __name__ == "__main__":
    with app.app_context():
        db.create_all()     # 首次运行时自动建表
        from app.utils.schema import ensure_schema
        added = ensure_schema(db)     # 旧库补齐新增字段
        if added:
            print("已补充字段：%s" % "、".join(added))
    print("=" * 58)
    print("   历史遗留废弃矿山自然恢复适宜性评价系统  Ｖ１．０")
    print("=" * 58)
```

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    print("   访问地址：%s" % URL)
    print("   演示账号：admin      密码：admin123")
    print("   停止运行：按 Ctrl+C，或直接关闭本窗口")
    print("=" * 58)
    if not NO_BROWSER and not DEV:
        print("   正在打开浏览器……")
    print()
    # 开发模式才启用热重载（重载会二次启动进程，故此时不自动开浏览器）
    if not NO_BROWSER and not DEV:
        open_browser_later()
    app.run(host="127.0.0.1", port=PORT, debug=DEV,
    use_reloader=DEV)
# ==== app/__init__.py ====
# -*- coding: utf-8 -*-
"""Flask 应用工厂。"""
from datetime import date
from flask import Flask
from flask_login import LoginManager
from app.config import Config
from app.models import db, User
from app.utils.constants import ROLE_LABELS
login_manager = LoginManager()
login_manager.login_view = "auth.login"
login_manager.login_message = "请先登录"
@login_manager.user_loader
def load_user(user_id):
    return db.session.get(User, int(user_id))
def create_app(config_class=Config):
    app = Flask(__name__)
    app.config.from_object(config_class)
    db.init_app(app)
    login_manager.init_app(app)
    # 注册视图（各模块路由）
    from app.views import auth, project, unit, standard,
    constraint, \
        ahp, indicator, evaluate, report, search
    app.register_blueprint(auth.bp)
    app.register_blueprint(project.bp)
    app.register_blueprint(unit.bp)
    app.register_blueprint(standard.bp)
    app.register_blueprint(constraint.bp)
    app.register_blueprint(ahp.bp)
    app.register_blueprint(indicator.bp)
    app.register_blueprint(evaluate.bp)
    app.register_blueprint(report.bp)
    app.register_blueprint(search.bp)
    # 全局登录保护：除登录页与静态资源外，均需登录
    from flask import request, redirect, url_for,
    send_from_directory
    from flask_login import current_user, login_user
```

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PUBLIC_ENDPOINTS = {"auth.login", "auth.register",
"static", "favicon"}
# 浏览器默认会请求根路径下的 /favicon.ico。
# 页面里虽已用 <link rel="icon"> 指向 static 目录，但出错页等场景仍会
# 回退到这个路径，补一条路由免得日志里一直刷 404。
@app.route("/favicon.ico")
def favicon():
    return send_from_directory(app.static_folder,
"favicon.ico",
                               mimetype="image/vnd.micro
soft.icon")
@app.before_request
def require_login():
    # 截图模式（仅用于生成说明书配图）：自动以管理员身份登录
    if app.config.get("SCREENSHOT_MODE"):
        if (request.endpoint not in ("auth.login",
"auth.register", "static")
                and not current_user.is_authenticated):
            admin = User.query.filter_by(role="admin").f
irst()
            if admin:
                login_user(admin)
        return None
    if request.endpoint in PUBLIC_ENDPOINTS or
request.endpoint is None:
        return None
    if not current_user.is_authenticated:
        return redirect(url_for("auth.login"))
    return None
# 侧边栏导航：按论文评价流程的顺序编排，分为「评价流程」与「系统管理」两组。
# 不使用模块编号，避免出现与实际流程不一致的代号。
# 每项带图标名（内联 SVG，见 templates/_macros.html 的 ICONS）
# 与可选的待办角标（该步骤还没做完的图斑数）。
FLOW_LINKS = [
    ("项目与矿山档案", "/project/", "folder"),
    ("政策与标准知识库", "/standard/", "book"),
    ("限制性指标判定", "/constraint/", "shield"),
    ("微地质单元", "/unit/", "layers"),
    ("指标体系与权重", "/ahp/", "sliders"),
    ("指标数据采集", "/indicator/", "database"),
    ("适宜性定量评价", "/evaluate/", "target"),
    ("报告与结果输出", "/report/", "file"),
]
# 「用户与操作日志」已并入「个人设置」，不再是独立菜单，
# 因此系统管理组只留一个入口。
ADMIN_LINKS = [
    ("个人设置", "/profile", "user"),
]
@app.context_processor
def inject_nav():
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    path = request.path or "/"
    def hit(prefix):
        """当前路径是否属于某个入口（/unit/ 也匹配 /unit/3）。"""
        return path.startswith(prefix.rstrip("/"))
    badges = {}
    if current_user.is_authenticated:
        from app.services import dashboard as dash
        badges = dash.nav_badges()
    def build(links):
        return [{"label": label, "href": href, "icon": icon,
                 "active": hit(href), "badge": badges.get(href, 0)}
                for label, href, icon in links]
    groups = [
        {"title": "评价流程", "links": build(FLOW_LINKS)},
        {"title": "系统管理", "links": build(ADMIN_LINKS)},
    ]
    # 面包屑用的当前页名：按路径命中的入口取，未命中时给个通用名
    if path == "/":
        page_title = "工作台"
    elif path.startswith("/profile"):
        page_title = "个人设置"
    elif path.startswith(("/users", "/logs")):
        page_title = "账号与日志"
    elif path.startswith("/search"):
        page_title = "搜索结果"
    else:
        page_title = "工作台"
        for label, href, _icon in FLOW_LINKS:
            if hit(href):
                page_title = label
                break
    # 层级引导用：项目是归档单位，图斑才是评价对象。
    # 下游各页面按图斑组织数据，因此需要提示「哪些项目下还没有图斑」。
    from app.models import Project, MinePatch
    projects = Project.query.order_by(Project.id).all()
    return {
        "nav_groups": groups,
        "home_active": path == "/",
        "page_title": page_title,
        "today": date.today().strftime("%Y-%m-%d"),
        "search_q": request.args.get("q", ""),
        "role_labels": ROLE_LABELS,
        "project_count": len(projects),
        "patch_count": MinePatch.query.count(),
        "projects_without_patch": [p for p in projects
                                   if p.patches.count() == 0],
    }
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    # 操作审计：统一在响应阶段记录成功的写操作（操作日志）
    from app.services import audit
    @app.after_request
    def audit_writes(response):
        if request.method in ("POST", "PUT", "PATCH",
    "DELETE"):
            audit.record(request.endpoint,
    response.status_code)
        return response
    # 错误页：给出可读提示，而不是默认的英文错误页
    from flask import render_template
    @app.errorhandler(403)
    def forbidden(e):
        return render_template("error.html", code=403,
                               message="没有权限执行该操作"), 403
    @app.errorhandler(404)
    def not_found(e):
        return render_template("error.html", code=404,
                               message="请求的页面不存在"), 404
    return app
# ==== app/config.py ====
# -*- coding: utf-8 -*-
"""应用配置。"""
import os
from datetime import timedelta
BASE_DIR = os.path.abspath(os.path.dirname(__file__))
class Config:
    SECRET_KEY = os.environ.get("SECRET_KEY",
    "mine-restoration-dev-key")
    # 单机部署，SQLite 数据库文件放在应用目录下
    SQLALCHEMY_DATABASE_URI = "sqlite:///" +
    os.path.join(BASE_DIR, "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    # 禁止跨站请求伪造（CSRF），登录与表单提交依赖此项
    WTF_CSRF_ENABLED = True
    # 「记住我」：登录时勾选则下发持久 cookie，关闭浏览器后仍可自动登录
    REMEMBER_COOKIE_DURATION = timedelta(days=30)
    REMEMBER_COOKIE_HTTPONLY = True
    REMEMBER_COOKIE_SAMESITE = "Lax"
    # 影像与数据文件上传（遥感影像挂接、批量导入）
    UPLOAD_FOLDER = os.path.join(BASE_DIR, "static",
    "uploads")
    MAX_CONTENT_LENGTH = 32 * 1024 * 1024          # 单次上传上限
    32 MB
    ALLOWED_IMAGE_EXT = {"png", "jpg", "jpeg", "gif",
    "webp", "bmp", "tif", "tiff"}
    ALLOWED_DATA_EXT = {"csv", "xlsx", "xls"}
    # 头像只用浏览器能直接显示的格式（不含 bmp / tif），并单独限制体积
    ALLOWED_AVATAR_EXT = {"png", "jpg", "jpeg", "gif",
    "webp"}
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    MAX_AVATAR_LENGTH = 4 * 1024 * 1024          # 4 MB
# ==== app/models.py ====
# -*- coding: utf-8 -*-
"""数据模型：系统全部数据表。
本文件由甲在阶段 0 一次性写完，作为两人独立开发的地基。
约定：两人只读此文件；确需改表时，由乙提需求、甲统一修改。
"""
import json
from datetime import datetime
from flask_login import UserMixin
from flask_sqlalchemy import SQLAlchemy
db = SQLAlchemy()
class User(UserMixin, db.Model):
    """用户（账户与权限）。"""
    __tablename__ = "users"
    id = db.Column(db.Integer, primary_key=True)
    username = db.Column(db.String(64), unique=True,
    nullable=False, index=True)
    password_hash = db.Column(db.String(256),
    nullable=False)
    # 职务只有两种：管理员 admin（可管理账号与他人的职务） / 操作员 operator
    role = db.Column(db.String(20), nullable=False,
    default="operator")
    active = db.Column(db.Boolean, nullable=False,
    default=True)
    created_at = db.Column(db.DateTime,
    default=datetime.utcnow)
    # ---- 个人资料（用户可在「个人设置」里自行维护）----
    display_name = db.Column(db.String(64), nullable=True)
    # 昵称
    email = db.Column(db.String(128), nullable=True)
    phone = db.Column(db.String(32), nullable=True)
    gender = db.Column(db.String(8), nullable=True)
    # 男 / 女 / 保密
    avatar = db.Column(db.String(256), nullable=True)
    # 头像，相对 static 的路径
    @property
    def is_admin(self):
        return self.role == "admin"
    @property
    def shown_name(self):
        """界面上显示的名字：优先用昵称，没填就用账号名。"""
        return self.display_name or self.username
    @property
    def initial(self):
        """没有头像时用首字占位。"""
        return (self.shown_name or "?")[0].upper()
    def to_dict(self):
        return {"id": self.id, "username": self.username,
                "display_name": self.display_name,
```

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        "gender": self.gender,                                          301
                    "role": self.role, "active": self.active}           302
class OperationLog(db.Model):                                           303
    """操作日志。"""                                                    304
    __tablename__ = "operation_logs"                                    305
    id = db.Column(db.Integer, primary_key=True)                        306
    user_id = db.Column(db.Integer, db.ForeignKey("users.id"            307
    ), nullable=True)                                                   308
    action = db.Column(db.String(128), nullable=False)                  309
    # 操作类型                                                          310
    target_type = db.Column(db.String(64), nullable=True)               311
    # 对象类型                                                          312
    target_id = db.Column(db.Integer, nullable=True)                    313
    # 对象主键                                                          314
    detail = db.Column(db.Text, nullable=True)                          315
    # 操作详情                                                          316
    created_at = db.Column(db.DateTime,                                 317
    default=datetime.utcnow, index=True)                                318
class Project(db.Model):                                                319
    """评价项目（项目与矿山档案）。"""                                  320
    __tablename__ = "projects"                                          321
    id = db.Column(db.Integer, primary_key=True)                        322
    name = db.Column(db.String(128), nullable=False)                    323
    code = db.Column(db.String(64), nullable=True)                      324
    # 项目编号                                                          325
    description = db.Column(db.Text, nullable=True)                     326
    status = db.Column(db.String(20), default="进行中")                 327
    # 进行中 / 已完成 / 已归档                                          328
    created_by = db.Column(db.Integer,                                  329
    db.ForeignKey("users.id"), nullable=True)                           330
    created_at = db.Column(db.DateTime,                                 331
    default=datetime.utcnow)                                            332
    patches = db.relationship("MinePatch",                              333
    backref="project", lazy="dynamic",                                  334
                                    cascade="all, delete-orphan")       335
class MinePatch(db.Model):                                              336
    """矿山图斑。"""                                                    337
    __tablename__ = "mine_patches"                                      338
    id = db.Column(db.Integer, primary_key=True)                        339
    project_id = db.Column(db.Integer,                                  340
    db.ForeignKey("projects.id"), nullable=False)                       341
    name = db.Column(db.String(128), nullable=False)                    342
    admin_region = db.Column(db.String(128), nullable=True)             343
    # 所在行政区                                                        344
    center_x = db.Column(db.Float, nullable=True)                       345
    # 中心经度 / 横坐标                                                 346
    center_y = db.Column(db.Float, nullable=True)                       347
    # 中心纬度 / 纵坐标                                                 348
    area_ha = db.Column(db.Float, nullable=True)                        349
    # 图斑面积（公顷）                                                  350
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    mineral_type = db.Column(db.String(64), nullable=True)
    # 开采矿种
    mining_method = db.Column(db.String(64), nullable=True)
    # 开采方式
    stop_years = db.Column(db.Float, nullable=True)
    # 停产年限
    responsible_entity_missing = db.Column(db.Boolean,
    default=True)  # 责任主体是否缺失
    # 遥感底图 / 无人机正射影像挂接
    image_path = db.Column(db.String(256), nullable=True)
    # 相对 static 的路径
    image_note = db.Column(db.String(256), nullable=True)
    # 影像说明（如来源、时相）
    image_uploaded_at = db.Column(db.DateTime,
    nullable=True)
    created_at = db.Column(db.DateTime,
    default=datetime.utcnow)
    units = db.relationship("MicroUnit", backref="patch",
    lazy="dynamic",
                            cascade="all, delete-orphan")
class ReferenceEcosystem(db.Model):
    """参照生态系统：标准化基准值，与图斑绑定。"""
    __tablename__ = "reference_ecosystems"
    id = db.Column(db.Integer, primary_key=True)
    patch_id = db.Column(db.Integer, db.ForeignKey("mine_pat
    ches.id"), nullable=False)
    coverage_benchmark = db.Column(db.Float, default=75.0)
    # 植被覆盖度基准 %
    shannon_benchmark = db.Column(db.Float, default=0.7)
    # Shannon-Wiener 基准
    dominant_species = db.Column(db.String(256),
    nullable=True)  # 建群种
    soil_benchmark = db.Column(db.String(256),
    nullable=True)    # 土壤基准
class Standard(db.Model):
    """地方标准 / 政策文件（政策与标准知识库）。"""
    __tablename__ = "standards"
    id = db.Column(db.Integer, primary_key=True)
    province = db.Column(db.String(32), nullable=False)
    # 省份
    name = db.Column(db.String(256), nullable=False)
    # 名称
    doc_number = db.Column(db.String(128), nullable=True)
    # 标准编号 / 文号
    publish_date = db.Column(db.String(32), nullable=True)
    implement_date = db.Column(db.String(32), nullable=True)
    # 类别：地方标准 / 政策文件
    category = db.Column(db.String(32), nullable=False,
    default="地方标准")
    summary = db.Column(db.Text, nullable=True)
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    # 特点概述
    rules = db.relationship("StandardRule",
    backref="standard",
                            cascade="all, delete-orphan")
class StandardRule(db.Model):
    """结构化规则：将标准条文拆解为可执行规则。"""
    __tablename__ = "standard_rules"
    id = db.Column(db.Integer, primary_key=True)
    standard_id = db.Column(db.Integer,
    db.ForeignKey("standards.id"), nullable=False)
    item = db.Column(db.String(128), nullable=False)
    # 判定项
    operator = db.Column(db.String(8), nullable=False)
    # 比较符：> >= < <= ==
    threshold = db.Column(db.String(64), nullable=True)
    # 阈值
    condition_type = db.Column(db.String(32),
    default="必要条件")  # 必要条件 / 充分条件
    region = db.Column(db.String(64), nullable=True)
    # 适用分区（宁夏分区差异化）
    note = db.Column(db.Text, nullable=True)
class ConstraintCheck(db.Model):
    """限制性判定记录。"""
    __tablename__ = "constraint_checks"
    id = db.Column(db.Integer, primary_key=True)
    patch_id = db.Column(db.Integer, db.ForeignKey("mine_pat
    ches.id"), nullable=False)
    policy_pass = db.Column(db.Boolean, nullable=True)
    # 政策约束
    location_pass = db.Column(db.Boolean, nullable=True)
    # 区位约束
    pollution_pass = db.Column(db.Boolean, nullable=True)
    # 土壤污染
    geo_pass = db.Column(db.Boolean, nullable=True)
    # 地质灾害
    farmland_pass = db.Column(db.Boolean, nullable=True)
    # 耕地属性
    passed = db.Column(db.Boolean, nullable=True)
    # 总体是否通过
    reason = db.Column(db.Text, nullable=True)
    # 不通过原因
    checked_at = db.Column(db.DateTime,
    default=datetime.utcnow)
class MicroUnit(db.Model):
    """微地质单元：岩质边坡 / 碎石边坡 / 平台。"""
    __tablename__ = "micro_units"
    id = db.Column(db.Integer, primary_key=True)
    patch_id = db.Column(db.Integer, db.ForeignKey("mine_pat
    ches.id"), nullable=False)
    unit_type = db.Column(db.String(32), nullable=False)
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    # 单元类型
    area = db.Column(db.Float, nullable=True)
    # 面积
    ratio = db.Column(db.Float, nullable=True)
    # 面积占比（自动计算）
    description = db.Column(db.Text, nullable=True)
    # 现场描述
class Indicator(db.Model):
    """指标定义：指标树，含角色与启用单元类型。"""
    __tablename__ = "indicators"
    id = db.Column(db.Integer, primary_key=True)
    code = db.Column(db.String(32), unique=True,
    nullable=False)  # 指标编码
    name = db.Column(db.String(64), nullable=False)
    # 指标名称
    level = db.Column(db.Integer, nullable=False)
    # 1/2/3 级
    parent_code = db.Column(db.String(32), nullable=True)
    # 父指标编码
    # 角色：土壤污染指标同时作为限制性指标与评价指标
    is_constraint = db.Column(db.Boolean, default=False)
    is_evaluative = db.Column(db.Boolean, default=True)
    unit = db.Column(db.String(32), nullable=True)
    # 单位
    ideal_value = db.Column(db.String(64), nullable=True)
    # 理想值（可配置）
    ideal_type = db.Column(db.String(32), nullable=True)
    # benefit/cost/mid/discrete
    # 启用该指标的微地质单元（逗号分隔）。岩质边坡不启用土壤指标、平台不启用地形指标
    enabled_units = db.Column(db.String(128), nullable=True)
    description = db.Column(db.Text, nullable=True)
    # 取值依据说明
    def units_list(self):
        return [s for s in (self.enabled_units or
    "").split(",") if s]
class WeightScheme(db.Model):
    """权重方案：按微地质单元类型分别保存。"""
    __tablename__ = "weight_schemes"
    id = db.Column(db.Integer, primary_key=True)
    unit_type = db.Column(db.String(32), nullable=False)
    # 所属单元类型
    name = db.Column(db.String(64), nullable=True)
    # 方案名称
    level = db.Column(db.Integer, nullable=False,
    default=1)        # 1=一级指标 2=二级指标
    is_default = db.Column(db.Boolean, default=False)
    # 是否默认（论文表 3）
    created_at = db.Column(db.DateTime,
    default=datetime.utcnow)
class WeightItem(db.Model):
```

```
    """权重明细：方案内的各项权重。"""                                    501
    __tablename__ = "weight_items"                                      502
    id = db.Column(db.Integer, primary_key=True)                        503
    scheme_id = db.Column(db.Integer,                                   504
    db.ForeignKey("weight_schemes.id"), nullable=False)                 505
    indicator_code = db.Column(db.String(32),                           506
    nullable=False)       # 指标编码                                    507
    weight = db.Column(db.Float, nullable=False)                        508
    # 权重值                                                            509
class ExpertMatrix(db.Model):                                           510
    """专家判断矩阵：多专家打分的原始矩阵。"""                            511
    __tablename__ = "expert_matrices"                                   512
    id = db.Column(db.Integer, primary_key=True)                        513
    scheme_id = db.Column(db.Integer,                                   514
    db.ForeignKey("weight_schemes.id"), nullable=True)                  515
    # 直接标注适用对象，便于无需先建权重方案即可录入专家矩阵             516
    unit_type = db.Column(db.String(32), nullable=True)                 517
    # 微地质单元类型                                                    518
    l1_name = db.Column(db.String(64), nullable=True)                   519
    # 所属一级指标（二级矩阵用）                                        520
    expert_name = db.Column(db.String(64), nullable=True)               521
    # 专家姓名                                                          522
    level = db.Column(db.Integer, nullable=False,                       523
    default=1)          # 矩阵对应指标层级                              524
    matrix_json = db.Column(db.Text, nullable=False)                    525
    # 判断矩阵（JSON 二维数组）                                         526
    cr = db.Column(db.Float, nullable=True)                             527
    # 该专家矩阵的一致性比例                                            528
    created_at = db.Column(db.DateTime,                                 529
    default=datetime.utcnow)                                            530
class IndicatorData(db.Model):                                          531
    """指标实测数据：按微地质单元组织。"""                               532
    __tablename__ = "indicator_data"                                    533
    id = db.Column(db.Integer, primary_key=True)                        534
    unit_id = db.Column(db.Integer, db.ForeignKey("micro_uni            535
    ts.id"), nullable=False, unique=True)                               536
    soil_ratio = db.Column(db.Float, nullable=True)                     537
    # 土石比                                                            538
    thickness = db.Column(db.Float, nullable=True)                      539
    # 土层厚度 cm                                                       540
    ph = db.Column(db.Float, nullable=True)                             541
    organic_matter = db.Column(db.Float, nullable=True)                 542
    # 有机质 g/kg                                                       543
    # 重金属含量，JSON 字符串：{"Hg": 0.05, ...}，未检测项缺失或为 null   544
    heavy_metals = db.Column(db.Text, nullable=True)                    545
    coverage = db.Column(db.Float, nullable=True)                       546
    # 植被覆盖度 %                                                      547
    shannon = db.Column(db.Float, nullable=True)                        548
    # Shannon-Wiener 多样性指数                                         549
    slope = db.Column(db.Float, nullable=True)                          550
```

```
    # 坡度 °
    aspect = db.Column(db.String(16), nullable=True)
    # 坡向：阳坡 / 阴阳坡 / 阴坡
    def heavy_metals_dict(self):
        return json.loads(self.heavy_metals) if
    self.heavy_metals else {}
class EvaluationResult(db.Model):
    """评价结果。"""
    __tablename__ = "evaluation_results"
    id = db.Column(db.Integer, primary_key=True)
    patch_id = db.Column(db.Integer, db.ForeignKey("mine_pat
    ches.id"), nullable=False)
    final_score = db.Column(db.Float, nullable=True)
    # 最终评价因子 F
    grade = db.Column(db.String(32), nullable=True)
    # 适宜性等级
    measures = db.Column(db.Text, nullable=True)
    # 推荐措施
    unit_scores = db.Column(db.Text, nullable=True)
    # 各单元得分 JSON
    created_at = db.Column(db.DateTime,
    default=datetime.utcnow)
class ScoreDetail(db.Model):
    """得分明细：每一层指标的权重、标准化值、加权贡献，供追溯与复核。"""
    __tablename__ = "score_details"
    id = db.Column(db.Integer, primary_key=True)
    result_id = db.Column(db.Integer,
    db.ForeignKey("evaluation_results.id"), nullable=False)
    unit_id = db.Column(db.Integer, nullable=True)
    l1_name = db.Column(db.String(64), nullable=True)
    # 一级指标
    l2_name = db.Column(db.String(64), nullable=True)
    # 二级指标
    raw_value = db.Column(db.String(64), nullable=True)
    # 实测值
    std_value = db.Column(db.Float, nullable=True)
    # 标准化值
    weight = db.Column(db.Float, nullable=True)
    # 权重
    contribution = db.Column(db.Float, nullable=True)
    # 加权贡献
# ==== app/forms.py ====
# -*- coding: utf-8 -*-
"""表单定义（共享，由甲统一维护）。
各模块的 Flask-WTF 表单集中于此。
"""
import re
from flask_wtf import FlaskForm
from wtforms import SelectField, SubmitField, StringField,
    PasswordField, BooleanField
```

```
from wtforms.validators import (DataRequired, Length,
    Optional, EqualTo,
                                ValidationError)
from app.utils.constants import UNIT_TYPES, ASPECT_TYPES,
    GENDERS
#  邮箱格式：够用的轻量校验。
#  WTForms 自带的 Email() 需要额外安装 email_validator，
#  为一个字段引入依赖不划算，这里自己用正则判断。
EMAIL_RE = re.compile(r"^[^@\s]+@[^@\s]+\.[A-Za-z]{2,}$")
def email_format(form, field):
    v = (field.data or "").strip()
    if v and not EMAIL_RE.match(v):
        raise ValidationError("邮箱格式不正确")
def email_ok(value):
    """邮箱格式判断，供不用 WTForms 的表单复用（如管理员编辑用户）。"""
    v = (value or "").strip()
    return (not v) or bool(EMAIL_RE.match(v))
class AHPSelectForm(FlaskForm):
    """选择微地质单元类型与指标层级，以确定判断矩阵规模。"""
    unit_type = SelectField("微地质单元类型",
    validators=[DataRequired()])
    level = SelectField("指标层级",
                        choices=[("1", "一级指标"), ("2",
    "二级指标")],
                        validators=[DataRequired()])
    l1 = SelectField("一级指标", validators=[],
    validate_choice=False)  # 动态填充，层级=2 时使用
    submit = SubmitField("生成判断矩阵")
    def __init__(self, *args, **kwargs):
        super().__init__(*args, **kwargs)
        self.unit_type.choices = [(t, t) for t in
    UNIT_TYPES]
class EmptyForm(FlaskForm):
    """仅用于承载 CSRF 令牌的空表单（矩阵录入等自定义表单使用）。"""
    pass
class LoginForm(FlaskForm):
    """登录表单。支持「记住我」以在关闭浏览器后保持登录状态。
    校验消息用中文写明：登录框带 novalidate，浏览器不会代填提示，
    字段为空时若不自己给消息，用户点了「登录」会以为没反应。
    """
    username = StringField("账户名称",
                           validators=[DataRequired(message=
    "请输入账户名称"),
                                       Length(max=64,
                                              message="账户名称不能超过 64 个字符")])
    password = PasswordField("密码",
                             validators=[DataRequired(messag
    e="请输入密码")])
    remember = BooleanField("记住我")
```

```
    submit = SubmitField("登录")
GENDER_CHOICES = [("", "不填")] + [(g, g) for g in GENDERS]
class ProfileForm(FlaskForm):
    """个人资料：昵称、性别与联系方式。字段都可留空。
    职务不在这里 —— 只能由管理员在「个人设置 · 账号」里改。
    """
    display_name = StringField("昵称", validators=[Optional(),
    Length(max=64)])
    gender = SelectField("性别", choices=GENDER_CHOICES,
    validators=[Optional()])
    email = StringField("邮箱", validators=[Optional(),
    email_format,
                                            Length(max=128)])
    phone = StringField("手机号", validators=[Optional(),
    Length(max=32)])
    submit_info = SubmitField("保存资料")
class RegisterForm(FlaskForm):
    """注册：账号与密码必填，其余选填。注册后默认是操作员。"""
    username = StringField("账号名称",
                           validators=[DataRequired("请填写账号名称
    "),
                                       Length(min=2, max=64,
                                              message="账号名称
    2–64 个字符")])
    password = PasswordField("密码",
                             validators=[DataRequired("请设置密码
    "),
                                         Length(min=6,
    max=64,

    message="密码至少 6 位")])
    confirm = PasswordField("确认密码",
                            validators=[DataRequired(),
                                        EqualTo("password",


    message="两次输入的密码不一致")])
    display_name = StringField("昵称", validators=[Optional(),
    Length(max=64)])
    gender = SelectField("性别", choices=GENDER_CHOICES,
    validators=[Optional()])
    email = StringField("邮箱", validators=[Optional(),
    email_format,
                                            Length(max=128)])
    phone = StringField("手机号", validators=[Optional(),
    Length(max=32)])
    submit_reg = SubmitField("注册并登录")
class PasswordForm(FlaskForm):
    """修改密码：需要先验证当前密码。"""
    old_password = PasswordField("当前密码",
    validators=[DataRequired()])
```

```
    new_password = PasswordField("新密码",
                                 validators=[DataRequired(),
    Length(min=6, max=64,

    message="新密码至少 6 位")])
    confirm = PasswordField("确认新密码",
                            validators=[DataRequired(),

    EqualTo("new_password",

    message="两次输入的新密码不一致")])
    submit_pwd = SubmitField("修改密码")
# ==== app/utils/constants.py ====
# -*- coding: utf-8 -*-
"""评价指标常量表。
全部数值取自论文，作为系统计算的唯一基准：
- 论文 4.2.2 节：指标标准化理想极值与取值依据
- 论文 4.2.3 节 / 表 4：适宜性等级划分
- 论文表 3：按微地质单元分别确定的一级、二级指标权重
- 论文 4.2.1 节：平均随机一致性指标 RI
"""
# ------------------------------------------------------------------
    ------------------
# 一、微地质单元类型
# ------------------------------------------------------------------
    ------------------
# 论文 4.1 节："按照地形地貌划分，将其划分为平台、碎石边坡和岩质边坡
# 三种地质微单元类型"
UNIT_TYPES = ("岩质边坡", "碎石边坡", "平台")
# 每个单元类型启用的指标类别（论文表 3 隐含：岩质边坡无土壤指标，平台无地形指标）
# 供数据录入界面与评价引擎判断"该单元是否需要某类指标"
UNIT_ENABLED_INDICATORS = {
    "岩质边坡": ("植被指标", "地形指标"),                # 不设土壤指标
    "碎石边坡": ("土壤指标", "植被指标", "地形指标"),
    "平台": ("土壤指标", "植被指标"),                    # 不设地形指标
}
# ------------------------------------------------------------------
    ------------------
# 二、指标标准化理想极值（论文 4.2.2 节）
# ------------------------------------------------------------------
    ------------------
IDEAL_SOIL_RATIO = 0.8          # 土石比理想值
IDEAL_THICKNESS = 20.0          # 土层厚度理想值（cm）
IDEAL_PH_RANGE = (5.0, 9.0)     # pH 理想区间（适中型指标）
IDEAL_ORGANIC = 30.0            # 有机质理想值
    （g/kg），全国第二次土壤普查养分等级划分标准二等最低值
IDEAL_COVERAGE = 75.0           # 植被覆盖度参照生态系统（%）
IDEAL_SHANNON = 0.7             # Shannon-Wiener 多样性指数参照生态系统
# 重金属极值（mg/kg），《绿化种植土壤》CJ/T 340—2016 土壤重金属含量 IV 级最高值
# 论文原文：镉、汞、铅、铬、砷、镍、铜、锌分别为 2、2、530、400、45、220、600、800
```

```
HEAVY_METALS = ("Cd", "Hg", "Pb", "Cr", "As", "Ni", "Cu",
    "Zn")
HEAVY_METAL_LIMIT = {
    "Cd": 2.0, "Hg": 2.0, "Pb": 530.0, "Cr": 400.0,
    "As": 45.0, "Ni": 220.0, "Cu": 600.0, "Zn": 800.0,
}
# 污染指标下 8 种重金属"重要性相当，权重平均分配"（论文 4.2.1 节）
HEAVY_METAL_EQUAL_WEIGHT = 1.0 / 8.0
# 坡度分界，论文 4.2.2 节引《新一轮退耕还林还草总体方案》发改西部〔2014〕1772 号
SLOPE_BREAK = 25.0
# 坡向离散赋值，论文 4.2.2 节："阳坡、阴阳坡、阴坡分别赋值 0.2、0.6、1"
ASPECT_SCORES = {"阳坡": 0.2, "阴阳坡": 0.6, "阴坡": 1.0}
ASPECT_TYPES = ("阳坡", "阴阳坡", "阴坡")
# pH 适中型标准化的容差边界（区间外线性衰减，避免得分突变）
PH_TOL_LOW = 4.0
PH_TOL_HIGH = 10.0
# ---------------------------------------------------------------------
    ------------------
# 三、适宜性等级划分（论文表 4）
# ---------------------------------------------------------------------
    ------------------
GRADE_LEVELS = (
    {"grade": "不适宜", "min": None, "max": 0.3,
     "meaning": "存在限制性条件或区域生态本底较差，自然恢复影响因子严重制约自然恢复",
     "measure": "生态重建、人工辅助"},
    {"grade": "基本适宜", "min": 0.3, "max": 0.7,
     "meaning": "生态本底良好，核心自然恢复影响因子基本满足自然恢复需求",
     "measure": "自然恢复为主，可辅以封闭场地和少量补种、补植、飞播等"},
    {"grade": "适宜", "min": 0.7, "max": None,
     "meaning": "各自然恢复影响因子均能满足自然恢复需求，自然演替速度快",
     "measure": "自然恢复"},
)
def grade_of(score):
    """按论文表 4 判定适宜性等级。"""
    for g in GRADE_LEVELS:
        lo, hi = g["min"], g["max"]
        if lo is None and score < hi:
            return g
        if hi is None and score >= lo:
            return g
        if lo is not None and hi is not None and lo <=
    score < hi:
            return g
    # score 恰好等于 0.7 时落入"适宜"
    return GRADE_LEVELS[-1]
# ---------------------------------------------------------------------
    ------------------
# 四、平均随机一致性指标 RI（论文 4.2.1 节给出 n=2,3,4,5，其余为常用扩展值）
# ---------------------------------------------------------------------
    ------------------
```

```
RI_TABLE = {1: 0.0, 2: 0.0, 3: 0.58, 4: 0.90, 5: 1.12, 6: 1.24,
            7: 1.32, 8: 1.41, 9: 1.45, 10: 1.49}
# 一致性比例阈值（论文：当 CR<0.1 时，判断矩阵满足一致性要求）
CR_THRESHOLD = 0.1
# ------------------------------------------------------------------
# 五、默认权重方案（论文表 3）
# ------------------------------------------------------------------
# 论文 4.2.2 节："划分岩质边坡、碎石土边坡、平台三类微地质单元开展独立评价，
# 分别计算评价指标权重"（论文摘要、图 2 流程与表 5 均作"碎石边坡"，
# 系统统一采用"碎石边坡"）。权重由 10 位专家打分后经 AHP 方法确定。
# 下表为论文表 3 所载的归一化权重，作为系统默认方案。
DEFAULT_WEIGHTS = {
    "岩质边坡": {
        "l1": {"植被指标": 0.45, "地形指标": 0.55},
        "l2": {
            "植被指标": {"植被覆盖度": 0.67, "生物多样性": 0.33},
            "地形指标": {"坡度": 0.64, "坡向": 0.36},
        },
    },
    "碎石边坡": {
        "l1": {"土壤指标": 0.44, "植被指标": 0.31, "地形指标": 0.25},
        "l2": {
            "土壤指标": {"土石比": 0.25, "土层厚度": 0.25, "pH": 0.17,
                         "有机质": 0.17, "污染指标": 0.17},
            "植被指标": {"植被覆盖度": 0.64, "生物多样性": 0.36},
            "地形指标": {"坡度": 0.67, "坡向": 0.33},
        },
    },
    "平台": {
        "l1": {"土壤指标": 0.54, "植被指标": 0.46},
        "l2": {
            "土壤指标": {"土石比": 0.25, "土层厚度": 0.25, "pH": 0.17,
                         "有机质": 0.17, "污染指标": 0.17},
            "植被指标": {"植被覆盖度": 0.61, "生物多样性": 0.39},
        },
    },
}
# ------------------------------------------------------------------
# 六、限制性指标（论文表 2 + 4.1 节流程）
# ------------------------------------------------------------------
# 论文摘要与 4.1 节：四项限制性指标；流程段另明确"是否为耕地"为约束条件
CONSTRAINT_ITEMS = ("政策", "区位约束", "土壤污染", "地质灾害", "耕地属性")
# ------------------------------------------------------------------
# 七、账号职务
```

```
# ------------------------------------------------------------------------ 851
    ----------------- 852
# 只分两种：管理员可管理账号、维护基础数据并查看操作日志，操作员用业务功能。 853
ROLE_LABELS = {"admin": "管理员", "operator": "操作员"} 854
ROLE_CODES = ("admin", "operator") 855
DEFAULT_ROLE = "operator"  # 新注册账号的默认职务 856
ROLE_ORDER = ("admin", "operator")  # 列表排序用：管理员在前 857
# 性别（注册与个人资料中选填） 858
GENDERS = ("男", "女", "保密") 859
# ==== app/utils/schema.py ==== 860
# -*- coding: utf-8 -*- 861
"""轻量结构迁移。 862
SQLite 下 `db.create_all()` 只建缺失的表，不会给已有表补列。 863
本模块在启动时检查目标列是否存在，缺失则 ALTER TABLE 补上， 864
使旧数据库无需重建即可用上新字段。 865
""" 866
from sqlalchemy import text 867
# （表名， 列名， 列定义） 868
REQUIRED_COLUMNS = [ 869
    # 遥感影像挂接 870
    ("mine_patches", "image_path", "VARCHAR(256)"), 871
    ("mine_patches", "image_note", "VARCHAR(256)"), 872
    ("mine_patches", "image_uploaded_at", "DATETIME"), 873
    # 指标维护 874
    ("indicators", "enabled_units", "VARCHAR(128)"), 875
    ("indicators", "description", "TEXT"), 876
    # 多专家矩阵 877
    ("expert_matrices", "unit_type", "VARCHAR(32)"), 878
    ("expert_matrices", "l1_name", "VARCHAR(64)"), 879
    ("expert_matrices", "cr", "FLOAT"), 880
    # 用户个人资料（账户页） 881
    ("users", "display_name", "VARCHAR(64)"), 882
    ("users", "email", "VARCHAR(128)"), 883
    ("users", "phone", "VARCHAR(32)"), 884
    ("users", "gender", "VARCHAR(8)"), 885
    ("users", "avatar", "VARCHAR(256)"), 886
] 887
# 职务体系由「管理员 / 评价员 / 查看者」三种收敛为「管理员 / 操作员」两种， 888
# 旧数据里的后两种统一归为操作员。 889
ROLE_MIGRATION = {"evaluator": "operator", "viewer": 890
    "operator"} 891
def _add_columns(db): 892
    added = [] 893
    with db.engine.connect() as conn: 894
        for table, column, ddl in REQUIRED_COLUMNS: 895
            rows = conn.execute(text("PRAGMA 896
    table_info(%s)" % table)).fetchall() 897
            if not rows: 898
                continue  # 899
    表还不存在，create_all 会建 900
```

```
                existing = {r[1] for r in rows}                          901
                if column in existing:                                   902
                    continue                                             903
                conn.execute(text(                                       904
                    "ALTER TABLE %s ADD COLUMN %s %s" % (table,          905
    column, ddl)))                                                       906
                conn.commit()                                            907
                added.append("%s.%s" % (table, column))                  908
    return added                                                         909
def _migrate_roles(db):                                                  910
    """把旧角色名统一为新的两种，返回改动的行数。"""                          911
    changed = 0                                                          912
    with db.engine.connect() as conn:                                    913
        rows = conn.execute(text("PRAGMA                                 914
    table_info(users)")).fetchall()                                      915
        if not rows:                                                     916
            return 0                                                     917
        for old, new in ROLE_MIGRATION.items():                          918
            r = conn.execute(text("UPDATE users SET role =               919
    :new "                                                               920
                                  "WHERE role = :old"),                  921
                                 {"new": new, "old": old})               922
            changed += r.rowcount or 0                                   923
        conn.commit()                                                    924
    return changed                                                       925
def ensure_schema(db):                                                   926
    """为已有表补齐缺失列，并做必要的数据迁移。返回变更清单。"""                 927
    try:                                                                 928
        changed = _add_columns(db)                                       929
        n = _migrate_roles(db)                                           930
        if n:                                                            931
            changed.append("用户角色归一为管理员/操作员 %d 条" % n)              932
        return changed                                                   933
    except Exception as e:                                    #          934
    迁移失败不应阻断启动                                                  935
        print("[schema] 结构检查跳过：%s" % e)                             936
        return []                                                        937
# ==== app/utils/uploads.py ====                                         938
# -*- coding: utf-8 -*-                                                  939
"""上传文件处理（遥感影像、批量导入）。"""                                    940
import os                                                                941
import re                                                                942
import uuid                                                              943
from datetime import datetime                                            944
from werkzeug.utils import secure_filename                               945
def _safe_ext(filename):                                                 946
    return os.path.splitext(filename)[1].lower().lstrip(".")             947
def ext_allowed(filename, allowed):                                      948
    return _safe_ext(filename) in allowed                                949
def save_upload(file_storage, subdir, allowed_ext,                       950
```

```
    upload_root):
    """保存上传文件，返回相对 static 的路径（如 uploads/patches/xxx.png）。
    文件名重命名为「时间戳_随机串.扩展名」，避免中文名与重名问题。
    """
    if file_storage is None or not file_storage.filename:
        return None, "未选择文件"
    ext = _safe_ext(file_storage.filename)
    if ext not in allowed_ext:
        return None, "不支持的文件类型：.%s" % ext
    target_dir = os.path.join(upload_root, subdir)
    os.makedirs(target_dir, exist_ok=True)
    # 仅保留安全字符，并加时间戳与随机串防重名
    base = secure_filename(os.path.splitext(file_storage.fil
    ename)[0]) or "file"
    base = re.sub(r"[^A-Za-z0-9_\-]", "", base)[:40] or
    "file"
    name = "%s_%s.%s" % (datetime.now().strftime("%Y%m%d%H%M
    %S"),
                         uuid.uuid4().hex[:6], ext)
    abs_path = os.path.join(target_dir, name)
    file_storage.save(abs_path)
    rel = os.path.join("uploads", subdir,
    name).replace("\\", "/")
    return rel, None
def remove_upload(rel_path, upload_root):
    """删除已保存的上传文件（相对 static 的路径）。"""
    if not rel_path:
        return
    abs_path = os.path.join(upload_root,
    rel_path.replace("uploads/", "", 1))
    try:
        if os.path.isfile(abs_path):
            os.remove(abs_path)
    except OSError:
        pass
# ==== app/utils/charts.py ====
# -*- coding: utf-8 -*-
"""轻量图表。
用纯 SVG 输出，不依赖任何绘图库或前端脚本：
- 服务端渲染，页面与打印（PDF）中都能稳定显示
- 每个图元都带数值标签，黑白打印时仍可读（不单靠颜色区分）
- 单色方案，符合界面「一个视图一个强调色」的约定
"""
from html import escape
ACCENT = "#1f3a5f"
ACCENT_SOFT = "#7f93ad"
INK = "#23262b"
MUTED = "#6b7280"
LINE = "#dfe3e8"
# 灰度梯度：黑白打印时相邻扇区 / 条柱仍可区分
```

```
GRAYS = ["#1f3a5f", "#456188", "#6d86a8", "#96a9c4",
    "#bcc9da", "#dbe3ec"]
def _fmt(v, digits=4):
    try:
        return ("%%.%df" % digits) % float(v)
    except (TypeError, ValueError):
        return str(v)
def hbars(items, width=620, bar_h=22, gap=12, label_w=150,
    value_w=76):
    """水平条形图。items: [(标签, 数值, 显示文本)]"""
    if not items:
        return '<p class="muted">暂无数据</p>'
    n = len(items)
    height = n * (bar_h + gap) + 10
    max_v = max((abs(float(v)) for _, v, _ in items),
    default=1) or 1
    body_w = width - label_w - value_w
    parts = ['<svg viewBox="0 0 %d %d" width="100%%"
    height="%d" '
             'role="img" aria-label="条形图" '
             'style="max-width:%dpx;font-family:inherit;">'
    % (width, height, height, width)]
    for i, (label, value, disp) in enumerate(items):
        y = i * (bar_h + gap) + 5
        frac = (abs(float(value)) / max_v) if max_v else 0
        w = max(2.0, body_w * frac)
        fill = GRAYS[i % len(GRAYS)]
        parts.append(
            '<text x="%d" y="%d" text-anchor="end"
    font-size="12" fill="%s">%s</text>'
            % (label_w - 10, y + bar_h * 0.72, INK,
    escape(str(label))))
        parts.append(
            '<rect x="%d" y="%d" width="%.1f" height="%d"
    rx="3" fill="%s"/>'
            % (label_w, y, w, bar_h, fill))
        parts.append(
            '<text x="%d" y="%d" font-size="12" fill="%s" '
            'style="font-variant-numeric:tabular-nums;">%s</
    text>'
            % (label_w + w + 8, y + bar_h * 0.72, MUTED,
    escape(str(disp))))
    parts.append('</svg>')
    return "".join(parts)
def donut(items, size=210, thickness=34):
    """环形图（权重分布）。items: [(标签, 数值, 显示文本)]
    图例中给出数值与占比，黑白打印时靠图例与灰度深浅区分。
    """
    vals = [(l, float(v), d) for l, v, d in items if
    float(v) > 0]
```

```
if not vals:
    return '<p class="muted">暂无数据</p>'
total = sum(v for _, v, _ in vals) or 1
r = (size - thickness) / 2
cx = cy = size / 2
import math
parts = ['<svg viewBox="0 0 %d %d" width="%d"
height="%d" role="img" '
         'aria-label="权重分布环形图">' % (size, size, size,
size)]
angle = -math.pi / 2
for i, (label, v, disp) in enumerate(vals):
    sweep = 2 * math.pi * (v / total)
    a0, a1 = angle, angle + sweep
    x0, y0 = cx + r * math.cos(a0), cy + r *
math.sin(a0)
    x1, y1 = cx + r * math.cos(a1), cy + r *
math.sin(a1)
    large = 1 if sweep > math.pi else 0
    parts.append(
        '<path d="M %.2f %.2f A %.2f %.2f 0 %d 1 %.2f
%.2f" '
        'fill="none" stroke="%s" stroke-width="%d"/>'
        % (x0, y0, r, r, large, x1, y1, GRAYS[i %
len(GRAYS)], thickness))
    angle = a1
parts.append('<text x="%d" y="%d" text-anchor="middle"
font-size="12" '
             'fill="%s">合计</text>' % (cx, cy - 2,
MUTED))
parts.append('<text x="%d" y="%d" text-anchor="middle"
font-size="15" '
             'fill="%s" style="font-variant-numeric:tabu
lar-nums;">%s</text>'
             % (cx, cy + 18, INK, _fmt(total, 4)))
parts.append('</svg>')
legend = ['<table style="width:auto;margin-top:10px;"><t
head><tr>'
          '<th scope="col">项目</th><th
scope="col">数值</th>'
          '<th scope="col">占比</th></tr></thead><tbody>']
for i, (label, v, disp) in enumerate(vals):
    legend.append(
        '<tr><td style="text-align:left;">'
        '<span style="display:inline-block;width:10px;he
ight:10px;'
        'border-radius:2px;background:%s;margin-right:6p
x;"></span>%s</td>'
        '<td class="mono">%s</td><td
class="mono">%.1f%%</td></tr>'
```

```
                % (GRAYS[i % len(GRAYS)], escape(str(label)),
        escape(str(disp)),
                    100.0 * v / total))
        legend.append('</tbody></table>')
        return ('<div style="display:flex;flex-wrap:wrap;gap:20p
        x;align-items:center;">'
                + "".join(parts) + "".join(legend) + '</div>')
# ==== app/utils/browser.py ====
# -*- coding: utf-8 -*-
"""HTML 转 PDF（报告导出）。
复用本机已安装的 Chrome/Edge 无头模式打印，避免引入 reportlab、
weasyprint 等额外依赖。生成过程使用系统临时目录中的 ASCII 文件名，
避免中文路径在 file:// URL 下出问题。
"""
import os
import shutil
import subprocess
import tempfile
import uuid
CANDIDATES = [
        r"C:\Program Files\Google\Chrome\Application\chrome.exe"
        ,
        r"C:\Program Files (x86)\Google\Chrome\Application\chrom
        e.exe",
        r"C:\Program Files (x86)\Microsoft\Edge\Application\msed
        ge.exe",
        r"C:\Program Files\Microsoft\Edge\Application\msedge.exe
        ",
]
# 允许通过环境变量覆盖，便于在其他机器上指定浏览器
ENV_KEY = "DSH_BROWSER_PATH"
def browser_path():
        """定位可用的浏览器可执行文件，找不到返回 None。"""
        override = os.environ.get(ENV_KEY)
        if override and os.path.isfile(override):
                return override
        for p in CANDIDATES:
                if os.path.isfile(p):
                        return p
        for name in ("chrome", "msedge", "chromium"):
                found = shutil.which(name)
                if found:
                        return found
        return None
def html_to_pdf(html, timeout=90):
        """把 HTML 字符串渲染为 PDF，返回字节；失败返回 (None, 原因)。"""
        exe = browser_path()
        if exe is None:
                return None, ("未找到可用的浏览器（Chrome/Edge），无法生成 PDF。"
                              "可设置环境变量 %s 指向浏览器可执行文件。" % ENV_KEY)
```

```
    tmpdir = tempfile.mkdtemp(prefix="dshpdf_")
    tag = uuid.uuid4().hex[:8]
    html_path = os.path.join(tmpdir, "r_%s.html" % tag)
    pdf_path = os.path.join(tmpdir, "r_%s.pdf" % tag)
    try:
        with open(html_path, "w", encoding="utf-8") as f:
            f.write(html)
        cmd = [
            exe, "--headless=new", "--disable-gpu",
"--no-sandbox",
            "--no-first-run", "--no-default-browser-check",
            "--no-pdf-header-footer",
            "--virtual-time-budget=4000",
            "--print-to-pdf=%s" % pdf_path,
            "file:///" + html_path.replace("\\", "/"),
        ]
        p = subprocess.run(cmd, capture_output=True,
text=True,
                           timeout=timeout,
encoding="utf-8", errors="replace")
        if not os.path.isfile(pdf_path) or
os.path.getsize(pdf_path) < 800:
            return None, "浏览器未能生成 PDF（%s）" % (p.stderr or
"").strip()[:200]
        with open(pdf_path, "rb") as f:
            return f.read(), None
    except subprocess.TimeoutExpired:
        return None, "生成 PDF 超时，请稍后重试"
    except Exception as e:
        return None, "生成 PDF 失败：%s" % e
    finally:
        shutil.rmtree(tmpdir, ignore_errors=True)
# ==== app/services/ahp_service.py ====
# -*- coding: utf-8 -*-
"""AHP 权重引擎。
严格按论文 4.2.1 节实现：
- 式(2)(3)(4)：特征根法求权重（行元素乘积 → n 次方根 → 归一化）
- 式(5)(6)(7)：一致性检验（λmax、CI、CR）
- 论文要求 CR < 0.1；RI 按阶数查表
"""
import numpy as np
from app.utils.constants import RI_TABLE, CR_THRESHOLD
def matrix_from_upper(upper):
    """由判断矩阵上三角构造完整互反矩阵。
    参数
    ----
    upper : list[list[float]] 或 np.ndarray，上三角元素（不含对角线）。
        例：3 阶时传入 [[a12, a13], [a23]]。
        对角线为 1，下三角取倒数（a_ji = 1/a_ij）。
    """
```

```
    upper = np.asarray(upper, dtype=float)
    k = upper.shape[0]
    n = int(round((1 + (1 + 8 * k) ** 0.5) / 2))
    A = np.ones((n, n), dtype=float)
    idx = 0
    for i in range(n):
        for j in range(i + 1, n):
            A[i, j] = upper[idx]
            A[j, i] = 1.0 / upper[idx]
            idx += 1
    return A
def weights_and_consistency(A):
    """由判断矩阵 A 计算权重与一致性指标。
    返回
    ----
    dict：{weights: np.ndarray, lambda_max, ci, cr, n,
    consistent: bool}
    """
    A = np.asarray(A, dtype=float)
    n = A.shape[0]
    # 式(2) 行元素乘积
    row_prod = np.prod(A, axis=1)
    # 式(3) 开 n 次方根
    root = row_prod ** (1.0 / n)
    # 式(4) 归一化
    weights = root / root.sum()
    # 式(5) 最大特征值 λmax = (1/n) Σ (A·W)_i / W_i
    aw = A.dot(weights)
    lambda_max = float(np.sum(aw / weights) / n)
    # 式(6) 一致性指标 CI
    ci = (lambda_max - n) / (n - 1) if n > 1 else 0.0
    # 式(7) 一致性比例 CR = CI / RI
    ri = RI_TABLE.get(n, 1.49)
    cr = ci / ri if ri else 0.0
    return {
        "weights": weights,
        "lambda_max": lambda_max,
        "ci": ci,
        "cr": cr,
        "n": n,
        "consistent": cr < CR_THRESHOLD,
    }
def find_most_inconsistent(A):
    """定位判断矩阵中最不一致的一对比较。
    依据：完全一致矩阵应满足 a_ij = w_i / w_j。
    计算每个上三角元素与其理想值的相对偏差，返回偏差最大者。
    返回
    ----
    dict：{i, j, value, expected, deviation}，偏差最大的 (i, j)
    及对应取值。
```

```
    """
    A = np.asarray(A, dtype=float)
    r = weights_and_consistency(A)
    w = r["weights"]
    n = A.shape[0]
    best = None
    for i in range(n):
        for j in range(i + 1, n):
            expected = w[i] / w[j]
            if expected == 0:
                continue
            deviation = abs(A[i, j] - expected) / expected
            if best is None or deviation >
    best["deviation"]:
                best = {"i": i, "j": j, "value": A[i, j],
                        "expected": expected, "deviation":
    deviation}
    return best
def aggregate_matrices(matrices):
    """多专家判断矩阵聚合：几何平均（保持互反性）。
    参数
    ----
    matrices : list[np.ndarray]，多个同阶判断矩阵。
    """
    stack = np.stack([np.asarray(m, dtype=float) for m in
    matrices])
    return np.prod(stack, axis=0) ** (1.0 / len(stack))
def judge_from_weights(weights):
    """由权重向量构造完全一致判断矩阵并检验。
    用途：验证给定权重向量能否作为某个完全一致判断矩阵的主特征向量
    （A_ij = w_i / w_j 时 λmax = n、CR = 0）。
    """
    w = np.asarray(weights, dtype=float)
    n = w.shape[0]
    A = np.outer(w, 1.0 / w)
    r = weights_and_consistency(A)
    r["matrix"] = A
    return r
# ==== app/services/standardization.py ====
# -*- coding: utf-8 -*-
"""指标标准化（的一部分）。
按论文 4.2.2 节式(8) 实现：
    x' = (x - x_min) / (x_max - x_min)
按指标性质分为四类：越大越优 / 越小越优 / 适中 / 离散赋值。
所有理想极值取自 app.utils.constants，可配置。
"""
from app.utils.constants import (
    IDEAL_SOIL_RATIO, IDEAL_THICKNESS, IDEAL_PH_RANGE,
    IDEAL_ORGANIC,
    IDEAL_COVERAGE, IDEAL_SHANNON, HEAVY_METAL_LIMIT,
```

```
    HEAVY_METAL_EQUAL_WEIGHT, SLOPE_BREAK, ASPECT_SCORES,
    PH_TOL_LOW, PH_TOL_HIGH,
)
def std_benefit(value, ideal):
    """越大越优指标标准化：以理想值为上限，按比例给分，饱和于 [0,1]。"""
    if value is None:
        return None
    return max(0.0, min(1.0, value / ideal))
def std_cost(value, extreme):
    """越小越优指标标准化：以极值为参考，线性折减至 [0,1]。"""
    if value is None:
        return None
    return max(0.0, min(1.0, 1.0 - value / extreme))
def std_mid_ph(value, lo, hi, tol_lo=PH_TOL_LOW,
    tol_hi=PH_TOL_HIGH):
    """适中型指标标准化（pH）：理想区间内取 1，区间外线性衰减。"""
    if value is None:
        return None
    if lo <= value <= hi:
        return 1.0
    if value > hi:
        return max(0.0, (tol_hi - value) / (tol_hi - hi))
    return max(0.0, (value - tol_lo) / (lo - tol_lo))
def std_slope(value, break_point=SLOPE_BREAK, upper=90.0):
    """坡度标准化。
    论文 4.2.2 节：以 25° 为陡坡重要分界。25° 及以下记满分 1，
    超过 25° 线性折减，90° 记 0。
    """
    if value is None:
        return None
    if value <= break_point:
        return 1.0
    return max(0.0, (upper - value) / (upper - break_point))
def aspect_score(aspect):
    """坡向离散赋值：阳坡 0.2 / 阴阳坡 0.6 / 阴坡 1.0（论文 4.2.2 节）。"""
    if aspect is None:
        return None
    return ASPECT_SCORES.get(aspect)
def pollution_score(heavy_metals):
    """污染指标：8 种重金属等权（各 0.125）合成。
    论文注：未检测 Cr、Zn 的，从矿山类型看未受污染，按无污染(=1)处理。
    heavy_metals : dict，键为金属名，值为含量 mg/kg；缺失键视为未检测。
    """
    total = 0.0
    for metal, extreme in HEAVY_METAL_LIMIT.items():
        value = heavy_metals.get(metal)
        total += std_cost(value, extreme) if value is not
    None else 1.0
    return total / len(HEAVY_METAL_LIMIT)
def standardize_unit(unit_type, data, weights=None):
```

```
    """对单个微地质单元的全部指标做标准化。
    参数
    ----
    unit_type : str，岩质边坡 / 碎石边坡 / 平台
    data : dict，实测数据，字段见接口契约：
        soil_ratio, thickness, ph, organic_matter,
    heavy_metals(dict),
        coverage, shannon, slope, aspect
    weights : dict，该单元类型的权重（app.utils.constants.DEFAULT_WEIG
    HTS 结构），
                未提供时用默认值。
    返回
    ----
    dict：{ 一级指标： { 二级指标： （实测值， 标准化值， 权重， 贡献） } }
    标准化值可能为 None，表示该指标不适用或方法待定。
    """
    from app.utils.constants import DEFAULT_WEIGHTS
    if weights is None:
        weights = DEFAULT_WEIGHTS[unit_type]
    l1_enabled = weights["l1"].keys()
    l2w = weights["l2"]
    result = {}
    # ---- 土壤指标 ----
    if "土壤指标" in l1_enabled:
        result["土壤指标"] = {
            "土石比": (data.get("soil_ratio"),
                       std_benefit(data.get("soil_ratio"),
    IDEAL_SOIL_RATIO),
                       l2w["土壤指标"]["土石比"]),
            "土层厚度": (data.get("thickness"),
                         std_benefit(data.get("thickness"),
    IDEAL_THICKNESS),
                         l2w["土壤指标"]["土层厚度"]),
            "pH": (data.get("ph"),
                   std_mid_ph(data.get("ph"),
    *IDEAL_PH_RANGE),
                   l2w["土壤指标"]["pH"]),
            "有机质": (data.get("organic_matter"),
                       std_benefit(data.get("organic_matter")
    , IDEAL_ORGANIC),
                       l2w["土壤指标"]["有机质"]),
            "污染指标": (data.get("heavy_metals") or {},
                         pollution_score(data.get("heavy_meta
    ls") or {}),
                         l2w["土壤指标"]["污染指标"]),
        }
    # ---- 植被指标 ----
    if "植被指标" in l1_enabled:
        result["植被指标"] = {
            "植被覆盖度": (data.get("coverage"),
```

```
                                    std_benefit(data.get("coverage"),  1401
    IDEAL_COVERAGE),                                                   1402
                                    l2w["植被指标"]["植被覆盖度"]),    1403
                "生物多样性": (data.get("shannon"),                    1404
                                    std_benefit(data.get("shannon"),   1405
    IDEAL_SHANNON),                                                    1406
                                    l2w["植被指标"]["生物多样性"]),    1407
            }                                                          1408
        # ---- 地形指标 ----                                          1409
        if "地形指标" in l1_enabled:                                   1410
            result["地形指标"] = {                                     1411
                "坡度": (data.get("slope"),                            1412
                         std_slope(data.get("slope")),                 1413
                         l2w["地形指标"]["坡度"]),                     1414
                "坡向": (data.get("aspect"),                           1415
                         aspect_score(data.get("aspect")),             1416
                         l2w["地形指标"]["坡向"]),                     1417
            }                                                          1418
        return result                                                  1419
# ==== app/services/weight_service.py ====                           1420
# -*- coding: utf-8 -*-                                              1421
"""权重方案的存取与生效权重解析。                                      1422
系统有两套权重来源：                                                  1423
1. 论文表 3 给出的默认权重（app.utils.constants.DEFAULT_WEIGHTS），   1424
   是**权威取值来源**，案例复算与报告结论以此为准；                   1425
2. 通过层次分析法在本系统中实际算出的权重，保存在                     1426
   weight_schemes / weight_items 两张表中，可标记为「生效」。         1427
本模块负责把这两者合成为计算引擎直接可用的嵌套权重结构：              1428
以论文表 3 为底，用生效方案中已保存的条目覆盖。                       1429
没有任何生效方案时，即完全使用论文表 3 的权重。                       1430
这样既保证默认行为与论文一致，又使 AHP 计算结果真正参与评价，         1431
而不是算完即弃。                                                      1432
"""                                                                   1433
import copy                                                           1434
from app.models import db, WeightScheme, WeightItem                   1435
from app.utils.constants import DEFAULT_WEIGHTS                       1436
from app.services.indicator_service import TREE                       1437
# 方案名称：每个微地质单元类型固定一个可生效的方案                    1438
ACTIVE_NAME = "自定义权重方案（AHP 计算结果）"                        1439
def _code_maps():                                                     1440
    """由指标树导出 编码 → 名称 的映射，避免名称在两处各写一遍。"""  1441
    l1, l2 = {}, {}                                                   1442
    for row in TREE:                                                  1443
        code, name, level, parent = row[0], row[1], row[2],           1444
    row[3]                                                            1445
        if level == 1:                                                1446
            l1[code] = name                                           1447
        elif level == 2:                                              1448
            l2[code] = (l1.get(parent, parent), name)                 1449
    return l1, l2                                                     1450
```

```
def effective_weights(unit_type):
    """返回该单元类型当前生效的权重。
    返回 (weights, source_label, scheme)
        weights      嵌套权重结构，可直接交给标准化与评价引擎
        source_label 权重来源说明，用于界面与报告标注
        scheme       生效的权重方案对象，无自定义方案时为 None
    """
    base = DEFAULT_WEIGHTS.get(unit_type)
    if base is None:
        return None, "未定义该微地质单元类型的权重", None
    scheme = WeightScheme.query.filter_by(unit_type=unit_type,
                                          name=ACTIVE_NAME).first()
    if scheme is None:
        return copy.deepcopy(base), "论文表 3 默认权重", None
    merged = copy.deepcopy(base)
    l1_map, l2_map = _code_maps()
    applied = 0
    for item in WeightItem.query.filter_by(scheme_id=scheme.id).all():
        code, w = item.indicator_code, item.weight
        if w is None or w <= 0:
            continue
        if code in l1_map and l1_map[code] in merged["l1"]:
            merged["l1"][l1_map[code]] = w
            applied += 1
        elif code in l2_map:
            l1_name, l2_name = l2_map[code]
            if l1_name in merged["l2"] and l2_name in merged["l2"][l1_name]:
                merged["l2"][l1_name][l2_name] = w
                applied += 1
    if applied == 0:
        return copy.deepcopy(base), "论文表 3 默认权重", None
    return merged, "自定义方案（%d 项覆盖论文表 3）" % applied, scheme
def save_weights(unit_type, code_weight_pairs, note=None):
    """把一组「指标编码 → 权重」写入生效方案（已存在则更新对应条目）。
    仅覆盖本次传入的条目，其余条目保留，便于分层次（一级 / 二级）
    分次录入而无需一次填完。
    """
    scheme = WeightScheme.query.filter_by(unit_type=unit_type,
                                          name=ACTIVE_NAME).first()
    if scheme is None:
        scheme = WeightScheme(unit_type=unit_type,
                              name=ACTIVE_NAME,
                              level=1, is_default=True)
        db.session.add(scheme)
```

```
            db.session.flush()                                          1501
        if note:                                                        1502
            scheme.level = scheme.level or 1                            1503
        if not scheme.is_default:                                       1504
            scheme.is_default = True                                    1505
        written = 0                                                     1506
        for code, w in code_weight_pairs:                               1507
            if w is None or w <= 0:                                     1508
                continue                                                1509
            item = WeightItem.query.filter_by(scheme_id=scheme.i        1510
        d,                                                              1511
                                                                        1512
        indicator_code=code).first()                                    1513
            if item is None:                                            1514
                item = WeightItem(scheme_id=scheme.id,                  1515
        indicator_code=code)                                            1516
                db.session.add(item)                                    1517
            item.weight = float(w)                                      1518
            written += 1                                                1519
        db.session.commit()                                             1520
        return scheme, written                                          1521
    def reset_weights(unit_type):                                       1522
        """删除该单元类型的自定义方案，恢复为论文表 3 的默认权重。"""                  1523
        schemes = WeightScheme.query.filter_by(unit_type=unit_ty        1524
        pe,                                                             1525
                                                                        1526
        name=ACTIVE_NAME).all()                                         1527
        n = 0                                                           1528
        for s in schemes:                                               1529
            WeightItem.query.filter_by(scheme_id=s.id).delete(          1530
                synchronize_session=False)                              1531
            db.session.delete(s)                                        1532
            n += 1                                                      1533
        db.session.commit()                                             1534
        return n                                                        1535
    def scheme_items(unit_type):                                        1536
        """返回该单元类型已保存的自定义权重条目 {指标编码： 权重}。"""                1537
        scheme = WeightScheme.query.filter_by(unit_type=unit_typ        1538
        e,                                                              1539
                                                                        1540
        name=ACTIVE_NAME).first()                                       1541
        if scheme is None:                                              1542
            return {}, None                                             1543
        items = {i.indicator_code: i.weight                             1544
                 for i in WeightItem.query.filter_by(scheme_id=s        1545
        cheme.id).all()}                                                1546
        return items, scheme                                            1547
    def summary():                                                      1548
        """各单元类型的权重来源概览，供权重首页展示。"""                          1549
        rows = []                                                       1550
```

```
    for unit_type in DEFAULT_WEIGHTS:
        weights, label, scheme = effective_weights(unit_type
    )
        rows.append({
            "unit_type": unit_type,
            "label": label,
            "custom": scheme is not None,
        })
    return rows
# ==== app/services/evaluation_service.py ====
# -*- coding: utf-8 -*-
"""适宜性评价引擎。
按论文式（９） 实现：
- 单元级：F' = Σ（二级指标权重 × 该二级指标下三级指标标准化值）
- 矿山级：F = Σ（Si × F'i），Si 为各微地质单元面积占比
- 等级判定按论文表 4。
"""
from app.utils.constants import DEFAULT_WEIGHTS, grade_of
from app.services import standardization as std
def score_unit(unit_type, data, weights=None):
    """计算单个微地质单元的评价因子 F'。
    返回
    ----
    dict：{ l1_scores: {一级指标: 得分}, unit_score: F', details:
    [...] }
    若某一级指标下存在"不适用"（标准化值为 None）的二级指标，
    则按其余项权重归一化后计算，保证单元得分始终有定义。
    """
    if weights is None:
        weights = DEFAULT_WEIGHTS[unit_type]
    std_result = std.standardize_unit(unit_type, data,
    weights)
    l1_weights = weights["l1"]
    l1_scores = {}
    details = []
    for l1_name, items in std_result.items():
        wsum = 0.0
        acc = 0.0
        for l2_name, (raw, std_value, w2) in items.items():
            if std_value is None:
                # 不适用项：不计入
                details.append({
                    "l1": l1_name, "l2": l2_name, "raw":
    raw,
                    "std_value": None, "weight": w2,
    "contribution": None,
                })
                continue
            contribution = std_value * w2
            acc += contribution
```

```
                wsum += w2                                                    1601
                details.append({                                              1602
                    "l1": l1_name, "l2": l2_name, "raw": raw,                 1603
                    "std_value": std_value, "weight": w2,                     1604
                    "contribution": contribution,                             1605
                })                                                            1606
            l1_scores[l1_name] = acc / wsum if wsum > 0 else 0.0              1607
        #  一级指标权重按实际参与计算的指标归一化。                            1608
        #  论文表 3 的权重本身已归一（和为 1），此处为恒等变换；               1609
        #  若某层指标缺失或自定义权重不完整，可避免单元得分被整体压低。        1610
        lw_sum = sum(l1_weights[k] for k in l1_scores if k in                 1611
        l1_weights)                                                           1612
        if lw_sum > 0:                                                        1613
            unit_score = sum(l1_weights[k] * l1_scores[k]                     1614
                             for k in l1_scores if k in                       1615
        l1_weights) / lw_sum                                                  1616
        else:                                                                 1617
            unit_score = 0.0                                                  1618
        return {"l1_scores": l1_scores, "unit_score":                         1619
        unit_score, "details": details}                                       1620
def score_mine(units, weights_map=None):                                      1621
    """计算整个矿山的评价因子 F 与适宜性等级。                                1622
    参数                                                                      1623
    ----                                                                      1624
    units : list[dict]，每个元素：                                            1625
        {unit_type, ratio, data}                                              1626
        ratio 为该单元面积占比（0~1），data 为实测数据 dict。                 1627
    weights_map : dict，{单元类型: 权重结构}。未提供的单元类型                1628
        使用论文表 3 的默认权重。权重可来自 AHP 计算并经                      1629
        weight_service 保存生效，从而真正参与评价。                          1630
    返回                                                                      1631
    ----                                                                      1632
    dict:{ final_score, grade, grade_info, unit_results }                    1633
    """                                                                       1634
    weights_map = weights_map or {}                                           1635
    total = 0.0                                                               1636
    unit_results = []                                                         1637
    for u in units:                                                           1638
        r = score_unit(u["unit_type"], u["data"],                             1639
                       weights_map.get(u["unit_type"]))                       1640
        weighted = u["ratio"] * r["unit_score"]                               1641
        total += weighted                                                     1642
        unit_results.append({                                                 1643
            "unit_type": u["unit_type"],                                      1644
            "ratio": u["ratio"],                                              1645
            "unit_score": r["unit_score"],                                    1646
            "weighted": weighted,                                             1647
            "l1_scores": r["l1_scores"],                                      1648
            "details": r["details"],                                          1649
        })                                                                    1650
```

```
    grade_info = grade_of(total)
    return {
        "final_score": total,
        "grade": grade_info["grade"],
        "grade_info": grade_info,
        "unit_results": unit_results,
    }
# ==== app/services/report_service.py ====
# -*- coding: utf-8 -*-
"""报告生成：生成 Word 评价报告。
依赖 python-docx。报告结构：
标题 → 图斑信息 → 评价结论 → 各微地质单元得分 → 完整得分明细 → 生成时间。
"""
import io
from datetime import datetime
from docx import Document
from docx.shared import Pt, Cm
from docx.oxml.ns import qn
from docx.enum.text import WD_ALIGN_PARAGRAPH
def _set_run(run, text=None, size=12, bold=False):
    if text is not None:
        run.text = text
    run.font.name = "Times New Roman"
    run.font.size = Pt(size)
    run.font.bold = bold
    # 中文字体
    run._element.rPr.rFonts.set(qn("w:eastAsia"), "宋体")
    return run
def _para(doc, text, size=12, bold=False, align=None,
    space_after=6):
    p = doc.add_paragraph()
    if align is not None:
        p.alignment = align
    p.paragraph_format.space_after = Pt(space_after)
    _set_run(p.add_run(), text, size, bold)
    return p
def _heading(doc, text, level=1):
    h = doc.add_heading(text, level=level)
    for run in h.runs:
        run.font.name = "Times New Roman"
        run._element.rPr.rFonts.set(qn("w:eastAsia"), "黑体")
    return h
def _table(doc, headers, rows, widths=None):
    t = doc.add_table(rows=1, cols=len(headers))
    t.style = "Table Grid"
    hdr = t.rows[0].cells
    for i, htext in enumerate(headers):
        _set_run(hdr[i].paragraphs[0].add_run(), htext,
    10.5, bold=True)
    for row in rows:
```

```
        cells = t.add_row().cells                                    1701
        for i, val in enumerate(row):                                1702
            _set_run(cells[i].paragraphs[0].add_run(),               1703
    str(val), 10.5)                                                  1704
    return t                                                         1705
def build_report(patch, mine_result):                                1706
    """生成评价报告 Word 文档，返回 BytesIO。                          1707
    参数                                                             1708
    ----                                                             1709
    patch : MinePatch，图斑对象                                      1710
    mine_result : evaluation_service.score_mine 的返回 dict          1711
    """                                                              1712
    doc = Document()                                                 1713
    # 页边距                                                         1714
    for section in doc.sections:                                     1715
        section.top_margin = Cm(2.5)                                 1716
        section.bottom_margin = Cm(2.5)                              1717
        section.left_margin = Cm(2.5)                                1718
        section.right_margin = Cm(2.5)                               1719
    _heading(doc, "历史遗留废弃矿山自然恢复适宜性评价报告", 0)          1720
    _para(doc, "软件：历史遗留废弃矿山自然恢复适宜性评价系统 Ｖ1．0",    1721
          size=10.5, align=WD_ALIGN_PARAGRAPH.CENTER)                1722
    _para(doc, "生成时间：%s" % datetime.now().strftime("%Y-%m-%d    1723
    %H：%M：%S"),                                                    1724
          size=10.5, align=WD_ALIGN_PARAGRAPH.CENTER,                1725
    space_after=12)                                                  1726
    # 图斑信息                                                       1727
    _heading(doc, "一、图斑基本信息", 1)                             1728
    info = [                                                         1729
        ("图斑名称", patch.name),                                    1730
        ("所在行政区", patch.admin_region or "—"),                   1731
        ("图斑面积（公顷）", str(patch.area_ha) if patch.area_ha is  1732
    not None else "—"),                                              1733
        ("开采矿种", patch.mineral_type or "—"),                     1734
        ("开采方式", patch.mining_method or "—"),                    1735
    ]                                                                1736
    _table(doc, ["项目", "内容"], info)                              1737
    # 评价结论                                                       1738
    _heading(doc, "二、评价结论", 1)                                 1739
    gi = mine_result["grade_info"]                                   1740
    concl = [                                                        1741
        ("最终评价因子 F", "%.4f" % mine_result["final_score"]),     1742
        ("适宜性等级", mine_result["grade"]),                        1743
        ("判定依据", gi["meaning"]),                                 1744
        ("推荐措施", gi["measure"]),                                 1745
    ]                                                                1746
    _table(doc, ["项目", "结果"], concl)                             1747
    # 各单元得分                                                     1748
    _heading(doc, "三、各微地质单元得分", 1)                         1749
    rows = []                                                        1750
```

```
    for u in mine_result["unit_results"]:
        rows.append((u["unit_type"], "%.2f%%" % (u["ratio"] * 100),
                     "%.4f" % u["unit_score"], "%.4f" % u["weighted"]))
    _table(doc, ["微地质单元", "面积占比", "单元评价因子 F'", "加权贡献"], rows)
    # 完整得分明细
    _heading(doc, "四、得分明细", 1)
    for u in mine_result["unit_results"]:
        _para(doc, "%s（占比 %.2f%%）" % (u["unit_type"], u["ratio"] * 100),
              size=11, bold=True, space_after=4)
        detail_rows = []
        for d in u["details"]:
            raw = "—" if d["raw"] is None else d["raw"]
            std = "—" if d["std_value"] is None else "%.4f" % d["std_value"]
            w = "%.4f" % d["weight"]
            c = "—" if d["contribution"] is None else "%.4f" % d["contribution"]
            detail_rows.append((d["l1"], d["l2"], raw, std, w, c))
        _table(doc, ["一级指标", "二级指标", "实测值", "标准化值", "权重", "贡献"],
               detail_rows)
        # 一级指标得分
        l1_row = []
        for k, v in u["l1_scores"].items():
            l1_row.append("%s=%.4f" % (k, v))
        _para(doc, "一级指标得分：" + "，".join(l1_row), size=10.5, space_after=10)
    buf = io.BytesIO()
    doc.save(buf)
    buf.seek(0)
    return buf
# ==== app/services/indicator_service.py ====
# -*- coding: utf-8 -*-
"""指标定义维护。
把论文表 2/表 3 的指标体系写入 indicators 表，使指标名称、单位、
理想值与取值依据可在界面上查看维护，而不再只存在于代码常量中。
代码常量（app.utils.constants）仍是计算时的**权威取值来源**；
本表用于展示、核对与记录取值依据。二者由 seed_indicators() 保持同步。
"""
from app.models import db, Indicator
from app.utils.constants import (
    UNIT_TYPES, UNIT_ENABLED_INDICATORS, IDEAL_SOIL_RATIO,
    IDEAL_THICKNESS,
    IDEAL_PH_RANGE, IDEAL_ORGANIC, IDEAL_COVERAGE,
    IDEAL_SHANNON,
```

```
    HEAVY_METAL_LIMIT, SLOPE_BREAK, ASPECT_SCORES,
)
ALL_UNITS = ",".join(UNIT_TYPES)
# (code, name, level, parent, unit, ideal_value,
    ideal_type, enabled_units, desc, is_constraint)
TREE = [
    # ---- 一级指标 ----
    ("soil", "土壤指标", 1, None, None, None, None, "碎石边坡,平台",
    "论文表 2 评价指标之一级指标", False),
    ("veg", "植被指标", 1, None, None, None, None, ALL_UNITS,
    "论文表 2 评价指标之一级指标", False),
    ("terrain", "地形指标", 1, None, None, None, None,
    "岩质边坡,碎石边坡", "论文表 2 评价指标之一级指标", False),
    # ---- 二级指标：土壤 ----
    ("soil_ratio", "土石比", 2, "soil", "—",
    str(IDEAL_SOIL_RATIO), "benefit", "碎石边坡,平台",
     "论文 4.2.2 节：理想值 0.8，根据矿山修复实际情况确定", False),
    ("thickness", "土层厚度", 2, "soil", "cm",
    str(IDEAL_THICKNESS), "benefit", "碎石边坡,平台",
     "论文 4.2.2 节：理想值 20 cm", False),
    ("ph", "pH", 2, "soil", "—", "%g~%g" % IDEAL_PH_RANGE,
    "mid", "碎石边坡,平台",
     "论文 4.2.2 节：pH 用适中指标的标准化方法，理想区间 %g~%g" %
    IDEAL_PH_RANGE, False),
    ("organic_matter", "有机质", 2, "soil", "g/kg",
    str(IDEAL_ORGANIC), "benefit", "碎石边坡,平台",
     "《全国第二次土壤普查养分等级划分标准》二等（较丰富）最低值 30 g/kg", False),
    ("pollution", "污染指标", 2, "soil", "mg/kg", "见三级指标",
    "cost", "碎石边坡,平台",
     "同时作为限制性指标与评价指标；8 种重金属等权合成（各 0.125）", True),
    # ---- 二级指标：植被 ----
    ("coverage", "植被覆盖度", 2, "veg", "%",
    str(IDEAL_COVERAGE), "benefit", ALL_UNITS,
     "论文 4.2.2 节：参照生态系统取值 75%", False),
    ("shannon", "生物多样性", 2, "veg", "—", str(IDEAL_SHANNON),
    "benefit", ALL_UNITS,
     "Shannon-Wiener 多样性指数，参照生态系统取值 0.7", False),
    # ---- 二级指标：地形 ----
    ("slope", "坡度", 2, "terrain", "°", "%g° 分界" %
    SLOPE_BREAK, "cost", "岩质边坡,碎石边坡",
     "论文 4.2.2 节引《新一轮退耕还林还草总体方案》发改西部〔2014〕1772 号：25°
    为陡坡重要分界", False),
    ("aspect", "坡向", 2, "terrain", "—",
    "阳坡0.2/阴阳坡0.6/阴坡1.0", "discrete", "岩质边坡,碎石边坡",
     "论文 4.2.2 节：阴坡及半阴坡水分条件更优，故赋值高于阳坡", False),
]
# 污染指标下的三级指标（8 种重金属）
for _m, _limit in HEAVY_METAL_LIMIT.items():
    TREE.append((
        "hm_" + _m, _m, 3, "pollution", "mg/kg",
```

```
        str(_limit), "cost",
            "碎石边坡,平台",
            "《绿化种植土壤》CJ/T 340—2016 土壤重金属含量 IV 级最高值 %g mg/kg" %
        _limit,
            False,
        ))
def seed_indicators():
    """写入/更新指标定义（幂等）。返回写入条数。"""
    code_map = {
        "benefit": "越大越优", "cost": "越小越优",
        "mid": "适中型", "discrete": "离散赋值",
    }
    n = 0
    for (code, name, level, parent, unit, ideal, itype,
    units, desc, is_con) in TREE:
        ind = Indicator.query.filter_by(code=code).first()
        if ind is None:
            ind = Indicator(code=code)
            db.session.add(ind)
        ind.name = name
        ind.level = level
        ind.parent_code = parent
        ind.unit = unit
        ind.ideal_value = ideal
        ind.ideal_type = itype
        ind.enabled_units = units
        ind.description = desc
        ind.is_constraint = is_con
        ind.is_evaluative = (level in (2, 3))
        n += 1
    db.session.commit()
    return n
def ideal_type_label(code):
    return {"benefit": "越大越优", "cost": "越小越优",
            "mid": "适中型", "discrete": "离散赋值"}.get(code,
    code or "—")
# ==== app/services/import_service.py ====
# -*- coding: utf-8 -*-
"""指标数据批量导入。
支持 CSV 与 Excel（.xlsx/.xls）。一行对应一个微地质单元，
按「单元类型」匹配图斑内已有的微地质单元后写入指标数据。
表头约定（缺失列按不适用处理）：
    单元类型， 土石比， 土层厚度， pH， 有机质，
    Hg, As, Ni, Cu, Cd, Pb, Cr, Zn,
    植被覆盖度， Shannon， 坡度， 坡向
"""
import csv
import io
import json
from app.utils.constants import HEAVY_METALS, ASPECT_TYPES,
```

```
    UNIT_TYPES
# 表头（顺序即导出模板顺序）
HEADERS = (["单元类型", "土石比", "土层厚度", "pH", "有机质"]
           + list(HEAVY_METALS)
           + ["植被覆盖度", "Shannon", "坡度", "坡向"])
FIELD_ALIASES = {
    "单元类型": "unit_type",
    "土石比": "soil_ratio",
    "土层厚度": "thickness",
    "pH": "ph", "PH": "ph", "ph": "ph",
    "有机质": "organic_matter",
    "植被覆盖度": "coverage",
    "Shannon": "shannon", "shannon": "shannon", "生物多样性":
    "shannon",
    "坡度": "slope",
    "坡向": "aspect",
}
for _m in HEAVY_METALS:
    FIELD_ALIASES[_m] = "hm_" + _m
def build_template_csv():
    """生成导入模板（含表头与示例行）。"""
    buf = io.StringIO()
    w = csv.writer(buf)
    w.writerow(HEADERS)
    # 示例：碎石边坡一行，说明每列含义
    row = {h: "" for h in HEADERS}
    row.update({"单元类型": "碎石边坡", "土石比": "0.34", "土层厚度": "10",
                "pH": "9.24", "有机质": "4.89", "Hg": "0.05",
    "As": "5.17",
                "Ni": "12.60", "Cu": "11.70", "Cd": "0.04",
    "Pb": "6.50",
                "植被覆盖度": "87.92", "Shannon": "0.60",
                "坡度": "28.75", "坡向": "阳坡"})
    w.writerow([row[h] for h in HEADERS])
    return buf.getvalue()
def _read_rows(filename, stream):
    """读取上传文件为「表头 -> 值」的字典列表。"""
    ext = filename.rsplit(".", 1)[-1].lower()
    if ext == "csv":
        text = stream.read().decode("utf-8-sig",
    errors="replace")
        reader = csv.DictReader(io.StringIO(text))
        return [dict(r) for r in reader]
    if ext in ("xlsx", "xls"):
        from openpyxl import load_workbook
        wb = load_workbook(io.BytesIO(stream.read()),
    data_only=True)
        ws = wb.active
        rows = list(ws.iter_rows(values_only=True))
        if not rows:
```

```
            return []
        header = [str(c).strip() if c is not None else ""
for c in rows[0]]
        out = []
        for r in rows[1:]:
            if all(c is None or str(c).strip() == "" for c
in r):
                continue
            out.append({header[i]: (r[i] if i < len(r) else
None)
                        for i in range(len(header)) if
header[i]})
        return out
    raise ValueError("不支持的文件类型：.%s" % ext)
def _num(value):
    if value is None:
        return None, None
    s = str(value).strip()
    if s == "":
        return None, None
    try:
        return float(s), None
    except ValueError:
        return None, "「%s」不是有效数值" % s
def parse_upload(filename, stream, unit_type_map):
    """解析上传文件。
    参数
    ----
    unit_type_map : dict,{单元类型: MicroUnit 对象}
    返回
    ----
    (parsed, errors)
      parsed : list[dict],每项 {unit_type, fields{...},
heavy_metals{}}
      errors : list[str],逐行问题说明
    """
    rows = _read_rows(filename, stream)
    parsed, errors = [], []
    if not rows:
        return [], ["文件为空或没有数据行"]
    # 表头校验
    unknown = [h for h in (rows[0].keys() if rows else [])
if h not in FIELD_ALIASES]
    if unknown:
        errors.append("存在无法识别的列，将被忽略：%s" %
"、".join(map(str, unknown)))
    for idx, raw in enumerate(rows, start=2):       # 第 1 行是表头
        rec = {}
        for k, v in raw.items():
            key = FIELD_ALIASES.get(str(k).strip())
```

```
            if key:                                                    2001
                rec[key] = v                                           2002
        unit_type = str(rec.get("unit_type") or "").strip()            2003
        if not unit_type:                                              2004
            errors.append("第 %d 行：缺少「单元类型」" % idx)              2005
            continue                                                   2006
        if unit_type not in unit_type_map:                             2007
            errors.append("第 %d 行：图斑内不存在单元类型「%s」（现有：%s）"  2008
                          % (idx, unit_type,                           2009
"、".join(unit_type_map) or "无"))                                      2010
            continue                                                   2011
        if unit_type not in UNIT_TYPES:                                2012
            errors.append("第 %d 行：单元类型「%s」不是系统支持的三种之一"    2013
                          % (idx, unit_type))                          2014
            continue                                                   2015
        fields, heavy = {}, {}                                         2016
        ok = True                                                      2017
        for key, val in rec.items():                                   2018
            if key in ("unit_type",):                                  2019
                continue                                               2020
            if key.startswith("hm_"):                                  2021
                n, err = _num(val)                                     2022
                if err:                                                2023
                    errors.append("第 %d 行：%s %s" % (idx,            2024
key[3:], err))                                                         2025
                    ok = False                                         2026
                elif n is not None:                                    2027
                    heavy[key[3:]] = n                                 2028
                continue                                               2029
            if key == "aspect":                                        2030
                s = str(val).strip() if val is not None                2031
else ""                                                                2032
                if s == "":                                            2033
                    fields["aspect"] = None                            2034
                elif s in ASPECT_TYPES:                                2035
                    fields["aspect"] = s                               2036
                else:                                                  2037
                    errors.append("第 %d 行：坡向「%s」无效（应为 %s）"  2038
                                  % (idx, s,                           2039
"、".join(ASPECT_TYPES)))                                              2040
                    ok = False                                         2041
                continue                                               2042
            n, err = _num(val)                                         2043
            if err:                                                    2044
                errors.append("第 %d 行：%s %s" % (idx, key,           2045
err))                                                                  2046
                ok = False                                             2047
            else:                                                      2048
                fields[key] = n                                        2049
        if ok:                                                         2050
```

```
                parsed.append({"unit_type": unit_type,                  2051
    "fields": fields,                                                   2052
                                      "heavy_metals": heavy})           2053
    return parsed, errors                                               2054
#  ==== app/services/workflow.py ====                                  2055
#  -*- coding: utf-8 -*-                                               2056
"""评价流程前置条件（定量评价与报告输出的流程门禁）。                  2057
论文 ４．１ 节确定的评价流程存在严格的先后关系：                       2058
    建立图斑 → 限制性条件判定（通过） → 划分微地质单元并计算面积占比  2059
    → 逐单元录入指标数据 → 计算评价因子并判定等级 → 输出报告           2060
论文原文：「先进行政策、位置约束地质灾害判断……对于满足限制性条件的   2061
废弃矿山，按照地形地貌划分，将其划分为平台、碎石边坡和岩质边坡三种   2062
地质微单元类型，并计算各种地形面积占比。」限制性条件判定（一票否决）  2063
在前，微地质单元划分在后，顺序不可颠倒。                             2064
本模块把该顺序固化为可检查的状态。后续模块在入口处调用它，           2065
前置步骤未完成时直接拦截并给出「缺什么、去哪补」的提示，             2066
避免在条件不具备时产出看似合理的评价结论。                           2067
"""                                                                     2068
from app.models import (db, MicroUnit, IndicatorData,                   2069
    ConstraintCheck,                                                    2070
                                  EvaluationResult)                     2071
#  指标数据非空字段（用于判断某单元是否已录入数据）                    2072
DATA_FIELDS = ("soil_ratio", "thickness", "ph",                         2073
    "organic_matter",                                                   2074
                "coverage", "shannon", "slope", "aspect")               2075
def _has_data(ind):                                                     2076
    if ind is None:                                                     2077
        return False                                                    2078
    if any(getattr(ind, f, None) is not None for f in                   2079
    DATA_FIELDS):                                                       2080
        return True                                                     2081
    return bool(ind.heavy_metals_dict())                                2082
def patch_progress(patch_id):                                           2083
    """检查某图斑的评价流程完成情况。                                  2084
    返回 dict：                                                         2085
        units        {done, count, ratio_sum, detail}                   2086
        constraint   {done, passed, detail}                             2087
        data         {done, ready, missing, detail}                     2088
        blocking     未满足的前置步骤列表（按流程顺序）                 2089
        ready        是否可进入定量评价                                 2090
    """                                                                 2091
    units = MicroUnit.query.filter_by(patch_id=patch_id).all2092
    ()                                                                  2093
    ratio_sum = sum(u.ratio or 0.0 for u in units)                      2094
    #  ---- 步骤一：微地质单元 ----                                     2095
    units_step = {                                                      2096
        "done": bool(units),                                            2097
        "count": len(units),                                            2098
        "ratio_sum": ratio_sum,                                         2099
        "detail": ("已划分 %d 个微地质单元，面积占比合计 %.2f%%"        2100
```

```
                    % (len(units), ratio_sum * 100))
                   if units else "尚未划分微地质单元",
}
#  ----  步骤二：限制性条件判定  ----
cc = ConstraintCheck.query.filter_by(patch_id=patch_id)
\
    .order_by(ConstraintCheck.id.desc()).first()
if cc is None:
    constraint_step = {"done": False, "passed": False,
                       "detail": "尚未执行限制性条件判定"}
elif cc.passed:
    constraint_step = {"done": True, "passed": True,
                       "detail": "限制性条件判定已通过"}
else:
    constraint_step = {"done": True, "passed": False,
                       "detail": "限制性条件判定未通过（%s）" %
(cc.reason or "")}
#  ----  步骤三：指标数据  ----
missing = []
for u in units:
    ind = IndicatorData.query.filter_by(unit_id=u.id).fi
rst()
    if not _has_data(ind):
        missing.append(u.unit_type)
if not units:
    data_step = {"done": False, "ready": False,
"missing": [],
                 "detail": "尚无可录入数据的微地质单元"}
elif missing:
    data_step = {"done": False, "ready": False,
"missing": missing,
                 "detail": "以下微地质单元尚未录入指标数据：%s"
                           % "、".join(missing)}
else:
    data_step = {"done": True, "ready": True,
"missing": [],
                 "detail": "全部 %d 个微地质单元的指标数据已录入" %
len(units)}
ratio_ok = ratio_sum > 0.999 if units else False
if units and not ratio_ok:
    units_step["detail"] += "（占比合计不等于 100%，请检查面积录入）"
#  ----  汇总未满足的前置条件（按论文 4.1 节的流程顺序：限制性判定优先）  ----
#  reason 只写「缺什么」这一个事实，不写为什么重要的说明 ——
#  缺少什么由 label 指明，怎么补由 fix_label 给出，页面不必再解释一遍。
blocking = []
if not constraint_step["done"]:
    blocking.append({
        "step": "constraint", "label": "限制性条件判定未执行",
        "reason": "五项限制性条件尚未逐项判定。",
        "fix_url": "/constraint/%d" % patch_id,
```

```
        "fix_label": "去执行判定",
        })
    elif not constraint_step["passed"]:
        blocking.append({
            "step": "constraint", "label": "限制性条件判定未通过",
            "reason": cc.reason or "",
            "fix_url": "/constraint/%d" % patch_id,
        "fix_label": "去复核判定结果",
        })
    if not units_step["done"]:
        blocking.append({
            "step": "units", "label": "未划分微地质单元",
            "reason": "该图斑下没有任何微地质单元。",
            "fix_url": "/unit/%d" % patch_id, "fix_label":
    "去划分",
        })
    elif not ratio_ok:
        blocking.append({
            "step": "units", "label": "面积占比合计不为 100%",
            "reason": "当前合计 %.2f%%。" % (ratio_sum * 100),
            "fix_url": "/unit/%d" % patch_id, "fix_label":
    "去核对面积",
        })
    if units_step["done"] and not data_step["ready"]:
        blocking.append({
            "step": "data", "label": "指标数据未录入完整",
            "reason": data_step["detail"],
            "fix_url": "/indicator/", "fix_label": "去录入",
        })
    return {
        "units": units_step,
        "constraint": constraint_step,
        "data": data_step,
        "blocking": blocking,
        "ready": not blocking,
    }
def latest_result(patch_id):
    """取该图斑最近一次的评价记录。"""
    return EvaluationResult.query.filter_by(patch_id=patch_i
    d) \
        .order_by(EvaluationResult.id.desc()).first()
def report_progress(patch_id):
    """报告输出的前置条件：评价流程齐备，且已有评价记录。
    报告本质上是评价结论的呈现，因此必须先真实执行过定量评价，
    导出时读取已保存的评价结果，而不是在导出环节重新计算。
    否则评价完成后再修改指标数据，报告会与系统记录的评价结论不一致，
    同一份评价出现两个数字，结论无法追溯。
    返回 dict：{ready, blocking, result, flow}
        result 为最近一次评价记录（EvaluationResult），无记录时为空。
    """
```

```
    flow = patch_progress(patch_id)
    blocking = list(flow["blocking"])
    record = latest_result(patch_id)
    if not blocking and record is None:
        blocking.append({
            "step": "evaluate", "label": "尚未执行适宜性评价",
            "reason": "该图斑还没有评价记录。",
            "fix_url": "/evaluate/%d" % patch_id,
            "fix_label": "去执行评价",
        })
    return {
        "ready": not blocking,
        "blocking": blocking,
        "result": record,
        "flow": flow,
    }
# ==== app/services/dashboard.py ====
# -*- coding: utf-8 -*-
"""工作台数据：统计数字、评价流程步骤状态、侧边栏待办角标。
首页（工作台）需要把「现在有哪些数据」「流程走到哪一步」「哪一步还差活」
一次算清楚，供模板直接渲染。集中在此，避免在视图与模板里各算一遍。
"""
from flask import g
from app.models import (db, Project, MinePatch, MicroUnit,
    IndicatorData,
                        ConstraintCheck, EvaluationResult,
    Standard)
from app.services import workflow as wf
#  评价流程的八个步骤（顺序即论文 4.1 节的作业顺序）
#    key          步骤标识
#    no           步骤编号（01~08）
#    title        名称
#    icon         图标名（见 templates/_macros.html 的 ICONS）
#    desc         一句话说明
#    href         入口地址
STEPS = [
    dict(key="project", no="01", title="项目与矿山档案",
    icon="folder",
          href="/project/",
          desc="建立评价项目、录入矿山图斑、挂接遥感影像、设置参照生态系统"),
    dict(key="standard", no="02", title="政策与标准知识库",
    icon="book",
          href="/standard/",
          desc="收录地方标准与政策文件，结构化判定规则，支持省际口径比对"),
    dict(key="constraint", no="03", title="限制性指标判定",
    icon="shield",
          href="/constraint/",
          desc="政策、区位约束、土壤污染、地质灾害、耕地属性五项一票否决"),
    dict(key="unit", no="04", title="微地质单元", icon="layers",
          href="/unit/",
```

```
            desc="划分岩质边坡、碎石边坡、平台三类单元，自动计算面积占比"),
        dict(key="ahp", no="05", title="指标体系与权重",
        icon="sliders",
            href="/ahp/",
            desc="指标定义维护，判断矩阵录入、权重计算与一致性检验、多专家汇总"),
        dict(key="indicator", no="06", title="指标数据采集",
        icon="database",
            href="/indicator/",
            desc="按微地质单元录入土壤、植被、地形指标，支持 CSV / Excel 批量导入"),
        dict(key="evaluate", no="07", title="适宜性定量评价",
        icon="target",
            href="/evaluate/",
            desc="指标标准化、分单元赋分、面积加权合成，判定适宜性等级"),
        dict(key="report", no="08", title="报告与结果输出",
        icon="file",
            href="/report/",
            desc="结果图表与逐层得分明细，导出 PDF 与 Word 评价报告"),
]
def _has_data(ind):
    return wf._has_data(ind)
#  系统的九个功能模块（登录页介绍用，含不在评价流程内的账号与权限管理）
MODULES = [
    ("01", "账号与权限管理", "自助注册与登录注销、管理员与操作员两种职务、写操作自动留痕"),
    ("02", "项目与矿山档案", "建立项目、录入图斑、挂接遥感影像、设置参照生态系统"),
    ("03", "政策与标准知识库", "标准判定条件结构化为规则，支持省际口径比对"),
    ("04", "限制性指标判定", "政策、区位、污染、地质灾害、耕地五项一票否决"),
    ("05", "微地质单元管理", "划分岩质边坡、碎石边坡、平台，自动算面积占比"),
    ("06", "指标体系与权重", "指标定义维护、AHP 判断矩阵与一致性检验、多专家汇总"),
    ("07", "指标数据采集", "按单元类型录入土壤、植被、地形指标，支持批量导入"),
    ("08", "适宜性定量评价", "四类标准化、分单元赋分、面积加权、等级判定"),
    ("09", "报告与结果输出", "结果图表与逐层得分明细，导出 PDF 与 Word 报告"),
]
def snapshot():
    """一次算出工作台需要的全部数据（同一请求内只算一次）。
    返回 dict：
        stats       {projects, patches, unit_types, standards}
        steps       八个步骤，附 available / pending / note
        badges      {key: 待办数量}，仅含有待办的项
        progress    {patches, units, data_units, ready,
    evaluated}
    """
    cached = getattr(g, "_dash_snapshot", None)
    if cached is None:
        cached = _compute()
        g._dash_snapshot = cached
    return cached
def _compute():
    projects = Project.query.count()
    patches = MinePatch.query.all()
    patch_ids = [p.id for p in patches]
```

```
    units = MicroUnit.query.filter(                                    2301
        MicroUnit.patch_id.in_(patch_ids)).all() if                    2302
patch_ids else []                                                      2303
    unit_by_patch = {}                                                 2304
    for u in units:                                                    2305
        unit_by_patch.setdefault(u.patch_id, []).append(u)             2306
    unit_ids = [u.id for u in units]                                   2307
    data_unit_ids = set()                                              2308
    if unit_ids:                                                       2309
        for ind in IndicatorData.query.filter(                         2310
                IndicatorData.unit_id.in_(unit_ids)).all():            2311
            if _has_data(ind):                                         2312
                data_unit_ids.add(ind.unit_id)                         2313
    passed_constraint = set()                                          2314
    if patch_ids:                                                      2315
        for c in ConstraintCheck.query.filter(                         2316
                ConstraintCheck.patch_id.in_(patch_ids)).all2317
():                                                                    2318
            if c.passed:                                               2319
                passed_constraint.add(c.patch_id)                      2320
    evaluated = set()                                                  2321
    if patch_ids:                                                      2322
        for r in EvaluationResult.query.filter(                        2323
                EvaluationResult.patch_id.in_(patch_ids)).al2324
l():                                                                   2325
            evaluated.add(r.patch_id)                                  2326
    # ---- 各步骤的待办数量 ----                                       2327
    pending = {                                                        2328
        "constraint": sum(1 for p in patches if p.id not in 2329
passed_constraint),                                                    2330
        "unit": sum(1 for p in patches if not                          2331
unit_by_patch.get(p.id)),                                              2332
        "indicator": 0,                                                2333
        "evaluate": 0,                                                 2334
    }                                                                  2335
    ready_patches = 0                                                  2336
    for p in patches:                                                  2337
        us = unit_by_patch.get(p.id, [])                               2338
        if us and any(u.id not in data_unit_ids for u in       2339
us):                                                                   2340
            pending["indicator"] += 1                                  2341
        pr = wf.patch_progress(p.id)                                   2342
        if pr["ready"]:                                                2343
            ready_patches += 1                                         2344
            if p.id not in evaluated:                                  2345
                pending["evaluate"] += 1                               2346
    # ---- 步骤可用性：前置数据不具备时置灰 ----                       2347
    avail = {                                                          2348
        "project": True,                                               2349
        "standard": True,                                              2350
```

```
        "constraint": bool(patches),
        "unit": bool(patches),
        "ahp": True,
        "indicator": bool(units),
        "evaluate": ready_patches > 0,
        "report": bool(evaluated),
    }
    notes = {
        "constraint": "需先建立矿山图斑",
        "unit": "需先建立矿山图斑",
        "indicator": "需先划分微地质单元",
        "evaluate": "需前置步骤齐备的图斑",
        "report": "需先执行适宜性评价",
    }
    steps = []
    for s in STEPS:
        steps.append(dict(
            s,
            available=avail[s["key"]],
            pending=pending.get(s["key"], 0),
            note=None if avail[s["key"]] else
    notes.get(s["key"], ""),
        ))
    unit_types = len({u.unit_type for u in units})
    stats = {
        "projects": projects,
        "patches": len(patches),
        "unit_types": unit_types,
        "standards": Standard.query.count(),
    }
    return {
        "stats": stats,
        "steps": steps,
        "badges": {k: v for k, v in pending.items() if v},
        "progress": {
            "patches": len(patches),
            "units": len(units),
            "data_units": len(data_unit_ids),
            "ready": ready_patches,
            "evaluated": len(evaluated),
        },
    }
def nav_badges():
    """侧边栏待办角标：{入口地址前缀： 数量}，只含有待办的项。"""
    snap = snapshot()
    return {
        "/constraint/": snap["badges"].get("constraint", 0),
        "/unit/": snap["badges"].get("unit", 0),
        "/indicator/": snap["badges"].get("indicator", 0),
        "/evaluate/": snap["badges"].get("evaluate", 0),
```

```
    }
# ==== app/services/audit.py ====
# -*- coding: utf-8 -*-
"""操作审计。
在应用工厂中以 after_request 钩子统一埋点：凡是成功的写操作（POST/PUT/DELETE）
且已登录，即写入 operation_logs 表，无需在每个视图里手工调用。
动作名称由「端点 → 中文动作」映射表给出，未收录的端点回退为方法 + 路径。
"""
from flask import request
from flask_login import current_user
from app.models import db, OperationLog
# 端点 → 可读动作名
ACTION_NAMES = {
    "auth.login": "登录系统",
    "auth.register": "注册账号",
    "auth.logout": "退出登录",
    "auth.add_user": "新增账号",
    "auth.edit_user": "编辑账号",
    "auth.delete_user": "删除账号",
    "auth.reset_password": "重置用户密码",
    "auth.profile": "修改个人资料",
    "auth.upload_avatar": "更新头像",
    "auth.delete_avatar": "移除头像",
    "project.add": "新建项目",
    "project.edit": "编辑项目",
    "project.delete": "删除项目",
    "project.patch_add": "新建矿山图斑",
    "project.patch_edit": "编辑矿山图斑",
    "project.patch_delete": "删除矿山图斑",
    "project.upload_image": "上传遥感影像",
    "unit.add": "添加微地质单元",
    "unit.edit": "编辑微地质单元",
    "unit.delete": "删除微地质单元",
    "standard.add": "添加标准/政策文件",
    "standard.edit": "编辑标准/政策文件",
    "standard.delete": "删除标准/政策文件",
    "standard.rules": "维护标准规则",
    "standard.rule_delete": "删除标准规则",
    "constraint.check": "执行限制性条件判定",
    "ahp.save_expert": "录入专家判断矩阵",
    "ahp.delete_expert": "删除专家判断矩阵",
    "ahp.aggregate": "汇总多专家矩阵",
    "indicator.edit": "保存指标数据",
    "indicator.import_data": "批量导入指标数据",
    "indicator.update_meta": "维护指标定义",
    "evaluate.run": "执行适宜性评价",
}
# 无需记录的操作（查看类或纯导航）
SKIP_ENDPOINTS = {"static", "auth.index"}
def action_name(endpoint):
```

```
    # 个人设置页的两个表单共用一个端点，按提交的动作区分
    if endpoint == "auth.profile":
        return "修改密码" if request.form.get("action") ==
    "password" \
               else "修改个人资料"
    if endpoint in ACTION_NAMES:
        return ACTION_NAMES[endpoint]
    return "写操作 %s %s" % (request.method, request.path)
def extract_target_id():
    """从视图参数中取出被操作对象的主键，取不到则返回 None。"""
    args = request.view_args or {}
    for key in ("pid", "patch_id", "unit_id", "sid", "rid",
    "result_id", "user_id"):
        if key in args:
            try:
                return int(args[key])
            except (TypeError, ValueError):
                return None
    return None
def record(endpoint, status_code):
    """记录一次写操作。失败（4xx/5xx）也记录，便于追溯。"""
    if endpoint in SKIP_ENDPOINTS:
        return
    try:
        detail = "路径 %s" % request.path
        if status_code >= 400:
            detail += "；结果 HTTP %d" % status_code
        db.session.add(OperationLog(
            user_id=current_user.id if
    current_user.is_authenticated else None,
            action=action_name(endpoint),
            target_type=(endpoint or "").split(".")[0] or
    None,
            target_id=extract_target_id(),
            detail=detail,
        ))
        db.session.commit()
    except Exception:
        # 审计失败不应影响业务响应
        db.session.rollback()
# ==== app/views/auth.py ====
# -*- coding: utf-8 -*-
"""账号与权限 —— 登录、注册、个人资料、用户维护、操作日志。
页面组织：
  /login          登录页（含「点我注册」入口）
  /register       自助注册（默认「操作员」）
  /profile        个人设置，分类切换：info / security / users / logs
                  后两个分类仅管理员可见，由本模块统一管控。
"""
from flask import (Blueprint, render_template, redirect,
```

```
    url_for, request,                                                    2501
                    flash, abort, current_app)                           2502
from flask_login import login_user, logout_user,                         2503
    login_required, current_user                                         2504
from werkzeug.security import check_password_hash,                       2505
    generate_password_hash                                               2506
from app.forms import (LoginForm, EmptyForm, ProfileForm,                2507
    PasswordForm,                                                        2508
                           RegisterForm, email_ok)                       2509
from app.models import db, User, OperationLog                            2510
from app.utils import uploads as up                                      2511
from app.utils.constants import ROLE_LABELS, DEFAULT_ROLE,               2512
    GENDERS                                                              2513
bp = Blueprint("auth", __name__)                                         2514
# 「个人设置」的四个分类；后两个是管理员专属                                2515
PROFILE_TABS = ("info", "security", "users", "logs")                     2516
ADMIN_TABS = ("users", "logs")                                           2517
def require_admin():                                                     2518
    """非管理员直接 403，避免误操作基础数据。"""                          2519
    if current_user.role != "admin":                                     2520
        abort(403)                                                       2521
def _intro_data():                                                       2522
    """登录页下方的系统介绍所用的静态内容。"""                            2523
    from app.services import dashboard as dash                           2524
    return {"modules": dash.MODULES}                                     2525
@bp.route("/")                                                           2526
def index():                                                             2527
    if current_user.is_authenticated:                                    2528
        from app.services import dashboard as dash                       2529
        return render_template("auth/home.html",                         2530
    dash=dash.snapshot())                                                2531
    return redirect(url_for("auth.login"))                               2532
@bp.route("/login", methods=["GET", "POST"])                             2533
def login():                                                             2534
    if current_user.is_authenticated:                                    2535
        return redirect(url_for("auth.index"))                           2536
    form = LoginForm()                                                   2537
    if form.validate_on_submit():                                        2538
        user = User.query.filter_by(username=form.username.d             2539
    ata).first()                                                         2540
        if user and check_password_hash(user.password_hash,              2541
    form.password.data):                                                 2542
            if not user.active:                                          2543
                flash("该账号已停用，请联系管理员", "toast-error")        2544
            else:                                                        2545
                login_user(user, remember=form.remember.data             2546
    )                                                                    2547
                # 用 toast 类别：渲染为页面中上方的浮层，淡入停留后淡出。 2548
                # 登录成功只是告知性反馈，不需要用户停下来阅读。          2549
                flash("登录成功", "toast")                               2550
```

```
                    return redirect(url_for("auth.index"))
               else:
                    flash("账户名称或密码错误", "toast-error")
          # 字段为空之类的校验失败，错误留在表单里逐项显示，不走浮层：
          # 那类消息要对着输入框看，一闪而过反而让人以为「点了没反应」。
          return render_template("auth/login.html", form=form,
          **_intro_data())
@bp.route("/register", methods=["GET", "POST"])
def register():
     """自助注册。新账号一律为「操作员」，权限只能由管理员事后调整。"""
     if current_user.is_authenticated:
          return redirect(url_for("auth.index"))
     form = RegisterForm()
     if form.validate_on_submit():
          username = form.username.data.strip()
          if User.query.filter_by(username=username).first():
               form.username.errors.append("该账户名称已被使用")
          else:
               user = User(username=username,
                           role=DEFAULT_ROLE,
                           password_hash=generate_password_hash
     (form.password.data),
                           display_name=(form.display_name.data
     or "").strip() or None,
                           gender=(form.gender.data or
     "").strip() or None,
                           email=(form.email.data or
     "").strip() or None,
                           phone=(form.phone.data or
     "").strip() or None)
               rel, err = _save_avatar(request.files.get("avata
     r"))
               if err:
                    flash(err, "error")
               else:
                    user.avatar = rel
                    db.session.add(user)
                    db.session.commit()
                    login_user(user)
                    flash("注册成功，欢迎使用", "toast")
                    return redirect(url_for("auth.index"))
     return render_template("auth/register.html", form=form)
@bp.route("/logout")
@login_required
def logout():
     logout_user()
     flash("已退出登录", "ok")
     return redirect(url_for("auth.login"))
# ------------------------------ 个人设置 ---------------------------
     ----
```

```
# 任何登录用户在这里维护自己；管理员额外能管理账号与查看日志。
# 「用户与操作日志」不再是独立菜单，而是本页的管理员分类。
@bp.route("/profile", methods=["GET", "POST"])
@login_required
def profile():
    is_admin = current_user.role == "admin"
    tab = request.values.get("tab") or "info"
    if tab not in PROFILE_TABS or (tab in ADMIN_TABS and
    not is_admin):
        tab = "info"
    # 表单放在同一页，用字段前缀区分，避免 submit / csrf_token 重名
    info_form = ProfileForm(prefix="info", obj=current_user)
    pwd_form = PasswordForm(prefix="pwd")
    if request.method == "POST":
        action = request.form.get("action")
        # 校验失败要重新渲染本页，此时必须回到提交表单所在的那个分类，
        # 否则错误提示会出现在另一个分类里，用户根本看不到。
        if action == "password":
            tab = "security"
        elif action == "info":
            tab = "info"
        if action == "info" and info_form.validate_on_submit
    ():
            current_user.display_name =
    (info_form.display_name.data or "").strip() or None
            current_user.gender = (info_form.gender.data or
    "").strip() or None
            current_user.email = (info_form.email.data or
    "").strip() or None
            current_user.phone = (info_form.phone.data or
    "").strip() or None
            db.session.commit()
            flash("个人资料已保存", "ok")
            return redirect(url_for("auth.profile",
    tab="info"))
        if action == "password" and pwd_form.validate_on_sub
    mit():
            if not check_password_hash(current_user.password
    _hash,

    pwd_form.old_password.data):
                pwd_form.old_password.errors.append("当前密码不正确
    ")
            else:
                current_user.password_hash =
    generate_password_hash(
                    pwd_form.new_password.data)
                db.session.commit()
                flash("密码已修改", "ok")
                return redirect(url_for("auth.profile",
```

```
    tab="security"))                                                    2651
    ctx = dict(info_form=info_form, pwd_form=pwd_form,                  2652
    avatar_form=EmptyForm(),                                            2653
                tab=tab, role_labels=ROLE_LABELS,                       2654
    genders=GENDERS)                                                    2655
    if tab == "users":                                                  2656
        ctx.update(_users_context())                                    2657
    elif tab == "logs":                                                 2658
        ctx.update(_logs_context())                                     2659
    return render_template("auth/profile.html", **ctx)                  2660
def _users_context():                                                   2661
    """管理员分类「账号」所需的数据。"""                                2662
    return {"users": User.query.order_by(User.id).all(),                2663
            "form": EmptyForm()}                                        2664
def _logs_context():                                                    2665
    """管理员分类「操作日志」所需的数据。"""                            2666
    page = request.args.get("page", 1, type=int)                        2667
    per_page = 50                                                       2668
    q = OperationLog.query.order_by(OperationLog.id.desc())             2669
    total = q.count()                                                   2670
    records = q.offset((page - 1) * per_page).limit(per_page2671
    ).all()                                                             2672
    users_map = {u.id: u.username for u in User.query.all()}2673
    pages = max(1, (total + per_page - 1) // per_page)                  2674
    return {"records": records, "page": page, "pages":                  2675
    pages,                                                              2676
            "total": total, "users_map": users_map}                     2677
# ---------------------------- 头像 ----------------------------2678
    --                                                                  2679
def _avatar_length_ok(file_storage):                                    2680
    """按字节检查头像大小。超限时 werkzeug 读到内存里的内容也能量出来。"""2681
    limit = current_app.config.get("MAX_AVATAR_LENGTH", 4 *  2682
    1024 * 1024)                                                        2683
    stream = file_storage.stream                                        2684
    pos = stream.tell()                                                 2685
    stream.seek(0, 2)                                                   2686
    size = stream.tell()                                                2687
    stream.seek(pos)                                                    2688
    return size <= limit, size                                          2689
def _save_avatar(file):                                                 2690
    """保存头像文件，返回 (相对路径, 错误信息)。没有文件时返回 (None, None)。"""2691
    if file is None or not file.filename:                               2692
        return None, None                                               2693
    ok, size = _avatar_length_ok(file)                                  2694
    if not ok:                                                          2695
        return None, ("头像不能超过 %.0f MB（当前 %.1f MB）"            2696
                      % (current_app.config["MAX_AVATAR_LENG2697
    TH"] / 1024 / 1024,                                                 2698
                         size / 1024 / 1024))                           2699
    return up.save_upload(file, "avatars",                              2700
```

```
                                current_app.config["ALLOWED_AVATAR
    _EXT"],
                                current_app.config["UPLOAD_FOLDER"
    ])
@bp.route("/profile/avatar", methods=["POST"])
@login_required
def upload_avatar():
    form = EmptyForm()
    back = url_for("auth.profile", tab=request.form.get("tab
    ") or "info")
    if not form.validate_on_submit():
        flash("表单校验失败，请重试", "error")
        return redirect(back)
    rel, err = _save_avatar(request.files.get("avatar"))
    if err:
        flash(err, "error")
        return redirect(back)
    if rel is None:
        flash("请选择头像图片", "error")
        return redirect(back)
    # 换新头像时删掉旧文件，避免残留
    up.remove_upload(current_user.avatar,
    current_app.config["UPLOAD_FOLDER"])
    current_user.avatar = rel
    db.session.commit()
    flash("头像已更新", "ok")
    return redirect(back)
@bp.route("/profile/avatar/delete", methods=["POST"])
@login_required
def delete_avatar():
    form = EmptyForm()
    back = url_for("auth.profile", tab=request.form.get("tab
    ") or "info")
    if form.validate_on_submit():
        up.remove_upload(current_user.avatar,
    current_app.config["UPLOAD_FOLDER"])
        current_user.avatar = None
        db.session.commit()
        flash("头像已移除", "ok")
    return redirect(back)
# ---------------------------- 账号维护（仅管理员）
    ----------------------------
@bp.route("/users")
@login_required
def users():
    """旧地址保留，统一并入「个人设置 · 账号」。"""
    return redirect(url_for("auth.profile", tab="users"))
@bp.route("/users/add", methods=["GET", "POST"])
@login_required
def add_user():
```

```
    require_admin()
    if request.method == "POST":
        username = request.form.get("username", "").strip()
        password = request.form.get("password", "")
        role = request.form.get("role", DEFAULT_ROLE)
        if role not in ROLE_LABELS:
            role = DEFAULT_ROLE
        if not username or not password:
            flash("账户名称和密码不能为空", "error")
            return redirect(url_for("auth.add_user"))
        if User.query.filter_by(username=username).first():
            flash("账户名称已存在", "error")
            return redirect(url_for("auth.add_user"))
        db.session.add(User(
            username=username, role=role,
            password_hash=generate_password_hash(password),
            display_name=request.form.get("display_name",
    "").strip() or None,
            gender=request.form.get("gender", "").strip()
    or None,
            email=request.form.get("email", "").strip() or
    None,
            phone=request.form.get("phone", "").strip() or
    None))
        db.session.commit()
        flash("账号已添加", "ok")
        return redirect(url_for("auth.profile",
    tab="users"))
    return render_template("auth/add_user.html",
    role_labels=ROLE_LABELS,
                           genders=GENDERS,
    default_role=DEFAULT_ROLE,
                           form=EmptyForm())
@bp.route("/users/<int:user_id>/edit", methods=["GET",
    "POST"])
@login_required
def edit_user(user_id):
    require_admin()
    user = db.get_or_404(User, user_id)
    if request.method == "POST":
        role = request.form.get("role", user.role)
        if role not in ROLE_LABELS:
            role = user.role
        active = request.form.get("active") == "1"
        # 保护：不能把最后一个管理员降级或停用，否则无人能管理系统
        if user.role == "admin" and (role != "admin" or not
    active):
            others = User.query.filter(User.role == "admin",
                                       User.id != user.id,
```

```
    User.active.is_(True)).count()
            if others == 0:
                flash("系统至少需保留一名启用状态的管理员", "error")
                return redirect(url_for("auth.edit_user",
    user_id=user.id))
        # 个人资料字段（用户自己也能在「个人设置」里改，管理员在此代改）
        email = request.form.get("email", "").strip()
        if not email_ok(email):
            flash("邮箱格式不正确", "error")
            return redirect(url_for("auth.edit_user",
    user_id=user.id))
        user.role = role
        user.active = active
        user.display_name = request.form.get("display_name",
    "").strip() or None
        user.gender = request.form.get("gender",
    "").strip() or None
        user.email = email or None
        user.phone = request.form.get("phone", "").strip()
    or None
        new_password = request.form.get("password",
    "").strip()
        if new_password:
            user.password_hash = generate_password_hash(new_
    password)
        db.session.commit()
        flash("账号信息已保存", "ok")
        return redirect(url_for("auth.profile",
    tab="users"))
    return render_template("auth/edit_user.html", user=user,
                           role_labels=ROLE_LABELS,
    genders=GENDERS,
                           form=EmptyForm())
@bp.route("/users/<int:user_id>/delete", methods=["POST"])
@login_required
def delete_user(user_id):
    require_admin()
    user = db.get_or_404(User, user_id)
    if user.id == current_user.id:
        flash("不能删除当前登录的账号", "error")
        return redirect(url_for("auth.profile",
    tab="users"))
    if user.role == "admin":
        others = User.query.filter(User.role == "admin",
                                   User.id != user.id,
                                   User.active.is_(True)).co
    unt()
        if others == 0:
            flash("系统至少需保留一名启用状态的管理员", "error")
            return redirect(url_for("auth.profile",
```

```
    tab="users"))
    db.session.delete(user)
    db.session.commit()
    flash("账号「%s」已删除" % user.username, "ok")
    return redirect(url_for("auth.profile", tab="users"))
# ---------------------------- 操作日志（仅管理员）
    ----------------------------
@bp.route("/logs")
@login_required
def logs():
    """旧地址保留，统一并入「个人设置 · 操作日志」。"""
    return redirect(url_for("auth.profile", tab="logs"))
# ==== app/views/project.py ====
# -*- coding: utf-8 -*-
"""项目与矿山档案 —— 项目、图斑、参照生态系统、遥感影像挂接。"""
from datetime import datetime
from flask import (Blueprint, render_template, request,
    redirect, url_for,
                   flash, current_app)
from flask_login import login_required, current_user
from app.forms import EmptyForm
from app.models import db, Project, MinePatch,
    ReferenceEcosystem
bp = Blueprint("project", __name__, url_prefix="/project")
@bp.route("/")
@login_required
def index():
    projects = Project.query.order_by(Project.id.desc()).all
    ()
    return render_template("project/index.html",
    projects=projects,
                           form=EmptyForm())
@bp.route("/add", methods=["GET", "POST"])
@login_required
def add():
    form = EmptyForm()
    if request.method == "POST" and form.validate_on_submit(
    ):
        p = Project(
            name=request.form.get("name", "").strip(),
            code=request.form.get("code", "").strip() or
    None,
            description=request.form.get("description",
    "").strip() or None,
            created_by=current_user.id,
        )
        if not p.name:
            flash("项目名称不能为空", "error")
        else:
            db.session.add(p)
```

```
            db.session.commit()                                    2901
            flash("项目已创建", "ok")                               2902
            return redirect(url_for("project.index"))              2903
    return render_template("project/edit.html", form=form,         2904
    project=None)                                                  2905
@bp.route("/<int:pid>/edit", methods=["GET", "POST"])              2906
@login_required                                                    2907
def edit(pid):                                                     2908
    p = db.get_or_404(Project, pid)                                2909
    form = EmptyForm()                                             2910
    if request.method == "POST" and form.validate_on_submit(       2911
    ):                                                             2912
        p.name = request.form.get("name", "").strip()              2913
        p.code = request.form.get("code", "").strip() or           2914
    None                                                           2915
        p.description = request.form.get("description",            2916
    "").strip() or None                                            2917
        p.status = request.form.get("status", "进行中")             2918
        db.session.commit()                                        2919
        flash("项目已保存", "ok")                                   2920
        return redirect(url_for("project.index"))                  2921
    return render_template("project/edit.html", form=form,         2922
    project=p)                                                     2923
@bp.route("/<int:pid>/delete", methods=["POST"])                   2924
@login_required                                                    2925
def delete(pid):                                                   2926
    p = db.get_or_404(Project, pid)                                2927
    db.session.delete(p)                                           2928
    db.session.commit()                                            2929
    flash("项目已删除", "ok")                                       2930
    return redirect(url_for("project.index"))                      2931
# ------------------------------ 矿山图斑 ---------------------------2932
    ----                                                           2933
@bp.route("/<int:pid>/patches")                                    2934
@login_required                                                    2935
def patches(pid):                                                  2936
    p = db.get_or_404(Project, pid)                                2937
    return render_template("project/patches.html",                 2938
    project=p,                                                     2939
                                form=EmptyForm())                  2940
@bp.route("/patch/<int:patch_id>/delete", methods=["POST"])        2941
@login_required                                                    2942
def patch_delete(patch_id):                                        2943
    """删除矿山图斑，及其微地质单元与指标数据。"""                    2944
    from app.models import MicroUnit, IndicatorData,               2945
    EvaluationResult, \                                            2946
        ScoreDetail, ConstraintCheck, ReferenceEcosystem           2947
    patch = db.get_or_404(MinePatch, patch_id)                     2948
    pid = patch.project_id                                         2949
    unit_ids = [u.id for u in MicroUnit.query.filter_by(patc       2950
```

```
    h_id=patch.id).all()]
    if unit_ids:
        IndicatorData.query.filter(IndicatorData.unit_id.in_
    (unit_ids)).delete(
            synchronize_session=False)
    for r in EvaluationResult.query.filter_by(patch_id=patch
    .id).all():
        ScoreDetail.query.filter_by(result_id=r.id).delete(
            synchronize_session=False)
    EvaluationResult.query.filter_by(patch_id=patch.id).dele
    te(
        synchronize_session=False)
    ConstraintCheck.query.filter_by(patch_id=patch.id).delet
    e(
        synchronize_session=False)
    MicroUnit.query.filter_by(patch_id=patch.id).delete(sync
    hronize_session=False)
    ReferenceEcosystem.query.filter_by(patch_id=patch.id).de
    lete(
        synchronize_session=False)
    db.session.delete(patch)
    db.session.commit()
    flash("图斑「%s」及其数据已删除" % patch.name, "ok")
    return redirect(url_for("project.patches", pid=pid))
@bp.route("/<int:pid>/patch/add", methods=["GET", "POST"])
@login_required
def patch_add(pid):
    p = db.get_or_404(Project, pid)
    form = EmptyForm()
    if request.method == "POST" and form.validate_on_submit(
    ):
        def fv(name, cast=str):
            v = request.form.get(name, "").strip()
            return cast(v) if v != "" else None
        patch = MinePatch(
            project_id=p.id,
            name=request.form.get("name", "").strip(),
            admin_region=fv("admin_region"),
            area_ha=fv("area_ha", float),
            mineral_type=fv("mineral_type"),
            mining_method=fv("mining_method"),
            stop_years=fv("stop_years", float),
            responsible_entity_missing=bool(request.form.get
    ("responsible_entity_missing")),
        )
        if not patch.name:
            flash("图斑名称不能为空", "error")
        else:
            db.session.add(patch)
            db.session.flush()
```

```
            # 参照生态系统默认取论文值
            db.session.add(ReferenceEcosystem(
                patch_id=patch.id, coverage_benchmark=75.0,
    shannon_benchmark=0.7))
            db.session.commit()
            flash("图斑已创建", "ok")
            return redirect(url_for("project.patches",
    pid=p.id))
    return render_template("project/patch_edit.html",
    form=form, project=p, patch=None)
@bp.route("/patch/<int:patch_id>/edit", methods=["GET",
    "POST"])
@login_required
def patch_edit(patch_id):
    patch = db.get_or_404(MinePatch, patch_id)
    form = EmptyForm()
    if request.method == "POST" and form.validate_on_submit(
    ):
        def fv(name, cast=str):
            v = request.form.get(name, "").strip()
            return cast(v) if v != "" else None
        patch.name = request.form.get("name", "").strip()
        patch.admin_region = fv("admin_region")
        patch.area_ha = fv("area_ha", float)
        patch.mineral_type = fv("mineral_type")
        patch.mining_method = fv("mining_method")
        patch.stop_years = fv("stop_years", float)
        patch.responsible_entity_missing =
    bool(request.form.get("responsible_entity_missing"))
        # 参照生态系统
        ref = ReferenceEcosystem.query.filter_by(patch_id=pa
    tch.id).first()
        if ref is None:
            ref = ReferenceEcosystem(patch_id=patch.id)
            db.session.add(ref)
        ref.coverage_benchmark = fv("coverage_benchmark",
    float) or 75.0
        ref.shannon_benchmark = fv("shannon_benchmark",
    float) or 0.7
        ref.dominant_species = fv("dominant_species")
        patch.image_note = fv("image_note")
        db.session.commit()
        flash("图斑已保存", "ok")
        return redirect(url_for("project.patches",
    pid=patch.project_id))
    return render_template("project/patch_edit.html",
    form=form,
                           project=patch.project,
    patch=patch,
                           reference=ReferenceEcosystem.quer
```

```
    y.filter_by(                                                    3051
                                        patch_id=patch.id).first())  3052
# ---------------------------- 遥感影像挂接                          3053
    ----------------------------                                    3054
@bp.route("/patch/<int:patch_id>/image", methods=["POST"])          3055
@login_required                                                     3056
def upload_image(patch_id):                                         3057
    """上传遥感底图 / 无人机正射影像并挂接到图斑。"""                  3058
    from app.utils import uploads as up                             3059
    patch = db.get_or_404(MinePatch, patch_id)                      3060
    form = EmptyForm()                                              3061
    if not form.validate_on_submit():                               3062
        flash("表单校验失败，请重试", "error")                      3063
        return redirect(url_for("project.patch_edit",               3064
    patch_id=patch.id))                                             3065
    rel, err = up.save_upload(request.files.get("image"),           3066
    "patches",                                                      3067
                                    current_app.config["ALLOWED_IM3068
    AGE_EXT"],                                                      3069
                                    current_app.config["UPLOAD_FOL3070
    DER"])                                                          3071
    if err:                                                         3072
        flash(err, "error")                                         3073
        return redirect(url_for("project.patch_edit",               3074
    patch_id=patch.id))                                             3075
    # 替换旧影像时清理原文件，避免残留                              3076
    if patch.image_path:                                            3077
        up.remove_upload(patch.image_path,                          3078
    current_app.config["UPLOAD_FOLDER"])                            3079
    patch.image_path = rel                                          3080
    patch.image_uploaded_at = datetime.utcnow()                     3081
    note = request.form.get("image_note", "").strip()               3082
    if note:                                                        3083
        patch.image_note = note                                     3084
    db.session.commit()                                             3085
    flash("遥感影像已挂接", "ok")                                   3086
    return redirect(url_for("project.patch_edit",                   3087
    patch_id=patch.id))                                             3088
@bp.route("/patch/<int:patch_id>/image/delete",                     3089
    methods=["POST"])                                               3090
@login_required                                                     3091
def delete_image(patch_id):                                         3092
    from app.utils import uploads as up                             3093
    patch = db.get_or_404(MinePatch, patch_id)                      3094
    form = EmptyForm()                                              3095
    if form.validate_on_submit():                                   3096
        up.remove_upload(patch.image_path,                          3097
    current_app.config["UPLOAD_FOLDER"])                            3098
        patch.image_path = None                                     3099
        patch.image_uploaded_at = None                              3100
```

```
            db.session.commit()                                          3101
            flash("遥感影像已移除", "ok")                                  3102
        return redirect(url_for("project.patch_edit",                    3103
        patch_id=patch.id))                                              3104
# ==== app/views/unit.py ====                                            3105
# -*- coding: utf-8 -*-                                                  3106
"""微地质单元管理 —— 岩质边坡 / 碎石边坡 / 平台的划分与面积占比。"""                3107
from flask import Blueprint, render_template, request,                   3108
    redirect, url_for, flash                                             3109
from flask_login import login_required                                   3110
from app.forms import EmptyForm                                          3111
from app.models import db, MinePatch, MicroUnit,                         3112
    IndicatorData                                                        3113
from app.utils.constants import UNIT_TYPES                               3114
bp = Blueprint("unit", __name__, url_prefix="/unit")                     3115
def recalc_ratio(patch_id):                                              3116
    """按面积重算该图斑各单元面积占比。"""                                       3117
    units = MicroUnit.query.filter_by(patch_id=patch_id).all3118
    ()                                                                   3119
    total = sum(u.area or 0.0 for u in units)                            3120
    if total > 0:                                                        3121
        for u in units:                                                  3122
            u.ratio = round((u.area or 0.0) / total, 6)                  3123
    return total                                                         3124
@bp.route("/")                                                           3125
@login_required                                                          3126
def list_patches():                                                      3127
    """微地质单元入口：列出各图斑及其单元划分情况。"""                           3128
    patches = MinePatch.query.order_by(MinePatch.id.desc()).3129
    all()                                                                3130
    rows = []                                                            3131
    for p in patches:                                                    3132
        units = MicroUnit.query.filter_by(patch_id=p.id).all3133
    ()                                                                   3134
        rows.append({                                                    3135
            "patch": p,                                                  3136
            "count": len(units),                                         3137
            "ratio_sum": sum(u.ratio or 0.0 for u in units),3138
        })                                                               3139
    return render_template("unit/list.html", rows=rows)                  3140
@bp.route("/<int:patch_id>")                                             3141
@login_required                                                          3142
def index(patch_id):                                                     3143
    patch = db.get_or_404(MinePatch, patch_id)                           3144
    units = MicroUnit.query.filter_by(patch_id=patch_id).all3145
    ()                                                                   3146
    total = sum(u.area or 0.0 for u in units)                            3147
    ratio_sum = sum(u.ratio or 0.0 for u in units)                       3148
    return render_template("unit/index.html", patch=patch,               3149
    units=units,                                                         3150
```

```
                                total_area=total,
    ratio_sum=ratio_sum,
                                unit_types=UNIT_TYPES,
    form=EmptyForm())
@bp.route("/<int:patch_id>/add", methods=["POST"])
@login_required
def add(patch_id):
    patch = db.get_or_404(MinePatch, patch_id)
    form = EmptyForm()
    if form.validate_on_submit():
        unit_type = request.form.get("unit_type")
        if unit_type not in UNIT_TYPES:
            flash("微地质单元类型不正确", "error")
            return redirect(url_for("unit.index",
    patch_id=patch_id))
        # 论文 4.1 节将图斑分为平台、碎石边坡、岩质边坡三类，每类至多一个。
        # 允许重复会让面积占比与按单元类型组织的指标数据对不上号。
        if MicroUnit.query.filter_by(patch_id=patch.id,
                                     unit_type=unit_type).fi
    rst():
            flash("该图斑已存在「%s」，同一类型只能划分一个" % unit_type,
    "error")
            return redirect(url_for("unit.index",
    patch_id=patch_id))
        area = request.form.get("area", "").strip()
        u = MicroUnit(patch_id=patch.id,
    unit_type=unit_type,
                      area=float(area) if area else None,
                      description=request.form.get("descript
    ion", "").strip() or None)
        db.session.add(u)
        db.session.flush()
        recalc_ratio(patch.id)
        db.session.commit()
        flash("微地质单元已添加", "ok")
    return redirect(url_for("unit.index",
    patch_id=patch_id))
@bp.route("/edit/<int:unit_id>", methods=["GET", "POST"])
@login_required
def edit(unit_id):
    """单个微地质单元的编辑页。"""
    u = db.get_or_404(MicroUnit, unit_id)
    form = EmptyForm()
    if request.method == "POST" and form.validate_on_submit(
    ):
        new_type = request.form.get("unit_type",
    u.unit_type)
        if new_type not in UNIT_TYPES:
            flash("微地质单元类型不正确", "error")
            return redirect(url_for("unit.edit",
```

```
    unit_id=u.id))                                                    3201
        # 改类型时同样不允许与本图斑已有的其他单元重复                  3202
        dup = MicroUnit.query.filter_by(patch_id=u.patch_id,          3203
                                                                      3204
    unit_type=new_type).first()                                       3205
        if dup is not None and dup.id != u.id:                        3206
            flash("该图斑已存在「%s」，同一类型只能划分一个" % new_type,  3207
    "error")                                                          3208
            return redirect(url_for("unit.edit",                      3209
    unit_id=u.id))                                                    3210
        u.unit_type = new_type                                        3211
        area = request.form.get("area", "").strip()                   3212
        u.area = float(area) if area else None                        3213
        u.description = request.form.get("description",               3214
    "").strip() or None                                               3215
        recalc_ratio(u.patch_id)                                      3216
        db.session.commit()                                           3217
        flash("微地质单元已保存", "ok")                               3218
        return redirect(url_for("unit.index",                         3219
    patch_id=u.patch_id))                                             3220
    return render_template("unit/edit.html", unit=u,                  3221
    patch=u.patch,                                                    3222
                                unit_types=UNIT_TYPES, form=form)     3223
@bp.route("/delete/<int:unit_id>", methods=["POST"])                  3224
@login_required                                                       3225
def delete(unit_id):                                                  3226
    u = db.get_or_404(MicroUnit, unit_id)                             3227
    pid = u.patch_id                                                  3228
    name = u.unit_type                                                3229
    # 一并删除该单元的指标数据，避免残留                              3230
    IndicatorData.query.filter_by(unit_id=u.id).delete(synch          3231
    ronize_session=False)                                             3232
    db.session.delete(u)                                              3233
    db.session.flush()                                                3234
    recalc_ratio(pid)                                                 3235
    db.session.commit()                                               3236
    flash("微地质单元「%s」已删除" % name, "ok")                      3237
    return redirect(url_for("unit.index", patch_id=pid))              3238
# ==== app/views/standard.py ====                                     3239
# -*- coding: utf-8 -*-                                               3240
"""政策与标准知识库 —— 标准 / 政策文件的结构化规则与省际口径比对。"""  3241
from flask import Blueprint, render_template, request,                3242
    redirect, url_for, flash                                          3243
from flask_login import login_required                                3244
from app.forms import EmptyForm                                       3245
from app.models import db, Standard, StandardRule                     3246
bp = Blueprint("standard", __name__, url_prefix="/standard")          3247
@bp.route("/")                                                        3248
@login_required                                                       3249
def index():                                                          3250
```

```
    standards = Standard.query.order_by(Standard.category,
    Standard.province).all()
    local = [s for s in standards if s.category == "地方标准"]
    policy = [s for s in standards if s.category != "地方标准"]
    return render_template("standard/index.html",
    local=local, policy=policy,
                           form=EmptyForm())
@bp.route("/add", methods=["GET", "POST"])
@login_required
def add():
    form = EmptyForm()
    if request.method == "POST" and form.validate_on_submit(
    ):
        s = Standard(
            province=request.form.get("province",
    "").strip(),
            name=request.form.get("name", "").strip(),
            doc_number=request.form.get("doc_number",
    "").strip() or None,
            publish_date=request.form.get("publish_date",
    "").strip() or None,
            implement_date=request.form.get("implement_date"
    , "").strip() or None,
            category=request.form.get("category", "地方标准"),
            summary=request.form.get("summary", "").strip()
    or None,
        )
        if not s.province or not s.name:
            flash("省份与名称不能为空", "error")
        else:
            db.session.add(s)
            db.session.commit()
            flash("标准已添加", "ok")
            return redirect(url_for("standard.index"))
    return render_template("standard/edit.html", form=form,
    standard=None)
@bp.route("/<int:sid>/edit", methods=["GET", "POST"])
@login_required
def edit(sid):
    s = db.get_or_404(Standard, sid)
    form = EmptyForm()
    if request.method == "POST" and form.validate_on_submit(
    ):
        s.province = request.form.get("province",
    "").strip()
        s.name = request.form.get("name", "").strip()
        s.doc_number = request.form.get("doc_number",
    "").strip() or None
        s.publish_date = request.form.get("publish_date",
    "").strip() or None
```

```
            s.implement_date = request.form.get("implement_date"3301
        , "").strip() or None                                       3302
            s.category = request.form.get("category", "地方标准")    3303
            s.summary = request.form.get("summary", "").strip()     3304
        or None                                                     3305
            db.session.commit()                                     3306
            flash("已保存", "ok")                                    3307
            return redirect(url_for("standard.rules", sid=s.id))3308
        return render_template("standard/edit.html", form=form,     3309
        standard=s)                                                 3310
@bp.route("/<int:sid>/delete", methods=["POST"])                    3311
@login_required                                                     3312
def delete(sid):                                                    3313
        s = db.get_or_404(Standard, sid)                            3314
        StandardRule.query.filter_by(standard_id=sid).delete()      3315
        db.session.delete(s)                                        3316
        db.session.commit()                                         3317
        flash("已删除", "ok")                                        3318
        return redirect(url_for("standard.index"))                  3319
@bp.route("/<int:sid>/rules", methods=["GET", "POST"])              3320
@login_required                                                     3321
def rules(sid):                                                     3322
        s = db.get_or_404(Standard, sid)                            3323
        form = EmptyForm()                                          3324
        if request.method == "POST" and form.validate_on_submit(3325
        ):                                                          3326
            r = StandardRule(                                       3327
                standard_id=s.id,                                   3328
                item=request.form.get("item", "").strip(),          3329
                operator=request.form.get("operator", ">="),        3330
                threshold=request.form.get("threshold",             3331
        "").strip() or None,                                        3332
                condition_type=request.form.get("condition_type"3333
        , "必要条件"),                                               3334
                region=request.form.get("region", "").strip()       3335
        or None,                                                    3336
                note=request.form.get("note", "").strip() or        3337
        None,                                                       3338
            )                                                       3339
            if not r.item:                                          3340
                flash("判定项不能为空", "error")                      3341
            else:                                                   3342
                db.session.add(r)                                   3343
                db.session.commit()                                 3344
                flash("规则已添加", "ok")                            3345
            return redirect(url_for("standard.rules", sid=s.id))3346
        rs = StandardRule.query.filter_by(standard_id=s.id).all(3347
        )                                                           3348
        return render_template("standard/rules.html",               3349
        form=form, standard=s, rules=rs)                            3350
```

```
@bp.route("/rule/<int:rid>/delete", methods=["POST"])
@login_required
def rule_delete(rid):
    r = db.get_or_404(StandardRule, rid)
    sid = r.standard_id
    db.session.delete(r)
    db.session.commit()
    flash("规则已删除", "ok")
    return redirect(url_for("standard.rules", sid=sid))
@bp.route("/compare")
@login_required
def compare():
    """省际口径比对：同一指标在不同省份的阈值横向对比。"""
    rules = StandardRule.query.join(Standard).all()
    groups = {}
    for r in rules:
        groups.setdefault(r.item, []).append(r)
    return render_template("standard/compare.html",
    groups=groups)
# ==== app/views/constraint.py ====
# -*- coding: utf-8 -*-
"""限制性指标判定 —— 一票否决判定。
按论文表 2 与 4.1 节流程，共 5 项：政策、区位约束、土壤污染、地质灾害、耕地属性。
任一项不通过即判定为"人工辅助或生态重建"，不再进入定量评价。
"""
from flask import Blueprint, render_template, request,
    redirect, url_for, flash
from flask_login import login_required
from app.forms import EmptyForm
from app.models import db, MinePatch, ConstraintCheck
from app.utils.constants import CONSTRAINT_ITEMS
bp = Blueprint("constraint", __name__,
    url_prefix="/constraint")
# 字段名 -> 中文名
FIELDS = [
    ("policy_pass", "政策约束"),
    ("location_pass", "区位约束"),
    ("pollution_pass", "土壤污染"),
    ("geo_pass", "地质灾害"),
    ("farmland_pass", "耕地属性"),
]
@bp.route("/")
@login_required
def index():
    patches = MinePatch.query.order_by(MinePatch.id.desc()).
    all()
    latest = {}
    for p in patches:
        latest[p.id] = ConstraintCheck.query.filter_by(patch
    _id=p.id) \
```

```
                .order_by(ConstraintCheck.id.desc()).first()
    return render_template("constraint/index.html",
    patches=patches, latest=latest)
@bp.route("/<int:patch_id>", methods=["GET", "POST"])
@login_required
def check(patch_id):
    patch = db.get_or_404(MinePatch, patch_id)
    form = EmptyForm()
    record = ConstraintCheck.query.filter_by(patch_id=patch_
    id) \
        .order_by(ConstraintCheck.id.desc()).first()
    if request.method == "POST" and form.validate_on_submit(
    ):
        if record is None:
            record = ConstraintCheck(patch_id=patch.id)
            db.session.add(record)
        for field, _ in FIELDS:
            setattr(record, field, request.form.get(field)
    == "1")
        record.passed = all(getattr(record, f) for f, _ in
    FIELDS)
        failed = [name for f, name in FIELDS if not
    getattr(record, f)]
        record.reason = ("全部通过" if record.passed
                         else "未通过：" + "、".join(failed))
        db.session.commit()
        flash("判定结果已保存", "ok")
        return redirect(url_for("constraint.check",
    patch_id=patch.id))
    # 预处理各条件的当前取值，供模板直接使用（Jinja 无 getattr）
    field_states = [(f, name, None if record is None else
    getattr(record, f))
                    for f, name in FIELDS]
    return render_template("constraint/check.html",
    patch=patch, form=form,
                           fields=FIELDS,
    field_states=field_states,
                           record=record)
# ==== app/views/ahp.py ====
# -*- coding: utf-8 -*-
"""指标体系与 AHP 权重 —— 路由与页面逻辑。"""
import json
from flask import Blueprint, render_template, request,
    redirect, url_for, flash
from app.forms import AHPSelectForm, EmptyForm
from app.models import db, Indicator, ExpertMatrix,
    WeightScheme
from app.utils.constants import DEFAULT_WEIGHTS, UNIT_TYPES
from app.services import ahp_service as ahp
from app.services import indicator_service as ind_svc
```

```
from app.services import weight_service as ws
bp = Blueprint("ahp", __name__, url_prefix="/ahp")
#  一次计算的权重结果对应的指标编码：名称 → 编码
_L1_CODE = {"土壤指标": "soil", "植被指标": "veg", "地形指标":
    "terrain"}
_L2_CODE = {
    ("土壤指标", "土石比"): "soil_ratio",
    ("土壤指标", "土层厚度"): "thickness",
    ("土壤指标", "pH"): "ph",
    ("土壤指标", "有机质"): "organic_matter",
    ("土壤指标", "污染指标"): "pollution",
    ("植被指标", "植被覆盖度"): "coverage",
    ("植被指标", "生物多样性"): "shannon",
    ("地形指标", "坡度"): "slope",
    ("地形指标", "坡向"): "aspect",
}
def to_code_pairs(unit_type, level, l1, names, weights):
    """把一次 AHP 计算的权重结果转成「指标编码 → 权重」，便于落库。"""
    pairs = []
    for name, w in zip(names, weights):
        if level == 1:
            code = _L1_CODE.get(name)
        else:
            code = _L2_CODE.get((l1, name))
        if code:
            pairs.append((code, w))
    return pairs
def normalize_l1(unit_type, level, l1):
    """把 l1 参数规整为该单元类型下确实存在的一级指标名。
    level=2 的页面（判断矩阵、多专家矩阵）都按 l1 取二级指标。
    直接带 ?level=2 访问而没给 l1 时，原先会 KeyError: None 直接 500；
    这里退回该单元类型的第一个一级指标，页面照常可用。
    """
    if level != 2:
        return None
    valid = list(DEFAULT_WEIGHTS.get(unit_type,
    {}).get("l2", {}))
    if not valid:
        return None
    return l1 if l1 in valid else valid[0]
def indicator_list(unit_type, level, l1=None):
    """返回 (指标名称列表, 默认权重列表)。
    level=1 时返回一级指标；level=2 时返回指定一级指标下的二级指标。
    l1 缺失或不属于该单元类型时，退回第一个一级指标，不抛异常。
    """
    weights = DEFAULT_WEIGHTS[unit_type]
    if level == 1:
        names = list(weights["l1"].keys())
        vec = [weights["l1"][n] for n in names]
        return names, vec
```

```
    l2_all = weights["l2"]
    if l1 not in l2_all:
        l1 = next(iter(l2_all))
    l2 = l2_all[l1]
    names = list(l2.keys())
    vec = [l2[n] for n in names]
    return names, vec
def l1_choices(unit_type):
    """某单元类型的一级指标名称列表（用于二级指标录入时选择父指标）。"""
    return list(DEFAULT_WEIGHTS[unit_type]["l1"].keys())
@bp.route("/", methods=["GET", "POST"])
def index():
    form = AHPSelectForm()
    # 先根据请求中的单元类型填充 l1 选项，再进行表单校验
    unit = request.form.get("unit_type") or
    form.unit_type.choices[0][0]
    form.l1.choices = [(n, n) for n in l1_choices(unit)]
    if form.validate_on_submit():
        unit_type = form.unit_type.data
        level = int(form.level.data)
        if level == 1:
            return redirect(url_for("ahp.matrix",
    unit_type=unit_type, level=1))
        if not form.l1.data:
            flash("选择「二级指标」时，请同时选择所属的一级指标", "error")
            return render_template("ahp/index.html",
    form=form,
                                   weight_rows=ws.summary())
        return redirect(url_for("ahp.matrix",
    unit_type=unit_type,
                                level=2, l1=form.l1.data))
    return render_template("ahp/index.html", form=form,
                           weight_rows=ws.summary())
@bp.route("/matrix", methods=["GET", "POST"])
def matrix():
    unit_type = request.values.get("unit_type")
    level = int(request.values.get("level", 1))
    l1 = normalize_l1(unit_type, level,
    request.values.get("l1"))
    if unit_type not in DEFAULT_WEIGHTS:
        flash("无效的微地质单元类型", "error")
        return redirect(url_for("ahp.index"))
    names, vec = indicator_list(unit_type, level, l1)
    n = len(names)
    form = EmptyForm()
    # 预填默认值：论文表 3 权重对应的完全一致矩阵 a_ij = w_i / w_j
    default = [[round(vec[i] / vec[j], 4) for j in
    range(n)] for i in range(n)]
    def compute(upper):
        A = ahp.matrix_from_upper(upper)
```

```
        r = ahp.weights_and_consistency(A)                                3551
        bad = ahp.find_most_inconsistent(A)                               3552
        return {                                                          3553
            "weights": [round(v, 4) for v in r["weights"]],               3554
            "lambda_max": round(r["lambda_max"], 4),                      3555
            "ci": r["ci"],                                                3556
            "cr": r["cr"],                                                3557
            "consistent": r["consistent"],                                3558
            "matrix": [[round(A[i, j], 4) for j in                        3559
range(n)] for i in range(n)],                                             3560
            "most_inconsistent": bad,                                     3561
        }                                                                 3562
result = None                                                             3563
if request.method == "POST" and form.validate_on_submit(                  3564
):                                                                        3565
        action = request.form.get("action", "calc")                       3566
        try:                                                              3567
            upper = [float(request.form.get("m_%d_%d" % (i,               3568
j), "1"))                                                                 3569
                     for i in range(n) for j in range(i +                 3570
1, n)]                                                                    3571
            result = compute(upper)                                       3572
        except (ValueError, ZeroDivisionError):                           3573
            flash("矩阵输入有误：请填入正数（对角线自动为 1，下三角自动取倒数）",          3574
"error")                                                                  3575
            return redirect(url_for("ahp.matrix",                         3576
unit_type=unit_type,                                                      3577
                                    level=level, l1=l1))                  3578
        if action == "save":                                              3579
            if not result["consistent"]:                                  3580
                flash("一致性比例 CR=%.4f ≥ 0.1，判断矩阵不满足一致性要求，"              3581
                      "请调整后再保存" % result["cr"], "error")                  3582
            else:                                                         3583
                pairs = to_code_pairs(unit_type, level, l1,               3584
names,                                                                    3585
                                      result["weights"])                  3586
                scheme, cnt = ws.save_weights(unit_type,                  3587
pairs)                                                                    3588
                flash("已保存并为「%s」生效 %d 项权重（CR=%.4f）。"                    3589
                      "此后该单元类型的评价将以此权重计算。"                              3590
                      % (unit_type, cnt, result["cr"]),                   3591
"ok")                                                                     3592
                return redirect(url_for("ahp.matrix",                     3593
unit_type=unit_type,                                                      3594
                                        level=level, l1=l1))              3595
elif request.values.get("calc") == "1":                                   3596
        # 按默认（论文表 3）矩阵直接试算，便于一键查看权重与一致性                          3597
        result = compute([default[i][j] for i in range(n)                 3598
for j in range(i + 1, n)])                                                3599
_, source_label, custom_scheme = ws.effective_weights(un                  3600
```

```
    it_type)                                                              3601
    return render_template(                                               3602
        "ahp/matrix.html",                                                3603
        form=form, unit_type=unit_type, level=level, l1=l1,               3604
        names=names, vec=vec, default=default,                            3605
    result=result,                                                        3606
        source_label=source_label, custom_scheme=custom_sche              3607
    me,                                                                   3608
    )                                                                     3609
@bp.route("/weights/reset", methods=["POST"])                             3610
def weights_reset():                                                      3611
    """删除自定义权重方案，恢复为该单元类型的论文表 3 默认权重。"""                    3612
    unit_type = request.form.get("unit_type")                             3613
    form = EmptyForm()                                                    3614
    if form.validate_on_submit():                                         3615
        if unit_type not in DEFAULT_WEIGHTS:                              3616
            flash("无效的微地质单元类型", "error")                             3617
        else:                                                             3618
            n = ws.reset_weights(unit_type)                               3619
            if n:                                                         3620
                flash("已恢复「%s」为论文表 3 默认权重" % unit_type,                3621
    "ok")                                                                 3622
            else:                                                         3623
                flash("「%s」本就是论文表 3 默认权重，无需恢复" % unit_type,          3624
    "ok")                                                                 3625
    return redirect(url_for("ahp.index"))                                 3626
# ---------------------------- 指标维护 ----------------------------      3627
    ----                                                                  3628
@bp.route("/indicators")                                                  3629
def indicators():                                                         3630
    """指标定义维护：查看指标树、单位、理想值与取值依据。"""                          3631
    if Indicator.query.count() == 0:                                      3632
        ind_svc.seed_indicators()                                         3633
    all_ind = Indicator.query.order_by(Indicator.level,                   3634
    Indicator.id).all()                                                   3635
    by_parent = {}                                                        3636
    for i in all_ind:                                                     3637
        by_parent.setdefault(i.parent_code, []).append(i)                 3638
    return render_template("ahp/indicators.html",                         3639
    by_parent=by_parent,                                                  3640
                                all_ind=all_ind,                          3641
    unit_types=UNIT_TYPES,                                                3642
                                type_label=ind_svc.ideal_type_lab         3643
    el,                                                                   3644
                                form=EmptyForm())                         3645
@bp.route("/indicators/<code>/edit", methods=["GET",                      3646
    "POST"])                                                              3647
def indicator_edit(code):                                                 3648
    ind = Indicator.query.filter_by(code=code).first_or_404(              3649
    )                                                                     3650
```

```
    form = EmptyForm()                                                    3651
    if request.method == "POST" and form.validate_on_submit(              3652
    ):                                                                    3653
        name = request.form.get("name", "").strip()                       3654
        if not name:                                                      3655
            flash("指标名称不能为空", "error")                                3656
            return redirect(url_for("ahp.indicator_edit",                 3657
    code=code))                                                           3658
        ind.name = name                                                   3659
        ind.unit = request.form.get("unit", "").strip() or                3660
    None                                                                  3661
        ind.ideal_value = request.form.get("ideal_value",                 3662
    "").strip() or None                                                   3663
        ind.ideal_type = request.form.get("ideal_type") or                3664
    ind.ideal_type                                                        3665
        units = request.form.getlist("enabled_units")                     3666
        ind.enabled_units = ",".join(units) if units else                 3667
    None                                                                  3668
        ind.is_constraint = request.form.get("is_constraint"              3669
    ) == "1"                                                              3670
        ind.description = request.form.get("description",                 3671
    "").strip() or None                                                   3672
        db.session.commit()                                               3673
        flash("指标「%s」已保存" % ind.name, "ok")                             3674
        return redirect(url_for("ahp.indicators"))                        3675
    return render_template("ahp/indicator_edit.html",                     3676
    ind=ind,                                                              3677
                           unit_types=UNIT_TYPES,                         3678
                           type_label=ind_svc.ideal_type_lab              3679
    el, form=form)                                                        3680
@bp.route("/indicators/reset", methods=["POST"])                          3681
def indicators_reset():                                                   3682
    """按论文取值重置指标定义。"""                                                3683
    form = EmptyForm()                                                    3684
    if form.validate_on_submit():                                         3685
        n = ind_svc.seed_indicators()                                     3686
        flash("已按论文取值重置 %d 项指标定义" % n, "ok")                          3687
    return redirect(url_for("ahp.indicators"))                            3688
# ---------------------------- 多专家矩阵 ------------------------------  3689
    -----                                                                 3690
def expert_scope():                                                       3691
    """从请求中取（并规整）单元类型、层级与所属一级指标。"""                              3692
    unit_type = request.values.get("unit_type") or                        3693
    UNIT_TYPES[0]                                                         3694
    if unit_type not in DEFAULT_WEIGHTS:                                  3695
        unit_type = UNIT_TYPES[0]                                         3696
    level = int(request.values.get("level", 1))                           3697
    l1 = normalize_l1(unit_type, level,                                   3698
    request.values.get("l1"))                                             3699
    return unit_type, level, l1                                           3700
```

```
def expert_query(unit_type, level, l1):
    q = ExpertMatrix.query.filter_by(unit_type=unit_type,
    level=level)
    if level == 2:
        q = q.filter_by(l1_name=l1)
    else:
        q = q.filter(ExpertMatrix.l1_name.is_(None))
    return q.order_by(ExpertMatrix.id)
def default_scheme(unit_type, level, l1=None):
    """取（或建）该范围的权重方案，专家矩阵统一挂到它下面。"""
    name = "%s · %s%s" % (unit_type,
                          "一级指标" if level == 1 else "二级指标",
                          "(%s)" % l1 if l1 else "")
    scheme = WeightScheme.query.filter_by(unit_type=unit_type,
                                          level=level,
    name=name).first()
    if scheme is None:
        scheme = WeightScheme(unit_type=unit_type,
    level=level, name=name,
                              is_default=False)
        db.session.add(scheme)
        db.session.flush()
    return scheme
@bp.route("/experts", methods=["GET", "POST"])
def experts():
    """多专家判断矩阵录入与汇总。"""
    unit_type, level, l1 = expert_scope()
    if unit_type not in DEFAULT_WEIGHTS:
        flash("无效的微地质单元类型", "error")
        return redirect(url_for("ahp.index"))
    names, vec = indicator_list(unit_type, level, l1)
    n = len(names)
    form = EmptyForm()
    if request.method == "POST" and form.validate_on_submit(
    ):
        action = request.form.get("action")
        if action == "add":
            expert_name = request.form.get("expert_name",
    "").strip() or "未具名专家"
            try:
                upper = [float(request.form.get("m_%d_%d" %
    (i, j), "1"))
                         for i in range(n) for j in range(i
    + 1, n)]
            except ValueError:
                flash("矩阵输入有误，请填入正数", "error")
                return redirect(url_for("ahp.experts",
    unit_type=unit_type,
                                        level=level, l1=l1))
```

```
            A = ahp.matrix_from_upper(upper)                                3751
            r = ahp.weights_and_consistency(A)                              3752
            scheme = default_scheme(unit_type, level, l1)                   3753
            db.session.add(ExpertMatrix(                                    3754
                scheme_id=scheme.id,                                        3755
                unit_type=unit_type, l1_name=(l1 if level                   3756
    == 2 else None),                                                        3757
                expert_name=expert_name, level=level,                       3758
                matrix_json=json.dumps([list(map(float,                     3759
    row)) for row in A]),                                                   3760
                cr=round(r["cr"], 6)))                                      3761
            db.session.commit()                                             3762
            flash("已录入专家「%s」的判断矩阵（CR=%.4f）" % (expert_name,    3763
    r["cr"]), "ok")                                                         3764
        elif action == "save_agg":                                          3765
            # 把汇总权重落库并生效，使多专家结果真正参与评价计算                3766
            weights = _aggregate(unit_type, level, l1)                      3767
            if weights is None:                                             3768
                flash("暂无可汇总的专家矩阵", "error")                        3769
            elif not weights["consistent"]:                                 3770
                flash("汇总矩阵一致性比例 CR=%.4f ≥ 0.1，不满足要求，"        3771
                      "请先检查或调整专家矩阵" % weights["cr"],               3772
    "error")                                                                3773
            else:                                                           3774
                names_, _ = indicator_list(unit_type,                       3775
    level, l1)                                                              3776
                pairs = to_code_pairs(unit_type, level, l1,                 3777
    names_,                                                                 3778
                                      weights["weights"])                   3779
                _, cnt = ws.save_weights(unit_type, pairs)                  3780
                flash("已保存并为「%s」生效 %d 项汇总权重（CR=%.4f）"          3781
                      % (unit_type, cnt, weights["cr"]),                    3782
    "ok")                                                                   3783
        return redirect(url_for("ahp.experts",                              3784
    unit_type=unit_type,                                                    3785
                                level=level, l1=l1))                        3786
    rows = expert_query(unit_type, level, l1).all()                         3787
    default = [[round(vec[i] / vec[j], 4) for j in                          3788
    range(n)] for i in range(n)]                                            3789
    return render_template("ahp/experts.html",                              3790
    unit_type=unit_type, level=level,                                       3791
                           l1=l1, names=names, rows=rows,                   3792
    default=default,                                                        3793
                           n=n, form=form)                                  3794
@bp.route("/experts/<int:mid>/delete", methods=["POST"])                    3795
def expert_delete(mid):                                                     3796
    row = db.get_or_404(ExpertMatrix, mid)                                  3797
    unit_type, level, l1 = row.unit_type, row.level,                        3798
    row.l1_name                                                             3799
    form = EmptyForm()                                                      3800
```

```
    if form.validate_on_submit():
        db.session.delete(row)
        db.session.commit()
        flash("已删除专家矩阵", "ok")
    return redirect(url_for("ahp.experts",
    unit_type=unit_type,
                            level=level, l1=l1))
def _aggregate(unit_type, level, l1):
    """按几何平均汇总多专家矩阵并做一致性检验。无矩阵时返回 None。"""
    rows = expert_query(unit_type, level, l1).all()
    mats = []
    for r in rows:
        try:
            mats.append([list(map(float, line))
                         for line in json.loads(r.matrix_jso
    n)])
        except Exception:
            continue
    if not mats:
        return None
    agg = ahp.aggregate_matrices(mats)
    res = ahp.weights_and_consistency(agg)
    return {
        "count": len(mats),
        "weights": res["weights"],
        "lambda_max": res["lambda_max"],
        "ci": res["ci"], "cr": res["cr"],
        "consistent": res["consistent"],
        "most_inconsistent": ahp.find_most_inconsistent(agg)
    ,
        "expert_crs": [r.cr for r in rows if r.cr is not
    None],
    }
@bp.route("/experts/aggregate")
def experts_aggregate():
    """按几何平均汇总多专家矩阵，并做一致性检验。"""
    unit_type, level, l1 = expert_scope()
    if unit_type not in DEFAULT_WEIGHTS:
        flash("无效的微地质单元类型", "error")
        return redirect(url_for("ahp.index"))
    names, vec = indicator_list(unit_type, level, l1)
    rows = expert_query(unit_type, level, l1).all()
    result = _aggregate(unit_type, level, l1)
    if result is not None:
        result["weights"] = [round(v, 4) for v in
    result["weights"]]
        result["lambda_max"] = round(result["lambda_max"],
    4)
    return render_template("ahp/experts_aggregate.html",
    unit_type=unit_type,
```

```
                                    level=level, l1=l1, names=names,
    rows=rows,
                                    result=result, form=EmptyForm())
# ==== app/views/indicator.py ====
# -*- coding: utf-8 -*-
"""指标数据采集与管理 —— 路由与页面逻辑。
按微地质单元组织录入；系统按单元类型自动决定应录入哪些指标
（岩质边坡无土壤指标、平台无地形指标）。
"""
import io
import json
from flask import (Blueprint, render_template, request,
    redirect, url_for,
                       flash, send_file)
from app.forms import EmptyForm
from app.models import db, MinePatch, MicroUnit,
    IndicatorData
from app.utils.constants import HEAVY_METALS, ASPECT_TYPES,
    UNIT_ENABLED_INDICATORS
from app.services import import_service as imp
bp = Blueprint("indicator", __name__,
    url_prefix="/indicator")
# 指标字段定义：字段名 -> (中文名, 单位, 是否可选)
SOIL_FIELDS = [
    ("soil_ratio", "土石比", ""),
    ("thickness", "土层厚度", "cm"),
    ("ph", "pH", ""),
    ("organic_matter", "有机质", "g/kg"),
]
VEG_FIELDS = [
    ("coverage", "植被覆盖度", "%"),
    ("shannon", "Shannon-Wiener 多样性指数", ""),
]
TERRAIN_FIELDS = [
    ("slope", "坡度", "°"),
    ("aspect", "坡向", ""),
]
def unit_field_config(unit_type):
    """根据单元类型返回需录入的字段分组。"""
    enabled = UNIT_ENABLED_INDICATORS.get(unit_type, ())
    groups = []
    if "土壤指标" in enabled:
        groups.append(("土壤指标", SOIL_FIELDS, True))    # True
    = 含重金属
    if "植被指标" in enabled:
        groups.append(("植被指标", VEG_FIELDS, False))
    if "地形指标" in enabled:
        groups.append(("地形指标", TERRAIN_FIELDS, False))
    return groups
@bp.route("/")
```

```
def index():
    patches = MinePatch.query.order_by(MinePatch.id.desc()).all()
    return render_template("indicator/index.html", patches=patches)
@bp.route("/<int:unit_id>", methods=["GET", "POST"])
def edit(unit_id):
    unit = db.get_or_404(MicroUnit, unit_id)
    ind = IndicatorData.query.filter_by(unit_id=unit.id).first()
    form = EmptyForm()
    groups = unit_field_config(unit.unit_type)
    if request.method == "POST" and form.validate_on_submit():
        def f(name, cast=float):
            v = request.form.get(name, "").strip()
            return cast(v) if v != "" else None
        heavy = {}
        for m in HEAVY_METALS:
            v = f("hm_" + m)
            if v is not None:
                heavy[m] = v
        if ind is None:
            ind = IndicatorData(unit_id=unit.id)
            db.session.add(ind)
        ind.soil_ratio = f("soil_ratio")
        ind.thickness = f("thickness")
        ind.ph = f("ph")
        ind.organic_matter = f("organic_matter")
        ind.heavy_metals = json.dumps(heavy, ensure_ascii=False) if heavy else None
        ind.coverage = f("coverage")
        ind.shannon = f("shannon")
        ind.slope = f("slope")
        ind.aspect = request.form.get("aspect", "").strip() or None
        db.session.commit()
        flash("指标数据已保存", "ok")
        return redirect(url_for("indicator.index"))
    # 组装当前值供模板回显
    current = {}
    if ind:
        current = {
            "soil_ratio": ind.soil_ratio, "thickness": ind.thickness,
            "ph": ind.ph, "organic_matter": ind.organic_matter,
            "coverage": ind.coverage, "shannon": ind.shannon,
            "slope": ind.slope, "aspect": ind.aspect,
```

```
        }                                                                    3951
        current["heavy"] = ind.heavy_metals_dict()                           3952
    return render_template("indicator/edit.html",                            3953
                           unit=unit, form=form,                             3954
    groups=groups,                                                           3955
                           heavy_metals=HEAVY_METALS,                        3956
                           aspect_types=ASPECT_TYPES,                        3957
    current=current)                                                         3958
# ---------------------------- 批量导入 ----------------------------        3959
    ----                                                                     3960
@bp.route("/import", methods=["GET", "POST"])                                3961
def import_data():                                                           3962
    """批量导入指标数据（CSV / Excel）。"""                                  3963
    patches = MinePatch.query.order_by(MinePatch.id.desc()).                 3964
    all()                                                                    3965
    form = EmptyForm()                                                       3966
    report = None                                                            3967
    if request.method == "POST" and form.validate_on_submit(                 3968
    ):                                                                       3969
        patch_id = request.form.get("patch_id", type=int)                    3970
        patch = db.session.get(MinePatch, patch_id) if                       3971
    patch_id else None                                                       3972
        if patch is None:                                                    3973
            flash("请选择矿山图斑", "error")                                 3974
            return redirect(url_for("indicator.import_data")                 3975
    )                                                                        3976
        file = request.files.get("datafile")                                 3977
        if file is None or not file.filename:                                3978
            flash("请选择要导入的文件", "error")                             3979
            return redirect(url_for("indicator.import_data")                 3980
    )                                                                        3981
        ext = file.filename.rsplit(".", 1)[-1].lower()                       3982
        if ext not in ("csv", "xlsx", "xls"):                                3983
            flash("仅支持 CSV 或 Excel（.xlsx/.xls）文件", "error")          3984
            return redirect(url_for("indicator.import_data")                 3985
    )                                                                        3986
        units = MicroUnit.query.filter_by(patch_id=patch.id)                 3987
    .all()                                                                   3988
        unit_map = {}                                                        3989
        for u in units:                                                      3990
            unit_map.setdefault(u.unit_type, u)                              3991
        if not unit_map:                                                     3992
            flash("该图斑尚未划分微地质单元，请先划分后再导入", "error")     3993
            return redirect(url_for("indicator.import_data")                 3994
    )                                                                        3995
        try:                                                                 3996
            parsed, errors = imp.parse_upload(file.filename,                 3997
    file.stream, unit_map)                                                   3998
        except Exception as e:                                               3999
            flash("文件解析失败：%s" % e, "error")                           4000
```

```
                return redirect(url_for("indicator.import_data")4001
    )                                                           4002
            saved = 0                                           4003
            for rec in parsed:                                  4004
                unit = unit_map[rec["unit_type"]]               4005
                ind = IndicatorData.query.filter_by(unit_id=unit4006
    .id).first()                                                4007
                if ind is None:                                 4008
                    ind = IndicatorData(unit_id=unit.id)        4009
                    db.session.add(ind)                         4010
                f = rec["fields"]                               4011
                for key in ("soil_ratio", "thickness", "ph",    4012
    "organic_matter",                                           4013
                            "coverage", "shannon", "slope",     4014
    "aspect"):                                                  4015
                    if key in f:                                4016
                        setattr(ind, key, f[key])               4017
                if rec["heavy_metals"]:                         4018
                    ind.heavy_metals = json.dumps(rec["heavy_met4019
    als"],                                                      4020
                                                                4021
    ensure_ascii=False)                                         4022
                saved += 1                                      4023
            db.session.commit()                                 4024
            report = {"patch": patch.name, "saved": saved,      4025
    "errors": errors,                                           4026
                      "total": len(parsed) + len(               4027
                          [e for e in errors if e.startswith("第 4028
    ")])}                                                       4029
            if saved:                                           4030
                flash("已导入 %d 个微地质单元的指标数据" % saved, "ok")  4031
            else:                                               4032
                flash("没有可导入的数据行，请检查文件内容", "error")    4033
        return render_template("indicator/import.html",         4034
        patches=patches,                                        4035
                                    form=form, report=report,   4036
        headers=imp.HEADERS)                                    4037
@bp.route("/import/template")                                   4038
def import_template():                                          4039
    """下载导入模板（CSV，含一行示例）。"""                          4040
    csv_text = imp.build_template_csv()                         4041
    data = csv_text.encode("utf-8-sig")       # 带 BOM，便于      4042
    Excel 正确识别中文                                            4043
    return send_file(io.BytesIO(data), as_attachment=True,      4044
                     download_name="指标数据导入模板.csv",          4045
                     mimetype="text/csv")                       4046
# ==== app/views/evaluate.py ====                               4047
# -*- coding: utf-8 -*-                                         4048
"""适宜性定量评价引擎 —— 路由与页面逻辑。                            4049
从数据库加载某图斑的微地质单元与指标数据，调用评价引擎计算，                  4050
```

```
展示各单元得分、最终评价因子　F、适宜性等级与推荐措施。
入口处按论文　４．１　节的流程顺序检查前置条件（见　app/services/workflow.py），
前置步骤未完成时不予计算，避免在条件不具备时产出评价结论。
"""
import json
from flask import Blueprint, render_template, request,
    redirect, url_for, flash
from app.forms import EmptyForm
from app.models import (db, MinePatch, MicroUnit,
    IndicatorData,
                        EvaluationResult, ScoreDetail)
from app.services import evaluation_service as ev
from app.services import workflow as wf
bp = Blueprint("evaluate", __name__, url_prefix="/evaluate")
def load_patch_input(patch_id):
    """从数据库组装评价引擎输入。"""
    units = MicroUnit.query.filter_by(patch_id=patch_id).all
    ()
    result = []
    for u in units:
        ind = IndicatorData.query.filter_by(unit_id=u.id).fi
    rst()
        data = {
            "soil_ratio": ind.soil_ratio if ind else None,
            "thickness": ind.thickness if ind else None,
            "ph": ind.ph if ind else None,
            "organic_matter": ind.organic_matter if ind
    else None,
            "heavy_metals": ind.heavy_metals_dict() if ind
    else {},
            "coverage": ind.coverage if ind else None,
            "shannon": ind.shannon if ind else None,
            "slope": ind.slope if ind else None,
            "aspect": ind.aspect if ind else None,
        }
        result.append({
            "unit_id": u.id,
            "unit_type": u.unit_type,
            "ratio": u.ratio if u.ratio is not None else
    0.0,
            "data": data,
        })
    return result
def _round(v, n=6):
    return None if v is None else round(v, n)
def load_weights(units):
    """解析各单元类型当前生效的权重与来源说明。
    权重可来自　ＡＨＰ　计算并保存的生效方案；未配置时用论文表　3　的默认值。
    来源一并记录，便于报告标注与事后复核。
    """
```

```
    from app.services import weight_service as ws
    weights, sources = {}, {}
    for u in units:
        ut = u["unit_type"]
        if ut not in weights:
            w, label, _ = ws.effective_weights(ut)
            weights[ut] = w
            sources[ut] = label
    return weights, sources
def pack_units(unit_results, sources=None):
    """把评价结果整理为可持久化的 ＪＳＯＮ。
    完整保留各单元得分、一级指标得分与逐条二级指标明细，
    使结果页与报告能够复现「已记录的那一次评价」，
    而不是每次打开时重新推算。
    """
    sources = sources or {}
    out = []
    for u in unit_results:
        out.append({
            "unit_type": u["unit_type"],
            "ratio": _round(u["ratio"]),
            "unit_score": _round(u["unit_score"]),
            "weighted": _round(u["weighted"]),
            "weight_source": sources.get(u["unit_type"]),
            "l1_scores": {k: _round(v) for k, v in
    u["l1_scores"].items()},
            "details": [{
                "l1": d["l1"], "l2": d["l2"],
                "raw": None if d["raw"] is None else
    str(d["raw"]),
                "std_value": _round(d["std_value"]),
                "weight": _round(d["weight"]),
                "contribution": _round(d["contribution"]),
            } for d in u["details"]],
        })
    return out
def save_evaluation(patch_id, units):
    """执行评价并把结果与逐层明细落库，返回评价记录。
    权重取各单元类型当前生效的方案（ＡＨＰ 计算结果），未配置时用
    论文表 3 的默认权重；来源一并记录。结果 ＪＳＯＮ 保留各单元得分、
    一级指标得分与逐条二级指标明细，结果页与报告据此复现
    记录下来的那次评价，不再重新推算。
    """
    weights, sources = load_weights(units)
    res = ev.score_mine(units, weights)
    saved = EvaluationResult(
        patch_id=patch_id,
        final_score=round(res["final_score"], 4),
        grade=res["grade"],
        measures=res["grade_info"]["measure"],
```

```
        unit_scores=json.dumps(pack_units(res["unit_results"
    ], sources),
                               ensure_ascii=False),
    )
    db.session.add(saved)
    db.session.flush()        # 先取得 result.id，再写明细
    unit_ids = {u["unit_type"]: u["unit_id"] for u in units}
    for u in res["unit_results"]:
        for d in u["details"]:
            db.session.add(ScoreDetail(
                result_id=saved.id,
                unit_id=unit_ids.get(u["unit_type"]),
                l1_name=d["l1"], l2_name=d["l2"],
                raw_value=None if d["raw"] is None else
    str(d["raw"]),
                std_value=d["std_value"],
    weight=d["weight"],
                contribution=d["contribution"]))
    db.session.commit()
    return saved
def block_reasons(progress):
    """把未满足的前置条件转成可读的拦截原因。"""
    return ["%s：%s" % (b["label"], b["reason"]) for b in
    progress["blocking"]]
@bp.route("/")
def index():
    patches = MinePatch.query.order_by(MinePatch.id.desc()).
    all()
    latest = {}
    progress = {}
    for p in patches:
        latest[p.id] = EvaluationResult.query.filter_by(patc
    h_id=p.id) \
            .order_by(EvaluationResult.id.desc()).first()
        progress[p.id] = wf.patch_progress(p.id)
    return render_template("evaluate/index.html",
    patches=patches,
                           latest=latest, progress=progress)
@bp.route("/<int:patch_id>", methods=["GET", "POST"])
def run(patch_id):
    patch = db.get_or_404(MinePatch, patch_id)
    form = EmptyForm()
    progress = wf.patch_progress(patch_id)
    if request.method == "POST" and form.validate_on_submit(
    ):
        if not progress["ready"]:
            for reason in block_reasons(progress):
                flash("无法执行评价 —— " + reason, "error")
            return redirect(url_for("evaluate.run",
    patch_id=patch_id))
```

```
            units = load_patch_input(patch_id)                          4201
            saved = save_evaluation(patch.id, units)                    4202
            return redirect(url_for("evaluate.result",                  4203
        result_id=saved.id))                                            4204
        # 展示将采用的权重来源，便于在执行前确认                                  4205
        units_now = load_patch_input(patch_id)                          4206
        _, sources = load_weights(units_now) if units_now else          4207
        ({}, {})                                                        4208
        return render_template("evaluate/run.html",                     4209
        patch=patch, form=form,                                         4210
                                    progress=progress,                  4211
        weight_sources=sources)                                         4212
@bp.route("/result/<int:result_id>")                                    4213
def result(result_id):                                                  4214
        r = db.get_or_404(EvaluationResult, result_id)                  4215
        patch = db.session.get(MinePatch, r.patch_id)                   4216
        unit_scores = json.loads(r.unit_scores) if                      4217
        r.unit_scores else []                                           4218
        return render_template("evaluate/result.html", r=r,             4219
        patch=patch,                                                    4220
                                    unit_scores=unit_scores)            4221
# ==== app/views/report.py ====                                         4222
# -*- coding: utf-8 -*-                                                 4223
"""报告与结果输出 —— 得分明细、图表、Word / PDF 导出。"""                          4224
import io                                                               4225
import json                                                             4226
from datetime import datetime                                           4227
from flask import Blueprint, render_template, send_file,                4228
        flash, redirect, url_for                                        4229
from app.models import db, MinePatch, ReferenceEcosystem                4230
from app.services import report_service as rs                           4231
from app.services import workflow as wf                                 4232
from app.utils import charts                                            4233
from app.utils import browser                                           4234
from app.utils.constants import grade_of                                4235
bp = Blueprint("report", __name__, url_prefix="/report")                4236
def _prepare(patch_id):                                                 4237
        """加载图斑并取出已记录的评价结果，供明细页与导出复用。                              4238
        报告是评价结论的呈现，因此这里读取最近一次保存的评价记录，                            4239
        而不是重新计算：评价完成后再改动指标数据，也不会让报告与                            4240
        系统记录的评价结论对不上。没有任何评价记录时，返回流程检查                          4241
        结果，由调用方拦截并提示先去执行评价。                                    4242
        """                                                             4243
        patch = db.get_or_404(MinePatch, patch_id)                      4244
        prog = wf.report_progress(patch_id)                             4245
        if not prog["ready"]:                                           4246
            return patch, None, None, None, prog                        4247
        rec = prog["result"]                                            4248
        result = {                                                      4249
            "final_score": rec.final_score,                             4250
```

```
            "grade": rec.grade,                                          4251
            "grade_info": grade_of(rec.final_score),                     4252
            "unit_results": json.loads(rec.unit_scores) if               4253
        rec.unit_scores else [],                                         4254
        }                                                                4255
        reference = ReferenceEcosystem.query.filter_by(patch_id=         4256
        patch_id).first()                                                4257
        return patch, reference, result, rec, prog                       4258
def _deny(progress, patch_id, endpoint="report.detail"):                 4259
    """前置条件未满足时的统一处理：逐条提示并退回报告列表。"""                4260
        for b in progress["blocking"]:                                   4261
            flash("暂不能生成报告 —— %s：%s" % (b["label"],                4262
        b["reason"]), "error")                                           4263
        if progress["blocking"]:                                         4264
            flash("请先完成上述步骤后再生成报告。", "error")               4265
        return redirect(url_for("report.index"))                         4266
def _charts(result):                                                     4267
    """生成两张图表：单元得分对比 + 一级指标平均得分分布。"""                  4268
        units = [(u["unit_type"], u["unit_score"],                       4269
                  "%.4f" % u["unit_score"]) for u in                     4270
        result["unit_results"]]                                          4271
        chart_units = charts.hbars(units)                                4272
        # 各一级指标在各单元上的平均得分，反映其对评价结果的实际影响              4273
        acc, cnt = {}, {}                                                4274
        for u in result["unit_results"]:                                 4275
            for k, v in (u.get("l1_scores") or {}).items():              4276
                acc[k] = acc.get(k, 0.0) + v                             4277
                cnt[k] = cnt.get(k, 0) + 1                               4278
        items = [(k, acc[k] / cnt[k], "%.4f" % (acc[k] /                 4279
        cnt[k])) for k in acc]                                           4280
        items.sort(key=lambda x: -x[1])                                  4281
        chart_weight = charts.donut(items)                               4282
        return chart_units, chart_weight                                 4283
@bp.route("/")                                                           4284
def index():                                                             4285
        patches = MinePatch.query.order_by(MinePatch.id.desc()).         4286
        all()                                                            4287
        progress = {p.id: wf.report_progress(p.id) for p in              4288
        patches}                                                         4289
        return render_template("report/index.html",                      4290
        patches=patches,                                                 4291
                                    progress=progress)                   4292
@bp.route("/<int:patch_id>")                                             4293
def detail(patch_id):                                                    4294
        patch, reference, result, record, prog =                         4295
        _prepare(patch_id)                                               4296
        if result is None:                                               4297
            return _deny(prog, patch_id)                                 4298
        chart_units, chart_weight = _charts(result)                      4299
        return render_template("report/detail.html",                     4300
```

```
    patch=patch, result=result,                                    4301
                                record=record,                     4302
    reference=reference,                                           4303
                                chart_units=chart_units,           4304
    chart_weight=chart_weight)                                     4305
@bp.route("/<int:patch_id>/word")                                  4306
def word(patch_id):                                                4307
    patch, reference, result, record, prog =                       4308
    _prepare(patch_id)                                             4309
    if result is None:                                             4310
        return _deny(prog, patch_id)                               4311
    buf = rs.build_report(patch, result)                           4312
    filename = "自然恢复适宜性评价报告_%s.docx" % patch.name          4313
    return send_file(buf, as_attachment=True,                      4314
    download_name=filename,                                        4315
                        mimetype="application/vnd.openxmlformat4316
    s-officedocument."                                             4317
                                    "wordprocessingml.document")   4318
@bp.route("/<int:patch_id>/pdf")                                   4319
def pdf(patch_id):                                                 4320
    """导出 PDF 报告：服务端渲染打印模板，再交浏览器无头打印。"""          4321
    patch, reference, result, record, prog =                       4322
    _prepare(patch_id)                                             4323
    if result is None:                                             4324
        return _deny(prog, patch_id)                               4325
    chart_units, chart_weight = _charts(result)                    4326
    html = render_template(                                        4327
        "report/print.html", patch=patch,                          4328
    reference=reference, result=result,                            4329
        chart_units=chart_units, chart_weight=chart_weight,        4330
        generated_at=datetime.now().strftime("%Y-%m-%d             4331
    %H:%M"))                                                       4332
    data, err = browser.html_to_pdf(html)                          4333
    if err:                                                        4334
        flash("PDF 导出失败：%s" % err, "error")                     4335
        return redirect(url_for("report.detail",                   4336
    patch_id=patch_id))                                            4337
    return send_file(io.BytesIO(data), as_attachment=True,         4338
                        download_name="自然恢复适宜性评价报告_%s.pdf" %  4339
    patch.name,                                                    4340
                        mimetype="application/pdf")                4341
# ==== app/views/search.py ====                                    4342
# -*- coding: utf-8 -*-                                            4343
"""全局搜索：在评价项目、矿山图斑、标准与政策文件中查找关键词。                  4344
顶栏的搜索框指向本模块。只做关键词包含匹配，不做分词，                          4345
但覆盖名称、编号、行政区、矿种、文号、摘要等常用检索字段。                      4346
"""                                                                4347
from sqlalchemy import or_                                         4348
from flask import Blueprint, render_template, request              4349
from app.models import db, Project, MinePatch, Standard            4350
```

```
bp = Blueprint("search", __name__)                                    4351
LIMIT = 50              #  每类结果上限，避免一次渲染过多                4352
@bp.route("/search")                                                  4353
def index():                                                          4354
    q = (request.args.get("q") or "").strip()                         4355
    projects, patches, standards = [], [], []                         4356
    if q:                                                             4357
        like = "%" + q + "%"                                          4358
        projects = Project.query.filter(or_(                          4359
            Project.name.like(like),                                  4360
            Project.code.like(like),                                  4361
            Project.description.like(like),                           4362
        )).order_by(Project.id.desc()).limit(LIMIT).all()             4363
        patches = MinePatch.query.filter(or_(                         4364
            MinePatch.name.like(like),                                4365
            MinePatch.admin_region.like(like),                        4366
            MinePatch.mineral_type.like(like),                        4367
        )).order_by(MinePatch.id.desc()).limit(LIMIT).all()           4368
        standards = Standard.query.filter(or_(                        4369
            Standard.name.like(like),                                 4370
            Standard.doc_number.like(like),                           4371
            Standard.province.like(like),                             4372
            Standard.summary.like(like),                              4373
        )).order_by(Standard.id).limit(LIMIT).all()                   4374
    total = len(projects) + len(patches) + len(standards)             4375
    return render_template("search.html", q=q,                        4376
    projects=projects,                                                4377
                                patches=patches,                      4378
    standards=standards, total=total)                                 4379
# ==== app/templates/base.html ====                                   4380
{% import "_macros.html" as ui %}<!DOCTYPE html>                      4381
<html lang="zh-CN">                                                   4382
<head>                                                                4383
  <meta charset="UTF-8">                                              4384
  <meta name="viewport" content="width=device-width,                  4385
    initial-scale=1.0">                                               4386
  <link rel="icon" href="{{ url_for('static',                         4387
    filename='favicon.ico') }}">                                      4388
  <title>{% block title %}历史遗留废弃矿山自然恢复适宜性评价系统{% endblock      4389
    %}</title>                                                        4390
  <style>                                                             4391
    /* ---------- 设计变量：绿色主色 + 中性灰阶 ---------- */          4392
    :root {                                                           4393
      --accent: #1f6b45;                  /* 主色：深绿，呼应自然恢复主题 */  4394
      --accent-hover: #185738;                                        4395
      --accent-soft: #e9f4ee;                                         4396
      --side-1: #17503a;                  /* 侧边栏渐变起止色 */      4397
      --side-2: #113c2b;                                              4398
      --danger: #b3261e;                                              4399
      --danger-soft: #fdecea;                                         4400
```

```
    --ok: #1e6b3a;                                                    4401
    --ok-soft: #eaf6ef;                                               4402
    --todo: #d98324;                        /* 待办角标 */            4403
    --ink: #23262b;                                                   4404
    --ink-soft: #6b7280;                                              4405
    --line: #e3e7ea;                                                  4406
    --surface: #ffffff;                                               4407
    --canvas: #f4f7f5;                                                4408
    --radius: 10px;                                                   4409
    --shadow: 0 1px 2px rgba(0, 0, 0, .05);                           4410
    --header-h: 60px;                                                 4411
    --nav-w: 226px;                                                   4412
}                                                                     4413
* { box-sizing: border-box; }                                         4414
html, body { margin: 0; padding: 0; }                                 4415
body {                                                                4416
    font-family: "Microsoft YaHei", "PingFang SC",                    4417
"SimSun", sans-serif;                                                 4418
    font-size: 14px;                                                  4419
    line-height: 1.6;                                                 4420
    color: var(--ink);                                                4421
    background: var(--canvas);                                        4422
}                                                                     4423
/* ---------- 跳到主要内容（键盘可达性） ---------- */                  4424
.skip-link {                                                          4425
    position: absolute;                                               4426
    left: 12px;                                                       4427
    top: -60px;                                                       4428
    z-index: 30;                                                      4429
    padding: 8px 14px;                                                4430
    background: var(--accent);                                        4431
    color: #fff;                                                      4432
    border-radius: var(--radius);                                     4433
    text-decoration: none;                                            4434
    transition: top .15s ease-out;                                    4435
}                                                                     4436
.skip-link:focus { top: 12px; }                                       4437
/* ---------- 整体骨架：左侧深绿导航 + 右侧内容 ----------            4438
   固定为一屏高，页面滚动交给右侧内容区，                             4439
   这样左侧导航栏（含底部用户区）不会随长页面一起被拉到下面去。 */    4440
.app-shell {                                                          4441
    display: flex;                                                    4442
    align-items: stretch;                                             4443
    height: 100vh;                                                    4444
    height: 100dvh;                                                   4445
    overflow: hidden;                                                 4446
}                                                                     4447
/* ---------- 侧边栏 ---------- */                                    4448
.app-side {                                                           4449
    flex: 0 0 var(--nav-w);                                           4450
```

```
  width: var(--nav-w);
  display: flex;
  flex-direction: column;
  background: linear-gradient(180deg, var(--side-1) 0%, var(--side-2) 100%);
  color: #fff;
}
.app-brand {
  display: flex;
  align-items: center;
  gap: 11px;
  padding: 17px 16px 15px;
  border-bottom: 1px solid rgba(255, 255, 255, .10);
}
.app-brand__mark {
  flex: none;
  display: grid;
  place-items: center;
  width: 36px;
  height: 36px;
  border-radius: 10px;
  background: rgba(255, 255, 255, .15);
}
.app-brand__name {
  font-size: 14.5px;
  font-weight: 600;
  line-height: 1.35;
  letter-spacing: .2px;
}
.app-brand__sub {
  font-size: 11.5px;
  line-height: 1.35;
  color: rgba(255, 255, 255, .58);
}
.app-nav { flex: 1; padding: 6px 12px 14px; overflow-y: auto; }
.app-nav__group {
  margin: 12px 0 5px;
  padding: 0 10px;
  font-size: 11.5px;
  letter-spacing: .05em;
  color: rgba(255, 255, 255, .48);
}
.app-nav ul { margin: 0; padding: 0; list-style: none; }
.app-nav a {
  display: flex;
  align-items: center;
  gap: 10px;
  padding: 9px 11px;
  margin-bottom: 2px;
```

```
  border-radius: 8px;
  color: rgba(255, 255, 255, .80);
  text-decoration: none;
  transition: background-color .16s ease, color .16s
ease;
}
.app-nav a:hover { background: rgba(255, 255, 255,
.10); color: #fff; }
.app-nav a:focus-visible {
  outline: 2px solid rgba(255, 255, 255, .60);
  outline-offset: 1px;
}
.app-nav a[aria-current="page"] {
  background: rgba(255, 255, 255, .17);
  color: #fff;
  font-weight: 600;
}
.app-nav .ico { flex: none; opacity: .92; }
.app-nav__badge {
  margin-left: auto;
  min-width: 20px;
  padding: 0 6px;
  font-size: 11px;
  font-weight: 600;
  line-height: 18px;
  text-align: center;
  border-radius: 9px;
  background: var(--todo);
  color: #fff;
  font-variant-numeric: tabular-nums;
}
.app-user {
  display: flex;
  align-items: center;
  gap: 8px;
  padding: 10px 14px;
  border-top: 1px solid rgba(255, 255, 255, .10);
}
.app-user__me {
  display: flex;
  flex: 1;
  min-width: 0;
  align-items: center;
  gap: 10px;
  padding: 4px 6px;
  border-radius: 8px;
  color: inherit;
  text-decoration: none;
  transition: background-color .16s ease;
}
```

```
.app-user__me:hover { background: rgba(255, 255, 255, .10); }
.app-user__me:focus-visible {
  outline: 2px solid rgba(255, 255, 255, .60);
  outline-offset: 1px;
}
.app-user__avatar {
  flex: none;
  display: grid;
  place-items: center;
  width: 32px;
  height: 32px;
  border-radius: 50%;
  background: rgba(255, 255, 255, .16);
  font-size: 13px;
  font-weight: 600;
  object-fit: cover;
}
.app-user__name {
  display: block;
  font-size: 13px;
  line-height: 1.3;
  white-space: nowrap;
  overflow: hidden;
  text-overflow: ellipsis;
}
.app-user__state {
  display: flex;
  align-items: center;
  gap: 5px;
  font-size: 11.5px;
  line-height: 1.3;
  color: rgba(255, 255, 255, .60);
}
.app-user__dot {
  width: 6px;
  height: 6px;
  border-radius: 50%;
  background: #6ed49a;
}
.app-user__out {
  flex: none;
  padding: 6px;
  border-radius: 6px;
  color: rgba(255, 255, 255, .68);
  text-decoration: none;
}
.app-user__out:hover { color: #fff; background: rgba(255, 255, 255, .12); }
/* ---------- 右侧内容区 ----------
```

```
    整个骨架固定一屏高，滚动只发生在内容区内部：
    左侧导航栏（含底部用户区）始终留在原位，右侧页面自己滚。 */
.app-content {
  flex: 1;
  min-width: 0;
  display: flex;
  flex-direction: column;
  overflow-y: auto;
  overflow-x: hidden;
  /* 细滚动条：默认半透明，悬停时加深，不抢视线 */
  scrollbar-width: thin;
  scrollbar-color: rgba(31, 107, 69, .28) transparent;
}
.app-content::-webkit-scrollbar { width: 12px; }
.app-content::-webkit-scrollbar-track { background:
transparent; }
.app-content::-webkit-scrollbar-thumb {
  background: rgba(31, 107, 69, .26);
  border: 3px solid transparent;
  background-clip: content-box;
  border-radius: 999px;
}
.app-content::-webkit-scrollbar-thumb:hover {
  background: rgba(31, 107, 69, .46);
  background-clip: content-box;
}
/* ---------- 顶栏：面包屑 + 搜索 ----------
    粘在内容区顶部，滚动时面包屑与搜索框始终可用 */
.app-header {
  position: sticky;
  top: 0;
  z-index: 20;
  display: flex;
  align-items: center;
  gap: 14px;
  height: var(--header-h);
  padding: 0 22px;
  background: var(--surface);
  border-bottom: 1px solid var(--line);
}
.crumb {
  display: flex;
  align-items: center;
  gap: 7px;
  font-size: 13px;
  color: var(--ink-soft);
  white-space: nowrap;
}
.crumb a { color: var(--ink-soft); text-decoration:
none; }
```

```
.crumb a:hover { color: var(--accent); }
.crumb strong { color: var(--ink); font-weight: 600; }
.crumb .ico { color: #c3c9d0; }
.app-header__spacer { flex: 1; }
.search-box {
  display: flex;
  align-items: center;
  gap: 8px;
  width: 268px;
  padding: 7px 12px;
  background: var(--canvas);
  border: 1px solid var(--line);
  border-radius: 9px;
  color: var(--ink-soft);
  transition: border-color .16s ease, background-color
.16s ease;
}
.search-box:focus-within { border-color: var(--accent);
background: #fff; }
.search-box input {
  flex: 1;
  min-width: 0;
  border: none;
  outline: none;
  background: transparent;
  font-family: inherit;
  font-size: 13px;
  color: var(--ink);
}
.search-box button {
  display: grid;
  place-items: center;
  padding: 0;
  border: none;
  background: transparent;
  color: inherit;
  cursor: pointer;
}
.icon-btn {
  display: grid;
  place-items: center;
  width: 34px;
  height: 34px;
  border-radius: 9px;
  border: 1px solid transparent;
  color: var(--ink-soft);
  text-decoration: none;
  transition: background-color .16s ease, color .16s
ease,
              border-color .16s ease;
```

```
}
.icon-btn:hover {
  color: var(--accent);
  background: var(--accent-soft);
  border-color: rgba(31, 107, 69, .18);
}
.app-main {
  flex: 1;
  min-width: 0;
  padding: 22px 26px 26px;
}
.container { max-width: 1120px; }
/* ---------- 页脚 ---------- */
.app-foot {
  display: flex;
  align-items: center;
  justify-content: space-between;
  gap: 14px;
  padding: 0 26px 18px;
  font-size: 12px;
  color: var(--ink-soft);
}
/* ---------- 未登录时的裸页面（登录页） ---------- */
.login-shell {
  display: flex;
  flex-direction: column;
  min-height: 100dvh;
}
/* ---------- 卡片 ---------- */
.card {
  background: var(--surface);
  border: 1px solid var(--line);
  border-radius: var(--radius);
  box-shadow: var(--shadow);
  padding: 22px 24px;
  margin-bottom: 18px;
}
.card > h2 {
  margin: 0 0 14px;
  font-size: 16px;
  font-weight: 600;
  color: var(--accent);
  text-wrap: balance;
}
.card > h3 {
  margin: 20px 0 10px;
  font-size: 14px;
  font-weight: 600;
  color: var(--accent);
}
```

```
/* ---------- 表格 ---------- */
table {
  width: 100%;
  border-collapse: collapse;
  margin: 10px 0 16px;
  font-variant-numeric: tabular-nums;
}
caption {
  text-align: left;
  color: var(--ink-soft);
  font-size: 12px;
  padding-bottom: 6px;
}
th, td {
  border: 1px solid var(--line);
  padding: 8px 12px;
  text-align: center;
  vertical-align: middle;
}
th {
  background: var(--accent-soft);
  font-weight: 600;
  white-space: nowrap;
}
tbody tr:hover { background: #fafbfc; }
/* ---------- 表单控件 ---------- */
label { display: inline-block; }
.field { margin-bottom: 12px; }
.field__label {
  display: block;
  margin-bottom: 4px;
  color: var(--ink-soft);
  font-size: 13px;
}
input[type="text"], input[type="number"],
input[type="password"],
select, textarea {
  padding: 6px 9px;
  font-family: inherit;
  font-size: 14px;
  color: var(--ink);
  background: #fff;
  border: 1px solid #c6ccd4;
  border-radius: 4px;
  max-width: 100%;
}
input[type="number"] {
  width: 104px;
  text-align: center;
  font-variant-numeric: tabular-nums;
```

```
}
textarea { width: 100%; min-height: 64px; resize: vertical; }
input:focus-visible, select:focus-visible, textarea:focus-visible {
  outline: 2px solid var(--accent);
  outline-offset: 1px;
  border-color: var(--accent);
}
.checkbox {
  display: inline-flex;
  align-items: center;
  gap: 7px;
  cursor: pointer;
  user-select: none;
}
.checkbox input { width: 16px; height: 16px; margin: 0; cursor: pointer; }
/* ---------- 按钮 ----------
   统一为圆角矩形：默认半透明底色，鼠标悬停时填充为实色。 */
.btn {
  display: inline-flex;
  align-items: center;
  justify-content: center;
  gap: 6px;
  padding: 8px 18px;
  font-family: inherit;
  font-size: 14px;
  line-height: 1.4;
  color: #fff;
  background: rgba(31, 107, 69, .80);
  border: 1px solid rgba(31, 107, 69, .30);
  border-radius: 8px;
  cursor: pointer;
  text-decoration: none;
  transition: background-color .16s ease, border-color .16s ease,
              color .16s ease, box-shadow .16s ease, transform .16s ease;
}
.btn:hover {
  background: var(--accent);
  border-color: var(--accent);
  box-shadow: 0 3px 12px rgba(31, 107, 69, .22);
}
.btn:active { transform: translateY(1px); box-shadow: none; }
.btn:focus-visible { outline: 2px solid var(--accent); outline-offset: 2px; }
.btn--ghost {
```

```
  color: var(--accent);                                                    4851
  background: rgba(31, 107, 69, .06);                                      4852
  border-color: rgba(31, 107, 69, .22);                                    4853
}                                                                          4854
.btn--ghost:hover {                                                        4855
  color: #fff;                                                             4856
  background: var(--accent);                                               4857
  border-color: var(--accent);                                             4858
}                                                                          4859
.btn--danger {                                                             4860
  color: var(--danger);                                                    4861
  background: rgba(179, 38, 30, .07);                                      4862
  border-color: rgba(179, 38, 30, .26);                                    4863
}                                                                          4864
.btn--danger:hover {                                                       4865
  color: #fff;                                                             4866
  background: var(--danger);                                               4867
  border-color: var(--danger);                                             4868
  box-shadow: 0 3px 12px rgba(179, 38, 30, .20);                           4869
}                                                                          4870
.btn--sm { padding: 5px 13px; font-size: 13px;                             4871
border-radius: 7px; }                                                      4872
.btn:disabled { opacity: .45; cursor: not-allowed; }                       4873
.btn:disabled:hover {                                                      4874
  background: rgba(31, 107, 69, .80);                                      4875
  border-color: rgba(31, 107, 69, .30);                                    4876
  box-shadow: none;                                                        4877
}                                                                          4878
.btn--ghost:disabled:hover { background: rgba(31, 107,                     4879
69, .06); color: var(--accent); }                                          4880
/* ---------- 表格操作列的行内小按钮 ---------- */                          4881
   取代原来「文字 | 文字 | 文字」的写法：                                    4882
   圆角矩形 + 半透明底色，悬停时变实色。 */                                  4883
.row-actions {                                                             4884
  display: inline-flex;                                                    4885
  flex-wrap: nowrap;          /* 按钮不折行；宁可让旁边长名称换行 */          4886
  align-items: center;                                                     4887
  gap: 6px;                                                                4888
  white-space: nowrap;                                                     4889
}                                                                          4890
.row-actions form { display: inline-flex; margin: 0; }                     4891
.link-btn {                                                                4892
  display: inline-flex;                                                    4893
  align-items: center;                                                     4894
  padding: 4px 11px;                                                       4895
  font-family: inherit;                                                    4896
  font-size: 13px;                                                         4897
  line-height: 1.5;                                                        4898
  color: var(--accent);                                                    4899
  background: rgba(31, 107, 69, .07);                                      4900
```

```
  border: 1px solid rgba(31, 107, 69, .20);
  border-radius: 7px;
  cursor: pointer;
  text-decoration: none;
  white-space: nowrap;
  transition: background-color .16s ease, border-color .16s ease,
              color .16s ease, box-shadow .16s ease, transform .16s ease;
}
.link-btn:hover {
  color: #fff;
  background: var(--accent);
  border-color: var(--accent);
  box-shadow: 0 2px 8px rgba(31, 107, 69, .20);
}
.link-btn:active { transform: translateY(1px); box-shadow: none; }
.link-btn:focus-visible { outline: 2px solid var(--accent); outline-offset: 2px; }
.link-btn--danger {
  color: var(--danger);
  background: rgba(179, 38, 30, .07);
  border-color: rgba(179, 38, 30, .24);
}
.link-btn--danger:hover {
  color: #fff;
  background: var(--danger);
  border-color: var(--danger);
  box-shadow: 0 2px 8px rgba(179, 38, 30, .22);
}
.actions { display: flex; flex-wrap: wrap; align-items: center; gap: 8px; }
.actions form { display: inline; }
/* ---------- 提示条 ---------- */
.flash {
  padding: 10px 14px;
  margin-bottom: 14px;
  border-radius: var(--radius);
  border-left: 4px solid;
}
.flash.error {
  color: var(--danger);
  background: var(--danger-soft);
  border-left-color: var(--danger);
}
.flash.ok {
  color: var(--ok);
  background: var(--ok-soft);
  border-left-color: var(--ok);
```

```
}
/*  ----------  浮层提示（登录成功、各类操作完成、登录失败）  ----------
    外层 toast-stack 固定在页面中上方，内层每条各自淡入 → 停留 → 淡出，
    多条提示会依次排开而不重叠。
    底色是浅绿半透明：既有一层底色把文字托住，又能透出后面的内容，
    不用高饱和色块，提示不该比页面内容更抢眼。
    pointer-events: none 让它不拦截鼠标，出现期间也能正常操作。 */
.toast-stack {
  position: fixed;
  top: 16px;
  left: 50%;
  z-index: 100;
  display: flex;
  flex-direction: column;
  align-items: center;
  gap: 6px;
  transform: translateX(-50%);
  pointer-events: none;
}
.toast-center {
  max-width: min(620px, 82vw);
  padding: 8px 22px;
  border-radius: 999px;
  /* 浅绿、半透明：不遮住背后的内容，也比纯透明更托得住文字 */
  background: rgba(223, 241, 231, .76);
  border: 1px solid rgba(31, 107, 69, .14);
  color: #2f4a3c;
  font-size: 16px;
  font-weight: 500;
  line-height: 1.65;
  letter-spacing: .03em;
  text-align: center;
  text-wrap: pretty;
  backdrop-filter: blur(6px) saturate(1.1);
  -webkit-backdrop-filter: blur(6px) saturate(1.1);
  box-shadow: 0 4px 16px rgba(23, 32, 27, .07);
  animation: toast-fade 2.6s ease-in-out forwards;
}
/* 失败类提示：同一套观感，换成浅红底，一眼能区分成败 */
.toast-center--error {
  background: rgba(252, 233, 230, .78);
  border-color: rgba(163, 43, 34, .16);
  color: #9c2b23;
}
@keyframes toast-fade {
  0%        { opacity: 0; transform: translateY(-6px); }
  14%, 74%  { opacity: 1; transform: none; }
  100%      { opacity: 0; transform: none; }
}
/* 系统设置里要求「减少动态效果」时，去掉位移，只做淡入淡出 */
```

```
@media (prefers-reduced-motion: reduce) {
  .toast-center { animation: toast-fade-plain 2.6s linear forwards; }
  @keyframes toast-fade-plain {
    0%          { opacity: 0; }
    14%, 74%    { opacity: 1; }
    100%        { opacity: 0; }
  }
}
/* ---------- 文字工具类 ---------- */
.muted { color: var(--ink-soft); font-size: 12.5px; }
.mono { font-family: Consolas, "Courier New", monospace; font-variant-numeric: tabular-nums; }
.num { font-variant-numeric: tabular-nums; }
.warn { color: var(--danger); font-weight: 600; }
.ok { color: var(--ok); font-weight: 600; }
.nowrap { white-space: nowrap; }
/* 表格操作列里常把删除按钮放在 <form> 中。form 默认是块级元素，
   会把「删除」挤到下一行，导致操作列参差不齐，这里让它留在行内。 */
.nowrap form { display: inline; margin: 0; }
.stack { display: flex; flex-direction: column; gap: 14px; }
/* 状态标签：不依赖颜色单独表意 */
.tag {
  display: inline-block;
  padding: 1px 9px;
  font-size: 12.5px;
  border-radius: 10px;
  border: 1px solid;
  white-space: nowrap;
}
.tag--ok { color: var(--ok); background: var(--ok-soft); border-color: #b8ddc6; }
.tag--warn { color: var(--danger); background: var(--danger-soft); border-color: #eec4c0; }
.tag--muted { color: var(--ink-soft); background: #f1f3f6; border-color: #dde1e6; }
/* ---------- 登录页 ---------- */
/* 仿 Apple 账户页：接近纯白的底，仅在顶部有一层极淡的绿色晕染，
   让彩色徽标成为整页唯一的视觉重心。排版见 auth/login.html。 */
.body--login {
  background:
    radial-gradient(1100px 640px at 50% 8%, #edf6f1 0%, rgba(237, 246, 241, 0) 72%),
    linear-gradient(180deg, #ffffff 0%, #fbfdfc 100%);
}
.body--login .app-main { padding: 0; }
.body--login .container { max-width: none; width: 100%; }
```

```
/*  仅供读屏软件的隐藏文本  */
.visually-hidden {
  position: absolute !important;
  width: 1px;
  height: 1px;
  margin: -1px;
  padding: 0;
  overflow: hidden;
  clip: rect(0 0 0 0);
  white-space: nowrap;
  border: 0;
}
/*  ---------- 工作台：页首横幅 ----------  */
.hero {
  position: relative;
  overflow: hidden;
  min-height: 176px;
  padding: 26px 28px;
  margin-bottom: 16px;
  border-radius: 12px;
  /*  左侧压深保证文字可读，右侧透出远山插画；
      背景按底部对齐，避免远山被居中裁掉  */
  background:
    linear-gradient(100deg, rgba(9, 40, 27, .92) 0%,
                            rgba(11, 48, 33, .68) 50%,
                            rgba(11, 48, 33, .06) 100%),
    url("{{ url_for('static', filename='hero.svg') }}")
    center bottom / cover no-repeat;
  color: #fff;
}
.hero__body { max-width: 720px; }
.hero__title {
  margin: 0 0 10px;
  font-size: 21px;
  font-weight: 700;
  line-height: 1.4;
  letter-spacing: .3px;
}
.hero__desc {
  margin: 0;
  max-width: 640px;
  font-size: 12.5px;
  line-height: 1.85;
  color: rgba(255, 255, 255, .86);
}
.hero__actions {
  display: flex;
  flex-wrap: wrap;
  gap: 10px;
  margin-top: 18px;
```

```
}
.hero-btn {
  display: inline-flex;
  align-items: center;
  gap: 6px;
  padding: 7px 15px;
  font-size: 13px;
  border-radius: 8px;
  text-decoration: none;
  transition: background-color .16s ease, border-color .16s ease,
              color .16s ease;
}
.hero-btn--solid {
  color: #14432f;
  background: rgba(255, 255, 255, .92);
  border: 1px solid rgba(255, 255, 255, .92);
}
.hero-btn--solid:hover { background: #fff; }
.hero-btn--ghost {
  color: #fff;
  background: rgba(255, 255, 255, .14);
  border: 1px solid rgba(255, 255, 255, .45);
}
.hero-btn--ghost:hover { background: rgba(255, 255, 255, .26); }
.hero-btn:focus-visible { outline: 2px solid #fff; outline-offset: 2px; }
/* ---------- 工作台：数据概览 ---------- */
.stat-cards {
  display: grid;
  grid-template-columns: repeat(auto-fit, minmax(190px, 1fr));
  gap: 12px;
  margin-bottom: 22px;
}
.stat-card {
  display: block;
  padding: 15px 17px;
  background: var(--surface);
  border: 1px solid var(--line);
  border-radius: var(--radius);
  box-shadow: var(--shadow);
  color: var(--ink);
  text-decoration: none;
  transition: border-color .16s ease, box-shadow .16s ease;
}
.stat-card:hover {
  border-color: rgba(31, 107, 69, .45);
```

```
    box-shadow: 0 3px 12px rgba(19, 60, 43, .08);
}
.stat-card__label {
    display: block;
    font-size: 12.5px;
    color: var(--ink-soft);
}
.stat-card__value {
    display: block;
    margin: 3px 0 2px;
    font-size: 26px;
    font-weight: 700;
    line-height: 1.25;
    color: var(--accent);
    font-variant-numeric: tabular-nums;
}
.stat-card__unit {
    margin-left: 3px;
    font-size: 13px;
    font-weight: 500;
    color: var(--ink-soft);
}
.stat-card__note {
    display: block;
    font-size: 12px;
    color: var(--ink-soft);
}
/* ---------- 工作台：分区标题 ---------- */
.section { margin-bottom: 22px; }
.section__head {
    display: flex;
    align-items: baseline;
    flex-wrap: wrap;
    gap: 6px 14px;
    margin-bottom: 12px;
}
.section__title { margin: 0; font-size: 16px; }
.section__desc { margin: 0; font-size: 12.5px; color:
var(--ink-soft); }
/* ---------- 工作台：流程步骤卡 ---------- */
.step-grid {
    display: grid;
    grid-template-columns: repeat(auto-fill,
minmax(238px, 1fr));
    gap: 12px;
}
.step-card {
    position: relative;
    display: block;
    padding: 16px 16px 15px;
```

```
  background: var(--surface);                                    5201
  border: 1px solid var(--line);                                 5202
  border-radius: var(--radius);                                  5203
  box-shadow: var(--shadow);                                     5204
  color: var(--ink);                                             5205
  text-decoration: none;                                         5206
  transition: border-color .16s ease, box-shadow .16s    5207
ease,                                                            5208
              transform .16s ease;                               5209
}                                                                5210
.step-card:hover {                                               5211
  border-color: rgba(31, 107, 69, .55);                          5212
  box-shadow: 0 5px 16px rgba(19, 60, 43, .10);                  5213
  transform: translateY(-1px);                                   5214
}                                                                5215
.step-card:focus-visible { outline: 2px solid                    5216
var(--accent); outline-offset: 2px; }                            5217
.step-card__no {                                                 5218
  position: absolute;                                            5219
  top: 13px;                                                     5220
  right: 15px;                                                   5221
  font-size: 17px;                                               5222
  font-weight: 700;                                              5223
  color: #d7e3db;                                                5224
  font-variant-numeric: tabular-nums;                            5225
}                                                                5226
.step-card__icon {                                               5227
  display: grid;                                                 5228
  place-items: center;                                           5229
  width: 32px;                                                   5230
  height: 32px;                                                  5231
  margin-bottom: 10px;                                           5232
  border-radius: 9px;                                            5233
  background: var(--accent-soft);                                5234
  color: var(--accent);                                          5235
}                                                                5236
.step-card__title {                                              5237
  display: block;                                                5238
  margin-bottom: 5px;                                            5239
  font-size: 14px;                                               5240
  font-weight: 600;                                              5241
  color: var(--accent);                                          5242
}                                                                5243
.step-card__desc {                                               5244
  display: block;                                                5245
  font-size: 12.5px;                                             5246
  line-height: 1.65;                                             5247
  color: var(--ink-soft);                                        5248
  text-wrap: pretty;                                             5249
}                                                                5250
```

```
.step-card__badge {
  display: inline-block;
  margin-top: 10px;
  padding: 1px 9px;
  font-size: 11.5px;
  border-radius: 9px;
  background: #fdf1e2;
  color: #9a5a12;
  border: 1px solid #f0d3ab;
}
/* 前置数据不具备时置灰，并说明缺什么 */
.step-card--locked {
  background: #fafbfa;
  border-color: #eceff0;
  box-shadow: none;
  cursor: default;
}
.step-card--locked .step-card__no { color: #e6e9e8; }
.step-card--locked .step-card__icon {
  background: #f1f3f2;
  color: #a9b2ae;
}
.step-card--locked .step-card__title { color: #9aa3a0; }
.step-card--locked .step-card__desc { color: #b3bab7; }
.step-card__lock {
  display: inline-block;
  margin-top: 10px;
  padding: 1px 9px;
  font-size: 11.5px;
  border-radius: 9px;
  background: #f1f3f2;
  color: #8b948f;
}
/* ---------- 工作台：整行入口 ---------- */
.row-card {
  display: flex;
  align-items: center;
  gap: 14px;
  padding: 15px 18px;
  background: var(--surface);
  border: 1px solid var(--line);
  border-radius: var(--radius);
  box-shadow: var(--shadow);
  color: var(--ink);
  text-decoration: none;
  transition: border-color .16s ease, box-shadow .16s
ease;
}
.row-card:hover {
  border-color: rgba(31, 107, 69, .45);
```

```
    box-shadow: 0 3px 12px rgba(19, 60, 43, .08);
}
.row-card__icon {
    display: grid;
    place-items: center;
    width: 36px;
    height: 36px;
    flex: none;
    border-radius: 10px;
    background: var(--accent-soft);
    color: var(--accent);
}
.row-card__main { flex: 1; min-width: 0; }
.row-card__title {
    display: block;
    font-size: 14px;
    font-weight: 600;
    color: var(--accent);
}
.row-card__desc {
    display: block;
    font-size: 12.5px;
    color: var(--ink-soft);
}
.row-card__arrow { color: #c3c9d0; flex: none; }
/* ---------- 结果摘要 ---------- */
.stat-row {
    display: flex;
    flex-wrap: wrap;
    gap: 28px;
    margin: 4px 0 6px;
}
.stat__label { font-size: 12.5px; color:
var(--ink-soft); }
.stat__value { font-size: 20px; font-weight: 700;
font-variant-numeric: tabular-nums; }
/* ---------- 流程步骤条 ---------- */
.steps { display: flex; flex-wrap: wrap; gap: 8px;
margin: 4px 0 18px; }
.step {
    display: flex; align-items: center; gap: 7px;
    padding: 8px 12px; font-size: 13px;
    background: #fff; border: 1px solid var(--line);
    border-radius: var(--radius);
}
.step__no {
    display: inline-flex; align-items: center;
justify-content: center;
    width: 20px; height: 20px; border-radius: 50%;
    background: #eef1f5; color: var(--ink-soft);
```

```
  font-size: 12px; font-variant-numeric: tabular-nums;
}
.step__name { font-weight: 600; }
.step__state { font-size: 12.5px; color:
var(--ink-soft); }
.step--done { border-color: #b8ddc6; background:
var(--ok-soft); }
.step--done .step__no { background: var(--ok); color:
#fff; }
.step--done .step__state { color: var(--ok); }
.step--fail { border-color: #eec4c0; background:
var(--danger-soft); }
.step--fail .step__no { background: var(--danger);
color: #fff; }
.step--fail .step__state { color: var(--danger);
font-weight: 600; }
.step--next { border-color: var(--accent); }
.step--next .step__no { background: var(--accent);
color: #fff; }
/* ---------- 提示块 ---------- */
.notice {
  padding: 14px 16px; margin: 0 0 16px;
  border-radius: var(--radius); border-left: 4px solid;
}
.notice--warn {
  background: var(--danger-soft);
  border-left-color: var(--danger);
  color: #7a1c17;
}
.notice__title { margin: 0 0 8px; font-weight: 600; }
.notice__list { margin: 0; padding-left: 20px;
line-height: 1.9; }
.notice__list a { color: var(--accent); }
.notice__hint { margin: 8px 0 0; font-size: 12.5px;
color: var(--ink-soft); }
/* ---------- 图斑面板（按图斑分块展示，如指标数据采集） ---------- */
.patch-panel {
  border: 1px solid var(--line);
  border-radius: var(--radius);
  background: var(--surface);
  padding: 14px 16px;
  margin: 0 0 14px;
}
.patch-panel:last-child { margin-bottom: 0; }
.patch-panel__head {
  display: flex;
  align-items: baseline;
  flex-wrap: wrap;
  gap: 6px 14px;
  margin: 0 0 12px;
```

```
      padding-bottom: 9px;
      border-bottom: 1px solid var(--line);
    }
    .patch-panel__name {
      margin: 0;
      font-size: 15px;
      font-weight: 600;
    }
    .patch-panel__meta {
      font-size: 12.5px;
      color: var(--ink-soft);
    }
    .patch-panel__meta a { color: var(--accent); }
    .patch-panel__tail { margin-left: auto; }
    .patch-panel .notice { margin: 0; }
    .patch-panel table { margin: 0; }
    /* ---------- 窄屏适配 ---------- */
    @media (max-width: 960px) {
      /* 上下布局时取消「一屏高 + 内容区内部滚动」，
         否则导航会占掉半屏、内容被挤在剩下的小格子里滚动 */
      .app-shell {
        flex-direction: column;
        height: auto;
        min-height: 100vh;
        min-height: 100dvh;
        overflow: visible;
      }
      .app-content { overflow: visible; }
      .app-header { position: static; }
      .app-side { width: auto; flex: none; }
      .app-brand { padding: 12px 14px; }
      .app-nav { padding: 4px 10px 10px; }
      .app-nav ul { display: flex; flex-wrap: wrap; gap:
    4px; }
      .app-nav a { margin-bottom: 0; }
      .app-nav__group { margin: 8px 0 4px; }
      .app-user { padding: 10px 14px; }
      .app-header { padding: 0 14px; gap: 10px; }
      .search-box { width: 180px; }
      .app-main { padding: 16px 14px 20px; }
      .app-foot { padding: 0 14px 14px; }
    }
    @media (max-width: 640px) {
      .crumb { display: none; }
      .search-box { width: auto; flex: 1; }
    }
  </style>
</head>
<body class="{% if not current_user.is_authenticated
    %}body--login{% endif %}">
```

```
<a class="skip-link" href="#main">跳到主要内容</a>
{# 取一次即可（同一请求内重复调用返回相同内容）。
   类别为 toast / ok / toast-error 的渲染成中上方浮层，出现后自动淡出；
   其余仍是页面顶部的提示条 ——
   需要对照着改表单的错误，留在页面上比一闪而过更有用。 #}
{% set flash_msgs = get_flashed_messages(with_categories=true) %}
<div class="toast-stack">
  {% for category, message in flash_msgs if category in ("toast", "ok", "toast-error") %}
  <div class="toast-center{% if category == 'toast-error' %} toast-center--error{% endif %}"
       role="status" aria-live="polite">{{ message }}</div>
  {% endfor %}
</div>
{% if current_user.is_authenticated %}
<div class="app-shell">
  <aside class="app-side">
    <div class="app-brand">
      <span class="app-brand__mark">{{ ui.icon("mountain", 21) }}</span>
      <span>
        <span class="app-brand__name">矿山自然恢复评价</span><br>
        <span class="app-brand__sub">适宜性评价系统 Ｖ1.0</span>
      </span>
    </div>
    <nav class="app-nav" aria-label="功能导航">
      <ul>
        <li>
          <a href="/"{% if home_active %} aria-current="page"{% endif %}>
            {{ ui.icon("home") }}<span>工作台</span>
          </a>
        </li>
      </ul>
      {% for g in nav_groups %}
      <p class="app-nav__group">{{ g.title }}</p>
      <ul>
        {% for item in g.links %}
        <li>
          <a href="{{ item.href }}"{% if item.active %} aria-current="page"{% endif %}>
            {{ ui.icon(item.icon) }}<span>{{ item.label }}</span>
            {% if item.badge %}<span class="app-nav__badge">{{ item.badge }}</span>{% endif %}
          </a>
        </li>
```

```
        {% endfor %}                                                    5501
      </ul>                                                             5502
      {% endfor %}                                                      5503
    </nav>                                                              5504
    <div class="app-user">                                              5505
      <a class="app-user__me" href="/profile"                           5506
title="个人设置">                                                       5507
        {% if current_user.avatar %}                                    5508
        <img class="app-user__avatar"                                   5509
             src="{{ url_for('static',                                  5510
filename=current_user.avatar) }}" alt="">                               5511
        {% else %}                                                      5512
        <span class="app-user__avatar"                                  5513
aria-hidden="true">{{ current_user.initial }}</span>                    5514
        {% endif %}                                                     5515
        <span style="min-width:0;">                                     5516
          <span class="app-user__name">{{                              5517
current_user.shown_name }}</span>                                       5518
          <span class="app-user__state">                               5519
            <span class="app-user__dot"                                 5520
aria-hidden="true"></span>                                              5521
            {{ role_labels.get(current_user.role,                       5522
current_user.role) }}                                                   5523
          </span>                                                       5524
        </span>                                                         5525
      </a>                                                              5526
      <a class="app-user__out" href="/logout" title="退出登录"          5527
         aria-label="退出登录">{{ ui.icon("logout", 17) }}</a>          5528
    </div>                                                              5529
</aside>                                                                5530
<div class="app-content">                                               5531
  <header class="app-header">                                           5532
    <nav class="crumb" aria-label="当前位置">                           5533
      <a href="/">首页</a>                                              5534
      {{ ui.icon("chevron", 13) }}                                      5535
      <strong>{{ page_title }}</strong>                                 5536
    </nav>                                                              5537
    <span class="app-header__spacer"></span>                            5538
    <form class="search-box" method="get"                               5539
action="/search" role="search">                                         5540
      {{ ui.icon("search", 16) }}                                       5541
      <label for="q" class="visually-hidden">搜索项目、图斑、标准<      5542
/label>                                                                 5543
      <input id="q" type="search" name="q"                              5544
autocomplete="off"                                                      5545
             placeholder="搜索项目、图斑、标准" value="{{               5546
search_q }}">                                                           5547
      <button type="submit" aria-label="搜索">{{                        5548
ui.icon("chevron", 15) }}</button>                                      5549
    </form>                                                             5550
```

```
          {% if current_user.is_admin %}                                    5551
          <a class="icon-btn" href="/profile?tab=logs"                      5552
    title="操作日志"                                                        5553
              aria-label="操作日志">{{ ui.icon("bell", 18) }}</a>           5554
          {% endif %}                                                       5555
        </header>                                                           5556
  {% else %}                                                                5557
  <div class="login-shell">                                                 5558
  {% endif %}                                                               5559
    <main id="main" class="app-main">                                       5560
      <div class="container">                                               5561
        {% for category, message in flash_msgs if category                  5562
    not in ("toast", "ok", "toast-error") %}                                5563
          <p class="flash {{ category }}" role="status">{{                  5564
    message }}</p>                                                          5565
        {% endfor %}                                                        5566
        {% block content %}{% endblock %}                                   5567
      </div>                                                                5568
    </main>                                                                 5569
  {% if current_user.is_authenticated %}                                    5570
      <footer class="app-foot">                                             5571
        <span>历史遗留废弃矿山自然恢复适宜性评价系统 Ｖ1.0</span>                5572
        <span>数据截止 {{ today }}</span>                                   5573
      </footer>                                                             5574
    </div>                                                                  5575
  </div>                                                                    5576
  {% else %}                                                                5577
  </div>                                                                    5578
  {% endif %}                                                               5579
  {% block scripts %}{% endblock %}                                         5580
</body>                                                                     5581
</html>                                                                     5582
#  ==== app/templates/_macros.html ====                                     5583
{#  共用宏：图标、流程状态面板 #}                                           5584
{#  ================================================================      5585
    ==================                                                      5586
    图标：内联 SVG 线性图标（24×24 视图框，随文字颜色走）                   5587
    不引图标字体或外部图片，避免额外请求与资源依赖；                        5588
    stroke 用 currentColor，在深色侧边栏与浅色内容区都能自适应。            5589
    ================================================================    5590
    ================== #}                                                  5591
{% set ICONS = {                                                            5592
  "dot": '<circle cx="12" cy="12" r="3"/>',                                 5593
  "home": '<path d="M3 9.6 12 3l9 6.6V20a2 2 0 0 1-2 2H5a2                  5594
    2 0 0 1-2-2z"/><path d="M9.5 22v-7h5v7"/>',                             5595
  "folder": '<path d="M3 7.5A2 2 0 0 1 5 5.5h3.6l2 2.2H19a2                 5596
    2 0 0 1 2 2v7.8a2 2 0 0 1-2 2H5a2 2 0 0 1-2-2z"/>',                     5597
  "book": '<path d="M4 4.8A2.8 2.8 0 0 1 6.8                                5598
    2H20v17.5H6.8A2.8 2.8 0 0 0 4 22z"/><path d="M4                         5599
    19.5A2.8 2.8 0 0 1 6.8 16.7H20"/>',                                     5600
```

```
  "shield": '<path d="M12 22s8-4 8-10V5.2L12 2 4 5.2V12c0 6
    8 10 8 10z"/><path d="m9 12 2.2 2.2L15.5 10"/>',
  "layers": '<path d="m12 2.8 9 4.7-9 4.7-9-4.7z"/><path
    d="m3 12.5 9 4.7 9-4.7"/><path d="m3 17 9 4.7 9-4.7"/>',
  "sliders": '<path d="M4 21v-6M4 11V3M12 21v-9M12 8V3M20
    21v-4M20 13V3"/><path d="M1.5 15h5M9.5 12h5M17.5
    17h5"/>',
  "database": '<ellipse cx="12" cy="5.5" rx="8"
    ry="3.2"/><path d="M4 5.5v13c0 1.8 3.6 3.2 8 3.2s8-1.4
    8-3.2v-13"/><path d="M4 12c0 1.8 3.6 3.2 8 3.2s8-1.4
    8-3.2"/>',
  "target": '<circle cx="12" cy="12" r="9"/><circle cx="12"
    cy="12" r="4.6"/><circle cx="12" cy="12" r="1"/>',
  "file": '<path d="M14 2.8H7a2 2 0 0 0-2 2v14.4a2 2 0 0 0
    2 2h10a2 2 0 0 0 2-2V7.8z"/><path d="M14
    2.8V7.8h5"/><path d="M8.8 13h6.4M8.8 17h6.4"/>',
  "users": '<circle cx="9.5" cy="8" r="3.6"/><path d="M3.5
    20.5v-1.4A3.6 3.6 0 0 1 7.1 15.5h4.8a3.6 3.6 0 0 1 3.6
    3.6v1.4"/><path d="M17 4.9a3.6 3.6 0 0 1 0 6.9"/><path
    d="M19 15.8a3.6 3.6 0 0 1 2.4 3.4v1.3"/>',
  "user": '<circle cx="12" cy="8" r="4"/><path d="M4.5
    21v-1.6A4.4 4.4 0 0 1 8.9 15h6.2a4.4 4.4 0 0 1 4.4
    4.4V21"/>',
  "mail": '<rect x="3" y="5" width="18" height="14"
    rx="2.5"/><path d="m3.8 7 8.2 5.8L20.2 7"/>',
  "phone": '<path d="M6.4 3h2.8l1.5 4.4-2 1.4a12.4 12.4 0 0
    0 6.5 6.5l1.4-2L21 14.8v2.8a2.4 2.4 0 0 1-2.6 2.4A16.6
    16.6 0 0 1 3.5 5.6 2.4 2.4 0 0 1 5.9 3z"/>',
  "lock": '<rect x="4" y="10.4" width="16" height="10.6"
    rx="2.6"/><path d="M8 10.4V7.6a4 4 0 0 1 8 0v2.8"/>',
  "briefcase": '<rect x="3" y="7.4" width="18"
    height="13.1" rx="2.6"/><path d="M8.4 7.4V6A2 2 0 0 1
    10.4 4h3.2a2 2 0 0 1 2 2v1.4"/><path d="M3 12.6h18"/>',
  "search": '<circle cx="11" cy="11" r="7"/><path d="m20.5
    20.5-4-4"/>',
  "bell": '<path d="M18 9.5a6 6 0 1 0-12 0c0 5-2.2 6.2-2.2
    6.2h16.4S18 14.5 18 9.5"/><path d="M13.8 19.5a2 2 0 0
    1-3.6 0"/>',
  "plus": '<path d="M12 5v14M5 12h14"/>',
  "upload": '<path d="M12 16V4"/><path d="m7.5 8.5 4.5-4.5
    4.5 4.5"/><path d="M4 16.5v2A2.5 2.5 0 0 0 6.5
    21h11a2.5 2.5 0 0 0 2.5-2.5v-2"/>',
  "logout": '<path d="M15 4h3.5A2.5 2.5 0 0 1 21 6.5v11A2.5
    2.5 0 0 1 18.5 20H15"/><path d="M10 16.5 5.5 12 10
    7.5"/><path d="M5.5 12H15"/>',
  "chevron": '<path d="m9.5 5.5 6.5 6.5-6.5 6.5"/>',
  "warn": '<path d="M12 3.5 21.5 20H2.5z"/><path d="M12
    10v4.2"/><path d="M12 17.4h.01"/>',
  "mountain": '<path d="m3 20 6.5-11L14 17l2.5-4L21
    20z"/><circle cx="17" cy="6" r="2.2"/>',
```

```
} %}
{% macro icon(name, size=18) %}
<svg class="ico" width="{{ size }}" height="{{ size }}"
     viewBox="0 0 24 24"
      fill="none" stroke="currentColor" stroke-width="1.7"
      stroke-linecap="round" stroke-linejoin="round"
      aria-hidden="true" focusable="false">{{
     ICONS.get(name, ICONS.dot)|safe }}</svg>
{% endmacro %}
{# 评价流程步骤条：按论文 4.1 节顺序显示各前置步骤的完成情况
   顺序为：限制性条件判定 → 划分微地质单元 → 录入指标数据 → 执行评价。
   论文原文：「先进行政策、位置约束地质灾害判断……对于满足限制性条件的
   废弃矿山，按照地形地貌划分……三种地质微单元类型」，故判定在前。 #}
{% macro workflow_steps(progress) %}
<div class="steps">
  <div class="step {% if progress.constraint.passed
    %}step--done{% elif progress.constraint.done
    %}step--fail{% else %}step--todo{% endif %}">
    <span class="step__no">1</span>
    <span class="step__name">限制性条件判定</span>
    <span class="step__state">
      {% if progress.constraint.passed %}已通过
      {% elif progress.constraint.done %}未通过
      {% else %}待完成{% endif %}
    </span>
  </div>
  <div class="step {% if progress.units.done %}step--done{%
    else %}step--todo{% endif %}">
    <span class="step__no">2</span>
    <span class="step__name">划分微地质单元</span>
    <span class="step__state">
      {% if progress.units.done %}已完成{% else %}待完成{% endif
    %}
    </span>
  </div>
  <div class="step {% if progress.data.ready %}step--done{%
    else %}step--todo{% endif %}">
    <span class="step__no">3</span>
    <span class="step__name">录入指标数据</span>
    <span class="step__state">
      {% if progress.data.ready %}已完成{% else %}待完成{% endif
    %}
    </span>
  </div>
  <div class="step step--next">
    <span class="step__no">4</span>
    <span class="step__name">执行适宜性评价</span>
    <span class="step__state">当前步骤</span>
  </div>
</div>
```

```
{% endmacro %}
{# 前置条件未满足时的提示块：只列缺什么，并给出处理入口 #}
{% macro workflow_blocking(progress) %}
{% if progress.blocking %}
<div class="notice notice--warn">
  <p class="notice__title">尚不能执行本步骤，需要先完成：</p>
  <ol class="notice__list">
    {% for b in progress.blocking %}
    <li>
      <strong>{{ b.label }}</strong>{% if b.reason %}：{{ b.reason }}{% endif %}
      <a class="link-btn" href="{{ b.fix_url }}">{{ b.fix_label }}</a>
    </li>
    {% endfor %}
  </ol>
</div>
{% endif %}
{% endmacro %}
{# 一行式状态摘要，用于列表页 #}
{% macro workflow_badge(progress) %}
{% if progress.ready %}
<span class="tag tag--ok">可执行评价</span>
{% elif progress.blocking %}
<span class="tag tag--warn">{{ progress.blocking[0].label }}</span>
<span class="muted">{{ progress.blocking[0].reason }}</span>
{% endif %}
{% endmacro %}
{# 层级引导：本系统是「项目 → 图斑」两级结构，评价针对的是图斑而非项目。
   下游各页按图斑组织数据，故只要存在「还没建图斑的项目」，就直接列出并给出入口。 #}
{% macro patch_guide(projects_without_patch) %}
{% if projects_without_patch %}
<div class="notice notice--warn">
  <p class="notice__title">
    以下 {{ projects_without_patch|length }}
    个项目下还没有图斑，本页不会出现它们：
  </p>
  <ul class="notice__list">
    {% for p in projects_without_patch %}
    <li>
      {{ p.name }}
      <a class="link-btn" href="/project/{{ p.id }}/patch/add">新建图斑</a>
    </li>
    {% endfor %}
  </ul>
</div>
{% endif %}
{% endmacro %}
```

```
# ==== app/templates/_login_badge.svg ====
<svg class="lp-badge__svg" xmlns="http://www.w3.org/2000/svg
    " viewBox="0 0 240 240" role="img"
    aria-label="历史遗留废弃矿山自然恢复适宜性评价系统徽标">
  <title>系统徽标</title>
  <g class="lp-ring" style="--k: 2.7">
    <circle cx="162.00" cy="120.00" r="2.60"
    fill="#c4f5ab"/>
    <circle cx="157.84" cy="138.22" r="2.97"
    fill="#b8f5ab"/>
    <circle cx="146.19" cy="152.84" r="3.06"
    fill="#acf5ab"/>
    <circle cx="129.35" cy="160.95" r="2.80"
    fill="#abf5b7"/>
    <circle cx="110.65" cy="160.95" r="2.40"
    fill="#abf5c2"/>
    <circle cx="93.81" cy="152.84" r="2.14" fill="#abf5ce"/>
    <circle cx="82.16" cy="138.22" r="2.23" fill="#abf5da"/>
    <circle cx="78.00" cy="120.00" r="2.60" fill="#abf5e6"/>
    <circle cx="82.16" cy="101.78" r="2.97" fill="#abf5da"/>
    <circle cx="93.81" cy="87.16" r="3.06" fill="#abf5ce"/>
    <circle cx="110.65" cy="79.05" r="2.80" fill="#abf5c2"/>
    <circle cx="129.35" cy="79.05" r="2.40" fill="#abf5b7"/>
    <circle cx="146.19" cy="87.16" r="2.14" fill="#acf5ab"/>
    <circle cx="157.84" cy="101.78" r="2.23"
    fill="#b8f5ab"/>
  </g>
  <g class="lp-ring" style="--k: 2.15">
    <circle cx="177.29" cy="129.07" r="3.92"
    fill="#a4e28d"/>
    <circle cx="171.68" cy="146.33" r="4.01"
    fill="#9be28d"/>
    <circle cx="161.01" cy="161.01" r="3.87"
    fill="#91e28d"/>
    <circle cx="146.33" cy="171.68" r="3.55"
    fill="#8de292"/>
    <circle cx="129.07" cy="177.29" r="3.17"
    fill="#8de29c"/>
    <circle cx="110.93" cy="177.29" r="2.88"
    fill="#8de2a6"/>
    <circle cx="93.67" cy="171.68" r="2.79" fill="#8de2af"/>
    <circle cx="78.99" cy="161.01" r="2.93" fill="#8de2b9"/>
    <circle cx="68.32" cy="146.33" r="3.25" fill="#8de2c2"/>
    <circle cx="62.71" cy="129.07" r="3.63" fill="#8de2cc"/>
    <circle cx="62.71" cy="110.93" r="3.92" fill="#8de2cc"/>
    <circle cx="68.32" cy="93.67" r="4.01" fill="#8de2c2"/>
    <circle cx="78.99" cy="78.99" r="3.87" fill="#8de2b9"/>
    <circle cx="93.67" cy="68.32" r="3.55" fill="#8de2af"/>
    <circle cx="110.93" cy="62.71" r="3.17" fill="#8de2a6"/>
    <circle cx="129.07" cy="62.71" r="2.88" fill="#8de29c"/>
```

```
    <circle cx="146.33" cy="68.32" r="2.79" fill="#8de292" />
    <circle cx="161.01" cy="78.99" r="2.93" fill="#91e28d" />
    <circle cx="171.68" cy="93.67" r="3.25" fill="#9be28d" />
    <circle cx="177.29" cy="110.93" r="3.63"
    fill="#a4e28d" />
  </g>
  <g class="lp-ring" style="--k: 1.85">
    <circle cx="194.00" cy="120.00" r="4.94"
    fill="#91cf72" />
    <circle cx="191.85" cy="137.71" r="4.92"
    fill="#88cf72" />
    <circle cx="185.52" cy="154.39" r="4.73"
    fill="#80cf72" />
    <circle cx="175.39" cy="169.07" r="4.42"
    fill="#78cf72" />
    <circle cx="162.04" cy="180.90" r="4.06"
    fill="#72cf73" />
    <circle cx="146.24" cy="189.19" r="3.73"
    fill="#72cf7b" />
    <circle cx="128.92" cy="193.46" r="3.51"
    fill="#72cf83" />
    <circle cx="111.08" cy="193.46" r="3.45"
    fill="#72cf8b" />
    <circle cx="93.76" cy="189.19" r="3.56" fill="#72cf93" />
    <circle cx="77.96" cy="180.90" r="3.81" fill="#72cf9c" />
    <circle cx="64.61" cy="169.07" r="4.16" fill="#72cfa4" />
    <circle cx="54.48" cy="154.39" r="4.51" fill="#72cfac" />
    <circle cx="48.15" cy="137.71" r="4.80" fill="#72cfb4" />
    <circle cx="46.00" cy="120.00" r="4.94" fill="#72cfbc" />
    <circle cx="48.15" cy="102.29" r="4.92" fill="#72cfb4" />
    <circle cx="54.48" cy="85.61" r="4.73" fill="#72cfac" />
    <circle cx="64.61" cy="70.93" r="4.42" fill="#72cfa4" />
    <circle cx="77.96" cy="59.10" r="4.06" fill="#72cf9c" />
    <circle cx="93.76" cy="50.81" r="3.73" fill="#72cf93" />
    <circle cx="111.08" cy="46.54" r="3.51" fill="#72cf8b" />
    <circle cx="128.92" cy="46.54" r="3.45" fill="#72cf83" />
    <circle cx="146.24" cy="50.81" r="3.56" fill="#72cf7b" />
    <circle cx="162.04" cy="59.10" r="3.81" fill="#72cf73" />
    <circle cx="175.39" cy="70.93" r="4.16" fill="#78cf72" />
    <circle cx="185.52" cy="85.61" r="4.51" fill="#80cf72" />
    <circle cx="191.85" cy="102.29" r="4.80"
    fill="#88cf72" />
  </g>
  <g class="lp-ring" style="--k: 1.62">
    <circle cx="209.57" cy="128.82" r="5.68"
    fill="#76bb59" />
    <circle cx="206.12" cy="146.13" r="5.40"
    fill="#6fbb59" />
    <circle cx="199.37" cy="162.43" r="5.06"
    fill="#68bb59" />
```

```
    <circle cx="189.57" cy="177.10" r="4.71"                    5851
    fill="#61bb59"/>                                            5852
    <circle cx="177.10" cy="189.57" r="4.41"                    5853
    fill="#5abb59"/>                                            5854
    <circle cx="162.43" cy="199.37" r="4.19"                    5855
    fill="#59bb5f"/>                                            5856
    <circle cx="146.13" cy="206.12" r="4.10"                    5857
    fill="#59bb66"/>                                            5858
    <circle cx="128.82" cy="209.57" r="4.15"                    5859
    fill="#59bb6d"/>                                            5860
    <circle cx="111.18" cy="209.57" r="4.32"                    5861
    fill="#59bb73"/>                                            5862
    <circle cx="93.87" cy="206.12" r="4.60" fill="#59bb7a"/>    5863
    <circle cx="77.57" cy="199.37" r="4.94" fill="#59bb81"/>    5864
    <circle cx="62.90" cy="189.57" r="5.29" fill="#59bb88"/>    5865
    <circle cx="50.43" cy="177.10" r="5.59" fill="#59bb8f"/>    5866
    <circle cx="40.63" cy="162.43" r="5.81" fill="#59bb96"/>    5867
    <circle cx="33.88" cy="146.13" r="5.90" fill="#59bb9d"/>    5868
    <circle cx="30.43" cy="128.82" r="5.85" fill="#59bba4"/>    5869
    <circle cx="30.43" cy="111.18" r="5.68" fill="#59bba4"/>    5870
    <circle cx="33.88" cy="93.87" r="5.40" fill="#59bb9d"/>     5871
    <circle cx="40.63" cy="77.57" r="5.06" fill="#59bb96"/>     5872
    <circle cx="50.43" cy="62.90" r="4.71" fill="#59bb8f"/>     5873
    <circle cx="62.90" cy="50.43" r="4.41" fill="#59bb88"/>     5874
    <circle cx="77.57" cy="40.63" r="4.19" fill="#59bb81"/>     5875
    <circle cx="93.87" cy="33.88" r="4.10" fill="#59bb7a"/>     5876
    <circle cx="111.18" cy="30.43" r="4.15" fill="#59bb73"/>    5877
    <circle cx="128.82" cy="30.43" r="4.32" fill="#59bb6d"/>    5878
    <circle cx="146.13" cy="33.88" r="4.60" fill="#59bb66"/>    5879
    <circle cx="162.43" cy="40.63" r="4.94" fill="#59bb5f"/>    5880
    <circle cx="177.10" cy="50.43" r="5.29" fill="#5abb59"/>    5881
    <circle cx="189.57" cy="62.90" r="5.59" fill="#61bb59"/>    5882
    <circle cx="199.37" cy="77.57" r="5.81" fill="#68bb59"/>    5883
    <circle cx="206.12" cy="93.87" r="5.90" fill="#6fbb59"/>    5884
    <circle cx="209.57" cy="111.18" r="5.85"                    5885
    fill="#76bb59"/>                                            5886
  </g>                                                          5887
  <g class="lp-ring" style="--k: 1.46">                         5888
    <circle cx="226.00" cy="120.00" r="5.94"                    5889
    fill="#65a843"/>                                            5890
    <circle cx="224.55" cy="137.45" r="5.62"                    5891
    fill="#5fa843"/>                                            5892
    <circle cx="220.26" cy="154.42" r="5.29"                    5893
    fill="#59a843"/>                                            5894
    <circle cx="213.22" cy="170.45" r="5.00"                    5895
    fill="#53a843"/>                                            5896
    <circle cx="203.65" cy="185.11" r="4.77"                    5897
    fill="#4da843"/>                                            5898
    <circle cx="191.79" cy="197.99" r="4.63"                    5899
    fill="#47a843"/>                                            5900
```

```
    <circle cx="177.98" cy="208.74" r="4.59"                    5901
    fill="#43a846"/>                                            5902
    <circle cx="162.58" cy="217.07" r="4.67"                    5903
    fill="#43a84c"/>                                            5904
    <circle cx="146.02" cy="222.76" r="4.85"                    5905
    fill="#43a852"/>                                            5906
    <circle cx="128.75" cy="225.64" r="5.10"                    5907
    fill="#43a858"/>                                            5908
    <circle cx="111.25" cy="225.64" r="5.42"                    5909
    fill="#43a85e"/>                                            5910
    <circle cx="93.98" cy="222.76" r="5.75" fill="#43a864"/>    5911
    <circle cx="77.42" cy="217.07" r="6.06" fill="#43a86a"/>    5912
    <circle cx="62.02" cy="208.74" r="6.33" fill="#43a870"/>    5913
    <circle cx="48.21" cy="197.99" r="6.52" fill="#43a876"/>    5914
    <circle cx="36.35" cy="185.11" r="6.60" fill="#43a87c"/>    5915
    <circle cx="26.78" cy="170.45" r="6.58" fill="#43a882"/>    5916
    <circle cx="19.74" cy="154.42" r="6.45" fill="#43a888"/>    5917
    <circle cx="15.45" cy="137.45" r="6.23" fill="#43a88e"/>    5918
    <circle cx="14.00" cy="120.00" r="5.94" fill="#43a894"/>    5919
    <circle cx="15.45" cy="102.55" r="5.62" fill="#43a88e"/>    5920
    <circle cx="19.74" cy="85.58" r="5.29" fill="#43a888"/>     5921
    <circle cx="26.78" cy="69.55" r="5.00" fill="#43a882"/>     5922
    <circle cx="36.35" cy="54.89" r="4.77" fill="#43a87c"/>     5923
    <circle cx="48.21" cy="42.01" r="4.63" fill="#43a876"/>     5924
    <circle cx="62.02" cy="31.26" r="4.59" fill="#43a870"/>     5925
    <circle cx="77.42" cy="22.93" r="4.67" fill="#43a86a"/>     5926
    <circle cx="93.98" cy="17.24" r="4.85" fill="#43a864"/>     5927
    <circle cx="111.25" cy="14.36" r="5.10" fill="#43a85e"/>    5928
    <circle cx="128.75" cy="14.36" r="5.42" fill="#43a858"/>    5929
    <circle cx="146.02" cy="17.24" r="5.75" fill="#43a852"/>    5930
    <circle cx="162.58" cy="22.93" r="6.06" fill="#43a84c"/>    5931
    <circle cx="177.98" cy="31.26" r="6.33" fill="#43a846"/>    5932
    <circle cx="191.79" cy="42.01" r="6.52" fill="#47a843"/>    5933
    <circle cx="203.65" cy="54.89" r="6.60" fill="#4da843"/>    5934
    <circle cx="213.22" cy="69.55" r="6.58" fill="#53a843"/>    5935
    <circle cx="220.26" cy="85.58" r="6.45" fill="#59a843"/>    5936
    <circle cx="224.55" cy="102.55" r="6.23"                    5937
    fill="#5fa843"/>                                            5938
  </g>                                                          5939
  <g class="lp-badge__mark">                                    5940
    <circle cx="120" cy="120" r="27" fill="#14432f"/>           5941
    <g transform="translate(120 120) scale(1.5)                 5942
    translate(-12 -12)" fill="none" stroke="#ffffff"            5943
    stroke-width="1.9" stroke-linecap="round"                   5944
    stroke-linejoin="round">                                    5945
      <path d="m3 20 6.5-11L14 17l2.5-4L21 20z"/>               5946
      <circle cx="17" cy="6" r="2.2"/>                          5947
    </g>                                                        5948
  </g>                                                          5949
</svg>                                                          5950
```

```
# ==== app/templates/error.html ====
{% extends "base.html" %}
{% block title %}{{ code }} · {{ message }}{% endblock %}
{% block content %}
<div class="card" style="max-width:560px;">
  <h2>{{ code }} · {{ message }}</h2>
  <p class="muted">
    请返回上一页重试；若确认该操作应当可用，请检查当前账号的角色权限。
  </p>
  <p class="actions" style="margin-top:16px;">
    <a class="btn" href="/">返回首页</a>
  </p>
</div>
{% endblock %}
# ==== app/templates/auth/login.html ====
{% extends "base.html" %}
{% from "_macros.html" import icon %}
{% block title %}登录 · 历史遗留废弃矿山自然恢复适宜性评价系统{% endblock %}
{% block content %}
<style>
  /* ================== 登录页（仿 Apple 账户页） ================== */
    纯白底、居中、大量留白；色彩只用在徽标与按钮上。
    首屏放徽标、标题、一句话说明与登录按钮；系统介绍在下方，
    随滚动逐条弹入、往回滚时弹回。 */
  .lp { color: #1d1d1f; }
  /* ---------- 首屏 ---------- */
  .lp-hero {
    display: flex;
    flex-direction: column;
    align-items: center;
    justify-content: center;
    text-align: center;
    min-height: 100dvh;
    padding: 56px 20px 72px;
  }
  /* 徽标：滚动时各圈向四周散开并淡出。
     --p 为 0~1 的滚动进度，由页面脚本写入；
     每圈自带 --k（散开倍数，由内向外递减），使各圈逐渐拉开距离 */
  .lp-badge {
    width: 212px;
    height: 212px;
    margin-bottom: 36px;
    --p: 0;
    animation: lp-in .7s ease both;
  }
  .lp-badge__svg {
    display: block;
    width: 100%;
    height: 100%;
    overflow: visible;                    /* 散开时圆点不被画布裁掉 */
```

```
    opacity: calc(1 - var(--p) * .94);                                  6001
    filter: blur(calc(var(--p) * 2.6px));                               6002
  }                                                                     6003
  .lp-ring {                                                            6004
    transform-box: fill-box;                                            6005
    transform-origin: center;                                           6006
    transform: scale(calc(1 + (var(--k) - 1) * var(--p)));              6007
  }                                                                     6008
  .lp-badge__mark {                                                     6009
    transform-box: fill-box;                                            6010
    transform-origin: center;                                           6011
    transform: scale(calc(1 + .18 * var(--p)));                         6012
  }                                                                     6013
  .lp-title {                                                           6014
    margin: 0 0 18px;                                                   6015
    font-size: clamp(28px, 4.2vw, 42px);                                6016
    font-weight: 700;                                                   6017
    line-height: 1.3;                                                   6018
    letter-spacing: -.4px;                                              6019
    animation: lp-in .7s ease .06s both;                                6020
  }                                                                     6021
  .lp-cta {                                                             6022
    padding: 13px 52px;                                                 6023
    font-family: inherit;                                               6024
    font-size: 17px;                                                    6025
    font-weight: 500;                                                   6026
    color: #fff;                                                        6027
    background: var(--accent);                                          6028
    border: none;                                                       6029
    border-radius: 980px;                                               6030
    cursor: pointer;                                                    6031
    animation: lp-in .7s ease .18s both;                                6032
    transition: background-color .2s ease, box-shadow .2s               6033
    ease,                                                               6034
                transform .2s ease;                                     6035
  }                                                                     6036
  .lp-cta:hover {                                                       6037
    background: var(--accent-hover);                                    6038
    box-shadow: 0 10px 26px rgba(31, 107, 69, .26);                     6039
  }                                                                     6040
  .lp-cta:active { transform: translateY(1px); box-shadow:              6041
    none; }                                                             6042
  .lp-cta:focus-visible { outline: 3px solid rgba(31, 107,              6043
    69, .40); outline-offset: 3px; }                                    6044
  .lp-more {                                                            6045
    margin-top: 38px;                                                   6046
    font-size: 13px;                                                    6047
    color: #86868b;                                                     6048
    text-decoration: none;                                              6049
    animation: lp-in .7s ease .24s both;                                6050
```

```
}
.lp-more:hover { color: var(--accent); }
@keyframes lp-in {
  from { opacity: 0; transform: translateY(12px); }
  to   { opacity: 1; transform: none; }
}
/* ---------- 滚动进场 ----------
   进入视口时按方向弹入，离开视口时弹回，可反复。
   交错延迟只加在「进入」的规则上，退出时不延迟，否则会显得拖沓。 */
.lp-reveal {
  opacity: 0;
  transition: opacity .62s ease,
              transform .62s cubic-bezier(.16, .84, .34, 1),
              border-color .62s ease, background-color .62s ease;
  transition-delay: 0s;
  will-change: opacity, transform;
}
.lp-reveal[data-dir="left"]  { transform: translate3d(-48px, 0, 0); }
.lp-reveal[data-dir="right"] { transform: translate3d(48px, 0, 0); }
.lp-reveal[data-dir="up"]    { transform: translate3d(0, 36px, 0); }
.lp-reveal.is-in {
  opacity: 1;
  transform: none;
  transition-delay: calc(var(--i, 0) * 85ms);
}
/* 标题颜色由淡到深，强化「浮现」的层次感 */
.lp-reveal h2,
.lp-reveal h3,
.lp-reveal .lp-mod__name { color: #c2d2c8; transition: color .7s ease; }
.lp-reveal.is-in h3,
.lp-reveal.is-in .lp-mod__name { color: var(--accent); }
.lp-reveal.is-in h2 { color: #1d1d1f; }
/* ---------- 系统介绍 ---------- */
.lp-intro {
  max-width: 980px;
  margin: 0 auto;
  padding: 20px 24px 72px;
}
.lp-intro__title {
  margin: 0 0 26px;
  font-size: clamp(22px, 2.6vw, 30px);
  text-align: center;
  letter-spacing: -.3px;
}
```

```
.lp-points {
  display: grid;
  grid-template-columns: repeat(auto-fit, minmax(240px, 1fr));
  gap: 16px;
  margin-bottom: 46px;
}
.lp-point {
  padding: 24px 22px;
  background: #f5f5f7;
  border-radius: 18px;
}
.lp-point__icon {
  display: grid;
  place-items: center;
  width: 38px;
  height: 38px;
  margin-bottom: 14px;
  border-radius: 11px;
  background: #fff;
  color: var(--accent);
}
.lp-point h3 { margin: 0 0 8px; font-size: 16px; }
.lp-point p {
  margin: 0;
  font-size: 13.5px;
  line-height: 1.85;
  color: #6e6e73;
  text-wrap: pretty;
}
.lp-sub-title { margin: 0 0 20px; font-size: 17px; }
.lp-mods {
  display: grid;
  grid-template-columns: repeat(auto-fit, minmax(258px, 1fr));
  gap: 12px;
  margin-bottom: 46px;
  align-items: start;
}
/* 模块卡：默认只显示编号与名称，鼠标悬停或键盘聚焦时才展开职责说明 */
.lp-mod {
  position: relative;
  padding: 17px 18px 17px 48px;
  background: #fff;
  border: 1px solid #e6e6e8;
  border-radius: 14px;
  outline: none;
  transition: border-color .3s ease, box-shadow .3s ease;
}
.lp-mod:hover,
```

```
.lp-mod:focus-within,
.lp-mod:focus {
  border-color: rgba(31, 107, 69, .42);
  box-shadow: 0 8px 22px rgba(19, 60, 43, .08);
}
.lp-mod__no {
  position: absolute;
  left: 17px;
  top: 17px;
  font-size: 13px;
  font-weight: 700;
  color: #c8d6cd;
  font-variant-numeric: tabular-nums;
}
.lp-mod__name {
  display: block;
  font-size: 14px;
  font-weight: 600;
}
/* 用 grid-template-rows 0fr→1fr 展开，高度过渡自然 */
.lp-mod__body {
  display: grid;
  grid-template-rows: 0fr;
  transition: grid-template-rows .34s ease, margin-top
  .34s ease;
}
.lp-mod:hover .lp-mod__body,
.lp-mod:focus-within .lp-mod__body {
  grid-template-rows: 1fr;
  margin-top: 7px;
}
.lp-mod__desc {
  overflow: hidden;
  margin: 0;
  font-size: 12.5px;
  line-height: 1.7;
  color: #6e6e73;
  opacity: 0;
  transition: opacity .3s ease;
  text-wrap: pretty;
}
.lp-mod:hover .lp-mod__desc,
.lp-mod:focus-within .lp-mod__desc { opacity: 1; }
/* 触摸设备没有悬停，直接把说明摊开 */
@media (hover: none) {
  .lp-mod__body { grid-template-rows: 1fr; margin-top:
  7px; }
  .lp-mod__desc { opacity: 1; }
}
.lp-foot {
```

```
    margin: 0;                                                    6201
    padding-top: 26px;                                            6202
    border-top: 1px solid #e6e6e8;                                6203
    text-align: center;                                           6204
    font-size: 12.5px;                                            6205
    color: #86868b;                                               6206
  }                                                               6207
  /* ---------- 登录面板：点击后渐变浮现 ---------- */              6208
  .lp-auth {                                                      6209
    position: fixed;                                              6210
    inset: 0;                                                     6211
    z-index: 60;                                                  6212
    display: grid;                                                6213
    place-items: center;                                          6214
    padding: 24px;                                                6215
    opacity: 0;                                                   6216
    visibility: hidden;                                           6217
    transition: opacity .42s ease, visibility 0s linear           6218
    .42s;                                                         6219
  }                                                               6220
  .lp-auth.is-open {                                              6221
    opacity: 1;                                                   6222
    visibility: visible;                                          6223
    transition: opacity .42s ease, visibility 0s;                 6224
  }                                                               6225
  /* 遮罩用绿色径向渐变，形成「渐变渗出来」的观感 */                  6226
  .lp-auth__scrim {                                               6227
    position: absolute;                                           6228
    inset: 0;                                                     6229
    background:                                                   6230
      radial-gradient(1000px 640px at 50% 0%,                     6231
                      rgba(31, 107, 69, .22), rgba(31, 107,       6232
    69, 0) 70%),                                                  6233
      linear-gradient(180deg, rgba(255, 255, 255, .84),           6234
                              rgba(246, 250, 247, .95));          6235
    backdrop-filter: blur(16px) saturate(1.25);                   6236
    -webkit-backdrop-filter: blur(16px) saturate(1.25);           6237
  }                                                               6238
  .lp-auth__panel {                                               6239
    position: relative;                                           6240
    width: 100%;                                                  6241
    max-width: 396px;                                             6242
    padding: 36px 36px 26px;                                      6243
    background: #fff;                                             6244
    border-radius: 20px;                                          6245
    box-shadow: 0 28px 80px rgba(14, 46, 32, .18);                6246
    opacity: 0;                                                   6247
    transform: translateY(18px) scale(.975);                      6248
    transition: transform .55s cubic-bezier(.16, .84, .34,        6249
    1) .05s,                                                      6250
```

```
                opacity .55s ease .05s;
}
.lp-auth.is-open .lp-auth__panel { opacity: 1; transform:
  none; }
.lp-auth__close {
  position: absolute;
  top: 14px;
  right: 14px;
  display: grid;
  place-items: center;
  width: 30px;
  height: 30px;
  color: #86868b;
  background: transparent;
  border: none;
  border-radius: 50%;
  cursor: pointer;
  transition: background-color .18s ease, color .18s ease;
}
.lp-auth__close:hover { background: #f5f5f7; color:
  #1d1d1f; }
.lp-auth__title {
  margin: 0 0 6px;
  text-align: center;
  font-size: 24px;
  font-weight: 700;
  letter-spacing: -.2px;
}
.lp-auth__sub {
  margin: 0 0 26px;
  text-align: center;
  font-size: 13px;
  color: #86868b;
}
/* 字段校验错误：贴着输入框显示，不做成会消失的浮层 ——
   这类消息要对着输入框看，一闪而过会让人以为「点了没反应」 */
.lp-field__error {
  margin: 7px 0 0;
  font-size: 12.5px;
  color: #9c2b23;
}
.lp-field input[aria-invalid="true"] {
  background: #fdeeec;
  border-color: rgba(156, 43, 35, .35);
}
.lp-field { margin-bottom: 15px; }
.lp-field label {
  display: block;
  margin-bottom: 7px;
  font-size: 13px;
```

```
    color: #6e6e73;                                                    6301
  }                                                                    6302
  .lp-field input {                                                    6303
    width: 100%;                                                       6304
    padding: 13px 16px;                                                6305
    font-family: inherit;                                              6306
    font-size: 16px;                /* 不小于 16px，避免手机端聚焦时自动放大 */    6307
    color: #1d1d1f;                                                    6308
    background: #f5f5f7;                                               6309
    border: 1px solid transparent;                                     6310
    border-radius: 12px;                                               6311
    outline: none;                                                     6312
    transition: background-color .18s ease, border-color               6313
    .18s ease,                                                         6314
                box-shadow .18s ease;                                  6315
  }                                                                    6316
  .lp-field input::placeholder { color: #a1a1a6; }                     6317
  .lp-field input:focus {                                              6318
    background: #fff;                                                  6319
    border-color: rgba(31, 107, 69, .55);                              6320
    box-shadow: 0 0 0 3px rgba(31, 107, 69, .14);                      6321
  }                                                                    6322
  .lp-remember {                                                       6323
    display: flex;                                                     6324
    align-items: center;                                               6325
    gap: 8px;                                                          6326
    margin: 4px 0 20px;                                                6327
    font-size: 13px;                                                   6328
    color: #6e6e73;                                                    6329
    cursor: pointer;                                                   6330
  }                                                                    6331
  .lp-remember input {                                                 6332
    width: 16px;                                                       6333
    height: 16px;                                                      6334
    margin: 0;                                                         6335
    accent-color: var(--accent);                                       6336
    cursor: pointer;                                                   6337
  }                                                                    6338
  .lp-submit {                                                         6339
    width: 100%;                                                       6340
    padding: 13px 20px;                                                6341
    font-family: inherit;                                              6342
    font-size: 16px;                                                   6343
    font-weight: 500;                                                  6344
    color: #fff;                                                       6345
    background: var(--accent);                                         6346
    border: none;                                                      6347
    border-radius: 980px;                                              6348
    cursor: pointer;                                                   6349
    transition: background-color .2s ease, box-shadow .2s              6350
```

```
      ease,                                                                    6351
                    transform .2s ease;                                        6352
    }                                                                          6353
    .lp-submit:hover {                                                         6354
      background: var(--accent-hover);                                         6355
      box-shadow: 0 10px 24px rgba(31, 107, 69, .26);                          6356
    }                                                                          6357
    .lp-submit:active { transform: translateY(1px);                            6358
      box-shadow: none; }                                                      6359
    .lp-submit:focus-visible { outline: 3px solid rgba(31,                     6360
      107, 69, .40); outline-offset: 3px; }                                    6361
    .lp-auth__hint {                                                           6362
      margin: 18px 0 0;                                                        6363
      text-align: center;                                                      6364
      font-size: 12.5px;                                                       6365
      color: #a1a1a6;                                                          6366
    }                                                                          6367
    .lp-auth__reg {                                                            6368
      color: var(--accent);                                                    6369
      font-weight: 500;                                                        6370
      text-decoration: none;                                                   6371
      border-bottom: 1px solid rgba(31, 107, 69, .35);                         6372
      transition: border-color .18s ease;                                      6373
    }                                                                          6374
    .lp-auth__reg:hover { border-bottom-color: var(--accent);                  6375
      }                                                                        6376
    /* 系统设置里要求减少动态效果时，去掉位移、缩放与滚动特效 */                  6377
    @media (prefers-reduced-motion: reduce) {                                  6378
      .lp-badge, .lp-title, .lp-cta, .lp-more { animation:                     6379
      none; }                                                                  6380
      .lp-badge__svg { opacity: 1; filter: none; }                             6381
      .lp-ring, .lp-badge__mark { transform: none; }                           6382
      .lp-reveal { opacity: 1; transform: none !important; }                   6383
      .lp-reveal h2, .lp-reveal h3, .lp-reveal .lp-mod__name                   6384
      { color: inherit; }                                                      6385
      .lp-reveal.is-in h2 { color: #1d1d1f; }                                  6386
      .lp-reveal.is-in h3, .lp-reveal.is-in .lp-mod__name {                    6387
      color: var(--accent); }                                                  6388
      .lp-auth, .lp-auth__panel { transition: opacity .2s                      6389
      linear; }                                                                6390
      .lp-auth__panel { transform: none; }                                     6391
    }                                                                          6392
    @media (max-width: 560px) {                                                6393
      .lp-hero { padding: 36px 18px 48px; min-height: auto; }                  6394
      .lp-badge { width: 160px; height: 160px; margin-bottom:                  6395
      26px; }                                                                  6396
      .lp-auth__panel { padding: 30px 24px 22px; }                             6397
    }                                                                          6398
</style>                                                                       6399
<div class="lp">                                                               6400
```

```
{# ---------- 首屏：徽标 + 标题 + 说明 + 登录入口 ---------- #}
<section class="lp-hero">
  <div class="lp-badge" id="lp-badge" aria-hidden="true">
    {% include "_login_badge.svg" %}
  </div>
  <h1 class="lp-title">历史遗留废弃矿山<br>自然恢复适宜性评价系统</h1>
  <button type="button" class="lp-cta" id="lp-open"
          aria-expanded="false" aria-controls="lp-auth">登录
  </button>
  <a class="lp-more" href="#intro">
    向下滑动，了解系统 <span aria-hidden="true">↓</span>
  </a>
</section>
{# ---------- 系统介绍：随滚动逐条弹入 ---------- #}
<section class="lp-intro" id="intro">
  <h2 class="lp-intro__title lp-reveal"
  data-dir="up">一套把评价流程固定下来的工具</h2>
  <div class="lp-points">
    {% set pdirs = ['left', 'up', 'right'] %}
    {% set points = [
      ('sliders', '方法可溯',
       '权重由层次分析法算出并做一致性检验，指标标准化、分单元赋分与面积加权合成的每一步都留有记录，可逐层回看。'),
      ('shield', '流程可控',
       '政策、区位约束、土壤污染、地质灾害、耕地属性五项限制性条件一票否决；未通过即终止评价，直接给出生态重建或人工辅助的结论。'),
      ('file', '结果可复算',
       '评价结果与逐层得分明细一并留存，报告取自已保存的评价记录，事后修改数据不会让报告与结论对不上。')
    ] %}
    {% for ic, title, text in points %}
    <div class="lp-point lp-reveal" data-dir="{{
  pdirs[loop.index0 % 3] }}"
         style="--i: {{ loop.index0 }}">
      <span class="lp-point__icon">{{ icon(ic, 20)
  }}</span>
      <h3>{{ title }}</h3>
      <p>{{ text }}</p>
    </div>
    {% endfor %}
  </div>
  <h3 class="lp-sub-title lp-reveal"
  data-dir="up">九个功能模块</h3>
  <div class="lp-mods">
    {% set mdirs = ['left', 'up', 'right'] %}
    {% for no, name, desc in modules %}
    <div class="lp-mod lp-reveal" data-dir="{{
  mdirs[loop.index0 % 3] }}"
         style="--i: {{ loop.index0 % 3 }}" tabindex="0">
      <span class="lp-mod__no">{{ no }}</span>
```

```
          <span class="lp-mod__name">{{ name }}</span>                    6451
          <div class="lp-mod__body"><p class="lp-mod__desc">{{            6452
    desc }}</p></div>                                                     6453
        </div>                                                            6454
        {% endfor %}                                                      6455
    </div>                                                                6456
    <p class="lp-foot lp-reveal" data-dir="up">                           6457
      历史遗留废弃矿山自然恢复适宜性评价系统 Ｖ1．0 ·                      6458
      运行于浏览器，数据保存在本机 ＳＱＬｉｔｅ 数据库                     6459
    </p>                                                                  6460
</section>                                                                6461
{# ---------------- 登录面板：点击「登录」后渐变浮现 ---------------- #}  6462
<div class="lp-auth" id="lp-auth" aria-hidden="true">                     6463
    <div class="lp-auth__scrim" data-lp-close></div>                      6464
    <div class="lp-auth__panel" role="dialog"                             6465
    aria-modal="true"                                                     6466
          aria-labelledby="lp-auth-title">                                6467
      <button type="button" class="lp-auth__close"                        6468
    data-lp-close                                                         6469
              aria-label="关闭登录框">                                    6470
        <svg width="16" height="16" viewBox="0 0 24 24"                   6471
    fill="none"                                                           6472
            stroke="currentColor" stroke-width="2"                        6473
    stroke-linecap="round"                                                6474
            aria-hidden="true"><path d="M6 6l12 12M18 6L6                 6475
    18"/></svg>                                                           6476
      </button>                                                           6477
      <h2 class="lp-auth__title" id="lp-auth-title">登录</h2>             6478
      <p class="lp-auth__sub">使用系统账号登录</p>                        6479
      <form method="post" novalidate>                                     6480
        {{ form.hidden_tag() }}                                           6481
        <div class="lp-field">                                            6482
          <label for="{{ form.username.id }}">账户名称</label>            6483
          {{ form.username(id=form.username.id,                           6484
    autocomplete="username",                                              6485
                           placeholder="请输入账户名称",                  6486
                           **({'aria-invalid': 'true'} if                 6487
    form.username.errors else {})) }}                                     6488
          {% for e in form.username.errors %}                             6489
          <p class="lp-field__error">{{ e }}</p>                          6490
          {% endfor %}                                                    6491
        </div>                                                            6492
        <div class="lp-field">                                            6493
          <label for="{{ form.password.id }}">密码</label>                6494
          {{ form.password(id=form.password.id,                           6495
    autocomplete="current-password",                                      6496
                           placeholder="请输入密码",                      6497
                           **({'aria-invalid': 'true'} if                 6498
    form.password.errors else {})) }}                                     6499
          {% for e in form.password.errors %}                             6500
```

```
            <p class="lp-field__error">{{ e }}</p>
            {% endfor %}
          </div>
          <label class="lp-remember" for="{{ form.remember.id
    }}">
            {{ form.remember(id=form.remember.id) }}
            <span>记住我（下次自动登录，一个月内有效）</span>
          </label>
          {{ form.submit(class_="lp-submit") }}
        </form>
        <p class="lp-auth__hint">
          还没有账号？<a class="lp-auth__reg"
                         href="{{ url_for('auth.register')
    }}">点我注册</a>
        </p>
        <p class="lp-auth__hint" style="margin-top:10px;">
          演示账号：admin 密码：admin123
        </p>
      </div>
    </div>
  </div>
{% endblock %}
{% block scripts %}
<script>
(function () {
  var reduce = window.matchMedia &&
    window.matchMedia('(prefers-reduced-motion:
    reduce)').matches;
  /* ---------- 登录面板 ---------- */
  var btn = document.getElementById('lp-open');
  var auth = document.getElementById('lp-auth');
  var lastFocus = null;
  function openPanel() {
    lastFocus = document.activeElement;
    auth.classList.add('is-open');
    auth.setAttribute('aria-hidden', 'false');
    btn.setAttribute('aria-expanded', 'true');
    document.body.style.overflow = 'hidden';
    // 记到地址栏，直接访问 /login#login 也能展开登录框
    if (location.hash !== '#login') {
      history.replaceState(null, '', location.pathname +
    '#login');
    }
    var first = auth.querySelector('input');
    if (first) window.setTimeout(function () {
    first.focus(); }, 300);
  }
  function closePanel() {
    auth.classList.remove('is-open');
    auth.setAttribute('aria-hidden', 'true');
```

```
    btn.setAttribute('aria-expanded', 'false');
    document.body.style.overflow = '';
    if (location.hash === '#login') {
      history.replaceState(null, '', location.pathname);
    }
    if (lastFocus && lastFocus.focus) lastFocus.focus();
  }
  if (btn && auth) {
    btn.addEventListener('click', openPanel);
    Array.prototype.forEach.call(
      auth.querySelectorAll('[data-lp-close]'),
      function (el) { el.addEventListener('click',
    closePanel); }
    );
    document.addEventListener('keydown', function (e) {
      if (e.key === 'Escape' && auth.classList.contains('is-
    open')) closePanel();
    });
    // 登录失败回到本页、或直接带 #login 访问时，直接展开登录框。
    // 失败可能落在两处：字段校验错误在表单里，凭据错误在浮层里。
    if (location.hash === '#login'
        || document.querySelector('.lp-field__error')
        || document.querySelector('.toast-center')) {
      openPanel();
    }
  }
  /* ---------- 内容进场 + 徽标散开：同一套滚动计算 ---------- */
     原先用 IntersectionObserver，实测在滚动到位置后回调并未生效，
     内容始终停在 opacity:0。改为直接按元素与视口的相对位置判断，
     与徽标用同一次滚动回调，行为可预期、也便于排查。 */
  var items = document.querySelectorAll('.lp-reveal');
  var badge = document.getElementById('lp-badge');
  var ticking = false;
  function applyScroll() {
    ticking = false;
    var vh = window.innerHeight || 800;
    /* 徽标：滚动前约三分之一屏内完成散开淡出 */
    if (badge) {
      var range = vh * 0.34;
      var p = range > 0 ? window.scrollY / range : 0;
      badge.style.setProperty('--p', Math.min(1,
    Math.max(0, p)).toFixed(3));
    }
    /* 内容：先批量测量，再改类名，避免读写交替触发反复重排 */
    if (reduce || !items.length) return;
    var box = [];
    for (var i = 0; i < items.length; i++) {
      box.push(items[i].getBoundingClientRect());
    }
    for (var j = 0; j < items.length; j++) {
```

```
        var r = box[j];                                                    6601
        // 上沿进入视口下部、下沿尚未越出视口上部 → 视为在视野内                6602
        var shown = r.top < vh * 0.88 && r.bottom > vh * 0.06;             6603
        if (shown) items[j].classList.add('is-in');                        6604
        else items[j].classList.remove('is-in');                           6605
      }                                                                    6606
    }                                                                      6607
    function onScroll() {                                                  6608
      if (!ticking) {                                                      6609
        ticking = true;                                                    6610
        window.requestAnimationFrame(applyScroll);                         6611
      }                                                                    6612
    }                                                                      6613
    // 减少动态效果时不做滚动特效，内容直接显示                             6614
    if (reduce) {                                                          6615
      Array.prototype.forEach.call(items, function (el) {                  6616
        el.classList.add('is-in');                                         6617
      });                                                                  6618
      if (badge) badge.style.setProperty('--p', '0');                      6619
    } else {                                                               6620
      window.addEventListener('scroll', onScroll, { passive:               6621
      true });                                                             6622
      window.addEventListener('resize', onScroll);                         6623
      applyScroll();                                                       6624
    }                                                                      6625
})();                                                                      6626
</script>                                                                  6627
{% endblock %}                                                             6628
# ==== app/templates/auth/register.html ====                               6629
{% extends "base.html" %}                                                  6630
{% block title %}注册账号 · 历史遗留废弃矿山自然恢复适宜性评价系统{% endblock %}   6631
{% block content %}                                                        6632
<style>                                                                    6633
  /* ================== 注册页 ================== */                        6634
     与登录页同一套观感：浅绿渐变底 + 白色卡片。                           6635
     必填只有账户名称与密码，其余收在「选填」分隔线以下，                  6636
     这样首次注册不会被一堆输入框挡住。 */                                 6637
  .rg {                                                                    6638
    min-height: 100dvh;                                                    6639
    display: grid;                                                         6640
    place-items: center;                                                   6641
    padding: 40px 20px 56px;                                               6642
    background:                                                            6643
      radial-gradient(900px 520px at 50% 0%,                               6644
                      rgba(31, 107, 69, .14), rgba(31, 107,                6645
    69, 0) 72%),                                                           6646
      #f6faf7;                                                             6647
  }                                                                        6648
  .rg-card {                                                               6649
    width: 100%;                                                           6650
```

```
    max-width: 470px;                                                    6651
    padding: 38px 40px 30px;                                             6652
    background: #fff;                                                    6653
    border-radius: 20px;                                                 6654
    box-shadow: 0 26px 70px rgba(14, 46, 32, .13);                       6655
    animation: rg-in .5s cubic-bezier(.16, .84, .34, 1)                  6656
    both;                                                                6657
}                                                                        6658
@keyframes rg-in {                                                       6659
    from { opacity: 0; transform: translateY(16px)                       6660
    scale(.98); }                                                        6661
    to   { opacity: 1; transform: none; }                                6662
}                                                                        6663
.rg-title {                                                              6664
    margin: 0 0 6px;                                                     6665
    text-align: center;                                                  6666
    font-size: 25px;                                                     6667
    font-weight: 700;                                                    6668
    letter-spacing: -.2px;                                               6669
}                                                                        6670
.rg-sub {                                                                6671
    margin: 0 0 28px;                                                    6672
    text-align: center;                                                  6673
    font-size: 13px;                                                     6674
    color: #86868b;                                                      6675
}                                                                        6676
.rg-field { margin-bottom: 15px; }                                       6677
.rg-field > label {                                                      6678
    display: block;                                                      6679
    margin-bottom: 7px;                                                  6680
    font-size: 13px;                                                     6681
    color: #6e6e73;                                                      6682
}                                                                        6683
.rg-field > label .rg-opt {                                              6684
    margin-left: 5px;                                                    6685
    font-size: 12px;                                                     6686
    color: #a1a1a6;                                                      6687
}                                                                        6688
.rg-field input[type="text"],                                            6689
.rg-field input[type="password"] {                                       6690
    width: 100%;                                                         6691
    padding: 12px 15px;                                                  6692
    font-family: inherit;                                                6693
    font-size: 16px;              /* 不小于 16px，避免手机端聚焦时自动放大 */   6694
    color: #1d1d1f;                                                      6695
    background: #f5f5f7;                                                 6696
    border: 1px solid transparent;                                       6697
    border-radius: 12px;                                                 6698
    outline: none;                                                       6699
    transition: background-color .18s ease, border-color                 6700
```

```
    .18s ease,                                                    6701
                box-shadow .18s ease;                             6702
}                                                                 6703
.rg-field input::placeholder { color: #a1a1a6; }                  6704
.rg-field input:focus {                                           6705
  background: #fff;                                               6706
  border-color: rgba(31, 107, 69, .55);                           6707
  box-shadow: 0 0 0 3px rgba(31, 107, 69, .14);                   6708
}                                                                 6709
.rg-field input[aria-invalid="true"] {                            6710
  border-color: rgba(180, 60, 50, .55);                           6711
  background: #fff;                                               6712
}                                                                 6713
.rg-field select {                                                6714
  width: 100%;                                                    6715
  padding: 12px 15px;                                             6716
  font-family: inherit;                                           6717
  font-size: 15px;                                                6718
  color: #1d1d1f;                                                 6719
  background: #f5f5f7;                                            6720
  border: 1px solid transparent;                                  6721
  border-radius: 12px;                                            6722
  outline: none;                                                  6723
}                                                                 6724
.rg-field select:focus {                                          6725
  background: #fff;                                               6726
  border-color: rgba(31, 107, 69, .55);                           6727
  box-shadow: 0 0 0 3px rgba(31, 107, 69, .14);                   6728
}                                                                 6729
.rg-err {                                                         6730
  margin: 6px 0 0;                                                6731
  font-size: 12.5px;                                              6732
  color: var(--danger);                                           6733
}                                                                 6734
/* 「选填」分隔线 */                                               6735
.rg-split {                                                       6736
  display: flex;                                                  6737
  align-items: center;                                            6738
  gap: 12px;                                                      6739
  margin: 24px 0 18px;                                            6740
  font-size: 12.5px;                                              6741
  color: #a1a1a6;                                                 6742
}                                                                 6743
.rg-split::before,                                                6744
.rg-split::after {                                                6745
  content: "";                                                    6746
  flex: 1;                                                        6747
  height: 1px;                                                    6748
  background: #ececee;                                            6749
}                                                                 6750
```

```
.rg-pair {
  display: grid;
  grid-template-columns: 1fr 1fr;
  gap: 0 14px;
}
@media (max-width: 480px) { .rg-pair {
  grid-template-columns: 1fr; } }
/* 头像：圆形预览 + 选择文件 */
.rg-avatar {
  display: flex;
  align-items: center;
  gap: 14px;
  margin-bottom: 15px;
}
.rg-avatar__face {
  display: grid;
  place-items: center;
  flex: none;
  width: 54px;
  height: 54px;
  border-radius: 50%;
  background: #eef2f0;
  color: #9aa8a1;
  font-size: 12px;
  overflow: hidden;
  background-size: cover;
  background-position: center;
}
.rg-avatar__pick { min-width: 0; }
.rg-avatar__pick input[type="file"] {
  max-width: 100%;
  font-family: inherit;
  font-size: 12.5px;
  color: #6e6e73;
}
.rg-avatar__tip {
  margin: 4px 0 0;
  font-size: 12px;
  color: #a1a1a6;
}
.rg-submit {
  width: 100%;
  margin-top: 24px;
  padding: 13px 20px;
  font-family: inherit;
  font-size: 16px;
  font-weight: 500;
  color: #fff;
  background: var(--accent);
  border: none;
```

```
      border-radius: 980px;                                          6801
      cursor: pointer;                                               6802
      transition: background-color .2s ease, box-shadow .2s          6803
      ease,                                                          6804
                  transform .2s ease;                                6805
    }                                                                6806
    .rg-submit:hover {                                               6807
      background: var(--accent-hover);                               6808
      box-shadow: 0 10px 24px rgba(31, 107, 69, .26);                6809
    }                                                                6810
    .rg-submit:active { transform: translateY(1px);                  6811
      box-shadow: none; }                                            6812
    .rg-submit:focus-visible { outline: 3px solid rgba(31,           6813
      107, 69, .40); outline-offset: 3px; }                          6814
    .rg-foot {                                                       6815
      margin: 20px 0 0;                                              6816
      text-align: center;                                            6817
      font-size: 13px;                                               6818
      color: #86868b;                                                6819
    }                                                                6820
    .rg-foot a { color: var(--accent); text-decoration: none;        6821
      }                                                              6822
    .rg-foot a:hover { text-decoration: underline; }                 6823
    @media (prefers-reduced-motion: reduce) {                        6824
      .rg-card { animation: none; }                                  6825
    }                                                                6826
</style>                                                             6827
<div class="rg">                                                     6828
    <div class="rg-card">                                            6829
      <h1 class="rg-title">创建账号</h1>                                6830
      <p class="rg-sub">新账号为「操作员」，职务由管理员调整</p>                    6831
      <form method="post" enctype="multipart/form-data"              6832
      novalidate>                                                    6833
        {{ form.hidden_tag() }}                                      6834
        <div class="rg-field">                                       6835
          <label for="{{ form.username.id }}">账户名称</label>           6836
          {{ form.username(id=form.username.id,                      6837
      autocomplete="username",                                       6838
                           placeholder="2–64 个字符",                  6839
                           **({'aria-invalid': 'true'} if              6840
      form.username.errors else {})) }}                              6841
          {% for e in form.username.errors %}<p                      6842
      class="rg-err">{{ e }}</p>{% endfor %}                         6843
        </div>                                                       6844
        <div class="rg-field">                                       6845
          <label for="{{ form.password.id }}">密码</label>               6846
          {{ form.password(id=form.password.id,                      6847
      autocomplete="new-password",                                   6848
                           placeholder="至少 6 位",                    6849
                           **({'aria-invalid': 'true'} if              6850
```

```
form.password.errors else {})) }}                                    6851
    {% for e in form.password.errors %}<p                            6852
class="rg-err">{{ e }}</p>{% endfor %}                               6853
  </div>                                                             6854
  <div class="rg-field">                                             6855
    <label for="{{ form.confirm.id }}">确认密码</label>              6856
    {{ form.confirm(id=form.confirm.id,                              6857
autocomplete="new-password",                                         6858
                    placeholder="再次输入密码",                      6859
                    **({'aria-invalid': 'true'} if                   6860
form.confirm.errors else {})) }}                                     6861
    {% for e in form.confirm.errors %}<p                             6862
class="rg-err">{{ e }}</p>{% endfor %}                               6863
  </div>                                                             6864
  <p class="rg-split">以下均为选填</p>                               6865
  <div class="rg-avatar">                                            6866
    <span class="rg-avatar__face" id="rg-face"                       6867
aria-hidden="true">头像</span>                                       6868
    <div class="rg-avatar__pick">                                    6869
      <label class="visually-hidden"                                 6870
for="rg-avatar">选择头像图片</label>                                 6871
      <input id="rg-avatar" type="file" name="avatar"                6872
             accept=".png,.jpg,.jpeg,.gif,.webp">                    6873
      <p class="rg-avatar__tip">PNG / JPG / GIF /                    6874
WebP，不超过 4 MB</p>                                                6875
    </div>                                                           6876
  </div>                                                             6877
  <div class="rg-pair">                                              6878
    <div class="rg-field">                                           6879
      <label for="{{ form.display_name.id }}">昵称</label>           6880
      {{ form.display_name(id=form.display_name.id,                  6881
                           placeholder="留空则显示账户名称") }}      6882
    </div>                                                           6883
    <div class="rg-field">                                           6884
      <label for="{{ form.gender.id }}">性别</label>                 6885
      {{ form.gender(id=form.gender.id) }}                           6886
    </div>                                                           6887
  </div>                                                             6888
  <div class="rg-field">                                             6889
    <label for="{{ form.email.id }}">邮箱</label>                    6890
    {{ form.email(id=form.email.id,                                  6891
autocomplete="email",                                                6892
                  placeholder="用于联系，可留空",                    6893
                  **({'aria-invalid': 'true'} if                     6894
form.email.errors else {})) }}                                       6895
    {% for e in form.email.errors %}<p                               6896
class="rg-err">{{ e }}</p>{% endfor %}                               6897
  </div>                                                             6898
  <div class="rg-field" style="margin-bottom:0;">                    6899
    <label for="{{ form.phone.id }}">手机号</label>                  6900
```

```
            {{ form.phone(id=form.phone.id, autocomplete="tel",
        placeholder="可留空") }}
          </div>
          {{ form.submit_reg(class_="rg-submit") }}
        </form>
        <p class="rg-foot">已有账号？<a href="{{
        url_for('auth.login') }}">返回登录</a></p>
      </div>
    </div>
  </div>
{% endblock %}
{% block scripts %}
<script>
//  选好头像后立刻在左侧圆框里预览，省得传上去才知道选错了
(function () {
  var input = document.getElementById('rg-avatar');
  var face = document.getElementById('rg-face');
  if (!input || !face || !window.FileReader) return;
  input.addEventListener('change', function () {
    var file = input.files && input.files[0];
    if (!file) return;
    var reader = new FileReader();
    reader.onload = function (e) {
      face.textContent = '';
      face.style.backgroundImage = 'url("' +
    e.target.result + '")';
    };
    reader.readAsDataURL(file);
  });
})();
</script>
{% endblock %}
#  ==== app/templates/auth/home.html ====
{% extends "base.html" %}
{% from "_macros.html" import icon %}
{% block title %}工作台 · 历史遗留废弃矿山自然恢复适宜性评价系统{% endblock %}
{% block content %}
{# ---------- 页首横幅 ---------- #}
<section class="hero">
  <div class="hero__body">
    <h1 class="hero__title">历史遗留废弃矿山自然恢复适宜性评价系统</h1>
    <p class="hero__desc">
      以论文《历史遗留废弃矿山自然恢复适宜性评价指标体系建设及典型案例》
      建立的评价方法为算法内核。下方按论文 4.1 节的评价流程顺序排列，
      系统会在每一步检查前置条件，前置步骤未完成时不会放行后续计算。
    </p>
    <div class="hero__actions">
      <a class="hero-btn hero-btn--ghost"
    href="/indicator/import">
        {{ icon("upload", 16) }}批量导入
      </a>
```

```
        <a class="hero-btn hero-btn--solid"
      href="/project/add">
          {{ icon("plus", 16) }}新建评价项目
        </a>
      </div>
    </div>
</section>
{# ---------- 数据概览 ---------- #}
<section class="stat-cards">
    <a class="stat-card" href="/project/">
      <span class="stat-card__label">评价项目</span>
      <span class="stat-card__value">{{ dash.stats.projects
      }}<span class="stat-card__unit">个</span></span>
      <span class="stat-card__note">
        {%- if projects_without_patch -%}
          其中 {{ projects_without_patch|length }} 个项目尚未建立图斑
        {%- elif dash.progress.patches -%}
          已建立 {{ dash.progress.patches }} 个矿山图斑
        {%- else -%}
          尚未建立矿山图斑
        {%- endif -%}
      </span>
    </a>
    <a class="stat-card" href="/project/">
      <span class="stat-card__label">矿山图斑</span>
      <span class="stat-card__value">{{ dash.stats.patches
      }}<span class="stat-card__unit">个</span></span>
      <span class="stat-card__note">
        {% if dash.progress.ready %}其中 {{ dash.progress.ready
      }} 个具备评价条件{% else %}暂无具备评价条件的图斑{% endif %}
      </span>
    </a>
    <a class="stat-card" href="/unit/">
      <span class="stat-card__label">微地质单元</span>
      <span class="stat-card__value">{{ dash.stats.unit_types
      }}<span class="stat-card__unit">类</span></span>
      <span class="stat-card__note">
        {% if dash.progress.units %}共划分 {{
      dash.progress.units }} 个单元{% else %}尚未划分单元{% endif %}
      </span>
    </a>
    <a class="stat-card" href="/standard/">
      <span class="stat-card__label">标准与政策</span>
      <span class="stat-card__value">{{ dash.stats.standards
      }}<span class="stat-card__unit">项</span></span>
      <span class="stat-card__note">地方标准与政策文件</span>
    </a>
</section>
{# ---------- 评价流程 ---------- #}
<section class="section">
```

```
  <div class="section__head">
    <h2 class="section__title">评价流程</h2>
    <p class="section__desc">
      按论文 4.1 节顺序排列，前置步骤未完成时不会执行后续计算
    </p>
  </div>
  <div class="step-grid">
    {% for s in dash.steps %}
    {% if s.available %}
    <a class="step-card" href="{{ s.href }}">
      <span class="step-card__no">{{ s.no }}</span>
      <span class="step-card__icon">{{ icon(s.icon, 18)
    }}</span>
      <span class="step-card__title">{{ s.title }}</span>
      <span class="step-card__desc">{{ s.desc }}</span>
      {% if s.pending %}
      <span class="step-card__badge">待处理 {{ s.pending
    }}</span>
      {% endif %}
    </a>
    {% else %}
    <div class="step-card step-card--locked"
    aria-disabled="true">
      <span class="step-card__no">{{ s.no }}</span>
      <span class="step-card__icon">{{ icon(s.icon, 18)
    }}</span>
      <span class="step-card__title">{{ s.title }}</span>
      <span class="step-card__desc">{{ s.desc }}</span>
      <span class="step-card__lock">{{ s.note }}</span>
    </div>
    {% endif %}
    {% endfor %}
  </div>
</section>
{# ---------- 系统管理 ---------- #}
<section class="section">
  <div class="section__head">
    <h2 class="section__title">系统管理</h2>
  </div>
  <a class="row-card" href="/profile">
    <span class="row-card__icon">{{ icon("user", 19)
    }}</span>
    <span class="row-card__main">
      <span class="row-card__title">个人设置</span>
      <span class="row-card__desc">维护昵称、性别、邮箱、手机号与登录密码</span
    >
    </span>
    <span class="row-card__arrow">{{ icon("chevron", 17)
    }}</span>
  </a>
```

```
  {% if current_user.is_admin %}
  <a class="row-card" href="/profile?tab=users">
    <span class="row-card__icon">{{ icon("users", 19)
    }}</span>
    <span class="row-card__main">
      <span class="row-card__title">账号与操作日志</span>
      <span class="row-card__desc">新增账号并设置职务；全部写操作自动留痕，可查询追溯
    </span>
    <span class="row-card__arrow">{{ icon("chevron", 17)
    }}</span>
  </a>
  {% endif %}
</section>
{% endblock %}
# ==== app/templates/auth/add_user.html ====
{% extends "base.html" %}
{% block title %}新增账号{% endblock %}
{% block content %}
<div class="card" style="max-width:560px;">
  <h2>新增账号</h2>
  <form method="post">
    {{ form.hidden_tag() }}
    <div class="field">
      <label class="field__label" for="username">账户名称</label
    >
      <input id="username" type="text" name="username"
    required
            maxlength="64" autocomplete="off"
    style="width:280px;">
    </div>
    <div class="field">
      <label class="field__label" for="password">密码</label>
      <input id="password" type="password" name="password"
    required
            autocomplete="new-password"
    style="width:280px;">
    </div>
    <div class="field">
      <label class="field__label" for="role">职务</label>
      <select id="role" name="role">
        {% for code, label in role_labels.items() %}
        <option value="{{ code }}" {% if code ==
    default_role %}selected{% endif %}>{{ label }}</option>
        {% endfor %}
      </select>
      <p class="muted" style="margin:6px 0 0;">
        管理员可管理系统、维护基础数据并查看操作日志；操作员可使用全部业务功能。
      </p>
    </div>
```

```
      <hr style="margin:22px 0;border:none;border-top:1px                    7101
      solid var(--line);">                                                  7102
      <h3 style="margin:0 0 12px;font-size:15px;">个人资料（选填）</h3       7103
      >                                                                     7104
      <div class="field">                                                   7105
        <label class="field__label" for="display_name">昵称</lab            7106
      el>                                                                   7107
        <input id="display_name" type="text"                                7108
      name="display_name" maxlength="64"                                    7109
              placeholder="留空则显示账户名称">                             7110
      </div>                                                                7111
      <div class="field">                                                   7112
        <label class="field__label" for="gender">性别</label>               7113
        <select id="gender" name="gender">                                  7114
          <option value="">不填</option>                                    7115
          {% for g in genders %}<option value="{{ g }}">{{ g                7116
      }}</option>{% endfor %}                                               7117
        </select>                                                           7118
      </div>                                                                7119
      <div class="field">                                                   7120
        <label class="field__label" for="email">邮箱</label>                7121
        <input id="email" type="text" name="email"                          7122
      maxlength="128">                                                      7123
      </div>                                                                7124
      <div class="field">                                                   7125
        <label class="field__label" for="phone">手机号</label>              7126
        <input id="phone" type="text" name="phone"                          7127
      maxlength="32">                                                       7128
      </div>                                                                7129
      <p class="actions" style="margin-top:18px;">                          7130
        <button type="submit" class="btn">保存</button>                     7131
        <a class="btn btn--ghost" href="{{                                  7132
      url_for('auth.profile', tab='users') }}">返回</a>                     7133
      </p>                                                                  7134
    </form>                                                                 7135
</div>                                                                      7136
{% endblock %}                                                              7137
# ==== app/templates/auth/edit_user.html ====                               7138
{% extends "base.html" %}                                                   7139
{% block title %}编辑账号 · {{ user.username }}{% endblock %}               7140
{% block content %}                                                         7141
<div class="card" style="max-width:560px;">                                 7142
  <h2>编辑账号 · {{ user.username }}</h2>                                   7143
  <h3 style="margin:0 0 12px;font-size:15px;">个人资料</h3>                 7144
  <p class="muted" style="margin:0 0 14px;font-size:12.5px;"                7145
    >                                                                       7146
    以下内容用户也可在「个人设置」中自行修改，此处为管理员代改。          7147
  </p>                                                                      7148
  <form method="post">                                                      7149
    {{ form.hidden_tag() }}                                                 7150
```

```
<div class="field">
  <label class="field__label" for="display_name">昵称</label>
  <input id="display_name" type="text" name="display_name" maxlength="64"
         value="{{ user.display_name or '' }}"
         placeholder="留空则显示账户名称">
</div>
<div class="field">
  <label class="field__label" for="gender">性别</label>
  <select id="gender" name="gender">
    <option value="">不填</option>
    {% for g in genders %}
    <option value="{{ g }}" {% if user.gender == g %}selected{% endif %}>{{ g }}</option>
    {% endfor %}
  </select>
</div>
<div class="field">
  <label class="field__label" for="email">邮箱</label>
  <input id="email" type="text" name="email" maxlength="128"
         value="{{ user.email or '' }}">
</div>
<div class="field">
  <label class="field__label" for="phone">手机号</label>
  <input id="phone" type="text" name="phone" maxlength="32"
         value="{{ user.phone or '' }}">
</div>
<hr style="margin:22px 0;border:none;border-top:1px solid var(--line);">
<h3 style="margin:0 0 12px;font-size:15px;">职务与权限</h3>
<p class="muted" style="margin:0 0 14px;font-size:12.5px;">
  职务只能由管理员设置：管理员可管理系统、维护基础数据并查看操作日志；
  操作员可使用全部业务功能。
</p>
<div class="field">
  <label class="field__label" for="role">职务</label>
  <select id="role" name="role">
    {% for code, label in role_labels.items() %}
    <option value="{{ code }}" {% if user.role == code %}selected{% endif %}>{{ label }}</option>
    {% endfor %}
  </select>
</div>
<div class="field">
  <label class="checkbox" for="active">
    <input id="active" type="checkbox" name="active"
```

```
      value="1"                                                                  7201
                    {% if user.active %}checked{% endif %}>                      7202
            <span>启用该账号</span>                                               7203
          </label>                                                               7204
          <p class="muted" style="margin:6px 0 0;">                              7205
            停用后该账号无法登录，但其历史操作记录会保留。                          7206
            系统至少需保留一名启用状态的管理员。                                   7207
          </p>                                                                   7208
      </div>                                                                     7209
      <div class="field">                                                        7210
        <label class="field__label" for="password">重置密码（留空则不修改7211
      ）</label>                                                                 7212
        <input id="password" type="password" name="password"                     7213
               autocomplete="new-password"                                       7214
      style="width:280px;"                                                       7215
               placeholder="留空表示保持原密码">                                 7216
      </div>                                                                     7217
      <p class="actions" style="margin-top:18px;">                               7218
        <button type="submit" class="btn">保存</button>                          7219
        <a class="btn btn--ghost" href="{{                                       7220
      url_for('auth.profile', tab='users') }}">返回</a>                          7221
      </p>                                                                       7222
    </form>                                                                      7223
</div>                                                                           7224
{% endblock %}                                                                   7225
# ==== app/templates/auth/profile.html ====                                      7226
{% extends "base.html" %}                                                        7227
{% from "_macros.html" import icon %}                                            7228
{% block title %}个人设置 · 历史遗留废弃矿山自然恢复适宜性评价系统{% endblock %}     7229
{% block content %}                                                              7230
<style>                                                                          7231
  /* ================== 个人设置（仿 Apple 账户页） ==================          7232
     左栏是身份与分类，右栏是该分类下的功能卡。                                  7233
     每张卡默认只显示当前值，点开才展开对应的修改表单 ——                        7234
     不点就不会误改，页面也不会一上来堆满输入框。 */                             7235
  .acct {                                                                        7236
    display: grid;                                                               7237
    grid-template-columns: 244px 1fr;                                            7238
    gap: 40px;                                                                   7239
    align-items: start;                                                          7240
  }                                                                              7241
  @media (max-width: 900px) {                                                    7242
    .acct { grid-template-columns: 1fr; gap: 24px; }                             7243
  }                                                                              7244
  /* ---------- 左栏 ---------- */                                               7245
  .acct-rail__avatar {                                                           7246
    display: grid;                                                               7247
    place-items: center;                                                         7248
    width: 88px;                                                                 7249
    height: 88px;                                                                7250
```

```
    border-radius: 50%;
    background: #e9edeb;
    color: #97a09b;
    font-size: 32px;
    font-weight: 500;
    object-fit: cover;
  }
  .acct-rail__name {
    margin: 16px 0 2px;
    font-size: 16px;
    font-weight: 600;
  }
  .acct-rail__sub {
    margin: 0 0 24px;
    font-size: 13px;
    color: #86868b;
    word-break: break-all;
  }
  .acct-tabs {
    display: flex;
    flex-direction: column;
    gap: 2px;
    margin-bottom: 24px;
  }
  .acct-tabs a {
    padding: 7px 12px;
    border: 1px solid transparent;
    border-radius: 7px;
    font-size: 15px;
    color: #1d1d1f;
    text-decoration: none;
    transition: background-color .16s ease, border-color
    .16s ease;
  }
  .acct-tabs a:hover { background: #f5f5f7; }
  .acct-tabs a.is-active {
    font-weight: 600;
    border-color: #bcc9c1;
    background: #fff;
  }
  /* 更换头像：折叠在左栏，点开才出现文件选择 */
  .acct-avatar-edit > summary {
    display: inline-block;
    padding: 5px 12px;
    font-size: 13px;
    color: var(--accent);
    background: var(--accent-soft);
    border-radius: 7px;
    cursor: pointer;
    list-style: none;
```

```
}
.acct-avatar-edit > summary::-webkit-details-marker {
  display: none; }
.acct-avatar-edit > summary:hover { background: #dcebe2; }
.acct-avatar-edit__body { margin-top: 12px; }
.acct-avatar-edit__body .btn { width: 100%;
  margin-bottom: 8px; }
.acct-avatar-edit__file {
  margin: 0 0 10px;
  font-size: 12px;
  color: var(--ink-soft);
  word-break: break-all;
}
/* ---------- 右栏 ---------- */
.acct-title {
  margin: 0 0 8px;
  font-size: 26px;
  font-weight: 700;
  letter-spacing: -.3px;
}
.acct-lead {
  margin: 0 0 26px;
  font-size: 14px;
  color: #6e6e73;
}
/* 固定两列，与 Apple 账户页的 2×2 一致；
   卡片也宽一些，邮箱之类的长值不至于被拆成两行 */
.acct-grid {
  display: grid;
  grid-template-columns: repeat(2, minmax(0, 1fr));
  gap: 16px;
  align-items: start;
}
@media (max-width: 760px) {
  .acct-grid { grid-template-columns: 1fr; }
}
/* 功能卡：details/summary 实现，点开才展开表单 */
.acct-card {
  background: #fff;
  border: 1px solid #e6e6e8;
  border-radius: 14px;
  transition: border-color .2s ease, box-shadow .2s ease;
}
.acct-card[open] {
  border-color: rgba(31, 107, 69, .45);
  box-shadow: 0 8px 22px rgba(19, 60, 43, .08);
}
.acct-card > summary {
  position: relative;
  display: block;
```

```
    padding: 18px 56px 18px 20px;
    border-radius: 14px;
    cursor: pointer;
    list-style: none;
}
.acct-card > summary::-webkit-details-marker { display:
    none; }
.acct-card > summary:hover { background: #fafbfa; }
.acct-card > summary:focus-visible {
    outline: 2px solid var(--accent);
    outline-offset: 2px;
}
.acct-card__title {
    display: block;
    font-size: 15px;
    font-weight: 600;
}
.acct-card__value {
    display: block;
    margin-top: 3px;
    font-size: 14px;
    color: #6e6e73;
    overflow-wrap: anywhere;
}
.acct-card__icon {
    position: absolute;
    top: 17px;
    right: 17px;
    display: grid;
    place-items: center;
    width: 27px;
    height: 27px;
    border-radius: 8px;
    background: var(--accent-soft);
    color: var(--accent);
}
.acct-card__body {
    padding: 4px 20px 20px;
    border-top: 1px solid #f0f1f2;
    margin-top: 2px;
}
.acct-card__body .field:first-child { margin-top: 16px; }
.acct-card__body .btn { width: 100%; }
/* 只读信息列表 */
.acct-kv {
    display: grid;
    grid-template-columns: 88px 1fr;
    gap: 7px 12px;
    margin: 16px 0 0;
    font-size: 13.5px;
```

```
  }
  .acct-kv dt { color: var(--ink-soft); }
  .acct-kv dd { margin: 0; }
  .field__error {
    margin: 5px 0 0;
    font-size: 12.5px;
    color: var(--danger);
  }
  /* 管理员分类（账号 / 操作日志）里的表格面板 */
  .acct-panel {
    background: #fff;
    border: 1px solid #e6e6e8;
    border-radius: 14px;
    overflow: hidden;
  }
  .acct-panel__head {
    display: flex;
    align-items: center;
    justify-content: space-between;
    gap: 12px;
    padding: 13px 20px;
    border-bottom: 1px solid #f0f1f2;
  }
  .acct-panel__count {
    margin: 0;
    font-size: 13.5px;
    color: #6e6e73;
  }
  .acct-panel table { margin: 0; }
  .acct-panel th,
  .acct-panel td { border: 0; border-bottom: 1px solid
    #f0f1f2; }
  .acct-panel thead th { background: #f7f9f8; }
  .acct-panel tbody tr:last-child td { border-bottom: 0; }
  .acct-panel tbody tr:hover { background: #fafbfa; }
</style>
<div class="acct">
  {# ---------------- 左栏：身份 + 分类 ---------------- #}
  <aside class="acct-rail">
    {% if current_user.avatar %}
    <img class="acct-rail__avatar"
         src="{{ url_for('static', filename=current_user.ava
    tar) }}"
         alt="{{ current_user.shown_name }} 的头像">
    {% else %}
    <span class="acct-rail__avatar" aria-hidden="true">{{
    current_user.initial }}</span>
    {% endif %}
    <p class="acct-rail__name">{{ current_user.shown_name
    }}</p>
```

```
<p class="acct-rail__sub">
  {{ role_labels.get(current_user.role, current_user.role) }}
  {% if current_user.email %} · {{ current_user.email }}{% endif %}
</p>
<nav class="acct-tabs" aria-label="个人设置分类">
  <a href="{{ url_for('auth.profile', tab='info') }}"
     {% if tab == 'info' %}class="is-active" aria-current="page"{% endif %}>个人信息</a>
  <a href="{{ url_for('auth.profile', tab='security') }}"
     {% if tab == 'security' %}class="is-active" aria-current="page"{% endif %}>登录与安全性</a>
  {% if current_user.is_admin %}
  <a href="{{ url_for('auth.profile', tab='users') }}"
     {% if tab == 'users' %}class="is-active" aria-current="page"{% endif %}>账号</a>
  <a href="{{ url_for('auth.profile', tab='logs') }}"
     {% if tab == 'logs' %}class="is-active" aria-current="page"{% endif %}>操作日志</a>
  {% endif %}
</nav>
<details class="acct-avatar-edit">
  <summary>更换头像</summary>
  <div class="acct-avatar-edit__body">
    <form method="post" action="/profile/avatar"
          enctype="multipart/form-data">
      {{ avatar_form.hidden_tag() }}
      <input type="hidden" name="tab" value="{{ tab }}">
      <label class="visually-hidden" for="avatar-file">选择头像图片</label>
      <input id="avatar-file" type="file" name="avatar"
             accept=".png,.jpg,.jpeg,.gif,.webp">
      <p class="acct-avatar-edit__file">PNG / JPG / GIF / WebP，不超过 4 MB</p>
      <button type="submit" class="btn btn--sm">上传头像</button>
    </form>
    {% if current_user.avatar %}
    <form method="post" action="/profile/avatar/delete"
          onsubmit="return confirm('确定移除当前头像吗？');">
      {{ avatar_form.hidden_tag() }}
      <input type="hidden" name="tab" value="{{ tab }}">
      <button type="submit" class="btn btn--sm btn--ghost">移除头像</button>
    </form>
    {% endif %}
  </div>
</details>
```

```
</aside>
{# ---------------- 右栏：功能卡 ---------------- #}
<div class="acct-main">
  {% if tab == 'security' %}
  <h1 class="acct-title">登录与安全性</h1>
  <p class="acct-lead">管理与登录账号、账号安全有关的设置。</p>
  <div class="acct-grid">
    <details class="acct-card"
             {% if pwd_form.old_password.errors or
  pwd_form.new_password.errors or pwd_form.confirm.errors
  %}open{% endif %}>
      <summary>
        <span class="acct-card__title">密码</span>
        <span class="acct-card__value">修改登录密码</span>
        <span class="acct-card__icon">{{ icon("lock", 15)
  }}</span>
      </summary>
      <div class="acct-card__body">
        <form method="post">
          {{ pwd_form.hidden_tag() }}
          <input type="hidden" name="action"
  value="password">
          <input type="hidden" name="tab"
  value="security">
          <div class="field">
            <label class="field__label" for="{{
  pwd_form.old_password.id }}">当前密码</label>
            {{ pwd_form.old_password(id=pwd_form.old_passw
  ord.id,

  autocomplete="current-password") }}
            {% for e in pwd_form.old_password.errors %}
            <p class="field__error">{{ e }}</p>
            {% endfor %}
          </div>
          <div class="field">
            <label class="field__label" for="{{
  pwd_form.new_password.id }}">新密码</label>
            {{ pwd_form.new_password(id=pwd_form.new_passw
  ord.id,

  autocomplete="new-password") }}
            {% for e in pwd_form.new_password.errors %}
            <p class="field__error">{{ e }}</p>
            {% endfor %}
          </div>
          <div class="field">
            <label class="field__label" for="{{
  pwd_form.confirm.id }}">确认新密码</label>
            {{ pwd_form.confirm(id=pwd_form.confirm.id,
```

```
                                    autocomplete="new-password
") }}
            {% for e in pwd_form.confirm.errors %}
            <p class="field__error">{{ e }}</p>
            {% endfor %}
          </div>
          <p class="actions">{{ pwd_form.submit_pwd(class_
="btn") }}</p>
        </form>
      </div>
    </details>
    <details class="acct-card">
      <summary>
        <span class="acct-card__title">登录账号</span>
        <span class="acct-card__value">{{
current_user.username }}</span>
        <span class="acct-card__icon">{{ icon("user", 15)
}}</span>
      </summary>
      <div class="acct-card__body">
        <dl class="acct-kv">
          <dt>登录账号</dt><dd class="mono">{{
current_user.username }}</dd>
          <dt>职务</dt>
          <dd>{{ role_labels.get(current_user.role,
current_user.role) }}</dd>
          <dt>状态</dt>
          <dd>
            {% if current_user.active %}
            <span class="tag tag--ok">启用</span>
            {% else %}
            <span class="tag tag--warn">停用</span>
            {% endif %}
          </dd>
          <dt>创建时间</dt>
          <dd class="mono">
            {{ current_user.created_at.strftime("%Y-%m-%d
%H:%M")
                if current_user.created_at else "—" }}
          </dd>
        </dl>
      </div>
    </details>
</div>
{% elif tab == 'users' %}
<h1 class="acct-title">账号</h1>
<p class="acct-lead">管理本系统的账号、职务与启用状态。新注册的账号默认为「操作员」。</p
>
{% include "auth/_users.html" %}
{% elif tab == 'logs' %}
```

```
<h1 class="acct-title">操作日志</h1>
<p class="acct-lead">记录各账号在本系统中的关键操作。</p>
{% include "auth/_logs.html" %}
{% else %}
<h1 class="acct-title">个人信息</h1>
<p class="acct-lead">管理你的昵称、性别以及可以联系到你的邮箱与手机号。</p>
<form method="post">
  {{ info_form.hidden_tag() }}
  <input type="hidden" name="action" value="info">
  <input type="hidden" name="tab" value="info">
  <div class="acct-grid">
    <details class="acct-card" {% if info_form.display_name.errors %}open{% endif %}>
      <summary>
        <span class="acct-card__title">昵称</span>
        <span class="acct-card__value">
          {{ current_user.display_name or "未设置" }}
        </span>
        <span class="acct-card__icon">{{ icon("user", 15) }}</span>
      </summary>
      <div class="acct-card__body">
        <div class="field">
          <label class="field__label" for="{{ info_form.display_name.id }}">
            昵称
          </label>
          {{ info_form.display_name(id=info_form.display_name.id,
placeholder="留空则显示账户名称") }}
          {% for e in info_form.display_name.errors %}
          <p class="field__error">{{ e }}</p>
          {% endfor %}
        </div>
        <p class="actions">{{ info_form.submit_info(class_="btn") }}</p>
      </div>
    </details>
    <details class="acct-card" {% if info_form.gender.errors %}open{% endif %}>
      <summary>
        <span class="acct-card__title">性别</span>
        <span class="acct-card__value">{{ current_user.gender or "未设置" }}</span>
        <span class="acct-card__icon">{{ icon("user", 15) }}</span>
      </summary>
      <div class="acct-card__body">
        <div class="field">
```

```
            <label class="field__label" for="{{                7651
info_form.gender.id }}">性别</label>                           7652
            {{ info_form.gender(id=info_form.gender.id) }}     7653
            {% for e in info_form.gender.errors %}             7654
            <p class="field__error">{{ e }}</p>                7655
            {% endfor %}                                       7656
          </div>                                               7657
          <p class="actions">{{ info_form.submit_info(clas     7658
s_="btn") }}</p>                                               7659
        </div>                                                 7660
      </details>                                               7661
      <details class="acct-card" {% if                         7662
info_form.email.errors %}open{% endif %}>                      7663
        <summary>                                              7664
          <span class="acct-card__title">邮箱</span>           7665
          <span class="acct-card__value">{{                    7666
current_user.email or "未设置" }}</span>                       7667
          <span class="acct-card__icon">{{ icon("mail",        7668
15) }}</span>                                                  7669
        </summary>                                             7670
        <div class="acct-card__body">                          7671
          <div class="field">                                  7672
            <label class="field__label" for="{{                7673
info_form.email.id }}">邮箱</label>                            7674
            {{ info_form.email(id=info_form.email.id) }}       7675
            {% for e in info_form.email.errors %}              7676
            <p class="field__error">{{ e }}</p>                7677
            {% endfor %}                                       7678
          </div>                                               7679
          <p class="actions">{{ info_form.submit_info(clas     7680
s_="btn") }}</p>                                               7681
        </div>                                                 7682
      </details>                                               7683
      <details class="acct-card" {% if                         7684
info_form.phone.errors %}open{% endif %}>                      7685
        <summary>                                              7686
          <span class="acct-card__title">手机号</span>         7687
          <span class="acct-card__value">{{                    7688
current_user.phone or "未设置" }}</span>                       7689
          <span class="acct-card__icon">{{ icon("phone",       7690
15) }}</span>                                                  7691
        </summary>                                             7692
        <div class="acct-card__body">                          7693
          <div class="field">                                  7694
            <label class="field__label" for="{{                7695
info_form.phone.id }}">手机号</label>                          7696
            {{ info_form.phone(id=info_form.phone.id) }}       7697
            {% for e in info_form.phone.errors %}              7698
            <p class="field__error">{{ e }}</p>                7699
            {% endfor %}                                       7700
```

```
                <div>                                                    7701
                <p class="actions">{{ info_form.submit_info(clas        7702
    s_="btn") }}</p>                                                     7703
            </div>                                                       7704
          </details>                                                     7705
        </div>                                                           7706
    </form>                                                              7707
    {% endif %}                                                          7708
  </div>                                                                 7709
</div>                                                                   7710
{% endblock %}                                                           7711
{% block scripts %}                                                      7712
<script>                                                                 7713
// 同时只展开一张卡，避免点开多张后页面被撑得很长                                 7714
(function () {                                                           7715
  var cards = document.querySelectorAll('.acct-card');                   7716
  Array.prototype.forEach.call(cards, function (card) {                  7717
    card.addEventListener('toggle', function () {                        7718
      if (!card.open) return;                                            7719
      Array.prototype.forEach.call(cards, function (other) {             7720
        if (other !== card && other.open) other.open =                   7721
    false;                                                               7722
      });                                                                7723
    });                                                                  7724
  });                                                                    7725
})();                                                                    7726
</script>                                                                7727
{% endblock %}                                                           7728
# ==== app/templates/auth/_users.html ====                               7729
{# 管理员的「账号」分类。由 auth/profile.html 内联引入，不单独成页。 #}                7730
<div class="acct-panel">                                                 7731
  <div class="acct-panel__head">                                         7732
    <p class="acct-panel__count">共 {{ users|length }}                   7733
    个账号</p>                                                           7734
    <a class="btn btn--sm" href="{{ url_for('auth.add_user')             7735
    }}">新增账号</a>                                                     7736
  </div>                                                                 7737
  <table>                                                                7738
    <thead>                                                              7739
      <tr>                                                               7740
        <th scope="col">ID</th>                                          7741
        <th scope="col">账户名称</th>                                     7742
        <th scope="col">昵称</th>                                        7743
        <th scope="col">职务</th>                                        7744
        <th scope="col">状态</th>                                        7745
        <th scope="col">操作</th>                                        7746
      </tr>                                                              7747
    </thead>                                                             7748
    <tbody>                                                              7749
      {% for u in users %}                                               7750
```

```
        <tr>                                                                    7751
          <td class="num">{{ u.id }}</td>                                       7752
          <td class="mono">                                                     7753
            {{ u.username }}                                                    7754
            {% if u.id == current_user.id %}<span                               7755
    class="muted">（当前登录）</span>{% endif %}                                7756
          </td>                                                                 7757
          <td style="text-align:left;">                                         7758
            {% if u.display_name %}{{ u.display_name }}                         7759
            {% else %}<span class="muted">—</span>{% endif %}                   7760
          </td>                                                                 7761
          <td>                                                                  7762
            <span class="tag {{ 'tag--ok' if u.role ==                          7763
    'admin' else 'tag--muted' }}">                                              7764
              {{ role_labels.get(u.role, u.role) }}                             7765
            </span>                                                             7766
          </td>                                                                 7767
          <td>                                                                  7768
            {% if u.active %}<span class="tag                                   7769
    tag--ok">启用</span>                                                        7770
            {% else %}<span class="tag tag--muted">停用</span>{%                7771
    endif %}                                                                    7772
          </td>                                                                 7773
          <td class="nowrap">                                                   7774
            <span class="row-actions">                                          7775
              <a class="link-btn" href="{{                                      7776
    url_for('auth.edit_user', user_id=u.id) }}">编辑</a>                        7777
              <form method="post"                                               7778
                    action="{{ url_for('auth.delete_user',                      7779
    user_id=u.id) }}"                                                           7780
                    onsubmit="return confirm('确定删除账号「{{                  7781
    u.username }}」吗？\n该操作无法恢复。');">                                  7782
                {{ form.hidden_tag() }}                                         7783
                <button type="submit" class="link-btn                           7784
    link-btn--danger"                                                           7785
                        {% if u.id == current_user.id                           7786
    %}disabled                                                                  7787
                        title="不能删除当前登录的账号"{% endif                  7788
    %}>删除</button>                                                            7789
              </form>                                                           7790
            </span>                                                             7791
          </td>                                                                 7792
        </tr>                                                                   7793
        {% endfor %}                                                            7794
      </tbody>                                                                  7795
    </table>                                                                    7796
</div>                                                                          7797
#  ==== app/templates/auth/_logs.html ====                                      7798
{# 管理员的「操作日志」分类。由 auth/profile.html 内联引入，不单独成页。 #}    7799
<div class="acct-panel">                                                        7800
```

```
<div class="acct-panel__head">
  <p class="acct-panel__count">
    共 {{ total }} 条记录，当前第 {{ page }} / {{ pages }} 页
  </p>
  {% if pages > 1 %}
  <nav class="row-actions" aria-label="日志分页">
    {% if page > 1 %}
    <a class="link-btn"
       href="{{ url_for('auth.profile', tab='logs',
  page=page - 1) }}">上一页</a>
    {% endif %}
    {% if page < pages %}
    <a class="link-btn"
       href="{{ url_for('auth.profile', tab='logs',
  page=page + 1) }}">下一页</a>
    {% endif %}
  </nav>
  {% endif %}
</div>
{% if not records %}
<p class="muted" style="padding:18px 20px;">暂无操作记录。</p>
{% else %}
<table>
  <thead>
    <tr>
      <th scope="col">时间</th>
      <th scope="col">操作人</th>
      <th scope="col">操作</th>
      <th scope="col">对象</th>
      <th scope="col">说明</th>
    </tr>
  </thead>
  <tbody>
    {% for r in records %}
    <tr>
      <td class="mono nowrap">{{ r.created_at.strftime("%Y
  -%m-%d %H:%M:%S") if r.created_at else "—" }}</td>
      <td>{{ users_map.get(r.user_id, "—") }}</td>
      <td style="text-align:left;">{{ r.action }}</td>
      <td>
        {% if r.target_type %}{{ r.target_type }}{% if
  r.target_id %}#{{ r.target_id }}{% endif %}
        {% else %}—{% endif %}
      </td>
      <td style="text-align:left;">{{ r.detail or "—"
  }}</td>
    </tr>
    {% endfor %}
  </tbody>
</table>
```

```
  {% endif %}
</div>
# ==== app/templates/project/index.html ====
{% extends "base.html" %}
{% block title %}项目与矿山档案{% endblock %}
{% block content %}
<div class="card">
  <h2>项目与矿山档案</h2>
  <p class="actions" style="margin-bottom:12px;">
    <a class="btn btn--sm" href="/project/add">新建项目</a>
  </p>
  {% if not projects %}
  <p class="muted">
    暂无项目。点击上方「新建项目」开始建立评价项目，
    或运行 <span class="mono">python seed_demo.py</span>
    载入论文案例。
  </p>
  {% else %}
  <table>
    <caption class="muted">共 {{ projects|length }}
    个项目</caption>
    <thead>
      <tr>
        <th scope="col">项目名称</th>
        <th scope="col">编号</th>
        <th scope="col">状态</th>
        <th scope="col">图斑数</th>
        <th scope="col">操作</th>
      </tr>
    </thead>
    <tbody>
      {% for p in projects %}
      <tr>
        <td style="text-align:left;">{{ p.name }}</td>
        <td class="mono">{{ p.code or "—" }}</td>
        <td><span class="tag tag--muted">{{ p.status
    }}</span></td>
        <td class="num">
          {% if p.patches.count() %}
            {{ p.patches.count() }}
          {% else %}
            <span class="tag tag--warn">0</span>
          {% endif %}
        </td>
        <td class="nowrap">
          <span class="row-actions">
            <a class="link-btn" href="/project/{{ p.id
    }}/patches">{% if p.patches.count() %}矿山图斑{% else %}＋
    新建图斑{% endif %}</a>
            <a class="link-btn" href="/project/{{ p.id
```

```
      }}/edit">编辑</a>                                                    7901
                <form method="post" action="/project/{{ p.id             7902
      }}/delete"                                                          7903
                        onsubmit="return confirm('确定删除项目「{{         7904
      p.name }}」吗？\n该项目下的图斑、微地质单元与评价数据将一并删除，且无法恢复。');">  7905
                  {{ form.hidden_tag() }}                                 7906
                  <button type="submit" class="link-btn                   7907
      link-btn--danger">删除</button>                                     7908
                </form>                                                   7909
              </span>                                                     7910
            </td>                                                         7911
          </tr>                                                           7912
          {% endfor %}                                                    7913
        </tbody>                                                          7914
      </table>                                                            7915
      {% endif %}                                                         7916
</div>                                                                    7917
{% endblock %}                                                            7918
# ==== app/templates/project/edit.html ====                               7919
{% extends "base.html" %}                                                 7920
{% block title %}{{ "编辑项目" if project else "新建项目" }}{%             7921
      endblock %}                                                         7922
{% block content %}                                                       7923
<div class="card" style="max-width:640px;">                               7924
      <h2>{{ "编辑项目" if project else "新建项目" }}</h2>                 7925
      <form method="post">                                                7926
        {{ form.hidden_tag() }}                                           7927
        <div class="field">                                               7928
          <label class="field__label" for="name">项目名称</label>         7929
          <input id="name" type="text" name="name"                        7930
        style="width:100%;" required                                      7931
                value="{{ project.name if project else '' }}">            7932
        </div>                                                            7933
        <div class="field">                                               7934
          <label class="field__label" for="code">项目编号</label>         7935
          <input id="code" type="text" name="code"                        7936
        style="width:240px;"                                              7937
                value="{{ project.code or '' if project else              7938
        '' }}">                                                           7939
        </div>                                                            7940
        {% if project %}                                                  7941
        <div class="field">                                               7942
          <label class="field__label" for="status">状态</label>           7943
          <select id="status" name="status">                              7944
            {% for s in ["进行中", "已完成", "已归档"] %}                 7945
            <option value="{{ s }}" {% if project.status == s             7946
        %}selected{% endif %}>{{ s }}</option>                            7947
            {% endfor %}                                                  7948
          </select>                                                       7949
        </div>                                                            7950
```

```
      {% endif %}                                                    7951
      <div class="field">                                            7952
        <label class="field__label" for="description">项目说明</la   7953
      bel>                                                           7954
        <textarea id="description" name="description">{{            7955
      project.description or '' if project else ''                   7956
      }}</textarea>                                                  7957
      </div>                                                         7958
      <p class="actions" style="margin-top:18px;">                   7959
        <button type="submit" class="btn">保存</button>              7960
        <a class="btn btn--ghost" href="/project/">返回</a>          7961
      </p>                                                           7962
    </form>                                                          7963
</div>                                                               7964
{% endblock %}                                                       7965
# ==== app/templates/project/patches.html ====                       7966
{% extends "base.html" %}                                            7967
{% block title %}矿山图斑 · {{ project.name }}{% endblock %}         7968
{% block content %}                                                  7969
<div class="card">                                                   7970
  <h2>矿山图斑 · {{ project.name }}</h2>                             7971
  <p class="actions" style="margin-bottom:12px;">                    7972
    <a class="btn btn--sm" href="/project/{{ project.id              7973
    }}/patch/add">新建图斑</a>                                       7974
    <a class="btn btn--sm btn--ghost"                                7975
    href="/project/">返回项目列表</a>                                7976
  </p>                                                               7977
  {% if not project.patches.count() %}                               7978
  <p class="muted">该项目下暂无图斑。点击上方「新建图斑」建立第一个矿山图斑。</p>   7979
  {% else %}                                                         7980
  <table>                                                            7981
    <caption class="muted">共 {{ project.patches.count() }}          7982
    个图斑</caption>                                                 7983
    <thead>                                                          7984
      <tr>                                                           7985
        <th scope="col">图斑名称</th>                                7986
        <th scope="col">行政区</th>                                  7987
        <th scope="col">面积（公顷）</th>                            7988
        <th scope="col">开采矿种</th>                                7989
        <th scope="col">微地质单元</th>                              7990
        <th scope="col">操作</th>                                    7991
      </tr>                                                          7992
    </thead>                                                         7993
    <tbody>                                                          7994
      {% for p in project.patches %}                                 7995
      <tr>                                                           7996
        <td style="text-align:left;">{{ p.name }}</td>               7997
        <td>{{ p.admin_region or "—" }}</td>                         7998
        <td class="num">{{ p.area_ha if p.area_ha is not             7999
    none else "—" }}</td>                                            8000
```

```
            <td>{{ p.mineral_type or "—" }}</td>                          8001
            <td class="num">{{ p.units.count() }}</td>                     8002
            <td class="nowrap">                                            8003
              <span class="row-actions">                                   8004
                <a class="link-btn" href="/unit/{{ p.id                    8005
      }}">微地质单元</a>                                                    8006
                <a class="link-btn" href="/constraint/{{ p.id              8007
      }}">限制性判定</a>                                                    8008
                <a class="link-btn" href="/project/patch/{{                8009
      p.id }}/edit">编辑</a>                                               8010
                <form method="post" action="/project/patch/{{              8011
      p.id }}/delete"                                                      8012
                      onsubmit="return confirm('确定删除图斑「{{           8013
      p.name }}」吗？\n该图斑的微地质单元、指标数据与评价结果将一并删除，且无法恢复。');">  8014
                  {{ form.hidden_tag() }}                                  8015
                  <button type="submit" class="link-btn                    8016
      link-btn--danger">删除</button>                                      8017
                </form>                                                    8018
              </span>                                                      8019
            </td>                                                          8020
          </tr>                                                            8021
          {% endfor %}                                                     8022
        </tbody>                                                           8023
      </table>                                                             8024
      {% endif %}                                                          8025
</div>                                                                     8026
{% endblock %}                                                             8027
# ==== app/templates/project/patch_edit.html ====                          8028
{% extends "base.html" %}                                                  8029
{% block title %}{{ "编辑图斑" if patch else "新建图斑" }}{%               8030
     endblock %}                                                           8031
{% block content %}                                                        8032
<div class="card" style="max-width:720px;">                                8033
  <h2>{{ "编辑图斑" if patch else "新建图斑" }} · {{ project.name          8034
    }}</h2>                                                                8035
  <form method="post">                                                     8036
    {{ form.hidden_tag() }}                                                8037
    <div class="field">                                                    8038
      <label class="field__label" for="patch-name">图斑名称</lab           8039
    el>                                                                    8040
      <input id="patch-name" type="text" name="name"                       8041
    style="width:100%;" required                                           8042
             value="{{ patch.name if patch else '' }}">                    8043
    </div>                                                                 8044
    <div class="field">                                                    8045
      <label class="field__label" for="admin-region">所在行政区</          8046
    label>                                                                 8047
      <input id="admin-region" type="text"                                 8048
    name="admin_region" style="width:100%;"                                8049
             value="{{ patch.admin_region or '' if patch                   8050
```

```
else '' }}">
</div>
<div class="field">
  <label class="field__label" for="area-ha">图斑面积（公顷）</label>
  <input id="area-ha" type="number" step="any" name="area_ha"
         value="{{ patch.area_ha if patch and patch.area_ha is not none else '' }}">
</div>
<div class="field">
  <label class="field__label" for="mineral-type">开采矿种</label>
  <input id="mineral-type" type="text" name="mineral_type" style="width:240px;"
         value="{{ patch.mineral_type or '' if patch else '' }}">
</div>
<div class="field">
  <label class="field__label" for="mining-method">开采方式</label>
  <input id="mining-method" type="text" name="mining_method" style="width:240px;"
         value="{{ patch.mining_method or '' if patch else '' }}">
</div>
<div class="field">
  <label class="field__label" for="stop-years">停产年限</label>
  <input id="stop-years" type="number" step="any" name="stop_years"
         value="{{ patch.stop_years if patch and patch.stop_years is not none else '' }}">
</div>
<div class="field">
  <label class="checkbox" for="responsible-missing">
    <input id="responsible-missing" type="checkbox" name="responsible_entity_missing" value="1"
           {% if not patch or patch.responsible_entity_missing %}checked{% endif %}>
    <span>责任主体缺失</span>
  </label>
</div>
<h3>参照生态系统（指标标准化基准）</h3>
<p class="muted">
  论文 4.2.2 节取值：植被覆盖度 75%、Shannon-Wiener 多样性指数 0.7。
  该基准与图斑绑定管理，可按不同参照生态系统分别设定。
</p>
<div class="field">
  <label class="field__label" for="coverage-benchmark">植
```

```
      被覆盖度基准（%）</label>
        <input id="coverage-benchmark" type="number"
      step="any" name="coverage_benchmark"
              value="{{ reference.coverage_benchmark if
      reference else 75 }}">
      </div>
      <div class="field">
        <label class="field__label" for="shannon-benchmark">Sh
      annon-Wiener 多样性指数基准</label>
        <input id="shannon-benchmark" type="number"
      step="any" name="shannon_benchmark"
              value="{{ reference.shannon_benchmark if
      reference else 0.7 }}">
      </div>
      <div class="field">
        <label class="field__label" for="dominant-species">建群种
      </label>
        <input id="dominant-species" type="text"
      name="dominant_species" style="width:100%;"
              value="{{ reference.dominant_species or '' if
      reference else '' }}">
      </div>
      <p class="actions" style="margin-top:18px;">
        <button type="submit" class="btn">保存</button>
        <a class="btn btn--ghost" href="/project/{{
      project.id }}/patches">返回</a>
      </p>
    </form>
</div>
{% if patch %}
<div class="card" style="max-width:720px;">
    <h2>遥感影像挂接</h2>
    <p class="muted">
      挂接遥感底图或无人机正射影像，用于辅助微地质单元划分与植被覆盖度判读。
      支持 PNG、JPG、GIF、WebP、BMP、TIFF，单个文件不超过 32 MB。
    </p>
    {% if patch.image_path %}
    <figure style="margin:0 0 16px;">
      <img src="{{ url_for('static', filename=patch.image_path
      ) }}"
           alt="图斑「{{ patch.name }}」已挂接的遥感影像"
           style="max-width:100%;border:1px solid
      var(--line);border-radius:6px;">
      <figcaption class="muted" style="margin-top:6px;">
        {{ patch.image_note or "遥感影像" }}
        {% if patch.image_uploaded_at %}
        · 上传于 {{ patch.image_uploaded_at.strftime("%Y-%m-%d
      %H:%M") }}
        {% endif %}
      </figcaption>
```

```
  </figure>
  <form method="post" action="/project/patch/{{ patch.id }}/image/delete"
        onsubmit="return confirm('确定移除该遥感影像吗？');">
    {{ form.hidden_tag() }}
    <button type="submit" class="btn btn--sm btn--danger">移除影像</button>
  </form>
  <hr style="margin:16px 0;border:none;border-top:1px solid var(--line);">
  {% endif %}
  <form method="post" action="/project/patch/{{ patch.id }}/image"
        enctype="multipart/form-data">
    {{ form.hidden_tag() }}
    <div class="field">
      <label class="field__label" for="image-file">
        {{ "替换影像文件" if patch.image_path else "选择影像文件" }}
      </label>
      <input id="image-file" type="file" name="image"
             accept=".png,.jpg,.jpeg,.gif,.webp,.bmp,.tif,.tiff" required>
    </div>
    <div class="field">
      <label class="field__label" for="image-note">影像说明（如来源、时相）</label>
      <input id="image-note" type="text" name="image_note"
             style="width:100%;"
             placeholder="如：2025 年 6 月无人机正射影像"
             value="{{ patch.image_note or '' }}">
    </div>
    <p class="actions">
      <button type="submit" class="btn">{{ "替换影像" if patch.image_path else "上传并挂接" }}</button>
    </p>
  </form>
</div>
{% endif %}
{% endblock %}
# ==== app/templates/unit/list.html ====
{% extends "base.html" %}
{% from "_macros.html" import patch_guide %}
{% block title %}微地质单元{% endblock %}
{% block content %}
<div class="card">
  <h2>微地质单元</h2>
  {{ patch_guide(projects_without_patch) }}
  {% if not rows %}
  <p class="muted">
    暂无矿山图斑。请先到「项目与矿山档案」建立图斑。
```

```
  </p>                                                                    8201
  {% else %}                                                              8202
  <table>                                                                 8203
    <thead>                                                               8204
      <tr>                                                                8205
        <th scope="col">所属项目</th>                                      8206
        <th scope="col">矿山图斑</th>                                      8207
        <th scope="col">已划分单元</th>                                    8208
        <th scope="col">占比合计</th>                                      8209
        <th scope="col">操作</th>                                          8210
      </tr>                                                               8211
    </thead>                                                              8212
    <tbody>                                                               8213
      {% for r in rows %}                                                 8214
      <tr>                                                                8215
        <td>{{ r.patch.project.name if r.patch.project else               8216
  "—" }}</td>                                                             8217
        <td style="text-align:left;">{{ r.patch.name }}</td>              8218
        <td class="num">{{ r.count }} 个</td>                             8219
        <td class="num">                                                  8220
          {% if r.count %}                                                8221
            {{ "%.2f%%"|format(r.ratio_sum * 100) }}                      8222
            {% if r.ratio_sum < 0.999 or r.ratio_sum >                    8223
  1.001 %}                                                                8224
              <span class="warn">（不等于 100%）</span>                   8225
            {% endif %}                                                   8226
          {% else %}—{% endif %}                                          8227
        </td>                                                             8228
        <td class="nowrap">                                               8229
          <span class="row-actions">                                      8230
            <a class="link-btn" href="/unit/{{ r.patch.id                 8231
  }}">划分与维护</a>                                                      8232
            <a class="link-btn" href="/indicator/">录入指标数据</a        8233
  >                                                                       8234
          </span>                                                         8235
        </td>                                                             8236
      </tr>                                                               8237
      {% endfor %}                                                        8238
    </tbody>                                                              8239
  </table>                                                                8240
  {% endif %}                                                             8241
</div>                                                                    8242
{% endblock %}                                                            8243
# ==== app/templates/unit/index.html ====                                 8244
{% extends "base.html" %}                                                 8245
{% block title %}微地质单元 · {{ patch.name }}{% endblock %}              8246
{% block content %}                                                       8247
<div class="card">                                                        8248
  <h2>微地质单元 · {{ patch.name }}</h2>                                  8249
  {% if units %}                                                          8250
```

```
<table>
  <caption class="muted">
    共 {{ units|length }} 个单元；
    面积合计 <span class="num">{{ "%.4f"|format(total_area) }}</span>；
    占比合计 <span class="num">{{ "%.2f%%"|format(ratio_sum * 100) }}</span>
    {% if ratio_sum and (ratio_sum < 0.999 or ratio_sum > 1.001) %}
    <span class="warn">（占比合计不等于 100%，请检查面积录入）</span>
    {% endif %}
  </caption>
  <thead>
    <tr>
      <th scope="col">微地质单元</th>
      <th scope="col">面积</th>
      <th scope="col">面积占比</th>
      <th scope="col">说明</th>
      <th scope="col">操作</th>
    </tr>
  </thead>
  <tbody>
    {% for u in units %}
    <tr>
      <td>{{ u.unit_type }}</td>
      <td class="num">{{ u.area if u.area is not none else "—" }}</td>
      <td class="num">
        {% if u.ratio is not none %}{{ "%.2f%%"|format(u.ratio * 100) }}{% else %}—{% endif %}
      </td>
      <td style="text-align:left;">{{ u.description or "—" }}</td>
      <td class="nowrap">
        <span class="row-actions">
          <a class="link-btn" href="/unit/edit/{{ u.id }}">编辑</a>
          <form method="post" action="/unit/delete/{{ u.id }}"
                onsubmit="return confirm('确定删除微地质单元「{{ u.unit_type }}」吗？\n该单元已录入的指标数据将一并删除，且无法恢复。');">
            {{ form.hidden_tag() }}
            <button type="submit" class="link-btn link-btn--danger">删除</button>
          </form>
        </span>
      </td>
    </tr>
    {% endfor %}
  </tbody>
```

```
  </table>                                                                    8301
  {% else %}                                                                  8302
  <p class="muted">尚未划分微地质单元。请在下方添加。</p>                    8303
  {% endif %}                                                                 8304
  <h3>添加微地质单元</h3>                                                     8305
  <form method="post" action="/unit/{{ patch.id }}/add">                      8306
    {{ form.hidden_tag() }}                                                   8307
    <div class="actions">                                                     8308
      <label class="field__label" for="add-unit-type"                         8309
    style="margin:0;">类型</label>                                            8310
      <select id="add-unit-type" name="unit_type">                            8311
        {% for t in unit_types %}<option value="{{ t }}">{{                   8312
    t }}</option>{% endfor %}                                                 8313
      </select>                                                               8314
      <label class="field__label" for="add-unit-area"                         8315
    style="margin:0;">面积</label>                                            8316
      <input id="add-unit-area" type="number" step="any"                      8317
    name="area">                                                              8318
      <label class="field__label" for="add-unit-desc"                         8319
    style="margin:0;">说明</label>                                            8320
      <input id="add-unit-desc" type="text"                                   8321
    name="description" style="width:220px;">                                  8322
      <button type="submit" class="btn btn--sm">添加</button>                 8323
    </div>                                                                    8324
  </form>                                                                     8325
  <p class="actions" style="margin-top:20px;">                                8326
    <a class="btn btn--sm btn--ghost" href="/project/{{                       8327
    patch.project_id }}/patches">返回图斑列表</a>                             8328
    <a class="btn btn--sm btn--ghost"                                         8329
    href="/indicator/">去录入指标数据</a>                                     8330
  </p>                                                                        8331
</div>                                                                        8332
{% endblock %}                                                                8333
# ==== app/templates/unit/edit.html ====                                      8334
{% extends "base.html" %}                                                     8335
{% block title %}编辑微地质单元{% endblock %}                                 8336
{% block content %}                                                           8337
<div class="card">                                                            8338
  <h2>编辑微地质单元 · {{ patch.name }}</h2>                                  8339
  <p class="muted">修改面积后，系统会自动重算本图斑各单元的面积占比。</p>    8340
  <form method="post">                                                        8341
    {{ form.hidden_tag() }}                                                   8342
    <div class="field">                                                       8343
      <label class="field__label" for="unit-type">微地质单元类型</l          8344
    abel>                                                                     8345
      <select id="unit-type" name="unit_type">                                8346
        {% for t in unit_types %}                                             8347
        <option value="{{ t }}" {% if unit.unit_type == t                     8348
    %}selected{% endif %}>{{ t }}</option>                                    8349
        {% endfor %}                                                          8350
```

```
        </select>
      </div>
      <div class="field">
        <label class="field__label" for="unit-area">面积</label>
        <input id="unit-area" type="number" step="any"
      name="area"
               value="{{ unit.area if unit.area is not none
      else '' }}">
      </div>
      <div class="field">
        <label class="field__label" for="unit-desc">说明</label>
        <input id="unit-desc" type="text" name="description"
      style="width:420px;"
               value="{{ unit.description or '' }}">
      </div>
      <p class="actions" style="margin-top:16px;">
        <button type="submit" class="btn">保存</button>
        <a class="btn btn--ghost" href="/unit/{{
      unit.patch_id }}">返回</a>
      </p>
    </form>
</div>
{% endblock %}
# ==== app/templates/standard/index.html ====
{% extends "base.html" %}
{% block title %}政策与标准知识库{% endblock %}
{% block content %}
<div class="card">
    <h2>政策与标准知识库</h2>  <p class="actions"
      style="margin-bottom:14px;">
      <a class="btn btn--sm" href="/standard/add">添加标准</a>
      <a class="btn btn--sm btn--ghost"
      href="/standard/compare">省际口径比对</a>
    </p>
    <h3>地方标准</h3>
    {% if local %}
    <table>
      <thead>
        <tr>
          <th scope="col">省份</th>
          <th scope="col">名称</th>
          <th scope="col">编号</th>
          <th scope="col">规则数</th>
          <th scope="col">操作</th>
        </tr>
      </thead>
      <tbody>
        {% for s in local %}
        <tr>
          <td>{{ s.province }}</td>
```

```
          <td style="text-align:left;">{{ s.name }}</td>
          <td class="mono">{{ s.doc_number or "—" }}</td>
          <td class="num">{{ s.rules|length }}</td>
          <td class="nowrap">
            <span class="row-actions">
              <a class="link-btn" href="/standard/{{ s.id
    }}/rules">规则</a>
              <a class="link-btn" href="/standard/{{ s.id
    }}/edit">编辑</a>
              <form method="post" action="/standard/{{ s.id
    }}/delete"
                    onsubmit="return confirm('确定删除「{{
    s.province }} {{ s.name }}」吗？\n该标准下的 {{ s.rules|length
    }} 条规则将一并删除，且无法恢复。');">
                {{ form.hidden_tag() }}
                <button type="submit" class="link-btn
    link-btn--danger">删除</button>
              </form>
            </span>
          </td>
        </tr>
        {% endfor %}
      </tbody>
    </table>
    {% else %}
    <p class="muted">暂无地方标准。运行 <span class="mono">python
      seed_demo.py</span> 可载入论文表 1 的 11 项标准与政策文件。</p>
    {% endif %}
    <h3>政策文件</h3>
    {% if policy %}
    <table>
      <thead>
        <tr>
          <th scope="col">省份</th>
          <th scope="col">名称</th>
          <th scope="col">文号</th>
          <th scope="col">规则数</th>
          <th scope="col">操作</th>
        </tr>
      </thead>
      <tbody>
        {% for s in policy %}
        <tr>
          <td>{{ s.province }}</td>
          <td style="text-align:left;">{{ s.name }}</td>
          <td class="mono">{{ s.doc_number or "—" }}</td>
          <td class="num">{{ s.rules|length }}</td>
          <td class="nowrap">
            <span class="row-actions">
              <a class="link-btn" href="/standard/{{ s.id
```

```
      }}/rules">规则</a>                                                    8451
                <a class="link-btn" href="/standard/{{ s.id                 8452
      }}/edit">编辑</a>                                                     8453
                <form method="post" action="/standard/{{ s.id               8454
      }}/delete"                                                            8455
                        onsubmit="return confirm('确定删除「{{              8456
      s.province }} {{ s.name }}」吗？\n该文件下的 {{ s.rules|length        8457
      }} 条规则将一并删除，且无法恢复。');">                                8458
                  {{ form.hidden_tag() }}                                   8459
                  <button type="submit" class="link-btn                     8460
      link-btn--danger">删除</button>                                       8461
                </form>                                                     8462
              </span>                                                       8463
            </td>                                                           8464
          </tr>                                                             8465
          {% endfor %}                                                      8466
        </tbody>                                                            8467
      </table>                                                              8468
      {% else %}                                                            8469
      <p class="muted">暂无政策文件。</p>                                   8470
      {% endif %}                                                           8471
</div>                                                                      8472
{% endblock %}                                                              8473
# ==== app/templates/standard/edit.html ====                                8474
{% extends "base.html" %}                                                   8475
{% block title %}{{ "编辑标准" if standard else "添加标准" }}{%             8476
    endblock %}                                                             8477
{% block content %}                                                         8478
<div class="card" style="max-width:800px;">                                 8479
  <h2>{{ "编辑标准" if standard else "添加标准" }}</h2>                     8480
  <form method="post">                                                      8481
    {{ form.hidden_tag() }}                                                 8482
    <div class="field">                                                     8483
      <label class="field__label" for="province">省份</label>               8484
      <input id="province" type="text" name="province"                      8485
    style="width:160px;" required                                           8486
             value="{{ standard.province if standard else                   8487
    '' }}">                                                                 8488
    </div>                                                                  8489
    <div class="field">                                                     8490
      <label class="field__label" for="std-name">名称</label>               8491
      <input id="std-name" type="text" name="name"                          8492
    style="width:100%;" required                                            8493
             value="{{ standard.name if standard else ''                    8494
    }}">                                                                    8495
    </div>                                                                  8496
    <div class="field">                                                     8497
      <label class="field__label" for="doc-number">标准编号 /               8498
    文号</label>                                                            8499
      <input id="doc-number" type="text" name="doc_number"                  8500
```

```
style="width:320px;"                                                    8501
           value="{{ standard.doc_number or '' if                       8502
standard else '' }}">                                                  8503
</div>                                                                  8504
<div class="field">                                                     8505
  <label class="field__label" for="category">类别</label>               8506
  <select id="category" name="category">                                8507
    {% for c in ["地方标准", "政策文件"] %}                             8508
    <option value="{{ c }}" {% if standard and                          8509
standard.category == c %}selected{% endif %}>{{ c                       8510
}}</option>                                                            8511
    {% endfor %}                                                        8512
  </select>                                                             8513
</div>                                                                  8514
<div class="field">                                                     8515
  <label class="field__label" for="publish-date">发布时间</l            8516
abel>                                                                   8517
  <input id="publish-date" type="text"                                  8518
name="publish_date" style="width:200px;"                                8519
           placeholder="如 2023-03-30"                                  8520
           value="{{ standard.publish_date or '' if                     8521
standard else '' }}">                                                  8522
</div>                                                                  8523
<div class="field">                                                     8524
  <label class="field__label" for="implement-date">实施时间<            8525
/label>                                                                 8526
  <input id="implement-date" type="text"                                8527
name="implement_date" style="width:200px;"                              8528
           placeholder="如 2023-04-30（未明确可留空）"                  8529
           value="{{ standard.implement_date or '' if                   8530
standard else '' }}">                                                  8531
</div>                                                                  8532
<div class="field">                                                     8533
  <label class="field__label" for="summary">特点 /                      8534
判定标准</label>                                                        8535
  <textarea id="summary" name="summary"                                 8536
style="min-height:96px;">{{ standard.summary or '' if                   8537
standard else '' }}</textarea>                                          8538
</div>                                                                  8539
<p class="actions" style="margin-top:18px;">                            8540
  <button type="submit" class="btn">保存</button>                       8541
  {% if standard %}                                                     8542
  <form method="post" action="/standard/{{ standard.id                  8543
}}/delete" style="display:inline;"                                      8544
          onsubmit="return confirm('确定删除「{{                        8545
standard.province }} {{ standard.name                                   8546
}}」吗？\n该标准下的规则将一并删除，且无法恢复。');">                  8547
    {{ form.hidden_tag() }}                                             8548
    <button type="submit" class="btn                                    8549
btn--danger">删除该标准</button>                                        8550
```

```
        </form>                                                          8551
        {% endif %}                                                      8552
        <a class="btn btn--ghost" href="/standard/">返回</a>             8553
      </p>                                                               8554
    </form>                                                              8555
</div>                                                                   8556
{% endblock %}                                                           8557
# ==== app/templates/standard/rules.html ====                            8558
{% extends "base.html" %}                                                8559
{% block title %}结构化规则 · {{ standard.province }}{% endblock         8560
    %}                                                                   8561
{% block content %}                                                      8562
<div class="card">                                                       8563
  <h2>结构化规则 · {{ standard.province }} {{ standard.name              8564
    }}</h2>                                                              8565
  {% if rules %}                                                         8566
  <table>                                                                8567
    <caption class="muted">共 {{ rules|length }}                         8568
    条规则</caption>                                                     8569
    <thead>                                                              8570
      <tr>                                                               8571
        <th scope="col">判定项</th>                                      8572
        <th scope="col">条件</th>                                        8573
        <th scope="col">条件类型</th>                                    8574
        <th scope="col">适用分区</th>                                    8575
        <th scope="col">备注</th>                                        8576
        <th scope="col">操作</th>                                        8577
      </tr>                                                              8578
    </thead>                                                             8579
    <tbody>                                                              8580
      {% for r in rules %}                                               8581
      <tr>                                                               8582
        <td style="text-align:left;">{{ r.item }}</td>                   8583
        <td class="mono nowrap">{{ r.operator }} {{                      8584
    r.threshold or "" }}</td>                                            8585
        <td>{{ r.condition_type }}</td>                                  8586
        <td>{{ r.region or "—" }}</td>                                   8587
        <td style="text-align:left;">{{ r.note or "—"                    8588
    }}</td>                                                              8589
        <td>                                                             8590
          <form method="post" action="/standard/rule/{{                  8591
    r.id }}/delete"                                                      8592
                onsubmit="return confirm('确定删除规则「{{ r.item        8593
    }} {{ r.operator }} {{ r.threshold or '' }}」吗？');">               8594
            {{ form.hidden_tag() }}                                      8595
            <button type="submit" class="btn btn--sm                     8596
    btn--danger">删除</button>                                           8597
          </form>                                                        8598
        </td>                                                            8599
      </tr>                                                              8600
```

```
      {% endfor %}
    </tbody>
  </table>
  {% else %}
  <p class="muted">该标准暂无规则。请在下方添加第一条规则。</p>
  {% endif %}
  <h3>添加规则</h3>
  <form method="post">
    {{ form.hidden_tag() }}
    <div class="actions">
      <div>
        <label class="field__label" for="r-item">判定项</label>
        <input id="r-item" type="text" name="item"
    placeholder="如：植被覆盖度" required>
      </div>
      <div>
        <label class="field__label" for="r-op">比较符</label>
        <select id="r-op" name="operator">
          {% for o in [">=", ">", "<=", "<", "=="]
    %}<option value="{{ o }}">{{ o }}</option>{% endfor %}
        </select>
      </div>
      <div>
        <label class="field__label" for="r-thr">阈值</label>
        <input id="r-thr" type="text" name="threshold"
    placeholder="如：50" style="width:110px;">
      </div>
      <div>
        <label class="field__label" for="r-cond">条件类型</label
    >
        <select id="r-cond" name="condition_type">
          <option value="必要条件">必要条件</option>
          <option value="充分条件">充分条件</option>
        </select>
      </div>
      <div>
        <label class="field__label" for="r-region">适用分区</lab
    el>
        <input id="r-region" type="text" name="region"
    placeholder="如宁夏分区" style="width:130px;">
      </div>
      <div>
        <label class="field__label" for="r-note">备注</label>
        <input id="r-note" type="text" name="note"
    style="width:220px;">
      </div>
      <div>
        <button type="submit" class="btn">添加规则</button>
      </div>
    </div>
```

```
  </form>
  <p class="actions" style="margin-top:20px;">
    <a class="btn btn--sm btn--ghost"
    href="/standard/">返回标准列表</a>
    <a class="btn btn--sm btn--ghost"
    href="/standard/compare">省际口径比对</a>
  </p>
</div>
{% endblock %}
# ==== app/templates/standard/compare.html ====
{% extends "base.html" %}
{% block title %}省际口径比对{% endblock %}
{% block content %}
<div class="card">
  <h2>省际口径比对</h2>
  {% if not groups %}
  <p class="muted">
    暂无规则数据。请先在「标准知识库」中为标准录入规则，
    或运行 <span class="mono">python seed_demo.py</span> 载入论文表
    1 数据。
  </p>
  {% else %}
  {% for item, rs in groups.items() %}
  <h3>{{ item }}</h3>
  <table>
    <thead>
      <tr>
        <th scope="col">省份</th>
        <th scope="col">标准 / 文件</th>
        <th scope="col">条件</th>
        <th scope="col">条件类型</th>
        <th scope="col">备注</th>
      </tr>
    </thead>
    <tbody>
      {% for r in rs %}
      <tr>
        <td>{{ r.standard.province }}</td>
        <td style="text-align:left;">{{ r.standard.name
    }}</td>
        <td class="mono nowrap">{{ r.operator }} {{
    r.threshold or "" }}</td>
        <td>{{ r.condition_type }}</td>
        <td style="text-align:left;">{{ r.note or "—"
    }}</td>
      </tr>
      {% endfor %}
    </tbody>
  </table>
  {% endfor %}
```

```
  {% endif %}
  <p class="actions" style="margin-top:16px;">
    <a class="btn btn--ghost" href="/standard/">返回标准列表</a>
  </p>
</div>
{% endblock %}
# ==== app/templates/constraint/index.html ====
{% extends "base.html" %}
{% from "_macros.html" import patch_guide %}
{% block title %}限制性指标判定{% endblock %}
{% block content %}
<div class="card">
  <h2>限制性指标判定</h2>
  {{ patch_guide(projects_without_patch) }}
  {% if not patches %}
  <p class="muted">暂无矿山图斑。请先在「项目与矿山档案」中建立图斑。</p>
  {% else %}
  <table>
    <thead>
      <tr>
        <th scope="col">所属项目</th>
        <th scope="col">矿山图斑</th>
        <th scope="col">判定结果</th>
        <th scope="col">操作</th>
      </tr>
    </thead>
    <tbody>
      {% for p in patches %}
      {% set r = latest.get(p.id) %}
      <tr>
        <td>{{ p.project.name if p.project else "—" }}</td>
        <td style="text-align:left;">{{ p.name }}</td>
        <td style="text-align:left;">
          {% if r %}
            {% if r.passed %}
            <span class="tag tag--ok">通过</span>
            {% else %}
            <span class="tag tag--warn">不通过</span>
            {% endif %}
            <span class="muted">{{ r.reason }}</span>
          {% else %}
          <span class="tag tag--muted">未判定</span>
          {% endif %}
        </td>
        <td class="nowrap">
          <span class="row-actions">
            <a class="link-btn" href="/constraint/{{ p.id
    }}">执行判定</a>
          </span>
        </td>
```

```
        </tr>                                                                    8751
        {% endfor %}                                                             8752
      </tbody>                                                                   8753
    </table>                                                                     8754
    {% endif %}                                                                  8755
</div>                                                                           8756
{% endblock %}                                                                   8757
# ==== app/templates/constraint/check.html ====                                  8758
{% extends "base.html" %}                                                        8759
{% block title %}限制性条件判定 · {{ patch.name }}{% endblock %}                  8760
{% block content %}                                                              8761
<div class="card" style="max-width:760px;">                                      8762
  <h2>限制性条件判定 · {{ patch.name }}</h2>                                     8763
  <form method="post">                                                           8764
    {{ form.hidden_tag() }}                                                      8765
    {% for field, name, val in field_states %}                                   8766
    <fieldset style="border:1px solid                                            8767
    var(--line);border-radius:6px;padding:10px                                   8768
    14px;margin:0 0 10px;">                                                      8769
      <legend style="padding:0 6px;font-weight:600;">{{                          8770
    name }}</legend>                                                             8771
      <div class="actions">                                                      8772
        <label class="checkbox">                                                 8773
          <input type="radio" name="{{ field }}" value="1"                       8774
                 {% if val is none or val %}checked{% endif                      8775
    %}>                                                                          8776
          <span>通过</span>                                                      8777
        </label>                                                                 8778
        <label class="checkbox">                                                 8779
          <input type="radio" name="{{ field }}" value="0"                       8780
                 {% if val is not none and not val                               8781
    %}checked{% endif %}>                                                        8782
          <span>不通过</span>                                                    8783
        </label>                                                                 8784
      </div>                                                                     8785
    </fieldset>                                                                  8786
    {% endfor %}                                                                 8787
    <p class="actions" style="margin-top:16px;">                                 8788
      <button type="submit" class="btn">保存判定</button>                        8789
      <a class="btn btn--ghost" href="/constraint/">返回</a>                     8790
    </p>                                                                         8791
  </form>                                                                        8792
  {% if record %}                                                                8793
  <div style="margin-top:18px;padding:14px;border-radius:6px                     8794
    ;border-left:4px solid;                                                      8795
                {% if record.passed %}background:var(--ok-soft                   8796
    );border-left-color:var(--ok);                                               8797
                {% else %}background:var(--danger-soft);border                   8798
    -left-color:var(--danger);{% endif %}">                                      8799
    <p style="margin:0 0 4px;font-weight:600;">                                  8800
```

```
        判定结论：                                                                    8801
        {% if record.passed %}                                                        8802
        <span class="ok">通过，可进入定量评价</span>                                  8803
        {% else %}                                                                    8804
        <span class="warn">不通过 —— 建议生态重建或人工辅助</span>                    8805
        {% endif %}                                                                   8806
      </p>                                                                            8807
      <p class="muted" style="margin:0;">{{ record.reason                             8808
      }}</p>                                                                          8809
    </div>                                                                            8810
    {% endif %}                                                                       8811
</div>                                                                                8812
{% endblock %}                                                                        8813
# ==== app/templates/ahp/index.html ====                                              8814
{% extends "base.html" %}                                                             8815
{% block title %}AHP 权重计算{% endblock %}                                           8816
{% block content %}                                                                   8817
<div class="card" style="max-width:720px;">                                           8818
    <h2>AHP 权重计算 · 选择指标层级</h2>                                             8819
    <form method="post">                                                              8820
      {{ form.hidden_tag() }}                                                         8821
      <div class="field">                                                             8822
        <label class="field__label" for="{{ form.unit_type.id                         8823
      }}">微地质单元类型</label>                                                      8824
        {{ form.unit_type(id=form.unit_type.id) }}                                    8825
      </div>                                                                          8826
      <div class="field">                                                             8827
        <label class="field__label" for="{{ form.level.id                             8828
      }}">指标层级</label>                                                            8829
        {{ form.level(id=form.level.id) }}                                            8830
      </div>                                                                          8831
      <div class="field">                                                             8832
        <label class="field__label" for="{{ form.l1.id }}">                           8833
          一级指标（层级选「二级指标」时需选择）                                      8834
        </label>                                                                      8835
        {{ form.l1(id=form.l1.id) }}                                                  8836
      </div>                                                                          8837
      <p class="actions" style="margin-top:16px;">                                    8838
        {{ form.submit(class_="btn") }}                                               8839
        <a class="btn btn--ghost" href="/ahp/indicators">指标定义维                   8840
      护</a>                                                                          8841
        <a class="btn btn--ghost" href="/ahp/experts">多专家矩阵</a                   8842
      >                                                                               8843
      </p>                                                                            8844
    </form>                                                                           8845
</div>                                                                                8846
<div class="card">                                                                    8847
    <h2>各微地质单元当前生效权重</h2>                                                8848
    <table>                                                                           8849
      <thead>                                                                         8850
```

```
        <tr>                                                                    8851
          <th scope="col">微地质单元</th>                                       8852
          <th scope="col">当前权重来源</th>                                     8853
          <th scope="col">操作</th>                                             8854
        </tr>                                                                   8855
      </thead>                                                                  8856
      <tbody>                                                                   8857
        {% for r in weight_rows %}                                              8858
        <tr>                                                                    8859
          <td style="text-align:left;">{{ r.unit_type }}</td>                   8860
          <td style="text-align:left;">                                         8861
            {% if r.custom %}                                                   8862
            <span class="tag tag--warn">{{ r.label }}</span>                    8863
            {% else %}                                                          8864
            <span class="tag tag--muted">{{ r.label }}</span>                   8865
            {% endif %}                                                         8866
          </td>                                                                 8867
          <td class="nowrap">                                                   8868
            <span class="row-actions">                                          8869
              <a class="link-btn" href="/ahp/matrix?unit_type=                  8870
      {{ r.unit_type }}&level=1">一级指标权重</a>                               8871
              <a class="link-btn" href="/ahp/matrix?unit_type=                  8872
      {{ r.unit_type }}&level=1&calc=1">查看一致性</a>                          8873
              {% if r.custom %}                                                 8874
              <form method="post" action="/ahp/weights/reset"                   8875
                    onsubmit="return confirm('确定把「{{                        8876
      r.unit_type }}」恢复为论文表 3 的默认权重吗？');">                        8877
                <input type="hidden" name="csrf_token"                          8878
      value="{{ csrf_token() }}">                                               8879
                <input type="hidden" name="unit_type"                           8880
      value="{{ r.unit_type }}">                                                8881
                <button type="submit" class="link-btn                           8882
      link-btn--danger">恢复默认</button>                                       8883
              </form>                                                           8884
              {% endif %}                                                       8885
            </span>                                                             8886
          </td>                                                                 8887
        </tr>                                                                   8888
        {% endfor %}                                                            8889
      </tbody>                                                                  8890
    </table>                                                                    8891
</div>                                                                          8892
{% endblock %}                                                                  8893
# ==== app/templates/ahp/matrix.html ====                                       8894
{% extends "base.html" %}                                                       8895
{% block title %}判断矩阵录入{% endblock %}                                     8896
{% block content %}                                                             8897
<div class="card">                                                              8898
  <h2>                                                                          8899
    判断矩阵录入                                                                8900
```

```
    · {{ unit_type }}                                                    8901
    · {% if level == 1 %}一级指标{% else %}二级指标（{{ l1 }}）{%       8902
    endif %}                                                             8903
</h2>                                                                    8904
<p class="muted">                                                        8905
    采用 1–9 标度：1 同等重要、3 稍微重要、5 明显重要、7 强烈重要、9 极端重要，  8906
    2、4、6、8 为相邻判断的中间值。只需录入上三角，对角线为 1，下三角自动取倒数。  8907
    矩阵已按论文表 3 的权重预填。                                        8908
</p>                                                                     8909
<form method="post">                                                     8910
    {{ form.hidden_tag() }}                                              8911
    <table style="width:auto;">                                          8912
      <thead>                                                            8913
        <tr>                                                             8914
          <th></th>                                                      8915
          {% for name in names %}<th scope="col">{{ name                 8916
    }}</th>{% endfor %}                                                  8917
        </tr>                                                            8918
      </thead>                                                           8919
      <tbody>                                                            8920
        {% for i in range(names|length) %}                               8921
        <tr>                                                             8922
          <th scope="row">{{ names[i] }}</th>                            8923
          {% for j in range(names|length) %}                             8924
            {% if i == j %}                                              8925
            <td class="muted num">1</td>                                 8926
            {% elif i < j %}                                             8927
            <td>                                                         8928
              <label class="muted" style="position:absolute;             8929
    left:-9999px;"                                                       8930
                     for="m-{{ i }}-{{ j }}">{{ names[i] }}              8931
    相对 {{ names[j] }} 的重要性</label>                                 8932
              <input id="m-{{ i }}-{{ j }}" type="number"                8933
    step="0.01" min="0.01"                                               8934
                     name="m_{{ i }}_{{ j }}" value="{{                  8935
    default[i][j] }}">                                                   8936
            </td>                                                        8937
            {% else %}                                                   8938
            <td class="muted num">{{ default[i][j] }}</td>               8939
            {% endif %}                                                  8940
          {% endfor %}                                                   8941
        </tr>                                                            8942
        {% endfor %}                                                     8943
      </tbody>                                                           8944
    </table>                                                             8945
    <p class="actions">                                                  8946
      <button type="submit" class="btn" name="action"                    8947
    value="calc">计算权重</button>                                       8948
      {% if result and result.consistent %}                              8949
      <button type="submit" class="btn" name="action"                    8950
```

```
    value="save">保存并生效</button>                                    8951
      {% endif %}                                                       8952
      <a class="btn btn--ghost" href="/ahp/">重新选择</a>               8953
    </p>                                                                8954
  </form>                                                               8955
  <p class="muted" style="margin-top:10px;">                            8956
    当前「{{ unit_type }}」生效权重：<strong>{{ source_label            8957
    }}</strong>。                                                       8958
    点击「保存并生效」后，本单元类型的适宜性评价将采用此处算出的权重；  8959
    如需回到论文取值，可在 <a href="/ahp/">权重首页</a> 恢复默认。      8960
  </p>                                                                  8961
  {% if result %}                                                       8962
  <hr style="margin:22px 0;border:none;border-top:1px solid             8963
    var(--line);">                                                      8964
  <h3>计算结果</h3>                                                     8965
  <table style="width:auto;">                                           8966
    <thead>                                                             8967
      <tr>                                                              8968
        <th scope="col">指标</th>                                       8969
        {% for name in names %}<th scope="col">{{ name                  8970
    }}</th>{% endfor %}                                                 8971
      </tr>                                                             8972
    </thead>                                                            8973
    <tbody>                                                             8974
      <tr>                                                              8975
        <th scope="row">权重</th>                                       8976
        {% for w in result.weights %}<td class="mono">{{ w              8977
    }}</td>{% endfor %}                                                 8978
      </tr>                                                             8979
    </tbody>                                                            8980
  </table>                                                              8981
  <div class="stat-row">                                                8982
    <div>                                                               8983
      <div class="stat__label">最大特征值 λmax</div>                    8984
      <div class="stat__value mono">{{ result.lambda_max                8985
    }}</div>                                                            8986
    </div>                                                              8987
    <div>                                                               8988
      <div class="stat__label">一致性指标 CI</div>                      8989
      <div class="stat__value mono">{{                                  8990
    "%.4f"|format(result.ci) }}</div>                                   8991
    </div>                                                              8992
    <div>                                                               8993
      <div class="stat__label">一致性比例 CR</div>                      8994
      <div class="stat__value mono">{{                                  8995
    "%.4f"|format(result.cr) }}</div>                                   8996
    </div>                                                              8997
    <div>                                                               8998
      <div class="stat__label">一致性检验（要求 CR &lt; 0.1）</div>     8999
      <div class="stat__value">                                         9000
```

```
          {% if result.consistent %}<span class="ok">通过</span>
          {% else %}<span class="warn">不通过</span>{% endif %}
        </div>
      </div>
    </div>
    {% if not result.consistent and result.most_inconsistent
      %}
    <p style="margin-top:14px;padding:12px;border-radius:6px;b
      ackground:var(--danger-soft);
              border-left:4px solid var(--danger);">
      <strong class="warn">建议优先调整：</strong>
      第 {{ result.most_inconsistent.i + 1 }} 行、第 {{
      result.most_inconsistent.j + 1 }} 列的比较
      （当前 <span class="mono num">{{ result.most_inconsistent.v
      alue|round(3) }}</span>，
      理想约 <span class="mono num">{{ result.most_inconsistent.e
      xpected|round(3) }}</span>）
    </p>
    {% endif %}
    {% endif %}
</div>
{% endblock %}
# ==== app/templates/ahp/indicators.html ====
{% extends "base.html" %}
{% block title %}指标定义维护{% endblock %}
{% block content %}
<div class="card">
    <h2>指标体系维护</h2>
    <p class="actions" style="margin-bottom:14px;">
      <a class="btn btn--sm btn--ghost"
      href="/ahp/">返回权重计算</a>
      <a class="btn btn--sm btn--ghost"
      href="/ahp/experts">多专家矩阵</a>
      <form method="post" action="/ahp/indicators/reset"
      style="display:inline;"
            onsubmit="return confirm('将按论文取值重置全部指标定义，自定义修改会丢失。
      确定继续？');">
        {{ form.hidden_tag() }}
        <button type="submit" class="btn btn--sm
      btn--danger">按论文取值重置</button>
      </form>
    </p>
    {% for level, title in [(1, "一级指标"), (2, "二级指标"), (3,
      "三级指标")] %}
    {% set level_items = all_ind|selectattr("level",
      "equalto", level)|list %}
    {% if level_items %}
    <h3>{{ title }}（{{ level_items|length }} 项）</h3>
    <table>
      <thead>
```

```
        <tr>
          <th scope="col">指标</th>
          <th scope="col">上级指标</th>
          <th scope="col">单位</th>
          <th scope="col">标准化类型</th>
          <th scope="col">理想取值</th>
          <th scope="col">启用单元</th>
          <th scope="col">限制性</th>
          <th scope="col">操作</th>
        </tr>
      </thead>
      <tbody>
        {% for i in level_items %}
        <tr>
          <td style="text-align:left;">
            <span class="mono muted">{{ i.code
      }}</span><br>{{ i.name }}
          </td>
          <td>{{ (all_ind|selectattr("code", "equalto",
      i.parent_code)|list|first).name if i.parent_code else
      "—" }}</td>
          <td>{{ i.unit or "—" }}</td>
          <td>{{ type_label(i.ideal_type) }}</td>
          <td class="mono">{{ i.ideal_value or "—" }}</td>
          <td style="text-align:left;">{{ i.enabled_units or
      "—" }}</td>
          <td>
            {% if i.is_constraint %}<span class="tag
      tag--warn">是</span>
            {% else %}<span class="tag tag--muted">否</span>{%
      endif %}
          </td>
          <td class="nowrap">
            <span class="row-actions">
              <a class="link-btn" href="/ahp/indicators/{{
      i.code }}/edit">编辑</a>
            </span>
          </td>
        </tr>
        {% endfor %}
      </tbody>
    </table>
    {% endif %}
    {% endfor %}
</div>
{% endblock %}
# ==== app/templates/ahp/indicator_edit.html ====
{% extends "base.html" %}
{% block title %}编辑指标 · {{ ind.name }}{% endblock %}
{% block content %}
```

```
<div class="card" style="max-width:680px;">
  <h2>编辑指标 · {{ ind.name }}</h2>
  <p class="muted">
    指标编码 <span class="mono">{{ ind.code }}</span>、层级 {{
    ind.level }} 级，
    由论文指标体系确定，不可修改。
  </p>
  <form method="post">
    {{ form.hidden_tag() }}
    <div class="field">
      <label class="field__label" for="name">指标名称</label>
      <input id="name" type="text" name="name"
    style="width:100%;" required
             value="{{ ind.name }}">
    </div>
    <div class="field">
      <label class="field__label" for="unit">计量单位</label>
      <input id="unit" type="text" name="unit"
    style="width:180px;"
             value="{{ ind.unit or '' }}">
    </div>
    <div class="field">
      <label class="field__label" for="ideal-type">标准化类型</la
    bel>
      <select id="ideal-type" name="ideal_type">
        {% for code, label in [("benefit","越大越优"),("cost","越
    小越优"),("mid","适中型"),("discrete","离散赋值")] %}
        <option value="{{ code }}" {% if ind.ideal_type ==
    code %}selected{% endif %}>{{ label }}</option>
        {% endfor %}
      </select>
    </div>
    <div class="field">
      <label class="field__label" for="ideal-value">理想取值</la
    bel>
      <input id="ideal-value" type="text"
    name="ideal_value" style="width:100%;"
             value="{{ ind.ideal_value or '' }}">
      <p class="muted" style="margin:4px 0 0;">
        如土石比 0.8、土层厚度 20、pH 5~9、植被覆盖度 75。
      </p>
    </div>
    <fieldset style="border:1px solid
    var(--line);border-radius:6px;padding:10px
    14px;margin:0 0 12px;">
      <legend style="padding:0 6px;font-weight:600;">启用该指标的微
    地质单元</legend>
      {% for t in unit_types %}
      <label class="checkbox" style="margin-right:18px;">
        <input type="checkbox" name="enabled_units"
```

```
      value="{{ t }}"                                                    9151
                    {% if t in ind.units_list() %}checked{%              9152
      endif %}>                                                          9153
          <span>{{ t }}</span>                                           9154
        </label>                                                         9155
        {% endfor %}                                                     9156
        <p class="muted" style="margin:8px 0 0;">                        9157
          论文设定：岩质边坡不启用土壤指标，平台不启用地形指标。                  9158
        </p>                                                             9159
      </fieldset>                                                        9160
      <div class="field">                                                9161
        <label class="checkbox" for="is-constraint">                     9162
          <input id="is-constraint" type="checkbox"                      9163
      name="is_constraint" value="1"                                     9164
                    {% if ind.is_constraint %}checked{% endif %}>        9165
          <span>同时作为限制性指标（参与一票否决判定）</span>                      9166
        </label>                                                         9167
      </div>                                                             9168
      <div class="field">                                                9169
        <label class="field__label" for="description">取值依据说明</      9170
      label>                                                             9171
        <textarea id="description" name="description"                    9172
      style="min-height:88px;">{{ ind.description or ''                  9173
      }}</textarea>                                                      9174
      </div>                                                             9175
      <p class="actions" style="margin-top:16px;">                       9176
        <button type="submit" class="btn">保存</button>                  9177
        <a class="btn btn--ghost" href="/ahp/indicators">返回指标列       9178
      表</a>                                                             9179
      </p>                                                               9180
    </form>                                                              9181
</div>                                                                   9182
{% endblock %}                                                           9183
# ==== app/templates/ahp/experts.html ====                               9184
{% extends "base.html" %}                                                9185
{% block title %}多专家判断矩阵{% endblock %}                                9186
{% block content %}                                                      9187
<div class="card">                                                       9188
  <h2>多专家判断矩阵 · 录入与汇总</h2>                                        9189
  <form method="get" class="actions" style="margin-bottom:14             9190
    px;">                                                                9191
    <div>                                                                9192
      <label class="field__label" for="unit-type">微地质单元类型</l      9193
    abel>                                                                9194
      <select id="unit-type" name="unit_type"                            9195
    onchange="this.form.submit()">                                       9196
        {% for t in ["岩质边坡", "碎石边坡", "平台"] %}                        9197
        <option value="{{ t }}" {% if unit_type == t                     9198
    %}selected{% endif %}>{{ t }}</option>                               9199
        {% endfor %}                                                     9200
```

```
        </select>                                                    9201
    </div>                                                           9202
    <div>                                                            9203
        <label class="field__label" for="level">指标层级</label>    9204
        <select id="level" name="level"                              9205
    onchange="this.form.submit()">                                   9206
            <option value="1" {% if level == 1 %}selected{%          9207
    endif %}>一级指标</option>                                       9208
            <option value="2" {% if level == 2 %}selected{%          9209
    endif %}>二级指标</option>                                       9210
        </select>                                                    9211
    </div>                                                           9212
</form>                                                              9213
<p class="muted">                                                    9214
    当前范围：<strong>{{ unit_type }}</strong> ·                     9215
    {% if level == 1 %}一级指标{% else %}二级指标（{{ l1 }}）{% endif 9216
    %}。                                                             9217
    若选二级指标，请先到                                             9218
    <a href="/ahp/">权重计算</a> 选择一级指标后进入本页。            9219
</p>                                                                 9220
{% if rows %}                                                        9221
<table>                                                              9222
    <caption class="muted">已录入 {{ rows|length }}                  9223
    位专家的矩阵</caption>                                           9224
    <thead>                                                          9225
        <tr>                                                         9226
            <th scope="col">专家</th>                                9227
            <th scope="col">矩阵规模</th>                            9228
            <th scope="col">该矩阵 CR</th>                           9229
            <th scope="col">一致性</th>                              9230
            <th scope="col">录入时间</th>                            9231
            <th scope="col">操作</th>                                9232
        </tr>                                                        9233
    </thead>                                                         9234
    <tbody>                                                          9235
        {% for r in rows %}                                          9236
        <tr>                                                         9237
            <td>{{ r.expert_name }}</td>                             9238
            <td class="num">{{ n }} × {{ n }}</td>                   9239
            <td class="mono">{{ "%.4f"|format(r.cr) if r.cr is       9240
    not none else "—" }}</td>                                        9241
            <td>                                                     9242
                {% if r.cr is not none and r.cr < 0.1 %}<span        9243
    class="tag tag--ok">通过</span>                                  9244
                {% else %}<span class="tag tag--warn">不通过</span>{%9245
    endif %}                                                         9246
            </td>                                                    9247
            <td class="mono nowrap">{{ r.created_at.strftime("%Y     9248
    -%m-%d %H:%M") if r.created_at else "—" }}</td>                  9249
            <td>                                                     9250
```

```
          <form method="post" action="/ahp/experts/{{ r.id 9251
    }}/delete" 9252
                onsubmit="return confirm('确定删除专家「{{ 9253
    r.expert_name }}」的矩阵吗？');"> 9254
            {{ form.hidden_tag() }} 9255
            <button type="submit" class="btn btn--sm 9256
    btn--danger">删除</button> 9257
          </form> 9258
        </td> 9259
      </tr> 9260
      {% endfor %} 9261
    </tbody> 9262
  </table> 9263
  <p class="actions" style="margin-bottom:10px;"> 9264
    <a class="btn" href="/ahp/experts/aggregate?unit_type={{ 9265
    unit_type }}&level={{ level }}{% if l1 %}&l1={{ l1 }}{% 9266
    endif %}"> 9267
      汇总计算（几何平均） 9268
    </a> 9269
  </p> 9270
  {% else %} 9271
  <p class="muted">尚未录入任何专家矩阵。请在下方录入第一位专家的判断矩阵。</p> 9272
  {% endif %} 9273
  <h3>录入一位专家的判断矩阵</h3> 9274
  <form method="post"> 9275
    {{ form.hidden_tag() }} 9276
    <input type="hidden" name="action" value="add"> 9277
    <input type="hidden" name="unit_type" value="{{ 9278
    unit_type }}"> 9279
    <input type="hidden" name="level" value="{{ level }}"> 9280
    {% if l1 %}<input type="hidden" name="l1" value="{{ l1 9281
    }}">{% endif %} 9282
    <div class="field"> 9283
      <label class="field__label" for="expert-name">专家姓名</la 9284
    bel> 9285
      <input id="expert-name" type="text" name="expert_name" 9286
             style="width:220px;" placeholder="如：张工"> 9287
    </div> 9288
    <table style="width:auto;"> 9289
      <thead> 9290
        <tr> 9291
          <th></th> 9292
          {% for name in names %}<th scope="col">{{ name 9293
    }}</th>{% endfor %} 9294
        </tr> 9295
      </thead> 9296
      <tbody> 9297
        {% for i in range(n) %} 9298
        <tr> 9299
          <th scope="row">{{ names[i] }}</th> 9300
```

```
            {% for j in range(n) %}                                          9301
              {% if i == j %}                                               9302
              <td class="muted num">1</td>                                  9303
              {% elif i < j %}                                              9304
              <td>                                                          9305
                <label class="muted" style="position:absolute;              9306
    left:-9999px;"                                                          9307
                       for="e-{{ i }}-{{ j }}">{{ names[i] }}               9308
    相对 {{ names[j] }} 的重要性</label>                                    9309
                <input id="e-{{ i }}-{{ j }}" type="number"                 9310
    step="0.01" min="0.01"                                                  9311
                       name="m_{{ i }}_{{ j }}" value="{{                   9312
    default[i][j] }}">                                                      9313
              </td>                                                         9314
              {% else %}                                                    9315
              <td class="muted num">{{ default[i][j] }}</td>                9316
              {% endif %}                                                   9317
            {% endfor %}                                                    9318
          </tr>                                                             9319
          {% endfor %}                                                      9320
        </tbody>                                                            9321
    </table>                                                                9322
    <p class="actions" style="margin-top:12px;">                            9323
      <button type="submit" class="btn">录入该专家矩阵</button>             9324
      <a class="btn btn--ghost" href="/ahp/indicators">指标维护<            9325
    /a>                                                                     9326
      <a class="btn btn--ghost" href="/ahp/">返回权重计算</a>               9327
    </p>                                                                    9328
  </form>                                                                   9329
</div>                                                                      9330
{% endblock %}                                                              9331
# ==== app/templates/ahp/experts_aggregate.html ====                        9332
{% extends "base.html" %}                                                   9333
{% block title %}多专家矩阵汇总{% endblock %}                               9334
{% block content %}                                                         9335
<div class="card">                                                          9336
  <h2>多专家矩阵汇总 · {{ unit_type }} · {% if level == 1 %}一级指标{%     9337
    else %}二级指标（{{ l1 }}）{% endif %}</h2>                             9338
  {% if not result %}                                                       9339
  <p class="muted">                                                         9340
    该范围内尚未录入专家矩阵。                                              9341
    <a href="/ahp/experts?unit_type={{ unit_type                            9342
    }}&level={{ level }}{% if l1 %}&l1={{ l1 }}{% endif                     9343
    %}">去录入</a>                                                          9344
  </p>                                                                      9345
  {% else %}                                                                9346
  <div class="stat-row">                                                    9347
    <div>                                                                   9348
      <div class="stat__label">参与汇总的专家</div>                         9349
      <div class="stat__value">{{ result.count }} 位</div>                  9350
```

```
    </div>                                                                  9351
    <div>                                                                   9352
      <div class="stat__label">λmax</div>                                   9353
      <div class="stat__value mono">{{ result.lambda_max                    9354
    }}</div>                                                                9355
    </div>                                                                  9356
    <div>                                                                   9357
      <div class="stat__label">CI</div>                                     9358
      <div class="stat__value mono">{{                                      9359
    "%.4f"|format(result.ci) }}</div>                                       9360
    </div>                                                                  9361
    <div>                                                                   9362
      <div class="stat__label">CR（要求 &lt; 0.1）</div>                    9363
      <div class="stat__value mono">{{                                      9364
    "%.4f"|format(result.cr) }}</div>                                       9365
    </div>                                                                  9366
    <div>                                                                   9367
      <div class="stat__label">一致性检验</div>                             9368
      <div class="stat__value">                                             9369
        {% if result.consistent %}<span class="ok">通过</span>              9370
        {% else %}<span class="warn">不通过</span>{% endif %}               9371
      </div>                                                                9372
    </div>                                                                  9373
  </div>                                                                    9374
  <h3>汇总后权重</h3>                                                       9375
  <table style="width:auto;">                                               9376
    <thead>                                                                 9377
      <tr>                                                                  9378
        <th scope="col">指标</th>                                           9379
        {% for name in names %}<th scope="col">{{ name                      9380
    }}</th>{% endfor %}                                                     9381
      </tr>                                                                 9382
    </thead>                                                                9383
    <tbody>                                                                 9384
      <tr>                                                                  9385
        <th scope="row">权重</th>                                           9386
        {% for w in result.weights %}<td class="mono">{{ w                  9387
    }}</td>{% endfor %}                                                     9388
      </tr>                                                                 9389
    </tbody>                                                                9390
  </table>                                                                  9391
  <h3>各专家矩阵的一致性</h3>                                               9392
  <table style="width:auto;">                                               9393
    <thead>                                                                 9394
      <tr><th scope="col">专家</th><th scope="col">该矩阵                   9395
    CR</th><th scope="col">判定</th></tr>                                   9396
    </thead>                                                                9397
    <tbody>                                                                 9398
      {% for r in rows %}                                                   9399
      <tr>                                                                  9400
```

```
            <td>{{ r.expert_name }}</td>                                    9401
            <td class="mono">{{ "%.4f"|format(r.cr) if r.cr is              9402
        not none else "—" }}</td>                                           9403
            <td>                                                            9404
              {% if r.cr is not none and r.cr < 0.1 %}<span                 9405
        class="ok">通过</span>                                              9406
              {% else %}<span class="warn">不通过</span>{% endif %}         9407
            </td>                                                           9408
          </tr>                                                             9409
          {% endfor %}                                                      9410
        </tbody>                                                            9411
      </table>                                                              9412
      {% if not result.consistent and result.most_inconsistent              9413
        %}                                                                  9414
      <p style="padding:12px;border-radius:6px;background:var(--            9415
        danger-soft);                                                       9416
                 border-left:4px solid var(--danger);">                     9417
        <strong class="warn">汇总矩阵一致性不通过，建议优先调整：</strong>          9418
        第 {{ result.most_inconsistent.i + 1 }} 行、第 {{                   9419
        result.most_inconsistent.j + 1 }} 列的比较                          9420
        （当前 <span class="mono">{{ result.most_inconsistent.value         9421
        |round(3) }}</span>，                                               9422
        理想约 <span class="mono">{{ result.most_inconsistent.expec         9423
        ted|round(3) }}</span>）。                                          9424
        该偏差通常来自个别专家打分，可先核对一致性较差的专家矩阵。                    9425
      </p>                                                                  9426
      {% endif %}                                                           9427
    {% endif %}                                                             9428
    <p class="actions" style="margin-top:16px;">                            9429
      <form method="post"                                                   9430
            action="/ahp/experts?unit_type={{ unit_type                     9431
        }}&level={{ level }}{% if l1 %}&l1={{ l1 }}{% endif %}"             9432
            style="display:inline;"                                         9433
            onsubmit="return confirm('保存后，「{{ unit_type                 9434
        }}」的适宜性评价将改用这组汇总权重计算。确定继续吗？');">                    9435
        {{ form.hidden_tag() }}                                             9436
        <input type="hidden" name="action" value="save_agg">                9437
        <button type="submit" class="btn" {% if not                         9438
        result.consistent %}disabled{% endif %}>                            9439
          保存为生效权重                                                        9440
        </button>                                                           9441
      </form>                                                               9442
      <a class="btn btn--ghost" href="/ahp/experts?unit_type={              9443
        { unit_type }}&level={{ level }}{% if l1 %}&l1={{ l1                9444
        }}{% endif %}">返回录入</a>                                         9445
      <a class="btn btn--ghost" href="/ahp/">返回权重计算</a>                   9446
    </p>                                                                    9447
</div>                                                                      9448
{% endblock %}                                                              9449
# ==== app/templates/indicator/index.html ====                              9450
```

```
{% extends "base.html" %}
{% from "_macros.html" import patch_guide %}
{% block title %}指标数据采集{% endblock %}
{% block content %}
<div class="card">
  <h2>指标数据采集与管理</h2>
  <p class="actions" style="margin-bottom:12px;">
    <a class="btn btn--sm" href="/indicator/import">批量导入（CSV / Excel）</a>
  </p>
  {{ patch_guide(projects_without_patch) }}
  {% if not patches %}
  <p class="muted">
    暂无矿山图斑。请先在「项目与矿山档案」中建立图斑，
    或运行 <span class="mono">python seed_demo.py</span>
    载入论文案例。
  </p>
  {% else %}
  {% for p in patches %}
  <div class="patch-panel">
    <div class="patch-panel__head">
      <h3 class="patch-panel__name">{{ p.name }}</h3>
      <span class="patch-panel__meta">
        所属项目：{{ p.project.name if p.project else "—" }}
      </span>
      <span class="patch-panel__meta">
        微地质单元：{{ p.units.count() }} 个
      </span>
      {% if p.units.count() %}
      <span class="patch-panel__meta patch-panel__tail">
        <a class="link-btn" href="/indicator/import">批量导入本图斑数据</a>
      </span>
      {% endif %}
    </div>
    {% if not p.units.count() %}
    <div class="notice notice--warn">
      <p class="notice__title">该图斑尚未划分微地质单元，暂不能录入指标数据。</p>
      <p style="margin:0;">
        指标数据是按微地质单元（岩质边坡 / 碎石边坡 / 平台）分别组织的，
        请先完成划分，再回到本页录入。
      </p>
      <p style="margin:10px 0 0;">
        <a class="btn btn--sm" href="/unit/{{ p.id }}">去划分微地质单元</a>
      </p>
    </div>
    {% else %}
    <table>
      <thead>
```

```
          <tr>                                                                    9501
            <th scope="col">微地质单元</th>                                       9502
            <th scope="col">面积占比</th>                                         9503
            <th scope="col">数据状态</th>                                         9504
            <th scope="col">操作</th>                                             9505
          </tr>                                                                   9506
        </thead>                                                                  9507
        <tbody>                                                                   9508
          {% for u in p.units %}                                                  9509
          <tr>                                                                    9510
            <td>{{ u.unit_type }}</td>                                            9511
            <td class="num">                                                      9512
              {% if u.ratio is not none %}{{                                      9513
    "%.2f%%"|format(u.ratio * 100) }}{% else %}—{% endif %}                       9514
            </td>                                                                 9515
            <td>                                                                  9516
              {% if u.indicator_data %}                                           9517
              <span class="tag tag--ok">已录入</span>                             9518
              {% else %}                                                          9519
              <span class="tag tag--muted">未录入</span>                          9520
              {% endif %}                                                         9521
            </td>                                                                 9522
            <td class="nowrap">                                                   9523
              <span class="row-actions">                                          9524
                <a class="link-btn" href="/indicator/{{ u.id                      9525
    }}">录入 / 编辑</a>                                                           9526
              </span>                                                             9527
            </td>                                                                 9528
          </tr>                                                                   9529
          {% endfor %}                                                            9530
        </tbody>                                                                  9531
      </table>                                                                    9532
      {% endif %}                                                                 9533
    </div>                                                                        9534
    {% endfor %}                                                                  9535
    {% endif %}                                                                   9536
</div>                                                                            9537
{% endblock %}                                                                    9538
# ==== app/templates/indicator/edit.html ====                                     9539
{% extends "base.html" %}                                                         9540
{% block title %}指标数据录入 · {{ unit.unit_type }}{% endblock %}                9541
{% block content %}                                                               9542
<div class="card" style="max-width:820px;">                                       9543
  <h2>指标数据录入 · {{ unit.unit_type }}</h2>                                    9544
  <p class="muted">                                                               9545
    输入框留空表示该指标「不适用」，与「实测值为 0」含义不同，计算时会被跳过。  9546
    岩质边坡不设土壤指标、平台不设地形指标，系统已按单元类型隐藏不相关字段。    9547
  </p>                                                                            9548
  <form method="post">                                                            9549
    {{ form.hidden_tag() }}                                                       9550
```

```
{% for group_name, fields, has_hm in groups %}
<h3>{{ group_name }}</h3>
{% for fname, cname, unit_ in fields %}
<div class="field">
  <label class="field__label" for="f-{{ fname }}">
    {{ cname }}{% if unit_ %} ({{ unit_ }}) {% endif %}
  </label>
  {% if fname == "aspect" %}
  <select id="f-{{ fname }}" name="aspect">
    <option value="">— 不适用 —</option>
    {% for a in aspect_types %}
    <option value="{{ a }}" {% if current.get("aspect")
== a %}selected{% endif %}>{{ a }}</option>
    {% endfor %}
  </select>
  {% else %}
  <input id="f-{{ fname }}" type="number" step="any"
name="{{ fname }}"
         value="{{ current.get(fname) if
current.get(fname) is not none else '' }}">
  {% endif %}
</div>
{% endfor %}
{% endfor %}
{% for group_name, fields, has_hm in groups %}
  {% if has_hm %}
  <h3>重金属含量（mg/kg）</h3>
  <p class="muted">未检测的金属可留空，系统按论文注释视为未受污染处理。</p>
  <table style="width:auto;">
    <thead>
      <tr>
        {% for m in heavy_metals %}<th scope="col">{{ m
}}</th>{% endfor %}
      </tr>
    </thead>
    <tbody>
      <tr>
        {% for m in heavy_metals %}
        <td>
          <label class="muted" style="position:absolute;
left:-9999px;"
                 for="hm-{{ m }}">{{ m }} 含量</label>
          <input id="hm-{{ m }}" type="number"
step="any" name="hm_{{ m }}"
                 value="{{ current.get('heavy',
{}).get(m) if current.get('heavy', {}).get(m) is not
none else '' }}">
        </td>
        {% endfor %}
      </tr>
```

```
          </tbody>                                                    9601
        </table>                                                      9602
        {% endif %}                                                   9603
      {% endfor %}                                                    9604
      <p class="actions" style="margin-top:18px;">                    9605
        <button type="submit" class="btn">保存</button>               9606
        <a class="btn btn--ghost" href="/indicator/">返回</a>         9607
      </p>                                                            9608
    </form>                                                           9609
</div>                                                                9610
{% endblock %}                                                        9611
# ==== app/templates/indicator/import.html ====                       9612
{% extends "base.html" %}                                             9613
{% block title %}批量导入指标数据{% endblock %}                        9614
{% block content %}                                                   9615
<div class="card" style="max-width:860px;">                           9616
  <h2>批量导入指标数据</h2>                                           9617
  <p class="muted">                                                   9618
    支持 CSV 与 Excel（.xlsx/.xls）。文件每行对应一个微地质单元，     9619
    系统按「单元类型」匹配图斑内已划分的单元后写入数据。             9620
    先下载模板按格式填写，可减少出错。                               9621
  </p>                                                                9622
  <p class="actions" style="margin-bottom:14px;">                     9623
    <a class="btn btn--sm btn--ghost"                                 9624
    href="/indicator/import/template">下载导入模板（CSV）</a>         9625
    <a class="btn btn--sm btn--ghost"                                 9626
    href="/indicator/">返回指标数据</a>                               9627
  </p>                                                                9628
  <form method="post" enctype="multipart/form-data">                  9629
    {{ form.hidden_tag() }}                                           9630
    <div class="field">                                               9631
      <label class="field__label" for="patch-id">目标矿山图斑</lab9632
    el>                                                               9633
      <select id="patch-id" name="patch_id" required>                 9634
        <option value="">— 请选择 —</option>                          9635
        {% for p in patches %}                                        9636
        <option value="{{ p.id }}">{{ p.name }}（{{                  9637
    p.units.count() }} 个单元）</option>                              9638
        {% endfor %}                                                  9639
      </select>                                                       9640
    </div>                                                            9641
    <div class="field">                                               9642
      <label class="field__label" for="datafile">数据文件</label9643
    >                                                                 9644
      <input id="datafile" type="file" name="datafile"                9645
             accept=".csv,.xlsx,.xls" required>                       9646
    </div>                                                            9647
    <p class="actions" style="margin-top:16px;">                      9648
      <button type="submit" class="btn">开始导入</button>             9649
    </p>                                                              9650
```

```
  </form>
  <h3>模板列说明</h3>
  <p class="muted">
    共 {{ headers|length }} 列。留空表示该指标「不适用」。
    坡向须填「阳坡 / 阴阳坡 / 阴坡」之一。未检测的重金属留空即可
    （按论文注释视为未受污染）。
  </p>
  <p class="mono muted" style="line-height:1.9;">{{
    headers|join(" , ") }}</p>
  <h3>导入规则</h3>
  <ul class="muted" style="padding-left:20px;line-height:1.9
    ;">
    <li>「单元类型」必须与图斑内已划分的单元一致，否则该行会被跳过并在结果中提示。</li>
    <li>同一单元类型若在文件中出现多行，以最后一行为准。</li>
    <li>已有数据的单元会被覆盖更新；没有数据的单元将新建记录。</li>
    <li>数值列请填纯数字，不要带单位（如填 <span class="mono">10</span> 而非
    <span class="mono">10cm</span>）。</li>
  </ul>
</div>
{% if report %}
<div class="card" style="max-width:860px;">
  <h2>导入结果</h2>
  <div class="stat-row">
    <div>
      <div class="stat__label">目标图斑</div>
      <div style="font-size:15px;">{{ report.patch }}</div>
    </div>
    <div>
      <div class="stat__label">成功导入</div>
      <div class="stat__value ok">{{ report.saved }}
    个单元</div>
    </div>
    <div>
      <div class="stat__label">问题行</div>
      <div class="stat__value {% if report.errors %}warn{%
    endif %}">{{ report.errors|length }}</div>
    </div>
  </div>
  {% if report.errors %}
  <h3>未导入的行</h3>
  <ul class="muted" style="padding-left:20px;line-height:1.9
    ;">
    {% for e in report.errors %}<li>{{ e }}</li>{% endfor %}
  </ul>
  {% else %}
  <p class="ok" style="margin-top:12px;">全部数据行均已成功导入。</p>
  {% endif %}
  <p class="actions" style="margin-top:16px;">
    <a class="btn btn--ghost" href="/indicator/">查看指标数据</a>
  </p>
```

```
</div>
{% endif %}
{% endblock %}
# ==== app/templates/evaluate/index.html ====
{% extends "base.html" %}
{% from "_macros.html" import patch_guide %}
{% block title %}适宜性定量评价{% endblock %}
{% block content %}
<div class="card">
  <h2>适宜性定量评价</h2>
  {{ patch_guide(projects_without_patch) }}
  {% if not patches %}
  <p class="muted">暂无矿山图斑。请先到「项目与矿山档案」建立图斑。</p>
  {% else %}
  <table>
    <thead>
      <tr>
        <th scope="col">所属项目</th>
        <th scope="col">矿山图斑</th>
        <th scope="col">前置条件</th>
        <th scope="col">最近评价结果</th>
        <th scope="col">操作</th>
      </tr>
    </thead>
    <tbody>
      {% for p in patches %}
      {% set pr = progress.get(p.id) %}
      <tr>
        <td>{{ p.project.name if p.project else "—" }}</td>
        <td style="text-align:left;">{{ p.name }}</td>
        <td style="text-align:left;">
          {% if pr and pr.ready %}
          <span class="tag tag--ok">已具备</span>
          {% elif pr %}
          <span class="tag tag--warn">缺：{{
    pr.blocking[0].label }}</span>
          <span class="muted">{{ pr.blocking[0].reason
    }}</span>
          {% endif %}
        </td>
        <td>
          {% if latest.get(p.id) %}
          F = <span class="mono num">{{
    latest[p.id].final_score }}</span>,
          <span class="tag tag--ok">{{ latest[p.id].grade
    }}</span>
          {% else %}
          <span class="tag tag--muted">未评价</span>
          {% endif %}
        </td>
```

```
            <td class="nowrap">                                                    9751
              <span class="row-actions">                                           9752
                <a class="link-btn" href="/evaluate/{{ p.id                        9753
      }}">执行评价</a>                                                             9754
                {% if latest.get(p.id) %}                                          9755
                <a class="link-btn" href="/evaluate/result/{{                      9756
      latest[p.id].id }}">查看结果</a>                                             9757
                {% endif %}                                                        9758
              </span>                                                              9759
            </td>                                                                  9760
          </tr>                                                                    9761
          {% endfor %}                                                             9762
        </tbody>                                                                   9763
      </table>                                                                     9764
      {% endif %}                                                                  9765
</div>                                                                             9766
{% endblock %}                                                                     9767
# ==== app/templates/evaluate/run.html ====                                        9768
{% extends "base.html" %}                                                          9769
{% import "_macros.html" as ui %}                                                  9770
{% block title %}执行评价 · {{ patch.name }}{% endblock %}                         9771
{% block content %}                                                                9772
<div class="card" style="max-width:880px;">                                        9773
  <h2>执行适宜性评价 · {{ patch.name }}</h2>                                       9774
  {{ ui.workflow_steps(progress) }}                                                9775
  {{ ui.workflow_blocking(progress) }}                                             9776
  {% if progress.ready %}                                                          9777
  <form method="post">                                                             9778
    {{ form.hidden_tag() }}                                                        9779
    <p class="actions">                                                            9780
      <button type="submit" class="btn">开始评价</button>                          9781
      <a class="btn btn--ghost" href="/evaluate/">返回</a>                         9782
    </p>                                                                           9783
  </form>                                                                          9784
  {% else %}                                                                       9785
  <p class="actions">                                                              9786
    <button type="button" class="btn" disabled>开始评价</button>                   9787
    <a class="btn btn--ghost" href="/evaluate/">返回</a>                           9788
  </p>                                                                             9789
  {% endif %}                                                                      9790
</div>                                                                             9791
<div class="card" style="max-width:880px;">                                        9792
  <h2>当前流程状态</h2>                                                           9793
  <table>                                                                          9794
    <thead>                                                                        9795
      <tr><th scope="col">步骤</th><th scope="col">状态</th><th                    9796
    scope="col">说明</th></tr>                                                     9797
    </thead>                                                                       9798
    <tbody>                                                                        9799
      <tr>                                                                         9800
```

```
            <td>限制性条件判定</td>                                                    9801
            <td>                                                                    9802
              {% if progress.constraint.passed %}<span                              9803
        class="tag tag--ok">已通过</span>                                           9804
              {% elif progress.constraint.done %}<span                              9805
        class="tag tag--warn">未通过</span>                                         9806
              {% else %}<span class="tag tag--muted">待完成</span>{                 9807
        % endif %}                                                                  9808
            </td>                                                                   9809
            <td style="text-align:left;">{{                                        9810
        progress.constraint.detail }}</td>                                          9811
          </tr>                                                                     9812
          <tr>                                                                      9813
            <td>划分微地质单元</td>                                                 9814
            <td>                                                                    9815
              {% if progress.units.done %}<span class="tag                          9816
        tag--ok">已完成</span>                                                      9817
              {% else %}<span class="tag tag--muted">待完成</span>{                 9818
        % endif %}                                                                  9819
            </td>                                                                   9820
            <td style="text-align:left;">{{                                        9821
        progress.units.detail }}</td>                                               9822
          </tr>                                                                     9823
          <tr>                                                                      9824
            <td>录入指标数据</td>                                                   9825
            <td>                                                                    9826
              {% if progress.data.ready %}<span class="tag                          9827
        tag--ok">已完成</span>                                                      9828
              {% else %}<span class="tag tag--muted">待完成</span>{                 9829
        % endif %}                                                                  9830
            </td>                                                                   9831
            <td style="text-align:left;">{{                                        9832
        progress.data.detail }}</td>                                                9833
          </tr>                                                                     9834
        </tbody>                                                                    9835
      </table>                                                                      9836
</div>                                                                              9837
{% if weight_sources %}                                                             9838
<div class="card" style="max-width:880px;">                                         9839
      <h2>本次将采用的指标权重</h2>  <table style="max-width:720px;">               9840
        <thead>                                                                     9841
          <tr><th scope="col">微地质单元</th><th                                    9842
        scope="col">权重来源</th></tr>                                              9843
        </thead>                                                                    9844
        <tbody>                                                                     9845
          {% for ut, label in weight_sources.items() %}                             9846
          <tr>                                                                      9847
            <td style="text-align:left;">{{ ut }}</td>                             9848
            <td style="text-align:left;">                                          9849
              {% if "自定义" in label %}                                           9850
```

```
            <span class="tag tag--warn">{{ label }}</span>
            {% else %}
            <span class="tag tag--muted">{{ label }}</span>
            {% endif %}
          </td>
        </tr>
        {% endfor %}
      </tbody>
    </table>
</div>
{% endif %}
{% endblock %}
# ==== app/templates/evaluate/result.html ====
{% extends "base.html" %}
{% block title %}评价结果{% endblock %}
{% block content %}
<div class="card">
  <h2>适宜性评价结果 · {{ patch.name if patch else "" }}</h2>
  <div class="stat-row">
    <div>
      <div class="stat__label">最终评价因子 F</div>
      <div class="stat__value mono">{{ r.final_score
    }}</div>
    </div>
    <div>
      <div class="stat__label">适宜性等级</div>
      <div class="stat__value ok">{{ r.grade }}</div>
    </div>
    <div style="flex:1;min-width:240px;">
      <div class="stat__label">推荐措施</div>
      <div style="font-size:15px;">{{ r.measures }}</div>
    </div>
  </div>
  {% if unit_scores %}
  <h3>各微地质单元得分</h3>
  <table style="max-width:860px;">
    <thead>
      <tr>
        <th scope="col">微地质单元</th>
        <th scope="col">面积占比</th>
        <th scope="col">单元评价因子 F'</th>
        <th scope="col">加权贡献</th>
        <th scope="col">权重来源</th>
      </tr>
    </thead>
    <tbody>
      {% for u in unit_scores %}
      <tr>
        <td>{{ u.unit_type }}</td>
        <td class="num">{{ "%.2f%%"|format(u.ratio * 100)
```

```
      }}</td>                                                                  9901
          <td class="mono">{{ u.unit_score }}</td>                             9902
          <td class="mono">{{ "%.4f"|format(u.ratio *                          9903
      u.unit_score) }}</td>                                                    9904
          <td class="muted">{{ u.weight_source or "论文表 3                    9905
      默认权重" }}</td>                                                        9906
        </tr>                                                                  9907
        {% endfor %}                                                           9908
      </tbody>                                                                 9909
    </table>                                                                   9910
    {% endif %}                                                                9911
    <p class="actions" style="margin-top:18px;">                               9912
      <a class="btn btn--ghost" href="/report/{{ r.patch_id                    9913
      }}">查看得分明细</a>                                                     9914
      <a class="btn btn--ghost" href="/evaluate/">返回评价列表</a>             9915
    </p>                                                                       9916
</div>                                                                         9917
{% endblock %}                                                                 9918
# ==== app/templates/report/index.html ====                                    9919
{% extends "base.html" %}                                                      9920
{% from "_macros.html" import patch_guide %}                                   9921
{% block title %}报告输出{% endblock %}                                        9922
{% block content %}                                                            9923
<div class="card">                                                             9924
    <h2>报告与结果输出</h2>                                                    9925
    {{ patch_guide(projects_without_patch) }}                                  9926
    {% if not patches %}                                                       9927
    <p class="muted">暂无矿山图斑。请先到「项目与矿山档案」建立图斑。</p>     9928
    {% else %}                                                                 9929
    <table>                                                                    9930
      <thead>                                                                  9931
        <tr>                                                                   9932
          <th scope="col">所属项目</th>                                        9933
          <th scope="col">矿山图斑</th>                                        9934
          <th scope="col">可生成报告</th>                                      9935
          <th scope="col">操作</th>                                            9936
        </tr>                                                                  9937
      </thead>                                                                 9938
      <tbody>                                                                  9939
        {% for p in patches %}                                                 9940
        {% set pr = progress.get(p.id) %}                                      9941
        <tr>                                                                   9942
          <td>{{ p.project.name if p.project else "—" }}</td>                  9943
          <td style="text-align:left;">{{ p.name }}</td>                       9944
          <td style="text-align:left;">                                        9945
            {% if pr and pr.ready %}                                           9946
            <span class="tag tag--ok">可以</span>                              9947
            {% elif pr %}                                                      9948
            <span class="tag tag--warn">暂不可</span>                          9949
            <span class="muted">{{ pr.blocking[0].reason                       9950
```

```
      }}</span>                                                          9951
            {% endif %}                                                  9952
          </td>                                                          9953
          <td class="nowrap">                                            9954
            {% if pr and pr.ready %}                                     9955
            <span class="row-actions">                                   9956
              <a class="link-btn" href="/report/{{ p.id                  9957
      }}">查看得分明细</a>                                                9958
              <a class="link-btn" href="/report/{{ p.id                  9959
      }}/pdf">导出 PDF</a>                                                9960
              <a class="link-btn" href="/report/{{ p.id                  9961
      }}/word">导出 Word</a>                                              9962
            </span>                                                      9963
            {% else %}                                                   9964
            <span class="muted">—</span>                                 9965
            {% endif %}                                                  9966
          </td>                                                          9967
        </tr>                                                            9968
        {% endfor %}                                                     9969
      </tbody>                                                           9970
    </table>                                                             9971
    {% endif %}                                                          9972
</div>                                                                   9973
{% endblock %}                                                           9974
# ==== app/templates/report/detail.html ====                             9975
{% extends "base.html" %}                                                9976
{% block title %}得分明细 · {{ patch.name }}{% endblock %}               9977
{% block content %}                                                      9978
<div class="card">                                                       9979
  <h2>得分明细 · {{ patch.name }}</h2>                                   9980
  <div class="stat-row">                                                 9981
    <div>                                                                9982
      <div class="stat__label">最终评价因子 F</div>                      9983
      <div class="stat__value mono">{{                                   9984
    "%.4f"|format(result.final_score) }}</div>                           9985
    </div>                                                               9986
    <div>                                                                9987
      <div class="stat__label">适宜性等级</div>                          9988
      <div class="stat__value ok">{{ result.grade }}</div>               9989
    </div>                                                               9990
    <div style="flex:1;min-width:240px;">                                9991
      <div class="stat__label">推荐措施</div>                            9992
      <div style="font-size:15px;">{{                                    9993
    result.grade_info.measure }}</div>                                   9994
    </div>                                                               9995
  </div>                                                                 9996
  <p class="muted" style="margin-top:12px;">                             9997
    本页与导出报告均取自                                                 9998
    <strong>{% if record %}{{ record.created_at.strftime("%Y             9999
    -%m-%d %H:%M:%S") }}{% endif %}                                      10000
```

```
      记录的那次评价结果</strong>（评价记录编号 {{ record.id if record else
      "—" }}），
      不会因事后修改指标数据而变化。
    </p>
    <p class="actions" style="margin-top:16px;">
      <a class="btn" href="/report/{{ patch.id }}/pdf">导出 PDF
      报告</a>
      <a class="btn btn--ghost" href="/report/{{ patch.id
      }}/word">导出 Word 报告</a>
      <a class="btn btn--ghost" href="/report/">返回</a>
    </p>
  </div>
  <div class="card">
    <h2>结果图表</h2>
    <h3>各微地质单元得分对比</h3>
    <div>{{ chart_units|safe }}</div>
    <h3>各一级指标平均得分</h3>
    <p class="muted">反映各一级指标对评价结果的实际影响，便于识别关键控制项。</p>
    <div>{{ chart_weight|safe }}</div>
  </div>
  {% for u in result.unit_results %}
  <div class="card">
    <h2>{{ u.unit_type }} · 占比 {{ "%.2f%%"|format(u.ratio *
      100) }} · 单元得分 {{ "%.4f"|format(u.unit_score) }}</h2>
    {% if u.weight_source %}
    <p class="muted" style="margin-top:-6px;">权重来源：{{
      u.weight_source }}</p>
    {% endif %}
    <table>
      <thead>
        <tr>
          <th scope="col">一级指标</th>
          <th scope="col">二级指标</th>
          <th scope="col">实测值</th>
          <th scope="col">标准化值</th>
          <th scope="col">权重</th>
          <th scope="col">贡献</th>
        </tr>
      </thead>
      <tbody>
        {% for d in u.details %}
        <tr>
          <td>{{ d.l1 }}</td>
          <td style="text-align:left;">{{ d.l2 }}</td>
          <td class="mono" style="text-align:left;">{{ "—" if
      d.raw is none else d.raw }}</td>
          <td class="mono">{{ "—" if d.std_value is none else
      "%.4f"|format(d.std_value) }}</td>
          <td class="mono">{{ "%.4f"|format(d.weight) }}</td>
          <td class="mono">{{ "—" if d.contribution is none
```

```
      else "%.4f"|format(d.contribution) }}</td>
        </tr>
        {% endfor %}
      </tbody>
    </table>
    <p class="muted" style="margin:0;">
      一级指标得分：
      {% for k, v in u.l1_scores.items() %}{{ k }} = <span
      class="num">{{ "%.4f"|format(v) }}</span>{% if not
      loop.last %};{% endif %}{% endfor %}
    </p>
  </div>
  {% endfor %}
  {% endblock %}
  # ==== app/templates/report/print.html ====
  <!DOCTYPE html>
  <html lang="zh-CN">
  <head>
  <meta charset="UTF-8">
  <title>自然恢复适宜性评价报告 · {{ patch.name }}</title>
  <style>
    @page { size: A4; margin: 18mm 16mm; }
    * { box-sizing: border-box; }
    body {
      margin: 0; color: #23262b; background: #fff;
      font-family: "Microsoft YaHei", "SimSun", sans-serif;
      font-size: 10.5pt; line-height: 1.65;
    }
    h1 { font-size: 17pt; text-align: center; margin: 0 0
      6px; }
    h2 { font-size: 13pt; color: #1f3a5f; margin: 20px 0 8px;
         border-bottom: 1px solid #dfe3e8; padding-bottom:
      4px; }
    h3 { font-size: 11.5pt; color: #1f3a5f; margin: 16px 0
      6px; }
    .sub { text-align: center; color: #6b7280; font-size:
      9.5pt; margin-bottom: 18px; }
    table { width: 100%; border-collapse: collapse; margin:
      8px 0 14px;
            font-variant-numeric: tabular-nums; }
    th, td { border: 1px solid #c9ccd1; padding: 4px 8px;
      text-align: center; }
    th { background: #eef2f7; font-weight: 600; }
    .left { text-align: left; }
    .mono { font-family: Consolas, "Courier New", monospace; }
    .kv { width: auto; }
    .kv th { width: 30%; text-align: left; }
    .kv td { text-align: left; }
    ul { margin: 6px 0 12px; padding-left: 20px; }
    .note { color: #6b7280; font-size: 9.5pt; }
```

```
  .chart { margin: 10px 0 16px; }                                  10101
  .sig { margin-top: 28px; font-size: 10pt; }                      10102
  .avoid-break { break-inside: avoid; }                            10103
  svg { max-width: 100%; }                                         10104
</style>                                                           10105
</head>                                                            10106
<body>                                                             10107
  <h1>历史遗留废弃矿山自然恢复适宜性评价报告</h1>                  10108
  <p class="sub">                                                  10109
    {{ patch.name }}                                               10110
    {% if patch.admin_region %} · {{ patch.admin_region            10111
    }}{% endif %}                                                  10112
    <br>历史遗留废弃矿山自然恢复适宜性评价系统 Ｖ1.0 ·             10113
    生成时间 {{ generated_at }}                                    10114
  </p>                                                             10115
  <h2>一、图斑基本信息</h2>                                        10116
  <table class="kv">                                               10117
    <tr><th>图斑名称</th><td>{{ patch.name }}</td></tr>            10118
    <tr><th>所在行政区</th><td>{{ patch.admin_region or "—"       10119
    }}</td></tr>                                                   10120
    <tr><th>图斑面积（公顷）</th><td>{{ patch.area_ha if           10121
    patch.area_ha is not none else "—" }}</td></tr>                10122
    <tr><th>开采矿种</th><td>{{ patch.mineral_type or "—"         10123
    }}</td></tr>                                                   10124
    <tr><th>开采方式</th><td>{{ patch.mining_method or "—"        10125
    }}</td></tr>                                                   10126
    <tr><th>责任主体缺失</th><td>{{ "是" if patch.responsible_entit10127
    y_missing else "否" }}</td></tr>                               10128
    {% if reference %}                                             10129
    <tr><th>参照生态系统</th>                                     10130
        <td>植被覆盖度基准 {{ reference.coverage_benchmark }}%,    10131
            多样性指数基准 {{ reference.shannon_benchmark          10132
    }}</td></tr>                                                   10133
    {% endif %}                                                    10134
  </table>                                                         10135
  <h2>二、评价结论</h2>                                            10136
  <table class="kv">                                               10137
    <tr><th>最终评价因子 F</th><td class="mono"><strong>{{        10138
    "%.4f"|format(result.final_score) }}</strong></td></tr>       10139
    <tr><th>适宜性等级</th><td><strong>{{ result.grade            10140
    }}</strong></td></tr>                                          10141
    <tr><th>判定依据</th><td>{{ result.grade_info.meaning        10142
    }}</td></tr>                                                   10143
    <tr><th>推荐措施</th><td>{{ result.grade_info.measure        10144
    }}</td></tr>                                                   10145
  </table>                                                         10146
  <h2>三、各微地质单元得分</h2>                                    10147
  <table>                                                          10148
    <thead>                                                        10149
      <tr><th>微地质单元</th><th>面积占比</th><th>单元评价因子     10150
```

```
    F'</th><th>加权贡献</th></tr>                                          10151
    </thead>                                                              10152
    <tbody>                                                               10153
      {% for u in result.unit_results %}                                  10154
      <tr>                                                                10155
        <td>{{ u.unit_type }}</td>                                        10156
        <td class="mono">{{ "%.2f%%"|format(u.ratio * 100)                10157
    }}</td>                                                               10158
        <td class="mono">{{ "%.4f"|format(u.unit_score)                   10159
    }}</td>                                                               10160
        <td class="mono">{{ "%.4f"|format(u.weighted)                     10161
    }}</td>                                                               10162
      </tr>                                                               10163
      {% endfor %}                                                        10164
    </tbody>                                                              10165
</table>                                                                  10166
<div class="chart">{{ chart_units|safe }}</div>                           10167
<h2>四、一级指标权重分布</h2>                                                10168
<div class="chart">{{ chart_weight|safe }}</div>                          10169
<h2>五、得分明细</h2>                                                       10170
{% for u in result.unit_results %}                                        10171
<h3>{{ u.unit_type }}（占比 {{ "%.2f%%"|format(u.ratio *                  10172
  100) }}，单元得分 {{ "%.4f"|format(u.unit_score) }}）</h3>               10173
<table>                                                                   10174
  <thead>                                                                 10175
    <tr>                                                                  10176
      <th>一级指标</th><th>二级指标</th><th>实测值</th>                          10177
      <th>标准化值</th><th>权重</th><th>贡献</th>                             10178
    </tr>                                                                 10179
  </thead>                                                                10180
  <tbody>                                                                 10181
    {% for d in u.details %}                                              10182
    <tr>                                                                  10183
      <td>{{ d.l1 }}</td>                                                 10184
      <td class="left">{{ d.l2 }}</td>                                    10185
      <td class="mono left">{{ "—" if d.raw is none else                  10186
  d.raw }}</td>                                                           10187
      <td class="mono">{{ "—" if d.std_value is none else                 10188
  "%.4f"|format(d.std_value) }}</td>                                      10189
      <td class="mono">{{ "%.4f"|format(d.weight) }}</td>                  10190
      <td class="mono">{{ "—" if d.contribution is none                   10191
  else "%.4f"|format(d.contribution) }}</td>                              10192
    </tr>                                                                 10193
    {% endfor %}                                                          10194
  </tbody>                                                                10195
</table>                                                                  10196
<p class="note">                                                          10197
  一级指标得分：                                                             10198
  {% for k, v in u.l1_scores.items() %}{{ k }} = {{                       10199
  "%.4f"|format(v) }}{% if not loop.last %};{% endif %}{%                 10200
```

```
      endfor %}
   </p>
   {% endfor %}
   <h2>六、评价依据说明</h2>
   <ul class="note">
      <li>指标标准化按论文式（８） 执行，按指标性质分为越大越优、越小越优、
          适中型与离散赋值四类。</li>
      <li>单元评价因子按论文式（９） 由二级指标权重加权求得；
          整矿评价因子由各微地质单元按面积占比加权合成。</li>
      <li>适宜性等级按论文表 4 划分：F&lt;0.3 不适宜、0.3≤F≤0.7 基本适宜、F≥0.7
      适宜。</li>
      <li>指标权重取自论文表 3，按岩质边坡、碎石边坡、平台三类微地质单元分别确定。</li>
   </ul>
   <p class="sig">
      评价人（签字）：_____ 日期：_____
   </p>
</body>
</html>
#  ==== app/templates/search.html ====
{% extends "base.html" %}
{% from "_macros.html" import icon %}
{% block title %}搜索 · 历史遗留废弃矿山自然恢复适宜性评价系统{% endblock %}
{% block content %}
<div class="card">
   <h2>搜索结果</h2>
   {% if not q %}
   <p class="muted">请输入关键词。可检索项目名称与编号、图斑名称与行政区/矿种、
      标准名称与文号。</p>
   {% else %}
   <p class="muted">
      关键词「<strong>{{ q }}</strong>」共找到
      <span class="num">{{ total }}</span> 条结果。
   </p>
   {% if not total %}
   <p class="muted">没有匹配的内容。换个词试试，或检查是否有错别字。</p>
   {% endif %}
   {% endif %}
</div>
{% if projects %}
<div class="card">
   <h2>评价项目（{{ projects|length }}）</h2>
   <table>
      <thead>
         <tr>
            <th scope="col">项目名称</th>
            <th scope="col">编号</th>
            <th scope="col">状态</th>
            <th scope="col">图斑数</th>
            <th scope="col">操作</th>
         </tr>
```

```
      </thead>                                                                  10251
      <tbody>                                                                   10252
        {% for p in projects %}                                                 10253
        <tr>                                                                    10254
          <td style="text-align:left;">{{ p.name }}</td>                        10255
          <td class="mono">{{ p.code or "—" }}</td>                             10256
          <td><span class="tag tag--muted">{{ p.status                          10257
      }}</span></td>                                                            10258
          <td class="num">{{ p.patches.count() }}</td>                          10259
          <td class="nowrap">                                                   10260
            <span class="row-actions">                                          10261
              <a class="link-btn" href="/project/{{ p.id                        10262
      }}/patches">矿山图斑</a>                                                  10263
              <a class="link-btn" href="/project/{{ p.id                        10264
      }}/edit">编辑</a>                                                         10265
            </span>                                                             10266
          </td>                                                                 10267
        </tr>                                                                   10268
        {% endfor %}                                                            10269
      </tbody>                                                                  10270
    </table>                                                                    10271
</div>                                                                          10272
{% endif %}                                                                     10273
{% if patches %}                                                                10274
<div class="card">                                                              10275
    <h2>矿山图斑（{{ patches|length }}）</h2>                                   10276
    <table>                                                                     10277
      <thead>                                                                   10278
        <tr>                                                                    10279
          <th scope="col">所属项目</th>                                         10280
          <th scope="col">图斑名称</th>                                         10281
          <th scope="col">行政区</th>                                           10282
          <th scope="col">矿种</th>                                             10283
          <th scope="col">操作</th>                                             10284
        </tr>                                                                   10285
      </thead>                                                                  10286
      <tbody>                                                                   10287
        {% for m in patches %}                                                  10288
        <tr>                                                                    10289
          <td>{{ m.project.name if m.project else "—" }}</td>                   10290
          <td style="text-align:left;">{{ m.name }}</td>                        10291
          <td>{{ m.admin_region or "—" }}</td>                                  10292
          <td>{{ m.mineral_type or "—" }}</td>                                  10293
          <td class="nowrap">                                                   10294
            <span class="row-actions">                                          10295
              <a class="link-btn" href="/constraint/{{ m.id                     10296
      }}">限制性判定</a>                                                        10297
              <a class="link-btn" href="/unit/{{ m.id                           10298
      }}">微地质单元</a>                                                        10299
              <a class="link-btn" href="/evaluate/{{ m.id                       10300
```

```
      }}">执行评价</a>                                                    10301
            </span>                                                      10302
          </td>                                                          10303
        </tr>                                                            10304
        {% endfor %}                                                     10305
      </tbody>                                                           10306
    </table>                                                             10307
</div>                                                                   10308
{% endif %}                                                              10309
{% if standards %}                                                       10310
<div class="card">                                                       10311
    <h2>标准与政策文件（{{ standards|length }}）</h2>                    10312
    <table>                                                              10313
      <thead>                                                            10314
        <tr>                                                             10315
          <th scope="col">类别</th>                                      10316
          <th scope="col">省份</th>                                      10317
          <th scope="col">名称</th>                                      10318
          <th scope="col">文号</th>                                      10319
          <th scope="col">操作</th>                                      10320
        </tr>                                                            10321
      </thead>                                                           10322
      <tbody>                                                            10323
        {% for s in standards %}                                         10324
        <tr>                                                             10325
          <td>{{ s.category }}</td>                                      10326
          <td>{{ s.province }}</td>                                      10327
          <td style="text-align:left;">{{ s.name }}</td>                 10328
          <td class="mono">{{ s.doc_number or "—" }}</td>                10329
          <td class="nowrap">                                            10330
            <span class="row-actions">                                   10331
              <a class="link-btn" href="/standard/{{ s.id                10332
      }}/rules">规则</a>                                                 10333
            </span>                                                      10334
          </td>                                                          10335
        </tr>                                                            10336
        {% endfor %}                                                     10337
      </tbody>                                                           10338
    </table>                                                             10339
</div>                                                                   10340
{% endif %}                                                              10341
{% endblock %}                                                           10342
# ==== seed_demo.py ====                                                 10343
# -*- coding: utf-8 -*-                                                  10344
"""内置玉田县案例：把论文第 5 节案例数据写入数据库。                     10345
用于功能验证与操作演示。数据逐字取自论文表 5 与第 5 节。                 10346
运行：python seed_demo.py                  幂等，只补写缺失的演示数据，  10347
                                                 不会删除用户自建的项目与图斑  10348
      python seed_demo.py --reset          连同标准库与指标定义一并重建  10349
"""                                                                      10350
```

```
import json
import sys
sys.stdout.reconfigure(encoding="utf-8")
from app import create_app
from app.models import (db, User, Project, MinePatch,
    ReferenceEcosystem,
                        MicroUnit, IndicatorData, Standard,
    StandardRule,
                        ConstraintCheck, EvaluationResult,
    ScoreDetail)
app = create_app()
#  内置演示案例的项目编号，用于区分「演示数据」与「用户自建数据」
DEMO_CODE = "DEMO-001"
#  论文表 1：11 项地方标准与政策文件
STANDARDS = [
    dict(category="地方标准", province="辽宁",
    name="矿山废弃地植被自然恢复技术规程",
        doc_number="DB21/T 3702—2023",
    publish_date="2023-03-30",
        implement_date="2023-04-30",
        summary="首个正式发布的矿山自然恢复地方标准；提出必要条件和充分条件",
        rules=[("植被覆盖度", ">=", "50", "必要条件", None),
               ("无地质灾害", "==", "是", "必要条件", None),
               ("无水土污染", "==", "是", "必要条件", None),
               ("与参照系差距", "<", "20", "充分条件", "%")]),
    dict(category="地方标准", province="湖北",
    name="废弃露天矿山生态环境自然恢复标准",
        doc_number="DB42/T 2421—2025",
    publish_date="2025-08-20",
        implement_date="2025-10-20",
        summary="结合矿山生态需求进行自然恢复适宜性分级判断",
        rules=[("自然恢复适宜性分级", "==", "是", "必要条件", None)]),
    dict(category="地方标准", province="内蒙古",
    name="历史遗留废弃矿山自然恢复判定规范",
        doc_number="DB15/T 4280—2026",
    publish_date="2026-01-30",
        implement_date="2026-03-01",
        summary="调查内容包含参照生态系统调查",
        rules=[("地质灾害发育程度", "<=", "弱发育", "必要条件", None),
               ("地形地貌与周边协调", "==", "是", "必要条件", None),
               ("有利于植被恢复的表层土壤", "==", "是", "必要条件", None),
               ("已形成稳定植被覆盖", "==", "是", "必要条件", None)]),
    dict(category="地方标准", province="江西",
    name="废弃矿山自然恢复技术规范",
        doc_number="DB36/T 2221—2026",
    publish_date="2026-02-02",
        implement_date="2026-08-01",
        summary="对参照生态系统进行了要求；提出自然恢复适宜性指数",
        rules=[("参照生态系统调查", "==", "是", "必要条件", None),
               ("自然恢复适宜性指数", ">=", "阈值", "充分条件", None)]),
```

```
    dict(category="地方标准", province="宁夏", name="矿山自然恢复认定规程",
        doc_number=None, publish_date="2026-06-18",
implement_date=None,
        summary="分区差异化：三个一级生态修复分区及 22 个二级分区",
        rules=[("无地质灾害风险", "==", "是", "必要条件", None),
               ("植被覆盖度趋势", "==", "稳定或正增长", "必要条件", None),
               ("与参照系差距", "<", "20", "充分条件", "%")]),
    dict(category="政策文件", province="安徽",
        name="安徽省自然资源厅关于规范废弃露天矿山采取自然恢复方式进行治理的通知",
        doc_number="皖自然资修函[2020]34号",
publish_date="2020-03-03", implement_date=None,
        summary="自然恢复条件：责任主体为政府、无地质灾害、无水土污染、不在三线三边、水土条件良好",
        rules=[("责任主体", "==", "政府", "必要条件", None),
               ("无地质灾害", "==", "是", "必要条件", None),
               ("无水土污染", "==", "是", "必要条件", None)]),
    dict(category="政策文件", province="山东",
        name="山东省自然资源厅关于进一步规范历史遗留废弃露天开采矿山采取自然恢复方式进行治理的通知",
        doc_number="鲁自然资字[2022]104号",
publish_date="2022-08-02", implement_date=None,
        summary="条件与安徽省相同，另增加自然恢复认定标准",
        rules=[("无地质灾害", "==", "是", "必要条件", None),
               ("无水土污染", "==", "是", "必要条件", None)]),
    dict(category="政策文件", province="湖南",
        name="关于印发《湖南省历史遗留矿山自然恢复指导意见（试行）》的通知",
        doc_number="湘自资办发[2023]58号",
publish_date="2023-07-20", implement_date=None,
        summary="自然恢复认定标准：无地质灾害或无威胁对象、无水土污染、设备拆除、符合规划、植被覆盖度
>75%、连续3年正向增长",
        rules=[("植被覆盖度", ">", "75", "必要条件", "%"),
               ("植被覆盖率趋势", "==", "连续3年正向增长", "必要条件", None),
               ("无地质灾害", "==", "是", "必要条件", None),
               ("符合国土空间规划", "==", "是", "必要条件", None)]),
    dict(category="政策文件", province="新疆",
        name="自治区历史遗留废弃矿山自然恢复方式销号图斑工作指南（试行）",
        doc_number="新自然资源办函〔2024〕219号",
publish_date="2024-12-20", implement_date=None,
        summary="荒漠区沙地/裸土地/裸岩石砾地满足无地质灾害、地形协调、无水土污染即可判定",
        rules=[("植被覆盖度", ">=", "周边30%", "必要条件", None),
               ("无地质灾害", "==", "是", "必要条件", None),
               ("地形平缓地貌稳定", "==", "是", "必要条件", None)]),
    dict(category="政策文件", province="广西",
        name="广西壮族自治区历史遗留矿山自然恢复技术指南",
        doc_number="桂自然资办〔2024〕250号",
publish_date="2024-12-30", implement_date=None,
        summary="除高陡边坡和裸露岩体外，水土水热条件良好、植被自然恢复趋势明显",
        rules=[("无地质灾害", "==", "是", "必要条件", None),
               ("植被自然恢复趋势", "==", "明显", "必要条件", None),
               ("生产设备拆除完毕", "==", "是", "必要条件", None)]),
    dict(category="政策文件", province="西藏",
        name="西藏自治区历史遗留废弃矿山生态自然恢复认定指导意见（试行）",
```

```
            doc_number="藏自然资发〔2025〕19号",
        publish_date="2025-09-24", implement_date=None,
            summary="含修复成本高、生态修复效益不显著等适用条件",
            rules=[("责任主体", "==", "政府", "必要条件", None),
                   ("地形地貌完整与周边协调", "==", "是", "必要条件", None),
                   ("无地质灾害", "==", "是", "必要条件", None)]),
]
CASE = {
    "name": "玉田县典型历史遗留废弃矿山案例",
    "patch_name": "玉田县孤树镇白云岩采石场",
    "area_ha": 5.6,
    "mineral_type": "白云岩",
    "units": [
        {
            "unit_type": "岩质边坡", "ratio": 0.1386,
            "data": {"coverage": 68.88, "shannon": 0.00,
                     "slope": 63.33, "aspect": "阴阳坡"},
        },
        {
            "unit_type": "碎石边坡", "ratio": 0.6295,
            "data": {"soil_ratio": 0.34, "thickness": 10,
    "ph": 9.24,
                     "organic_matter": 4.89,
                     "heavy_metals": {"Hg": 0.05, "As":
    5.17, "Ni": 12.60,
                                      "Cu": 11.70, "Cd":
    0.04, "Pb": 6.50},
                     "coverage": 87.92, "shannon": 0.60,
                     "slope": 28.75, "aspect": "阳坡"},
        },
        {
            "unit_type": "平台", "ratio": 0.2319,
            "data": {"soil_ratio": 0.56, "thickness": 30,
    "ph": 8.72,
                     "organic_matter": 10.92,
                     "heavy_metals": {"Hg": 0.02, "As":
    3.13, "Ni": 12.33,
                                      "Cu": 11.14, "Cd":
    0.09, "Pb": 9.98},
                     "coverage": 58.42, "shannon": 0.72},
        },
    ],
}
def clean_case(reset=False):
    """清理数据。
    默认只清理内置演示案例（code = DEMO-001）自身的数据，**不触碰
    用户自己建立的项目与图斑**。加 --reset 时才额外清空标准库与指标定义，
    以便把演示环境恢复到初始状态。
    注意：不要在此处执行 Project.query.delete()，否则用户新建的项目
    会在每次启动时被删除。
```

```
    """
    from app.models import EvaluationResult, ScoreDetail,
ConstraintCheck
    demo_projects = Project.query.filter_by(code=DEMO_CODE).
all()
    demo_ids = [p.id for p in demo_projects]
    patch_ids = []
    if demo_ids:
        patch_ids = [p.id for p in MinePatch.query.filter(
            MinePatch.project_id.in_(demo_ids)).all()]
    if patch_ids:
        unit_ids = [u.id for u in MicroUnit.query.filter(
            MicroUnit.patch_id.in_(patch_ids)).all()]
        if unit_ids:
            IndicatorData.query.filter(
                IndicatorData.unit_id.in_(unit_ids)).delete(
                synchronize_session=False)
        result_ids = [r.id for r in EvaluationResult.query.f
ilter(
            EvaluationResult.patch_id.in_(patch_ids)).all()]
        if result_ids:
            ScoreDetail.query.filter(
                ScoreDetail.result_id.in_(result_ids)).delet
e(
                synchronize_session=False)
        EvaluationResult.query.filter(
            EvaluationResult.patch_id.in_(patch_ids)).delete
(
            synchronize_session=False)
        ConstraintCheck.query.filter(
            ConstraintCheck.patch_id.in_(patch_ids)).delete(
            synchronize_session=False)
        ReferenceEcosystem.query.filter(
            ReferenceEcosystem.patch_id.in_(patch_ids)).dele
te(
            synchronize_session=False)
        MicroUnit.query.filter(MicroUnit.patch_id.in_(patch_
ids)).delete(
            synchronize_session=False)
        MinePatch.query.filter(MinePatch.id.in_(patch_ids))
delete(
            synchronize_session=False)
    if demo_ids:
        Project.query.filter(Project.id.in_(demo_ids)).delet
e(
            synchronize_session=False)
    if reset:
        StandardRule.query.delete()
        Standard.query.delete()
    # 兜底：清理指向已不存在评价记录的孤儿得分明细，
```

```
    # 避免长期运行后 score_details 表堆积无效数据。
    _purge_orphan_details()
    db.session.commit()
    return len(demo_ids)
def demo_patch():
    """取内置演示案例的图斑。
    库中可能存在用户自建项目或验证算例（编号 VERIFY-），
    故不能简单用 MinePatch.query.first()，否则取到的对象不确定。
    没有演示案例时才退回第一个图斑。
    """
    proj = Project.query.filter_by(code=DEMO_CODE).first()
    if proj is not None:
        m = MinePatch.query.filter_by(project_id=proj.id).first()
        if m is not None:
            return m
    return MinePatch.query.first()
def _purge_orphan_details():
    """清理指向已不存在评价记录的孤儿得分明细。
    长期运行后 score_details 表可能堆积无效明细，启动时顺带清一次。
    """
    live_ids = [r.id for r in EvaluationResult.query.all()]
    if live_ids:
        ScoreDetail.query.filter(~ScoreDetail.result_id.in_(
    live_ids)).delete(
            synchronize_session=False)
    else:
        ScoreDetail.query.delete()
    db.session.commit()
def seed(reset=False):
    with app.app_context():
        db.create_all()          # 全新环境下先建表，保证本脚本可独立运行
        _purge_orphan_details()
        admin = User.query.filter_by(username="admin").first(
    )
        if not admin:
            from werkzeug.security import
    generate_password_hash
            admin = User(username="admin",
                         password_hash=generate_password_has
    h("admin123"),
                         role="admin")
            db.session.add(admin)
            db.session.commit()
        # 幂等：演示案例已存在就不重复写入，更不清空用户自己的项目
        existing = Project.query.filter_by(code=DEMO_CODE).f
    irst()
        if existing and not reset:
            print("演示案例已存在，跳过写入（用户自建项目不受影响）")
            _report()
```

```
            _seed_reference_data(reset=False)                                   10601
            return                                                              10602
        clean_case(reset=reset)                                                 10603
        project = Project(name=CASE["name"], code=DEMO_CODE,                    10604
                          description="论文第 5 节典型案例",                    10605
created_by=admin.id)                                                            10606
        db.session.add(project)                                                 10607
        db.session.flush()                                                      10608
        patch = MinePatch(project_id=project.id,                                10609
name=CASE["patch_name"],                                                        10610
                          admin_region="河北省唐山市玉田县",                    10611
                          area_ha=CASE["area_ha"],                              10612
                          mineral_type=CASE["mineral_type"],                    10613
                          mining_method="露天开采",                             10614
stop_years=None,                                                                10615
                          responsible_entity_missing=True)                      10616
        db.session.add(patch)                                                   10617
        db.session.flush()                                                      10618
        ref = ReferenceEcosystem(patch_id=patch.id,                             10619
                                 coverage_benchmark=75.0,                       10620
shannon_benchmark=0.7)                                                          10621
        db.session.add(ref)                                                     10622
        for u in CASE["units"]:                                                 10623
            unit = MicroUnit(patch_id=patch.id,                                 10624
unit_type=u["unit_type"],                                                       10625
                             ratio=u["ratio"], area=None)                       10626
            db.session.add(unit)                                                10627
            db.session.flush()                                                  10628
            d = u["data"]                                                       10629
            ind = IndicatorData(                                                10630
                unit_id=unit.id,                                                10631
                soil_ratio=d.get("soil_ratio"),                                 10632
                thickness=d.get("thickness"),                                   10633
                ph=d.get("ph"),                                                 10634
                organic_matter=d.get("organic_matter"),                         10635
                heavy_metals=json.dumps(d["heavy_metals"],                      10636
ensure_ascii=False)                                                             10637
                                if d.get("heavy_metals") else                   10638
None,                                                                           10639
                coverage=d.get("coverage"),                                     10640
                shannon=d.get("shannon"),                                       10641
                slope=d.get("slope"),                                           10642
                aspect=d.get("aspect"),                                         10643
            )                                                                   10644
            db.session.add(ind)                                                 10645
        #  限制性条件判定记录：演示案例按论文第 5 节描述属「无地质灾害隐患点」、  10646
        #  具备自然恢复条件，故五项全部通过。缺此记录会使后续评价被流程门禁拦住。  10647
        db.session.add(ConstraintCheck(                                         10648
            patch_id=patch.id,                                                  10649
            policy_pass=True, location_pass=True,                               10650
```

```
        pollution_pass=True,                                                    10651
                    geo_pass=True, farmland_pass=True,                          10652
                    passed=True, reason="全部通过"))                             10653
            db.session.commit()                                                 10654
            print("已写入案例：", CASE["name"])                                  10655
            print("  图斑：", CASE["patch_name"])                               10656
            print("  微地质单元：岩质边坡 13.86% / 碎石边坡 62.95% / 平台          10657
        23.19%")                                                                10658
            print("  管理员账号：admin / admin123")                             10659
            _seed_reference_data(reset=reset)                                   10660
            _report()                                                           10661
    def _seed_reference_data(reset=False):                                      10662
        """写入标准库与指标定义。                                                10663
        参考资料只在库为空（或显式 --reset）时写入，避免覆盖用户自行维护          10664
        的标准条目与指标定义。                                                   10665
        """                                                                     10666
        if reset or Standard.query.count() == 0:                                10667
            seed_standards()                                                    10668
        from app.services.indicator_service import                              10669
        seed_indicators                                                         10670
        n = seed_indicators()                                                   10671
        print("指标定义：%d 项" % n)                                            10672
    def _report():                                                              10673
        total_p = Project.query.count()                                         10674
        total_patch = MinePatch.query.count()                                   10675
        print()                                                                 10676
        print("当前库内：项目 %d 个、图斑 %d 个" % (total_p, total_patch))       10677
        if total_p > total_patch:                                               10678
            print("  提示：项目是归档单位，评价针对的是图斑。")                   10679
            print("        新项目需先建立图斑，才会出现在后续各模块中。")         10680
    def seed_standards():                                                       10681
        """写入论文表 1 的 11 项标准与政策文件及其结构化规则。"""                10682
        StandardRule.query.delete()                                             10683
        Standard.query.delete()                                                 10684
        db.session.commit()                                                     10685
        for s in STANDARDS:                                                     10686
            std = Standard(category=s["category"],                              10687
        province=s["province"],                                                 10688
                           name=s["name"],                                      10689
        doc_number=s["doc_number"],                                             10690
                           publish_date=s["publish_date"],                      10691
                           implement_date=s["implement_date"],                  10692
                           summary=s["summary"])                                10693
            db.session.add(std)                                                 10694
            db.session.flush()                                                  10695
            for item, op, thr, ctype, note in s["rules"]:                       10696
                db.session.add(StandardRule(standard_id=std.id,                 10697
        item=item,                                                              10698
                                            operator=op,                        10699
        threshold=thr,                                                          10700
```

```
                                                                    10701
        condition_type=ctype, note=note))                           10702
        db.session.commit()                                         10703
        print("已写入标准/政策文件：%d 项，规则 %d 条"                    10704
              % (len(STANDARDS), sum(len(s["rules"]) for s in       10705
        STANDARDS)))                                                10706
if __name__ == "__main__":                                          10707
        #  --reset：把演示环境恢复到初始状态（清空标准库与指标定义后重建）。      10708
        #  无论是否 --reset，都不会删除用户自建的项目与图斑。                 10709
        seed(reset="--reset" in sys.argv[1:])                       10710
#  ==== 测试/test_ahp.py ====                                       10711
#  -*- coding: utf-8 -*-                                            10712
"""地基自检：验证 AHP 引擎与应用可启动。"""                                10713
import sys                                                          10714
sys.stdout.reconfigure(encoding="utf-8")                            10715
import numpy as np                                                  10716
import os                                                           10717
#  本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，                    10718
#  使 `from app import ...` 与 app/app.db 的定位不受影响。             10719
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__fil        10720
        e__)))                                                      10721
if ROOT not in sys.path:                                            10722
        sys.path.insert(0, ROOT)                                    10723
from app.services import ahp_service as ahp                         10724
LINE = "=" * 66                                                     10725
#  ------------------------------------------------------------     10726
        ------------------                                          10727
#  1） 完全一致矩阵 → 权重应精确还原，CR=0                                10728
#  ------------------------------------------------------------     10729
        ------------------                                          10730
print(LINE)                                                         10731
print("测试 1  完全一致判断矩阵的权重还原与一致性")                          10732
print(LINE)                                                         10733
#  由权重构造完全一致矩阵：a_ij = w_i / w_j                             10734
w_true = np.array([0.44, 0.31, 0.25])                               10735
A = np.outer(w_true, 1.0 / w_true)                                  10736
r = ahp.weights_and_consistency(A)                                  10737
print("  判断矩阵 A：")                                                10738
for row in A:                                                       10739
        print("      " + "  ".join(f"{v:8.4f}" for v in row))       10740
print()                                                             10741
print("  计算权重 :", "  ".join(f"{v:.4f}" for v in                  10742
        r["weights"]))                                              10743
print("  真实权重 :", "  ".join(f"{v:.4f}" for v in w_true))          10744
print("  λmax = %.6f   CI = %.3e   CR = %.3e   一致: %s"              10745
        % (r["lambda_max"], r["ci"], r["cr"],                       10746
        r["consistent"]))                                           10747
assert np.allclose(r["weights"], w_true, atol=1e-6),                10748
        "权重未精确还原"                                                10749
assert r["consistent"], "完全一致矩阵应通过检验"                          10750
```

```
# ----------------------------------------------------------------------
    ------------------
# 2） 上三角输入 → 完整矩阵 → 一致性定位
# ----------------------------------------------------------------------
    ------------------
print()
print(LINE)
print("测试 2  上三角录入与最不一致项定位")
print(LINE)
# 一个不太一致的 3 阶矩阵：上三角 [2, 5, 3]（a12=2, a13=5, a23=3）
M = ahp.matrix_from_upper([2, 5, 3])
r2 = ahp.weights_and_consistency(M)
bad = ahp.find_most_inconsistent(M)
print("  由 [a12=2, a13=5, a23=3] 构造矩阵：")
for row in M:
    print("    " + "  ".join(f"{v:8.4f}" for v in row))
print("  权重：", "  ".join(f"{v:.4f}" for v in r2["weights"]),
      "  CR = %.3f  一致：%s" % (r2["cr"], r2["consistent"]))
print("  最不一致比较项：第(%d,%d) 项，当前 %.3f，理想 %.3f"
      % (bad["i"] + 1, bad["j"] + 1, bad["value"],
    bad["expected"]))
# ----------------------------------------------------------------------
    ------------------
# 3） 多专家聚合（几何平均）
# ----------------------------------------------------------------------
    ------------------
print()
print(LINE)
print("测试 3  多专家判断矩阵聚合（几何平均）")
print(LINE)
m1 = ahp.matrix_from_upper([2, 4, 3])
m2 = ahp.matrix_from_upper([3, 5, 2])
agg = ahp.aggregate_matrices([m1, m2])
print("  专家1 上三角 [2,4,3]，专家2 上三角 [3,5,2]")
print("  聚合后上三角：a12=%.4f a13=%.4f a23=%.4f"
      % (agg[0, 1], agg[0, 2], agg[1, 2]))
assert abs(agg[0, 1] - np.sqrt(2 * 3)) < 1e-9, "几何平均错误"
# ----------------------------------------------------------------------
    ------------------
# 4） 论文表 3 权重可还原性
# ----------------------------------------------------------------------
    ------------------
print()
print(LINE)
print("测试 4  论文表 3 碎石边坡一级权重 (0.44, 0.31, 0.25)")
print(LINE)
j = ahp.judge_from_weights([0.44, 0.31, 0.25])
print("  λmax = %.6f  CR = %.3e  一致：%s" %
    (j["lambda_max"], j["cr"], j["consistent"]))
assert j["consistent"], "论文权重应能作为完全一致矩阵主特征向量"
```

```
# ------------------------------------------------------------------------
    ------------------
# 5） 应用可启动 + 建表
# ------------------------------------------------------------------------
    ------------------
print()
print(LINE)
print("测试 5  Flask 应用启动与建表")
print(LINE)
from app import create_app
from app.models import db
app = create_app()
with app.app_context():
    db.create_all()
    tables = sorted(db.metadata.tables.keys())
print("  已创建 %d 张表：" % len(tables))
for t in tables:
    print("    -", t)
print()
print(LINE)
print("全部测试通过 √")
print(LINE)
# ==== 测试/test_evaluate.py ====
# -*- coding: utf-8 -*-
"""评价引擎自检：用论文玉田县案例跑通完整计算链。"""
import sys
sys.stdout.reconfigure(encoding="utf-8")
import os
# 本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
# 使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app.services import evaluation_service as ev
from app.services import standardization as std
LINE = "=" * 70
# ------------------------------------------------------------------------
    ------------------
# 论文第 5 节案例数据（表 5，逐字抄录）
# ------------------------------------------------------------------------
    ------------------
# 三类单元面积占比：岩质边坡 １３．８６%、碎石边坡 ６２．９５%、平台 ２３．１９%
# 注：岩质边坡无土壤指标、平台无地形指标，对应"不适用"。
case = [
    {
        "unit_type": "岩质边坡",
        "ratio": 0.1386,
        "data": {
            "coverage": 68.88, "shannon": 0.00,
```

```
                "slope": 63.33, "aspect": "阴阳坡",
            },
        },
        {
            "unit_type": "碎石边坡",
            "ratio": 0.6295,
            "data": {
                "soil_ratio": 0.34, "thickness": 10, "ph":
    9.24, "organic_matter": 4.89,
                "heavy_metals": {"Hg": 0.05, "As": 5.17, "Ni":
    12.60,
                                 "Cu": 11.70, "Cd": 0.04, "Pb":
    6.50},
                "coverage": 87.92, "shannon": 0.60,
                "slope": 28.75, "aspect": "阳坡",
            },
        },
        {
            "unit_type": "平台",
            "ratio": 0.2319,
            "data": {
                "soil_ratio": 0.56, "thickness": 30, "ph":
    8.72, "organic_matter": 10.92,
                "heavy_metals": {"Hg": 0.02, "As": 3.13, "Ni":
    12.33,
                                 "Cu": 11.14, "Cd": 0.09, "Pb":
    9.98},
                "coverage": 58.42, "shannon": 0.72,
            },
        },
]
# ------------------------------------------------------------------
    ------------------
# 逐单元输出标准化与得分
# ------------------------------------------------------------------
    ------------------
print(LINE)
print("玉田县案例 · 评价引擎完整计算")
print(LINE)
for u in case:
    r = ev.score_unit(u["unit_type"], u["data"])
    print()
    print("  【%s】  (面积占比 %.2f%%)" % (u["unit_type"],
    u["ratio"] * 100))
    print("  " + "-" * 60)
    for l1, score in r["l1_scores"].items():
        print("    %s 得分 = %.4f" % (l1, score))
    print("    单元评价因子 F' = %.4f" % r["unit_score"])
print()
print(LINE)
```

```
print("面积加权合成（式(9)）")
print(LINE)
res = ev.score_mine(case)
total = res["final_score"]
print()
for ur in res["unit_results"]:
    print("  %s  %.2f%% × %.4f = %.4f"
          % (ur["unit_type"], ur["ratio"] * 100,
    ur["unit_score"], ur["weighted"]))
print("  " + "-" * 60)
print("  最终评价因子 F = %.4f" % total)
print()
print("  适宜性等级：%s" % res["grade"])
print("  判定依据  ：%s" % res["grade_info"]["meaning"])
print("  推荐措施  ：%s" % res["grade_info"]["measure"])
print()
print(LINE)
print("结论")
print(LINE)
print("  - 评价引擎按论文式(8)(9) 与表 3 权重完整跑通")
print("  - 坡度标准化采用论文 25° 分界的线性折减默认值（见 standardization.py）")
print("  - 最终 F ≈ %.2f，判定为「%s」，与论文「最终适宜自然恢复」的结论一致"
      % (total, res["grade"]))
print("  - 论文表 5 所载最终得分为 0.76；差异集中在坡度、坡向的标准化口径，")
print("    该口径在 standardization.std_slope / aspect_score
    中集中定义、可配置")
print(LINE)
# ==== 测试/test_web_ahp.py ====
# -*- coding: utf-8 -*-
"""AHP 网页界面自检：走通选择→录入→计算 全流程。"""
import sys
sys.stdout.reconfigure(encoding="utf-8")
import os
# 本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
# 使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file
    __)))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app import create_app
LINE = "=" * 66
# 测试配置：关闭 CSRF 以简化测试客户端（真实运行仍开启 CSRF）
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///:memory:"
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
app = create_app(TestConfig)
client = app.test_client()
```

```
# 建立表并创建测试用户，登录后再访问业务页面
with app.app_context():
    from app.models import db, User
    from werkzeug.security import generate_password_hash
    db.create_all()
    if not User.query.filter_by(username="admin").first():
        db.session.add(User(username="admin",
                            password_hash=generate_password_hash("admin123"),
                            role="admin"))
        db.session.commit()
client.post("/login", data={"username": "admin",
    "password": "admin123"})
print(LINE)
print("测试 1  首页与 AHP 入口")
print(LINE)
r = client.get("/")
print("  首页状态码：", r.status_code, "（应 200）")
assert r.status_code == 200
r = client.get("/ahp/")
print("  AHP 入口状态码：", r.status_code, "（应 200）")
assert r.status_code == 200
assert "微地质单元类型" in r.get_data(as_text=True)
print()
print(LINE)
print("测试 2  选择碎石边坡一级指标 → 跳转矩阵页")
print(LINE)
r = client.post("/ahp/", data={
    "unit_type": "碎石边坡", "level": "1", "submit": "生成判断矩阵",
}, follow_redirects=True)
body = r.get_data(as_text=True)
print("  跳转后页面含「判断矩阵录入」：", "判断矩阵录入" in body)
print("  含三个一级指标（土壤/植被/地形）：", all(n in body for n in
    ("土壤指标", "植被指标", "地形指标")))
assert "判断矩阵录入" in body
assert all(n in body for n in ("土壤指标", "植被指标", "地形指标"))
print()
print(LINE)
print("测试 3  提交论文表 3 一致矩阵 → 还原权重")
print(LINE)
# 碎石边坡一级权重 (0.44, 0.31, 0.25) 对应的完全一致矩阵上三角：
# a12 = 0.44/0.31 = 1.4194, a13 = 0.44/0.25 = 1.76, a23 =
    0.31/0.25 = 1.24
r = client.post("/ahp/matrix?unit_type=碎石边坡&level=1", data={
    "m_0_1": "1.4194", "m_0_2": "1.76", "m_1_2": "1.24",
}, follow_redirects=True)
body = r.get_data(as_text=True)
print("  结果页含「计算结果」：", "计算结果" in body)
print("  含权重 0.44：", "0.44" in body, " 含 0.31：", "0.31" in
    body, " 含 0.25：", "0.25" in body)
```

```
print("   含一致性检验通过：", "通过" in body)
assert "计算结果" in body
assert "0.44" in body and "0.31" in body and "0.25" in body
print()
print(LINE)
print("测试 4   不一致矩阵应提示不通过并定位")
print(LINE)
# 明显不一致：a12=9, a13=1, a23=9
r = client.post("/ahp/matrix?unit_type=碎石边坡&level=1", data={
    "m_0_1": "9", "m_0_2": "1", "m_1_2": "9",
}, follow_redirects=True)
body = r.get_data(as_text=True)
print("   含「不通过」：", "不通过" in body)
print("   含「建议优先调整」：", "建议优先调整" in body)
assert "不通过" in body
assert "建议优先调整" in body
print()
print(LINE)
print("网页界面全部测试通过 √")
print(LINE)
# ==== 测试/test_web_modules.py ====
# -*- coding: utf-8 -*-
"""模块自检：登录、项目档案、微地质单元、标准库、限制性判定。"""
import os
import sys
sys.stdout.reconfigure(encoding="utf-8")
# 本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
# 使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app import create_app
LINE = "=" * 66
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///" + os.path.join(
        ROOT, "app", "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
app = create_app(TestConfig)
client = app.test_client()
print(LINE)
print("测试 1   未登录应被重定向到登录页")
print(LINE)
r = client.get("/project/")
print("   /project/ 状态码：", r.status_code, " 重定向到：",
    r.headers.get("Location"))
assert r.status_code == 302 and "login" in
```

```
        r.headers.get("Location", "")
print()
print(LINE)
print("测试 2  登录")
print(LINE)
r = client.post("/login", data={"username": "admin",
        "password": "admin123"},
                        follow_redirects=True)
body = r.get_data(as_text=True)
#  工作台：页首横幅 + 评价流程分区 + 数据概览
print("  登录后含「评价流程」：", "评价流程" in body)
print("  含账户 admin：", "admin" in body)
print("  含页首横幅：", 'class="hero"' in body)
print("  含数据概览：", 'class="stat-card"' in body)
assert "评价流程" in body and 'class="hero"' in body
print()
print(LINE)
print("测试 3  项目与矿山档案")
print(LINE)
r = client.get("/project/")
body = r.get_data(as_text=True)
print("  项目列表状态码：", r.status_code)
print("  含案例项目：", "玉田县典型历史遗留废弃矿山案例" in body)
assert "玉田县典型历史遗留废弃矿山案例" in body
#  新建项目
client.post("/project/add", data={"name": "测试项目A", "code":
        "T-001"})
r = client.get("/project/")
print("  新建项目后含「测试项目A」：", "测试项目A" in r.get_data(as_text=True
        ))
assert "测试项目A" in r.get_data(as_text=True)
#  清理测试数据，避免污染演示库
with app.app_context():
        from app.models import db, Project
        for p in Project.query.filter_by(name="测试项目A").all():
                db.session.delete(p)
        db.session.commit()
print("  已清理测试项目")
print()
print(LINE)
print("测试 4  微地质单元（面积占比自动计算）")
print(LINE)
from app.models import MinePatch
with app.app_context():
        patch = MinePatch.query.filter_by(name="玉田县孤树镇白云岩采石场").f
        irst()
        pid = patch.id
r = client.get("/unit/%d" % pid)
body = r.get_data(as_text=True)
print("  单元列表含三类单元：", all(t in body for t in ("岩质边坡",
```

```
        "碎石边坡", "平台")))                                                    11101
print("  含占比合计:", "占比合计" in body)                                      11102
assert "岩质边坡" in body and "占比合计" in body                                 11103
print()                                                                        11104
print(LINE)                                                                    11105
print("测试 5  标准知识库与省际口径比对")                                       11106
print(LINE)                                                                    11107
r = client.get("/standard/")                                                   11108
body = r.get_data(as_text=True)                                                11109
print("  含辽宁/湖南/新疆标准:", all(p in body for p in ("辽宁", "湖南",        11110
        "新疆")))                                                              11111
assert "辽宁" in body                                                          11112
r = client.get("/standard/compare")                                            11113
body = r.get_data(as_text=True)                                                11114
print("  口径比对页含「植被覆盖度」分组:", "植被覆盖度" in body)                11115
print("  含多省份并列:", "湖南" in body and "新疆" in body)                     11116
assert "植被覆盖度" in body                                                    11117
print()                                                                        11118
print(LINE)                                                                    11119
print("测试 6  限制性指标判定（一票否决）")                                     11120
print(LINE)                                                                    11121
r = client.get("/constraint/%d" % pid)                                         11122
body = r.get_data(as_text=True)                                                11123
print("  含五项条件:", all(x in body for x in ("政策约束", "区位约束",         11124
        "土壤污染", "地质灾害", "耕地属性")))                                  11125
assert all(x in body for x in ("政策约束", "区位约束", "土壤污染",             11126
        "地质灾害", "耕地属性"))                                               11127
# 全部通过                                                                     11128
client.post("/constraint/%d" % pid, data={                                     11129
        "policy_pass": "1", "location_pass": "1",                              11130
        "pollution_pass": "1",                                                 11131
        "geo_pass": "1", "farmland_pass": "1"})                                11132
r = client.get("/constraint/%d" % pid)                                         11133
body = r.get_data(as_text=True)                                                11134
print("  全通过后显示「通过，可进入定量评价」:", "可进入定量评价" in body)      11135
assert "可进入定量评价" in body                                                11136
# 一项不通过                                                                   11137
client.post("/constraint/%d" % pid, data={                                     11138
        "policy_pass": "1", "location_pass": "0",                              11139
        "pollution_pass": "1",                                                 11140
        "geo_pass": "1", "farmland_pass": "1"})                                11141
r = client.get("/constraint/%d" % pid)                                         11142
body = r.get_data(as_text=True)                                                11143
print("  一项不通过后显示「不通过」:", "不通过" in body)                        11144
print("  含失败项「区位约束」:", "区位约束" in body)                            11145
assert "不通过" in body                                                        11146
# 恢复为全部通过，避免把演示数据留在「不通过」状态影响后续步骤                 11147
client.post("/constraint/%d" % pid, data={                                     11148
        "policy_pass": "1", "location_pass": "1",                              11149
        "pollution_pass": "1",                                                 11150
```

```
        "geo_pass": "1", "farmland_pass": "1"})
r = client.get("/constraint/%d" % pid)
print("   已恢复为全部通过：", "可进入定量评价" in r.get_data(as_text=True))
print()
print(LINE)
print("全部测试通过 √")
print(LINE)
#  ==== 测试/test_web_evaluate.py ====
#  -*- coding: utf-8 -*-
"""端到端自检：录数据 → 执行评价 → 查看结果。"""
import os
import sys
sys.stdout.reconfigure(encoding="utf-8")
#  本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
#  使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app import create_app
LINE = "=" * 66
#  与真实配置共用同一个 app.db，但关闭 CSRF 便于测试客户端
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///" + os.path.join(
        ROOT, "app", "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
app = create_app(TestConfig)
client = app.test_client()
#  登录（案例种子已创建 admin 用户）
client.post("/login", data={"username": "admin",
    "password": "admin123"})
print(LINE)
print("测试 1  指标数据列表")
print(LINE)
r = client.get("/indicator/")
body = r.get_data(as_text=True)
print("   状态码：", r.status_code)
print("   含案例图斑：", "玉田县孤树镇白云岩采石场" in body)
print("   含三个单元：", all(n in body for n in ("岩质边坡", "碎石边坡",
    "平台")))
assert r.status_code == 200
assert "岩质边坡" in body and "碎石边坡" in body and "平台" in body
print()
print(LINE)
print("测试 2  指标数据录入页（按单元类型显示字段）")
print(LINE)
from app.models import MicroUnit, MinePatch
```

```
with app.app_context():
    unit_rock = MicroUnit.query.filter_by(unit_type="岩质边坡").first()
    unit_gravel = MicroUnit.query.filter_by(unit_type="碎石边坡").first()
r = client.get("/indicator/%d" % unit_rock.id)
b_rock = r.get_data(as_text=True)
print("  岩质边坡页：无「土石比」字段：", "土石比" not in b_rock, "（应 True）")
print("  岩质边坡页：含「植被覆盖度」：", "植被覆盖度" in b_rock)
assert "土石比" not in b_rock and "植被覆盖度" in b_rock
r = client.get("/indicator/%d" % unit_gravel.id)
b_gravel = r.get_data(as_text=True)
print("  碎石边坡页：含「土石比」字段：", "土石比" in b_gravel)
print("  碎石边坡页：含 8 种重金属：", all(m in b_gravel for m in
    ("Cd", "Hg", "Pb", "Cr", "As", "Ni", "Cu", "Zn")))
assert "土石比" in b_gravel and all(m in b_gravel for m in
    ("Cd", "Hg", "Pb", "Cr", "As", "Ni", "Cu", "Zn"))
print()
print(LINE)
print("测试 3  执行评价")
print(LINE)
with app.app_context():
    import seed_demo
    patch = seed_demo.demo_patch()
r = client.post("/evaluate/%d" % patch.id, data={},
    follow_redirects=True)
body = r.get_data(as_text=True)
print("  跳转至结果页：", "适宜性评价结果" in body)
print("  含最终评价因子 F：", "最终评价因子 F" in body)
print("  含等级「适宜」：", "适宜" in body)
print("  含推荐措施「自然恢复」：", "自然恢复" in body)
assert "最终评价因子 F" in body and "自然恢复" in body
print()
print(LINE)
print("测试 4  结果持久化")
print(LINE)
from app.models import EvaluationResult
with app.app_context():
    res = EvaluationResult.query.order_by(EvaluationResult.id.desc()).first()
    print("  最终得分 F =", res.final_score)
    print("  等级 =", res.grade)
    assert res.grade == "适宜"
    assert 0.7 <= res.final_score < 1.0
print()
print(LINE)
print("端到端全部测试通过 √")
print(LINE)
# ==== 测试/test_web_report.py ====
# -*- coding: utf-8 -*-
```

```
"""报告输出自检：得分明细页 + Word 导出。"""
import os
import sys
import io
sys.stdout.reconfigure(encoding="utf-8")
# 本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
# 使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app import create_app
LINE = "=" * 66
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///" + os.path.join(
        ROOT, "app", "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
app = create_app(TestConfig)
client = app.test_client()
# 登录（案例种子已创建 admin 用户）
client.post("/login", data={"username": "admin",
    "password": "admin123"})
from app.models import MinePatch, EvaluationResult
with app.app_context():
    import seed_demo
    patch = seed_demo.demo_patch()
    pid = patch.id
# 报告取自已保存的评价记录，因此先按网页路径真实执行一次评价
from app.views.evaluate import load_patch_input,
    save_evaluation
with app.app_context():
    units = load_patch_input(pid)
    rec = save_evaluation(pid, units)
    rid = rec.id
    print("前置：已执行评价并保存记录 id=%d，F=%.4f（%s）"
          % (rid, rec.final_score, rec.grade))
    from app.models import ScoreDetail
    n_detail = ScoreDetail.query.filter_by(result_id=rid).co
    unt()
    print("        逐层得分明细已落库：%d 条" % n_detail)
print()
print(LINE)
print("测试 1  报告列表")
print(LINE)
r = client.get("/report/")
print("  状态码：", r.status_code)
body = r.get_data(as_text=True)
```

```
print("   含「所属项目」列：", "所属项目" in body)                    11301
assert r.status_code == 200                                          11302
print()                                                              11303
print(LINE)                                                          11304
print("测试 2   得分明细页")                                          11305
print(LINE)                                                          11306
r = client.get("/report/%d" % pid)                                   11307
body = r.get_data(as_text=True)                                      11308
print("   状态码：", r.status_code)                                   11309
print("   含「得分明细」：", "得分明细" in body)                       11310
print("   含「最终评价因子 F」：", "最终评价因子 F" in body)           11311
print("   含二级指标明细（植被覆盖度 / 坡度 / 坡向）：", all(n in body for n in  11312
    ("植被覆盖度", "坡度", "坡向")))                                  11313
print("   含权重 / 标准化值列：", "标准化值" in body and "贡献" in body)  11314
assert r.status_code == 200                                          11315
assert "植被覆盖度" in body and "标准化值" in body                    11316
print()                                                              11317
print(LINE)                                                          11318
print("测试 3  Word 报告导出")                                        11319
print(LINE)                                                          11320
r = client.get("/report/%d/word" % pid)                              11321
print("   状态码：", r.status_code)                                   11322
print("   Content-Type:", r.headers.get("Content-Type",               11323
    "")[:60])                                                        11324
data = r.data                                                        11325
print("   文件大小：", len(data), "字节")                             11326
print("   是否为有效 docx（PK 头）：", data[:2] == b"PK")             11327
assert r.status_code == 200 and data[:2] == b"PK"                    11328
# 用 python-docx 打开验证内容                                        11329
import docx                                                          11330
d = docx.Document(io.BytesIO(data))                                  11331
text = "\n".join(p.text for p in d.paragraphs)                       11332
print("   含报告标题：", "自然恢复适宜性评价报告" in text)            11333
print("   含评价结论章节：", "评价结论" in text)                      11334
print("   含得分明细章节：", "得分明细" in text)                      11335
print("   段落数：", len(d.paragraphs), " 表格数：", len(d.tables))   11336
assert "自然恢复适宜性评价报告" in text and len(d.tables) >= 3        11337
print()                                                              11338
print(LINE)                                                          11339
print("报告输出全部测试通过 √")                                      11340
print(LINE)                                                          11341
# ==== 测试/test_ui_fixes.py ====                                    11342
# -*- coding: utf-8 -*-                                              11343
"""界面修复自检：「记住我」与各模块删除功能。"""                      11344
import os                                                            11345
import re                                                            11346
import sys                                                           11347
# 本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，           11348
# 使 `from app import ...` 与 app/app.db 的定位不受影响。            11349
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__fil        11350
```

```
    e__)))                                                          11351
if ROOT not in sys.path:                                            11352
    sys.path.insert(0, ROOT)                                        11353
from app import create_app                                          11354
sys.stdout.reconfigure(encoding="utf-8")                            11355
LINE = "=" * 70                                                     11356
class TestConfig:                                                   11357
    SECRET_KEY = "test"                                             11358
    SQLALCHEMY_DATABASE_URI = "sqlite:///" + os.path.join(          11359
        ROOT, "app", "app.db")                                      11360
    SQLALCHEMY_TRACK_MODIFICATIONS = False                          11361
    WTF_CSRF_ENABLED = False                                        11362
    TESTING = True                                                  11363
    REMEMBER_COOKIE_DURATION = __import__("datetime").timede        11364
    lta(days=30)                                                    11365
app = create_app(TestConfig)                                        11366
client = app.test_client()                                          11367
# ------------------------------------------------------------      11368
    ----------------                                                11369
print(LINE)                                                         11370
print("一、「记住我」")                                              11371
print(LINE)                                                         11372
html = client.get("/login").get_data(as_text=True)                  11373
# 1) 复选框必须有可点击关联的 label（这是原先失效的根因）           11374
#    只判断「复选框与文字是否落在同一个 <label> 内」，不绑定具体类名 ——  11375
#    否则换个样式类就会误报，而真正的回归点其实与类名无关。         11376
labels = re.findall(r"<label[^>]*>(.*?)</label>", html,             11377
    re.S)                                                           11378
wrapped = any(('name="remember"' in b and "记住我" in b) for b      11379
    in labels)                                                      11380
print("  %s 「记住我」文字与复选框包在同一 label 内" % ("[符合]" if wrapped  11381
    else "[不符合]"))                                               11382
assert wrapped, "复选框未与文字关联，点击文字不会勾选"              11383
has_for = 'for="remember"' in html                                  11384
print("  %s label 带 for 属性指向复选框" % ("[符合]" if has_for else  11385
    "[不符合]"))                                                    11386
# 2) 勾选后应下发带过期时间的 remember_token                       11387
c = app.test_client()                                               11388
r = c.post("/login", data={"username": "admin", "password":         11389
    "admin123",                                                     11390
                               "remember": "y"},                    11391
    follow_redirects=False)                                         11392
cookies = r.headers.getlist("Set-Cookie")                           11393
rt = [h for h in cookies if h.startswith("remember_token=")]        11394
print("  %s 勾选后下发 remember_token" % ("[符合]" if rt else       11395
    "[不符合]"))                                                    11396
assert rt, "未下发 remember_token"                                  11397
exp = re.search(r"Expires=([^;]+)", rt[0])                          11398
print("  %s 该 cookie 带过期时间：%s"                              11399
      % ("[符合]" if exp else "[不符合]", exp.group(1) if exp       11400
```

```
        else "无（会话 cookie）"))                                          11401
assert exp, "remember_token 无过期时间，关浏览器即失效"                    11402
# 3) 不勾选时不应下发 remember_token                                        11403
c2 = app.test_client()                                                      11404
r2 = c2.post("/login", data={"username": "admin",                           11405
        "password": "admin123"},                                            11406
                 follow_redirects=False)                                    11407
rt2 = [h for h in r2.headers.getlist("Set-Cookie") if                       11408
        h.startswith("remember_token=")]                                    11409
print("  %s 未勾选时不下发 remember_token" % ("[符合]" if not rt2           11410
        else "[不符合]"))                                                   11411
assert not rt2                                                              11412
# 4) 携带 remember_token 的新会话应自动登录                                 11413
c3 = app.test_client()                                                      11414
c3.set_cookie("remember_token",                                             11415
              rt[0].split(";")[0].split("=", 1)[1],                         11416
        domain="localhost")                                                 11417
r3 = c3.get("/", follow_redirects=False)                                    11418
print("  %s 仅凭 remember_token 访问首页可自动登录（状态码 %d）"           11419
      % ("[符合]" if r3.status_code == 200 else "[不符合]",                 11420
        r3.status_code))                                                    11421
assert r3.status_code == 200                                                11422
# ------------------------------------------------------------------------11423
        ------------------                                                  11424
print()                                                                     11425
print(LINE)                                                                 11426
print("二、删除功能")                                                       11427
print(LINE)                                                                 11428
client.post("/login", data={"username": "admin",                            11429
        "password": "admin123"})                                            11430
# 项目：列表页须有删除按钮                                                  11431
html = client.get("/project/").get_data(as_text=True)                       11432
has_del = "/delete" in html and "btn--danger" in html                       11433
print("  %s 项目列表含删除按钮" % ("[符合]" if has_del else "[不符合]"))   11434
assert has_del, "项目列表缺少删除按钮"                                      11435
print("  %s 删除前有二次确认" % ("[符合]" if "confirm(" in html else       11436
        "[不符合]"))                                                        11437
assert "confirm(" in html                                                   11438
# 图斑列表：地址用的是项目主键，不是图斑主键                               11439
with app.app_context():                                                     11440
        from app.models import MinePatch, Project                           11441
        import seed_demo                                                    11442
        patch = seed_demo.demo_patch()                                      11443
        patch_id = patch.id                                                 11444
        project_id = patch.project_id                                       11445
html = client.get("/project/%d/patches" %                                   11446
        project_id).get_data(as_text=True)                                  11447
print("  %s 图斑列表含删除按钮" % ("[符合]" if "/project/patch/" in        11448
        html and "btn--danger" in html else "[不符合]"))                    11449
assert "/project/patch/" in html and "btn--danger" in html                  11450
```

```
# 微地质单元
html = client.get("/unit/%d" % patch_id).get_data(as_text=True)
print("  %s 微单元列表含删除按钮" % ("[符合]" if "/unit/delete/" in html else "[不符合]"))
assert "/unit/delete/" in html
print("  %s 微单元编辑为独立页面（表单不再嵌在表格行内）"
      % ("[符合]" if "/unit/edit/" in html else "[不符合]"))
# 标准
html = client.get("/standard/").get_data(as_text=True)
print("  %s 标准列表含删除按钮" % ("[符合]" if "btn--danger" in html else "[不符合]"))
assert "btn--danger" in html
# 实际执行一次删除：新建项目 → 删除 → 确认消失
client.post("/project/add", data={"name": "待删除测试项目", "code": "DEL-1"})
with app.app_context():
    from app.models import db, Project
    tmp = Project.query.filter_by(name="待删除测试项目").first()
    tmp_id = tmp.id
html = client.get("/project/").get_data(as_text=True)
print("  %s 新建项目成功" % ("[符合]" if "待删除测试项目" in html else "[不符合]"))
r = client.post("/project/%d/delete" % tmp_id, follow_redirects=True)
html = r.get_data(as_text=True)
gone = "待删除测试项目" not in html
print("  %s 删除项目后列表中不再出现" % ("[符合]" if gone else "[不符合]"))
assert gone
print("  %s 有删除成功提示" % ("[符合]" if "已删除" in html else "[不符合]"))
print()
print(LINE)
print("界面修复自检全部通过 √")
print(LINE)
# ==== 测试/test_new_features.py ====
# -*- coding: utf-8 -*-
"""补齐模块自检：操作日志与用户维护、遥感影像、指标与多专家矩阵、
批量导入、图表与 PDF 导出。
对应计划书功能清单与系统实现的差异核对结论。
"""
import io
import json
import os
import re
import sys
sys.stdout.reconfigure(encoding="utf-8")
import fitz
# 本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
# 使 `from app import ...` 与 app/app.db 的定位不受影响。
```

```
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app import create_app
from app.models import (db, User, Project, MinePatch,
    MicroUnit, IndicatorData,
                        Indicator, ExpertMatrix,
    OperationLog)
from app.utils.schema import ensure_schema
from app.utils import browser, charts
LINE = "=" * 72
BASE = ROOT
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///" +
    os.path.join(BASE, "app", "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
    UPLOAD_FOLDER = os.path.join(BASE, "app", "static",
    "uploads")
    MAX_CONTENT_LENGTH = 32 * 1024 * 1024
    ALLOWED_IMAGE_EXT = {"png", "jpg", "jpeg", "gif",
    "webp", "bmp", "tif", "tiff"}
app = create_app(TestConfig)
with app.app_context():
    db.create_all()
    ensure_schema(db)
    ExpertMatrix.query.delete()
    db.session.commit()
    # 固定取内置演示案例的图斑，避免受库中用户自建项目 / 验证算例的影响
    import seed_demo
    patch_id = seed_demo.demo_patch().id
c = app.test_client()
c.post("/login", data={"username": "admin", "password":
    "admin123"})
def ok(flag, label, extra=""):
    print("  %s %-44s %s" % ("[符合]" if flag else "[不符合]",
    label, extra))
    assert flag, label
# ------------------------------------------------------------
    ----------------
print(LINE)
print("操作日志与用户维护")
print(LINE)
before = None
with app.app_context():
    before = OperationLog.query.count()
c.post("/project/add", data={"name": "日志核对项目", "code":
```

```
        "LOG-T"})
with app.app_context():
        after = OperationLog.query.count()
        last = OperationLog.query.order_by(OperationLog.id.desc(
        )).first()
ok(after == before + 1, "写操作被写入操作日志", "%d -> %d" % (before,
        after))
ok(last.action == "新建项目", "日志记录了可读动作名", last.action)
ok("project" in (last.target_type or ""), "日志记录了对象类型",
        last.target_type)
r = c.get("/profile?tab=logs")
body = r.get_data(as_text=True)
ok(r.status_code == 200 and "操作日志" in body, "操作日志分类可访问")
ok("新建项目" in body, "日志页显示操作记录")
# 账号编辑 / 删除（职务只有管理员与操作员两种）
with app.app_context():
        u = User(username="tmp_user", role="operator",
                 password_hash=User.query.first().password_hash)
        db.session.add(u); db.session.commit()
        uid = u.id
r = c.get("/users/%d/edit" % uid)
ok(r.status_code == 200, "账号编辑页可访问")
# 保护校验要趁系统里只有一名管理员时做：
# 先降级唯一的 admin 应被拒绝，再把临时账号提为管理员看修改是否生效
with app.app_context():
        admin_id = User.query.filter_by(role="admin").first().id
c.post("/users/%d/edit" % admin_id,
           data={"role": "operator", "active": "1"})
with app.app_context():
        ok(db.session.get(User, admin_id).role == "admin",
        "拒绝把最后一个管理员降级")
c.post("/users/%d/edit" % uid, data={"role": "admin",
        "active": "1"})
with app.app_context():
        ok(db.session.get(User, uid).role == "admin",
        "职务已修改并保存")
c.post("/users/%d/delete" % uid, follow_redirects=True)
with app.app_context():
        ok(db.session.get(User, uid) is None, "账号删除生效")
# 清理日志核对用的项目
with app.app_context():
        for p in Project.query.filter_by(name="日志核对项目").all():
            db.session.delete(p)
        db.session.commit()
# ------------------------------------------------------------------
        ------------------
print()
print(LINE)
print("遥感影像挂接")
print(LINE)
```

```
doc = fitz.open()
pg = doc.new_page(width=320, height=200)
pg.insert_text((20, 60), "RS image", fontname="china-s",
    fontsize=16)
png = doc[0].get_pixmap(dpi=100).tobytes("png")
doc.close()
r = c.post("/project/patch/%d/image" % patch_id,
           data={"image": (io.BytesIO(png), "影像.png"),
                 "image_note": "２０２５年6月无人机正射影像"},
           content_type="multipart/form-data",
    follow_redirects=True)
body = r.get_data(as_text=True)
ok("已挂接" in body and "<img" in body, "影像上传并显示")
with app.app_context():
    rel = db.session.get(MinePatch, patch_id).image_path
ok(bool(rel), "数据库记录影像路径", rel)
abs_path = os.path.join(TestConfig.UPLOAD_FOLDER,
    rel.replace("uploads/", "", 1))
ok(os.path.isfile(abs_path), "影像文件落盘")
ok(c.get("/static/" + rel).status_code == 200, "影像可经 URL
    访问")
r = c.post("/project/patch/%d/image" % patch_id,
           data={"image": (io.BytesIO(b"x"), "bad.txt")},
           content_type="multipart/form-data",
    follow_redirects=True)
ok("不支持" in r.get_data(as_text=True), "拒绝非图片格式")
c.post("/project/patch/%d/image/delete" % patch_id,
    follow_redirects=True)
with app.app_context():
    ok(db.session.get(MinePatch, patch_id).image_path is
    None, "影像移除后字段清空")
ok(not os.path.isfile(abs_path), "影像文件已删除")
# ------------------------------------------------------------
    ------------------
print()
print(LINE)
print("指标定义维护与多专家矩阵")
print(LINE)
r = c.get("/ahp/indicators")
body = r.get_data(as_text=True)
with app.app_context():
    n_ind = Indicator.query.count()
ok(r.status_code == 200 and n_ind > 0, "指标定义已入库并可查看", "%d
    项" % n_ind)
ok(all(k in body for k in ("一级指标", "二级指标", "三级指标")),
   "指标树按层级展示")
ok("岩质边坡" in body and "平台" in body, "显示各指标启用的微地质单元")
c.post("/ahp/indicators/coverage/edit",
       data={"name": "植被覆盖度", "unit": "%", "ideal_type":
    "benefit",
```

```
                "ideal_value": "75", "enabled_units": ["岩质边坡"],
                "description": "自检修改"})
with app.app_context():
    ok(Indicator.query.filter_by(code="coverage").first().de
    scription == "自检修改",
       "指标定义可编辑")
c.post("/ahp/indicators/reset")
with app.app_context():
    ok("自检修改" not in (Indicator.query.filter_by(code="covera
    ge").first().description or ""),
       "可按论文取值重置")
base = {"m_0_1": "1.4194", "m_0_2": "1.76", "m_1_2": "1.24"}
for name in ("张工", "李工", "王工"):
    d = dict(base); d.update({"action": "add", "unit_type":
    "碎石边坡",
                              "level": "1", "expert_name":
    name})
    c.post("/ahp/experts", data=d)
with app.app_context():
    ok(ExpertMatrix.query.filter_by(unit_type="碎石边坡",
    level=1).count() == 3,
       "录入 3 位专家的判断矩阵")
r = c.get("/ahp/experts/aggregate?unit_type=碎石边坡&level=1")
body = r.get_data(as_text=True)
ok("汇总后权重" in body and "各专家矩阵的一致性" in body,
   "多专家矩阵几何平均汇总可用")
ws = re.search(r"<th scope=\"row\">权重</th>\s*(.*?)</tr>",
    body, re.S)
vals = re.findall(r">([0-9.]+)</td>", ws.group(1)) if ws
    else []
ok(len(vals) == 3 and abs(sum(map(float, vals)) - 1.0) <
    0.01,
   "汇总权重归一", "、".join(vals))
ok(abs(float(vals[0]) - 0.44) < 0.02, "汇总权重接近论文表 3 取值",
    vals[0] if vals else "")
# ------------------------------------------------------------
    ----------------
print()
print(LINE)
print("批量导入")
print(LINE)
r = c.get("/indicator/import/template")
tpl = r.data.decode("utf-8-sig")
ok(r.status_code == 200 and "单元类型" in tpl, "导入模板可下载")
ok("土石比" in tpl and "植被覆盖度" in tpl, "模板包含指标列")
csv_text = (
    "单元类型,土石比,土层厚度,pH,有机质,Hg,As,Ni,Cu,Cd,Pb,植被覆盖度,Shannon,坡度
    ,坡向\n"
    "碎石边坡,0.34,10,9.24,4.89,0.05,5.17,12.60,11.70,0.04,6.50,
    87.92,0.60,28.75,阳坡\n"
```

```
        "平台,0.56,30,8.72,10.92,0.02,3.13,12.33,11.14,0.09,9.98,5
        8.42,0.72,,,\n"
        "不存在单元,1,1,7,1,,,,,,,,,,\n"
)
r = c.post("/indicator/import",
           data={"patch_id": str(patch_id),
                 "datafile": (io.BytesIO(csv_text.encode("ut
        f-8")), "d.csv")},
           content_type="multipart/form-data",
        follow_redirects=True)
body = r.get_data(as_text=True)
ok("导入结果" in body, "导入结果页展示")
ok("不存在单元" in body, "错误行被报告")
with app.app_context():
        u = MicroUnit.query.filter_by(patch_id=patch_id,
        unit_type="碎石边坡").first()
        ind = IndicatorData.query.filter_by(unit_id=u.id).first(
        )
        ok(abs(ind.soil_ratio - 0.34) < 1e-6 and
        abs(ind.coverage - 87.92) < 1e-6,
            "导入数据已写入数据库")
r = c.post("/indicator/import",
           data={"patch_id": str(patch_id),
                 "datafile": (io.BytesIO(b"x"), "d.pdf")},
           content_type="multipart/form-data",
        follow_redirects=True)
ok("仅支持" in r.get_data(as_text=True), "拒绝不支持的导入格式")
# ------------------------------------------------------------------
        ------------------
print()
print(LINE)
print("图表与 PDF 导出")
print(LINE)
svg = charts.hbars([("岩质边坡", 0.54, "0.5401"), ("平台", 0.84,
        "0.8376")])
ok("<svg" in svg and "0.8376" in svg, "条形图生成且带数值标签")
svg2 = charts.donut([("植被指标", 0.6, "0.6000"), ("土壤指标", 0.4,
        "0.4000")])
ok("<svg" in svg2 and "占比" in svg2, "环形图生成且带图例")
#  报告是评价结论的呈现，需先真实执行一次评价才能打开
with app.app_context():
        from app.views.evaluate import load_patch_input,
        save_evaluation
        save_evaluation(patch_id, load_patch_input(patch_id))
r = c.get("/report/%d" % patch_id)
body = r.get_data(as_text=True)
ok(body.count("<svg") >= 2, "得分明细页含图表", "%d 个 SVG" %
        body.count("<svg"))
exe = browser.browser_path()
if exe is None:
```

```
        print("  [跳过] 未找到浏览器，PDF 导出无法验证")
else:
        r = c.get("/report/%d/pdf" % patch_id)
        data = r.data
        ok(data[:4] == b"%PDF", "PDF 报告导出成功", "%d 字节" %
        len(data))
        d = fitz.open(stream=data, filetype="pdf")
        txt = "".join(d[i].get_text() for i in
        range(d.page_count))
        ok("自然恢复适宜性评价报告" in txt, "PDF 含报告标题")
        ok("评价结论" in txt and "得分明细" in txt, "PDF 含结论与明细章节")
        ok(d.page_count >= 2, "PDF 页数合理", "%d 页" % d.page_count)
        d.close()
print()
print(LINE)
print("补齐模块自检全部通过 √")
print(LINE)
#  ==== 测试/test_workflow.py ====
#  -*- coding: utf-8 -*-
"""流程门禁自检：前置步骤未完成时，后续模块必须拦得住。
论文 4.1 节确定的评价流程存在先后关系，缺任何一环都不应产出评价结论。
本测试逐个构造「缺前置」的场景，验证系统不会放行。
"""
import os
import sys
sys.stdout.reconfigure(encoding="utf-8")
#  本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
#  使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file
    __)))
if ROOT not in sys.path:
        sys.path.insert(0, ROOT)
from app import create_app
from app.models import (db, Project, MinePatch, MicroUnit,
    IndicatorData,
                        ConstraintCheck, EvaluationResult)
from app.services import workflow as wf
LINE = "=" * 72
BASE = ROOT
class TestConfig:
        SECRET_KEY = "test"
        SQLALCHEMY_DATABASE_URI = "sqlite:///" +
        os.path.join(BASE, "app", "app.db")
        SQLALCHEMY_TRACK_MODIFICATIONS = False
        WTF_CSRF_ENABLED = False
        TESTING = True
app = create_app(TestConfig)
c = app.test_client()
c.post("/login", data={"username": "admin", "password":
    "admin123"})
```

```
_fails = []
def ok(flag, label, extra=""):
    print("  %s %-46s %s" % ("[符合]" if flag else "[不符合]",
    label, extra))
    if not flag:
        _fails.append(label)
def new_patch(tag):
    with app.app_context():
        p = Project(name="门禁-" + tag, code="WF-" + tag)
        db.session.add(p); db.session.flush()
        patch = MinePatch(project_id=p.id, name="门禁图斑-" +
    tag)
        db.session.add(patch); db.session.flush()
        pid = patch.id
        db.session.commit()
    return pid
def add_units(pid):
    with app.app_context():
        db.session.add(MicroUnit(patch_id=pid,
    unit_type="碎石边坡",
                                 area=1.0, ratio=1.0))
        db.session.commit()
def add_constraint(pid, passed=True):
    with app.app_context():
        db.session.add(ConstraintCheck(
            patch_id=pid, policy_pass=True,
    location_pass=passed,
            pollution_pass=True, geo_pass=True,
    farmland_pass=True,
            passed=passed, reason="全部通过" if passed else
    "未通过：区位约束"))
        db.session.commit()
def add_data(pid):
    with app.app_context():
        u = MicroUnit.query.filter_by(patch_id=pid).first()
        db.session.add(IndicatorData(unit_id=u.id,
    soil_ratio=0.34, thickness=10,
                                     ph=9.24,
    organic_matter=4.89, coverage=87.92,
                                     shannon=0.60,
    slope=28.75, aspect="阳坡"))
        db.session.commit()
def cleanup(pid, tag):
    with app.app_context():
        from app.models import ScoreDetail
        # 先删得分明细，再删评价记录，避免留下孤儿明细
        rids = [r.id for r in EvaluationResult.query.filter_
    by(patch_id=pid)]
        if rids:
            ScoreDetail.query.filter(ScoreDetail.result_id.i
```

```
        n_(rids)).delete(                                                  11851
                        synchronize_session=False)                         11852
            EvaluationResult.query.filter_by(patch_id=pid).delet           11853
        e()                                                                11854
            ConstraintCheck.query.filter_by(patch_id=pid).delete           11855
        ()                                                                 11856
            uids = [u.id for u in MicroUnit.query.filter_by(pat            11857
        h_id=pid)]                                                         11858
            if uids:                                                       11859
                IndicatorData.query.filter(IndicatorData.unit_id           11860
        .in_(uids)).delete(                                                11861
                        synchronize_session=False)                         11862
            MicroUnit.query.filter_by(patch_id=pid).delete()               11863
            m = db.session.get(MinePatch, pid)                             11864
            if m:                                                          11865
                db.session.delete(m)                                       11866
            pr = Project.query.filter_by(name="门禁-" +                    11867
        tag).first()                                                       11868
            if pr:                                                         11869
                db.session.delete(pr)                                      11870
            db.session.commit()                                            11871
    def results_of(pid):                                                   11872
        with app.app_context():                                            11873
            return EvaluationResult.query.filter_by(patch_id=pid           11874
        ).count()                                                          11875
    def post_eval(pid):                                                    11876
        return c.post("/evaluate/%d" % pid, data={},                       11877
        follow_redirects=False)                                            11878
    def get_report(pid):                                                   11879
        return c.get("/report/%d" % pid, follow_redirects=False)           11880
    # ------------------------------------------------------------------   11881
        ------------------                                                 11882
    print(LINE)                                                            11883
    print("一、前置条件缺失时，评价必须被拦住")                                11884
    print(LINE)                                                            11885
    print()                                                                11886
    print("场景 1：连微地质单元都没划分")                                      11887
    pid = new_patch("1")                                                   11888
    r = post_eval(pid)                                                     11889
    n = results_of(pid)                                                    11890
    ok(r.status_code == 302 and n == 0, "未划分单元 → 不产生评价结果",          11891
       "HTTP %d，结果数 %d" % (r.status_code, n))                           11892
    with app.app_context():                                                11893
        pr = wf.patch_progress(pid)                                        11894
    steps = [b["step"] for b in pr["blocking"]]                            11895
    ok(not pr["ready"] and "units" in steps,                               11896
       "拦原因包含「划分微地质单元」", "、".join(steps))                      11897
    ok(steps == ["constraint", "units"],                                   11898
       "拦截顺序符合论文 4.1 节：限制性判定在前，划分单元在后",               11899
       "、".join(steps))                                                   11900
```

```
cleanup(pid, "1")
print()
print("场景 2：有单元，但未做限制性条件判定")
pid = new_patch("2"); add_units(pid); add_data(pid)
r = post_eval(pid)
n = results_of(pid)
ok(r.status_code == 302 and n == 0, "未做限制性判定 → 不产生评价结果",
   "结果数 %d" % n)
with app.app_context():
    pr = wf.patch_progress(pid)
ok(any(b["step"] == "constraint" for b in pr["blocking"]),
   "拦原因指向「执行限制性条件判定」")
cleanup(pid, "2")
print()
print("场景 3：限制性条件判定【未通过】")
pid = new_patch("3"); add_units(pid); add_data(pid);
    add_constraint(pid, passed=False)
r = post_eval(pid)
n = results_of(pid)
ok(r.status_code == 302 and n == 0, "判定未通过 → 不进入定量评价",
   "结果数 %d" % n)
with app.app_context():
    pr = wf.patch_progress(pid)
ok(not pr["constraint"]["passed"], "流程状态显示判定未通过")
blocked = [b for b in pr["blocking"] if b["step"] ==
    "constraint"]
# 提示只写「缺什么」：label 说明未通过，reason 直接引用判定记录里没过的项
ok(bool(blocked) and "未通过" in blocked[0]["label"],
   "拦提示指明限制性条件判定未通过", blocked[0]["label"] if blocked else
    "")
ok(bool(blocked) and "区位约束" in (blocked[0]["reason"] or ""),
   "并把具体没通过的条件带出来", blocked[0]["reason"] if blocked else "")
cleanup(pid, "3")
print()
print("场景 4：判定通过，但指标数据全空")
pid = new_patch("4"); add_units(pid); add_constraint(pid,
    passed=True)
r = post_eval(pid)
n = results_of(pid)
ok(r.status_code == 302 and n == 0, "数据未录入 → 不产生评价结果",
   "结果数 %d" % n)
with app.app_context():
    pr = wf.patch_progress(pid)
ok(any(b["step"] == "data" for b in pr["blocking"]),
   "拦原因指向「录入指标数据」")
ok(pr["data"]["missing"] == ["碎石边坡"],
   "指出具体缺哪个单元的数据", "、".join(pr["data"]["missing"]))
cleanup(pid, "4")
print()
print("场景 5：单元面积占比合计不为 100%")
```

```
pid = new_patch("5"); add_constraint(pid, passed=True)
with app.app_context():
    u = MicroUnit(patch_id=pid, unit_type="碎石边坡", area=1.0,
    ratio=0.5)
    db.session.add(u); db.session.commit()
    db.session.add(IndicatorData(unit_id=u.id,
    soil_ratio=0.3, thickness=10,
                                 ph=8.0,
    organic_matter=5.0, coverage=80.0,
                                 shannon=0.6, slope=20.0,
    aspect="阳坡"))
    db.session.commit()
r = post_eval(pid)
ok(results_of(pid) == 0, "占比不足 100% → 不产生评价结果")
cleanup(pid, "5")
print()
print(LINE)
print("二、前置条件齐备时才放行")
print(LINE)
pid = new_patch("OK"); add_units(pid); add_constraint(pid,
    passed=True); add_data(pid)
with app.app_context():
    pr = wf.patch_progress(pid)
ok(pr["ready"], "单元 + 判定通过 + 数据齐备 → 判定为可执行")
r = post_eval(pid)
n = results_of(pid)
ok(n == 1, "此时执行评价成功产生结果", "结果数 %d" % n)
print("   重定向到:", r.headers.get("Location"))
print()
print(LINE)
print("三、报告不得先于评价产生")
print(LINE)
pid2 = new_patch("R")    # 什么都不做
r = get_report(pid2)
ok(r.status_code == 302 and "/report/" ==
    (r.headers.get("Location") or ""),
   "前置缺失时 /report/<id> 被退回列表", "HTTP %d" % r.status_code)
for suffix in ("/pdf", "/word"):
    rr = c.get("/report/%d%s" % (pid2, suffix),
    follow_redirects=False)
    ok(rr.status_code == 302, "导出 %s 同样被拦住" % suffix, "HTTP
    %d" % rr.status_code)
cleanup(pid2, "R")
cleanup(pid, "OK")
print()
print(LINE)
print("四、前置齐备但从未执行评价时，报告仍不得产生")
print(LINE)
#  报告是评价结论的呈现：即使前置步骤都做完了，只要没真正执行过定量评价，
#  也不应凭空推算出结论。
```

```
pid3 = new_patch("N")
add_units(pid3); add_constraint(pid3, passed=True);
    add_data(pid3)
with app.app_context():
    pr = wf.report_progress(pid3)
ok(not pr["ready"], "无评价记录 → 报告流程判定为未就绪")
ok(bool(pr["blocking"]) and pr["blocking"][0]["step"] ==
    "evaluate",
   "拦原因指向「执行适宜性评价」")
r = get_report(pid3)
ok(r.status_code == 302 and results_of(pid3) == 0,
   "直接访问 /report/<id> 被退回，且不产生任何评价结果",
   "HTTP %d" % r.status_code)
#  真正执行评价之后，报告才放行，且取值与评价记录一致
r = post_eval(pid3)
ok(results_of(pid3) == 1, "先执行评价 → 产生评价记录")
with app.app_context():
    pr = wf.report_progress(pid3)
    rec = pr["result"]
ok(pr["ready"] and rec is not None, "有评价记录后，报告流程就绪")
rr = get_report(pid3)
body = rr.get_data(as_text=True)
ok(rr.status_code == 200, "此时可打开得分明细页", "HTTP %d" %
    rr.status_code)
ok(("%.4f" % rec.final_score) in body,
   "报告数值取自评价记录，而非重新计算",
   "F=%.4f" % rec.final_score)
cleanup(pid3, "N")
print()
print(LINE)
print("五、内置演示数据的清理不得波及用户自建数据")
print(LINE)
#  回归保护：seed_demo.clean_case() 曾执行 Project.query.delete(),
#  导致用户新建的项目在每次启动时被连带删除。
import seed_demo as sd
with app.app_context():
    keep_p = Project(name="用户自建项目-守护", code="KEEP-001")
    db.session.add(keep_p); db.session.flush()
    keep_patch = MinePatch(project_id=keep_p.id,
    name="用户自建图斑-守护")
    db.session.add(keep_patch); db.session.flush()
    keep_pid = keep_p.id
    keep_patch_id = keep_patch.id
    db.session.commit()
with app.app_context():
    sd.clean_case(reset=False)
with app.app_context():
    still_p = db.session.get(Project, keep_pid)
    still_patch = db.session.get(MinePatch, keep_patch_id)
ok(still_p is not None, "清理演示数据后，用户自建项目仍存在")
```

```
ok(still_patch is not None, "清理演示数据后，用户自建图斑仍存在")
#  演示数据清掉后应能重新写入，保证演示环境可恢复
sd.seed()
with app.app_context():
    demo_back = Project.query.filter_by(code=sd.DEMO_CODE).first()
    irst()
ok(demo_back is not None, "演示案例可重新写入（幂等恢复）")
with app.app_context():
    m = db.session.get(MinePatch, keep_patch_id)
    if m:
        db.session.delete(m)
    p = db.session.get(Project, keep_pid)
    if p:
        db.session.delete(p)
    db.session.commit()
print()
print(LINE)
print("六、同一图斑不得重复划分同类型微地质单元")
print(LINE)
#  论文 4.1 节把图斑分为平台、碎石边坡、岩质边坡三类，每类至多一个。
#  允许重复会让面积占比、按类型组织的指标数据与批量导入对不上号。
pid4 = new_patch("D")
c.post("/unit/%d/add" % pid4, data={"unit_type": "平台",
    "area": "1.0"},
       follow_redirects=True)
with app.app_context():
    n1 = MicroUnit.query.filter_by(patch_id=pid4).count()
ok(n1 == 1, "首次添加「平台」成功", "单元数 %d" % n1)
r = c.post("/unit/%d/add" % pid4, data={"unit_type": "平台",
    "area": "2.0"},
           follow_redirects=True)
with app.app_context():
    n2 = MicroUnit.query.filter_by(patch_id=pid4).count()
ok(n2 == 1, "重复添加同类型被拒绝", "单元数 %d" % n2)
ok("同一类型只能划分一个" in r.get_data(as_text=True), "给出重复提示")
r = c.post("/unit/%d/add" % pid4, data={"unit_type": "火山口",
    "area": "1.0"},
           follow_redirects=True)
with app.app_context():
    n3 = MicroUnit.query.filter_by(patch_id=pid4).count()
ok(n3 == 1, "非法单元类型被拒绝", "单元数 %d" % n3)
c.post("/unit/%d/add" % pid4, data={"unit_type": "碎石边坡",
    "area": "3.0"},
       follow_redirects=True)
with app.app_context():
    uid = MicroUnit.query.filter_by(patch_id=pid4,
                                    unit_type="碎石边坡").first()
    ).id
r = c.post("/unit/edit/%d" % uid,
           data={"unit_type": "平台", "area": "3.0"},
```

```
    follow_redirects=True)
with app.app_context():
    types = sorted(u.unit_type for u in
                   MicroUnit.query.filter_by(patch_id=pid4)
    all())
ok(types == ["平台", "碎石边坡"], "改类型时同样拦截重复", "、".join(types))
cleanup(pid4, "D")
print()
print(LINE)
if _fails:
    print("存在未通过项：")
    for f in _fails:
        print("  -", f)
    sys.exit(1)
print("流程门禁自检全部通过 √")
print(LINE)
#  ==== 测试/test_weights.py ====
#  -*- coding: utf-8 -*-
"""权重生效链路自检：AHP 算出的权重必须真正参与评价计算。
此前 weight_schemes / weight_items 两张表从未被写入，评价引擎也只用
代码里的论文表 3 默认权重，相当于权重模块算完即弃。本测试守住这条链路：
    默认 → 论文表 3
    保存 AHP 权重 → 评价改用新权重，结果随之变化
    恢复默认 → 结果回到论文取值
"""
import os
import sys
sys.stdout.reconfigure(encoding="utf-8")
#  本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
#  使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app import create_app
from app.models import (db, Project, MinePatch, MicroUnit,
    IndicatorData,
                        ConstraintCheck, EvaluationResult,
    WeightScheme,
                        WeightItem)
from app.services import workflow as wf
from app.services import weight_service as ws
LINE = "=" * 72
BASE = ROOT
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///" +
    os.path.join(BASE, "app", "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
```

```
    TESTING = True
app = create_app(TestConfig)
c = app.test_client()
c.post("/login", data={"username": "admin", "password":
    "admin123"})
_fails = []
def ok(flag, label, extra=""):
    print("  %s %-46s %s" % ("[符合]" if flag else "[不符合]",
    label, extra))
    if not flag:
        _fails.append(label)
TAG = "W"
UNIT_TYPE = "碎石边坡"
def build_case():
    """建一个只含单一微地质单元的图斑，便于观察权重变化对得分的影响。"""
    with app.app_context():
        p = Project(name="权重测试-%s" % TAG, code="WEIGHT-%s"
    % TAG)
        db.session.add(p); db.session.flush()
        patch = MinePatch(project_id=p.id, name="权重测试图斑-%s"
    % TAG)
        db.session.add(patch); db.session.flush()
        db.session.add(MicroUnit(patch_id=patch.id,
    unit_type=UNIT_TYPE,
                                 area=1.0, ratio=1.0))
        db.session.flush()
        u = MicroUnit.query.filter_by(patch_id=patch.id).fi
    st()
        db.session.add(IndicatorData(
            unit_id=u.id, soil_ratio=0.34, thickness=10,
    ph=9.24,
            organic_matter=4.89, coverage=87.92,
    shannon=0.60,
            slope=28.75, aspect="阳坡"))
        db.session.add(ConstraintCheck(
            patch_id=patch.id, policy_pass=True,
    location_pass=True,
            pollution_pass=True, geo_pass=True,
    farmland_pass=True,
            passed=True, reason="全部通过"))
        db.session.commit()
        return patch.id
def cleanup(pid):
    with app.app_context():
        from app.models import ScoreDetail
        rids = [r.id for r in EvaluationResult.query.filter_
    by(patch_id=pid)]
        if rids:
            ScoreDetail.query.filter(ScoreDetail.result_id.i
    n_(rids)).delete(
```

```
                    synchronize_session=False)                    12201
        EvaluationResult.query.filter_by(patch_id=pid).delet12202
    e()                                                           12203
        ConstraintCheck.query.filter_by(patch_id=pid).delete12204
    ()                                                            12205
        uids = [u.id for u in MicroUnit.query.filter_by(patd12206
    h_id=pid)]                                                    12207
        if uids:                                                  12208
            IndicatorData.query.filter(IndicatorData.unit_id12209
    .in_(uids)).delete(                                           12210
                    synchronize_session=False)                    12211
        MicroUnit.query.filter_by(patch_id=pid).delete()          12212
        m = db.session.get(MinePatch, pid)                        12213
        if m:                                                     12214
            db.session.delete(m)                                  12215
        p = Project.query.filter_by(code="WEIGHT-%s" %            12216
    TAG).first()                                                  12217
        if p:                                                     12218
            db.session.delete(p)                                  12219
        db.session.commit()                                       12220
def run_eval(pid):                                                12221
    c.post("/evaluate/%d" % pid, data={},                         12222
    follow_redirects=False)                                       12223
    with app.app_context():                                       12224
        return EvaluationResult.query.filter_by(patch_id=pid12225
    ) \                                                           12226
            .order_by(EvaluationResult.id.desc()).first().fi12227
    nal_score                                                     12228
print(LINE)                                                       12229
print("一、默认应使用论文表 3 的权重")                                  12230
print(LINE)                                                       12231
with app.app_context():                                           12232
    ws.reset_weights(UNIT_TYPE)                                   12233
    w, label, scheme = ws.effective_weights(UNIT_TYPE)            12234
ok(scheme is None and "论文表" in label, "未配置时权重来源为论文表 3",      12235
    label)                                                        12236
ok(abs(w["l1"]["土壤指标"] - 0.44) < 1e-9, "一级权重取自论文表 3",          12237
   "土壤指标 %.2f" % w["l1"]["土壤指标"])                               12238
pid = build_case()                                                12239
base_score = run_eval(pid)                                        12240
print("  默认权重下的评价值 F = %.4f" % base_score)                      12241
print()                                                           12242
print(LINE)                                                       12243
print("二、保存 AHP 算出的权重后，评价应改用新权重")                          12244
print(LINE)                                                       12245
with app.app_context():                                           12246
    # 模拟 AHP 计算结果：把土壤指标权重提高、植被与地形降低（仍和为 1）          12247
    ws.save_weights(UNIT_TYPE, [("soil", 0.70), ("veg",           12248
    0.20),                                                        12249
                                ("terrain", 0.10)])               12250
```

```
    n_items = WeightItem.query.count()
ok(n_items >= 3, "weight_items 表已写入权重条目", "%d 条" % n_items)
with app.app_context():
    w2, label2, scheme2 = ws.effective_weights(UNIT_TYPE)
ok(scheme2 is not None and "自定义" in label2, "权重来源变为自定义方案",
    label2)
ok(abs(w2["l1"]["土壤指标"] - 0.70) < 1e-9, "新权重已生效",
   "土壤指标 %.2f" % w2["l1"]["土壤指标"])
ok(abs(w2["l1"]["植被指标"] - 0.20) < 1e-9, "同批保存的其他项也已生效")
ok(abs(w2["l2"]["土壤指标"]["土石比"] - 0.25) < 1e-9,
   "未覆盖的二级权重仍保留论文取值",
   "土石比 %.2f" % w2["l2"]["土壤指标"]["土石比"])
new_score = run_eval(pid)
print("  新权重下的评价值 F = %.4f" % new_score)
ok(abs(new_score - base_score) > 1e-4,
   "权重变化确实改变了评价结果",
   "%.4f → %.4f" % (base_score, new_score))
print()
print(LINE)
print("三、结果应记录所用权重来源")
print(LINE)
import json
with app.app_context():
    rec = EvaluationResult.query.filter_by(patch_id=pid) \
        .order_by(EvaluationResult.id.desc()).first()
    stored = json.loads(rec.unit_scores)
ok(any(u.get("weight_source") for u in stored),
   "评价记录中标注了权重来源",
   stored[0].get("weight_source") or "")
print()
print(LINE)
print("四、恢复默认后应回到论文取值")
print(LINE)
with app.app_context():
    ws.reset_weights(UNIT_TYPE)
    w3, label3, scheme3 = ws.effective_weights(UNIT_TYPE)
ok(scheme3 is None and "论文表" in label3, "已恢复为论文表 3 权重",
    label3)
back_score = run_eval(pid)
print("  恢复后评价值 F = %.4f（默认时 %.4f）" % (back_score,
    base_score))
ok(abs(back_score - base_score) < 1e-9, "评价结果与默认时一致")
print()
print(LINE)
print("五、权重首页概览")
print(LINE)
with app.app_context():
    rows = ws.summary()
ok(len(rows) == 3, "三种微地质单元均有权重来源说明", "%d 行" % len(rows))
r = c.get("/ahp/")
```

```
body = r.get_data(as_text=True)
ok(r.status_code == 200 and "当前生效权重" in body, "权重首页正常展示")
print()
print(LINE)
print("六、一致性不通过的矩阵不得保存生效")
print(LINE)
with app.app_context():
    ws.reset_weights(UNIT_TYPE)
n = 3
# 明显矛盾的矩阵：a12=9, a13=1/9, a23=9 —— CR 必然超限
mats = [(0, 1, "9"), (0, 2, "0.1111"), (1, 2, "9")]
data = {"csrf_token": "", "action": "save"}
for i, j, v in mats:
    data["m_%d_%d" % (i, j)] = v
r = c.post("/ahp/matrix?unit_type=%s&level=1" % UNIT_TYPE,
    data=data,
           follow_redirects=True)
with app.app_context():
    _, label4, scheme4 = ws.effective_weights(UNIT_TYPE)
ok(scheme4 is None, "CR 超限的矩阵未被保存生效", label4)
ok("一致性" in r.get_data(as_text=True), "页面给出了未通过一致性的提示")
cleanup(pid)
print()
print(LINE)
if _fails:
    print("存在未通过项：")
    for f in _fails:
        print("  -", f)
    sys.exit(1)
print("权重生效链路自检全部通过 √")
print(LINE)
# ==== app/services/__init__.py ====
# -*- coding: utf-8 -*-
# ==== app/utils/__init__.py ====
# -*- coding: utf-8 -*-
# ==== app/views/__init__.py ====
# -*- coding: utf-8 -*-
```